



US012403269B2

(12) **United States Patent**
Hunter et al.

(10) **Patent No.:** **US 12,403,269 B2**
(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **DROPLET DELIVERY DEVICE WITH PUSH EJECTION**

(71) Applicant: **PNEUMA RESPIRATORY, INC.**,
Boone, NC (US)

(72) Inventors: **Charles Eric Hunter**, Boone, NC (US);
Michael Scoggin, Boone, NC (US);
Jeffrey Miller, Boone, NC (US); **Jose Salazar**, Boone, NC (US); **Brian Beach**, Boone, NC (US); **Caley Modlin**, Boone, NC (US); **Matthew Culpepper**, Boone, NC (US);
Jianqiang Li, Shenzhen (CN); **Shi Bo Chengjie Li**, Shenzhen (CN); **Shi Bo Wang**, Shenzhen (CN); **Chao-Ping Lee**, Kaohsiung (TW); **Gregory Rapp**, Boone, NC (US); **Judson Sidney Clements**, Boone, NC (US)

(73) Assignee: **PNEUMA RESPIRATORY, INC.**,
Boone, NC (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/382,637**

(22) Filed: **Oct. 23, 2023**

(65) **Prior Publication Data**

US 2024/0269397 A1 Aug. 15, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/846,902, filed on Jun. 22, 2022, now Pat. No. 11,793,945.
(Continued)

(51) **Int. Cl.**

A61M 11/00 (2006.01)

A61M 15/00 (2006.01)

(52) **U.S. Cl.**

CPC **A61M 11/003** (2014.02); **A61M 11/005** (2013.01); **A61M 15/0001** (2014.02);
(Continued)

(58) **Field of Classification Search**

CPC **A61M 11/003**; **A61M 11/005**; **A61M 15/0001**; **A61M 15/001**; **A61M 15/0033**;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,934,585 A 1/1976 Maurice
3,970,250 A 7/1976 Drows
(Continued)

FOREIGN PATENT DOCUMENTS

AU 2012258488 1/2013
CA 2364248 8/2006
(Continued)

OTHER PUBLICATIONS

Copley, "Understanding cascade impaction and its importance for inhaler testing," Copley Scientific, Copley White Paper [serial online], Jul. 2007 [retrieved on May 7, 2017]. Retrieved from the Internet: URL: <http://www.copleyscientific.com/files/ww/articles/Understanding%20Cascade%20Impaction%20White%20Paper.pdf>; 6 pp.

(Continued)

Primary Examiner — Joseph D. Boecker

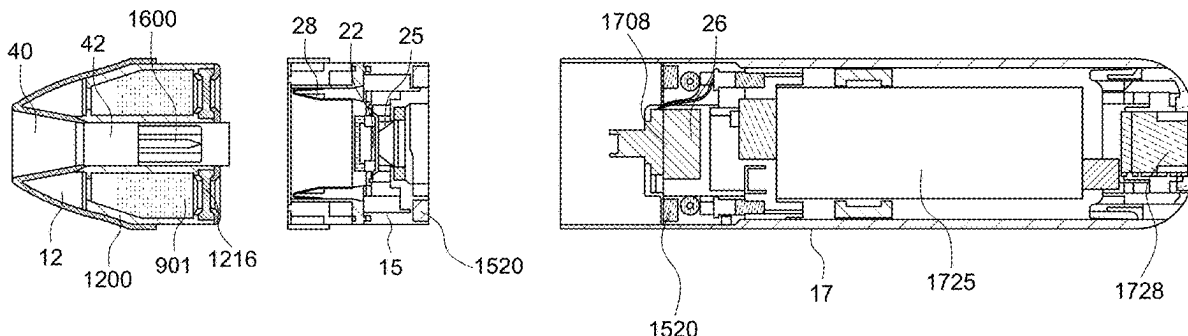
Assistant Examiner — Brian T Khong

(74) *Attorney, Agent, or Firm* — Polsinelli PC

(57) **ABSTRACT**

A droplet delivery device includes a housing with a mouth-piece port or outlet from a nasal device for releasing fluid droplets, a fluid reservoir, and an ejector bracket having a membrane positioned between a mesh with a plurality of openings and a vibrating member that is coupled to an electronic transducer, such as an ultrasonic transducer. The

(Continued)



transducer vibrates the vibrating member which causes the membrane to push fluid supplied by the reservoir through the mesh to generate droplets in an ejected stream released through the outlet.

18 Claims, 125 Drawing Sheets

Related U.S. Application Data

- (60) Provisional application No. 63/280,643, filed on Nov. 18, 2021, provisional application No. 63/256,546, filed on Oct. 16, 2021, provisional application No. 63/256,245, filed on Oct. 15, 2021, provisional application No. 63/213,634, filed on Jun. 22, 2021.

(52) U.S. Cl.

CPC *A61M 15/002* (2014.02); *A61M 15/0085* (2013.01); *A61M 2206/11* (2013.01)

(58) Field of Classification Search

CPC *A61M 15/0035*; *A61M 15/0036*; *A61M 15/004*; *A61M 15/0041*; *A61M 15/002*; *A61M 15/0085*; *A24F 40/42*

See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

5,021,701 A 6/1991 Takahashi et al.
5,164,740 A 11/1992 Ivri
5,239,993 A 8/1993 Evans
5,363,842 A 11/1994 Mishelevich et al.
5,487,378 A 1/1996 Robertson et al.
5,497,763 A 3/1996 Lloyd et al.
5,586,550 A 12/1996 Ivri et al.
5,607,410 A 3/1997 Branch
5,630,793 A 5/1997 Rowe
5,724,959 A 3/1998 McAughy et al.
5,758,637 A 6/1998 Ivri et al.
5,826,570 A 10/1998 Goodman et al.
5,828,394 A 10/1998 Khuri-Yakub et al.
5,881,716 A 3/1999 Wirsch et al.
5,884,620 A 3/1999 Gonda et al.
5,894,841 A 4/1999 Voges
5,906,202 A 5/1999 Schuster et al.
5,938,117 A 8/1999 Ivri
5,979,247 A 11/1999 Kizawa
6,011,062 A 1/2000 Schneider et al.
6,026,809 A 2/2000 Abrams et al.
6,062,212 A 5/2000 Davison et al.
6,071,498 A 6/2000 Narodnylo et al.
6,085,740 A 7/2000 Ivri et al.
6,158,431 A 12/2000 Poole
6,196,219 B1 3/2001 Hess et al.
6,235,177 B1 5/2001 Borland et al.
6,358,058 B1 3/2002 Strupat et al.
6,443,146 B1 9/2002 Voges
6,511,718 B1 1/2003 Paz de Araujo et al.
6,523,762 B1 2/2003 Luginbuhl et al.
6,615,826 B1 9/2003 Gabrio et al.
6,629,524 B1 10/2003 Goodall et al.
6,637,430 B1 10/2003 Voges et al.
6,896,910 B2 5/2005 Kim et al.
6,978,941 B2 12/2005 Litherland et al.
6,981,499 B2 1/2006 Anderson et al.
7,131,599 B2 11/2006 Katase
7,191,777 B2 3/2007 Brand et al.
7,198,044 B2 4/2007 Trueba
7,219,664 B2 5/2007 Ruckdeschel et al.
7,628,339 B2 12/2009 Ivri et al.
7,648,957 B2 1/2010 Heyden et al.
7,708,011 B2 5/2010 Hochrainer et al.

7,883,031 B2 2/2011 Collins, Jr. et al.
7,900,625 B2 3/2011 Kleinstreuer et al.
7,954,486 B2 6/2011 Papania et al.
7,976,140 B2 7/2011 Umeda
8,012,136 B2 9/2011 Collins, Jr. et al.
8,367,734 B1 2/2013 Gao et al.
8,474,452 B2 7/2013 Gumaste et al.
8,545,463 B2 10/2013 Collins, Jr. et al.
8,555,874 B2 10/2013 Fink et al.
8,616,195 B2 12/2013 Power et al.
8,684,980 B2 4/2014 Hunter et al.
8,733,935 B2 5/2014 Ballou, Jr. et al.
8,753,308 B2 6/2014 Palmer et al.
8,936,021 B2 1/2015 Collins, Jr.
8,985,100 B2 3/2015 Minocchieri et al.
9,022,027 B2 5/2015 Addington et al.
9,087,145 B2 7/2015 Ballou, Jr. et al.
9,227,029 B2 1/2016 Addington et al.
9,242,054 B2 1/2016 Fink et al.
9,352,108 B1 5/2016 Reed et al.
9,452,274 B2 9/2016 Addington et al.
9,463,486 B2 10/2016 Wilkerson et al.
9,539,604 B2 1/2017 Wilkerson et al.
9,956,360 B2 5/2018 Germinario et al.
9,962,507 B2 5/2018 Germinario et al.
10,449,314 B2 10/2019 Germinario et al.
10,525,220 B2 1/2020 Hunter et al.
10,568,543 B2 2/2020 Yan
10,857,310 B2 12/2020 Muellinger et al.
10,898,666 B2 1/2021 Germinario et al.
2002/0002975 A1 1/2002 Power
2002/0032387 A1 3/2002 Geva et al.
2002/0046750 A1 4/2002 Gonda et al.
2002/0071871 A1 6/2002 Snyder et al.
2002/0077369 A1 6/2002 Noolandi et al.
2002/0121274 A1 9/2002 Borland et al.
2003/0062042 A1 4/2003 Wensley et al.
2003/0072717 A1 4/2003 Reinhold et al.
2003/0098022 A1 5/2003 Nakao et al.
2003/0101991 A1 6/2003 Trueba
2003/0127538 A1 7/2003 Patel et al.
2003/0140921 A1 7/2003 Smith et al.
2003/0150445 A1 8/2003 Power et al.
2003/0196654 A1 10/2003 Stein
2003/0205229 A1 11/2003 Crockford et al.
2004/0009231 A1 1/2004 Jackson et al.
2004/0045547 A1 3/2004 Yamamoto et al.
2004/0084044 A1 5/2004 Childers et al.
2004/0139963 A1 7/2004 Ivri et al.
2004/0195403 A1 10/2004 Atterybury et al.
2004/0215157 A1 10/2004 Peclat et al.
2004/0256487 A1 12/2004 Collins, Jr. et al.
2005/0011514 A1 1/2005 Power et al.
2005/0077315 A1 4/2005 Pavlu et al.
2005/0121025 A1 6/2005 Gamard et al.
2005/0150489 A1 7/2005 Dunfield et al.
2005/0172476 A1 8/2005 Stone et al.
2005/0172958 A1 8/2005 Singer et al.
2005/0217666 A1 10/2005 Fink et al.
2005/0224075 A1 10/2005 Childers et al.
2005/0236501 A1 10/2005 Zimlich, Jr. et al.
2007/0023036 A1 2/2007 Grychowski et al.
2007/0044793 A1 3/2007 Kleinstreuer et al.
2007/0083677 A1 4/2007 Cecka et al.
2007/0119968 A1 5/2007 Collins, Jr. et al.
2007/0119969 A1 5/2007 Collins et al.
2007/0125370 A1 6/2007 Denyer et al.
2007/0157931 A1 7/2007 Parker et al.
2007/0240714 A1 10/2007 Dunne et al.
2007/0248645 A1 10/2007 Bague et al.
2007/0267010 A1 11/2007 Fink et al.
2008/0000470 A1 1/2008 Minocchieri et al.
2008/0059228 A1 3/2008 Bossi et al.
2008/0142010 A1 6/2008 Weaver et al.
2008/0243050 A1 10/2008 Power et al.
2008/0271732 A1 11/2008 Weaver et al.
2008/0283049 A1 11/2008 Mahoney et al.
2008/0283057 A1 11/2008 Rohrschneider et al.
2008/0295827 A1 12/2008 Kobayashi

(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0308096	A1	12/2008	Borgschulte et al.
2009/0038610	A1	2/2009	Bogh et al.
2009/0093772	A1	4/2009	Genosar et al.
2009/0101144	A1	4/2009	Gamard et al.
2009/0107492	A1	4/2009	Ooida
2009/0114218	A1	5/2009	Veatch
2009/0114742	A1	5/2009	Collins, Jr.
2009/0118243	A1	5/2009	Gjorstrup
2009/0134235	A1	5/2009	Ivri
2009/0167812	A1	7/2009	Asai et al.
2009/0192443	A1	7/2009	Collins, Jr.
2009/0212133	A1	8/2009	Collins, Jr.
2009/0235925	A1	9/2009	Power et al.
2009/0270752	A1	10/2009	Coifman
2009/0272818	A1	11/2009	Valpey, III et al.
2009/0314292	A1	12/2009	Overfield et al.
2009/0317496	A1	12/2009	Park et al.
2010/0037894	A1	2/2010	Rouse et al.
2010/0078013	A1	4/2010	Power et al.
2010/0089395	A1	4/2010	Power et al.
2010/0156995	A1	6/2010	Kanda et al.
2010/0181387	A1	7/2010	Zaffaroni et al.
2010/0222752	A1	9/2010	Collins, Jr. et al.
2010/0273895	A1	10/2010	Stinchcomb et al.
2010/0282247	A1	11/2010	Kadrichu et al.
2011/0094523	A1	4/2011	Thorens et al.
2011/0108025	A1	5/2011	Fink et al.
2011/0114090	A1	5/2011	Piper
2011/0230820	A1	9/2011	Lillis et al.
2011/0253139	A1	10/2011	Guthrie et al.
2011/0253140	A1	10/2011	Smyth et al.
2011/0253805	A1	10/2011	Lee
2012/0037154	A1	2/2012	Gallem et al.
2012/0048265	A1	3/2012	Smaldone
2012/0092416	A1	4/2012	Platt et al.
2012/0266878	A1	10/2012	Watanabe et al.
2012/0291781	A1	11/2012	Kaufmann et al.
2013/0064777	A1	3/2013	Tamarkin et al.
2013/0079732	A1	3/2013	Burt et al.
2013/0150812	A1	6/2013	Hunter et al.
2013/0172830	A1	7/2013	Hunter et al.
2013/0239956	A1	9/2013	Schulz et al.
2013/0267864	A1	10/2013	Addington et al.
2013/0284165	A1	10/2013	Krimsky
2013/0299607	A1	11/2013	Wilkerson et al.
2013/0319404	A1	12/2013	Feriani et al.
2013/0327323	A1	12/2013	Rubin
2013/0330400	A1	12/2013	Perkins et al.
2013/0334335	A1	12/2013	Wilkerson et al.
2013/0334339	A1	12/2013	Xu
2014/0037735	A1	2/2014	Montgomery
2014/0116426	A1	5/2014	Mullinger et al.
2014/0187969	A1	7/2014	Hunter et al.
2014/0190496	A1	7/2014	Wensley et al.
2014/0213925	A1	7/2014	Chan et al.
2014/0230817	A1	8/2014	Richardson
2014/0231538	A1	8/2014	Tabata et al.
2014/0283859	A1	9/2014	Minskoff et al.
2014/0336618	A1	11/2014	Wilkerson et al.
2015/0018694	A1	1/2015	Gomo
2015/0101596	A1	4/2015	Hogan
2015/0136129	A1	5/2015	Mehadevan et al.
2015/0136155	A1	5/2015	Verleur et al.
2015/0164375	A1	6/2015	Schindhelm et al.
2015/0174348	A1	6/2015	Tunnell et al.
2015/0196060	A1	7/2015	Wensley et al.
2015/0273165	A1	10/2015	Hadash
2015/0283339	A1	10/2015	Mahadevan et al.
2015/0328151	A1	11/2015	Ballou, Jr. et al.
2015/0352301	A1	12/2015	Stedman et al.
2016/0001018	A1	1/2016	Fink et al.
2016/0001019	A1	1/2016	Fink et al.
2016/0082208	A1	3/2016	Ballam et al.
2016/0106341	A1	4/2016	Adam et al.
2016/0129182	A1	5/2016	Schuster et al.
2016/0166768	A1	6/2016	Edwards et al.
2016/0213864	A1	7/2016	Eilat et al.
2016/0213866	A1	7/2016	Tan
2016/0245830	A1	8/2016	Mace et al.
2016/0310982	A1	10/2016	Von Hollen
2016/0325055	A1	11/2016	Cameron
2016/0354557	A1	12/2016	McPherson Allnutt et al.
2017/0007449	A1	1/2017	Nielsen
2017/0035924	A1	2/2017	Yang et al.
2017/0039344	A1	2/2017	Bitran et al.
2017/0106153	A1	4/2017	Davidson et al.
2017/0106155	A1	4/2017	Reed et al.
2017/0128677	A1	5/2017	Eilat et al.
2017/0158776	A1	6/2017	Feltquate et al.
2017/0203058	A1	7/2017	Davidson et al.
2017/0203323	A1	7/2017	Gschwind et al.
2017/0224706	A1	8/2017	Surber
2017/0270260	A1	9/2017	Shetty et al.
2017/0274163	A1	9/2017	Oliveras et al.
2017/0304565	A1	10/2017	Allosery
2017/0304566	A1	10/2017	Allosery
2017/0319796	A1	11/2017	Germinario et al.
2017/0319797	A1	11/2017	Germinario et al.
2017/0333646	A1	11/2017	Hemy et al.
2018/0021528	A1*	1/2018	Hsieh A61M 11/005 128/200.16
2018/0021530	A1	1/2018	Fink et al.
2018/0056018	A1	3/2018	Canvin et al.
2018/0116871	A1	5/2018	Hunter et al.
2018/0193175	A1	7/2018	Bluecher et al.
2018/0317557	A1	11/2018	Monsees et al.
2018/0344955	A1	12/2018	Germinario et al.
2018/0369515	A1	12/2018	Germinario et al.
2019/0117907	A1	4/2019	Germinario et al.
2019/0125985	A1	5/2019	Germinario et al.
2019/0125986	A1	5/2019	Germinario et al.
2019/0125987	A1	5/2019	Germinario et al.
2019/0134330	A1	5/2019	Germinario et al.
2019/0224426	A1	7/2019	Farina et al.
2019/0343793	A1	11/2019	Gunther et al.
2019/0358420	A1	11/2019	Hunter et al.
2020/0060346	A1	2/2020	Danek
2020/0147325	A1	5/2020	Wilson et al.
2020/0230329	A1	7/2020	Danek
2020/0246556	A1*	8/2020	Osoegawa A61M 15/0085
2020/0276398	A1	9/2020	Hebrank et al.
2020/0289770	A1	9/2020	Hebrank et al.
2020/0315842	A1	10/2020	Palanker et al.
2020/0330267	A1*	10/2020	Li A61M 35/00
2020/0345588	A1	11/2020	Merrell et al.
2020/0353186	A1	11/2020	Hebrank et al.
2021/0106772	A1	4/2021	Hebrank et al.
2021/0236745	A1	8/2021	Germinario et al.
2021/0275760	A1	9/2021	Hunter et al.
2022/0001122	A1	1/2022	Hunter et al.
2022/0296823	A1	9/2022	Hebrank et al.

FOREIGN PATENT DOCUMENTS

CH	715947	11/2020
CN	1788806	6/2006
CN	104511072	4/2015
CN	204995458	1/2016
CN	205019058	2/2016
EP	0923957	10/2001
EP	2724741	4/2014
EP	3127616	2/2017
JP	H11-042219	2/1999
JP	2003-265994	9/2003
JP	2006-68508	3/2006
KR	10-2019-122453	10/2019
WO	WO 93/12823	7/1993
WO	WO 96/09846	4/1996
WO	WO 96/14163	5/1996
WO	WO 98/48873	11/1998
WO	WO 00/10634	3/2000
WO	WO 00/47335	8/2000
WO	WO 00/50112	8/2000
WO	WO 01/85244	11/2001

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	WO 01/87378	11/2001
WO	WO 02/068128	9/2002
WO	WO 03/020349	3/2003
WO	WO 03/059413	7/2003
WO	WO 2004/078025	9/2004
WO	WO 2006/013952	2/2006
WO	WO 2006/083014	8/2006
WO	WO 2006/102345	9/2006
WO	WO 2006/108558	10/2006
WO	WO 2007/107160	9/2007
WO	WO 2008/056986	5/2008
WO	WO 2008/058941	5/2008
WO	WO 2008/116165	9/2008
WO	WO 2009/012371	1/2009
WO	WO 2009/111612	9/2009
WO	WO 2010/065452	6/2010
WO	WO 2011/042212	4/2011
WO	WO 2011/083377	7/2011
WO	WO 2011/091268	7/2011
WO	WO 2011/163272	12/2011
WO	WO 2012/026963	3/2012
WO	WO 2013/098334	7/2013
WO	WO 2013/158352	10/2013
WO	WO 2013/158967	10/2013
WO	WO 2013/173321	11/2013
WO	WO 2014/147550	9/2014
WO	WO 2015/004554	1/2015
WO	WO 2015/106150	7/2015
WO	WO 2015/136529	9/2015
WO	WO 2015/176033	11/2015
WO	WO 2015/191478	12/2015
WO	WO 2015/191481	12/2015
WO	WO 2016/001923	1/2016
WO	WO 2016/001924	1/2016
WO	WO 2016/003738	1/2016
WO	WO 2017/015303	1/2017
WO	WO 2017/056103	4/2017
WO	WO 2018/213834	11/2018
WO	WO 2019/071008	4/2019
WO	WO 2019/079461	4/2019
WO	WO 2019/136437	7/2019
WO	WO 2019/219865	11/2019
WO	WO 2019/219873	11/2019
WO	WO 2020/072478	4/2020
WO	WO 2020/141424	7/2020

WO	WO 2020/154497	7/2020	
WO	WO 2020/227717	11/2020	
WO	WO-2020227717	A1 *	11/2020 A61M 11/005
WO	WO 2020/264501	12/2020	
WO	WO-2020264501	A1 *	12/2020 A24F 40/05
WO	WO 2021/090135	5/2021	
WO	WO 2021/203038	10/2021	
WO	WO 2022/051496	3/2022	
WO	WO 2022/226407	10/2022	
WO	WO 2023/278551	1/2023	
WO	WO 2023/064477	4/2023	
WO	WO 2023/091637	5/2023	

OTHER PUBLICATIONS

Kharitonov, "Exhaled markers of inflammatory lung diseases: ready for routine monitoring?" *SWISS MED WKLY*, 2004; 134: 175-192.

Broeders et al., "Inhalation Profiles in Asthmatics and COPD Patients: Reproducibility and Effect of Instruction," *Journal of Aerosol Medicine*, vol. 16, No. 2, 2003, 131-141.

Taube et al., "Use of a portable device to record maximum inspiratory flow in relation to dyspnea in patients with COPD," *Respiratory Medicine*, 2011, 105, 316-312.

Steller, "Microcontroller Based Diagnostic Smart Inhaler," University of Cincinnati, Dec. 7, 2014, 63 pages.

Carvalho et al., "The function and performance of aqueous aerosol devices for inhalation therapy," *Journal of Pharmacy and Pharmacology*, vol. 68, No. 5, Apr. 8, 2016, pp. 556-578.

Pneuma Respiratory, Digitally breath-actuated inhaler device with precision droplet ejector technology and digital dose confidence. Available on Mar. 18, 2017 [retrieved on Jun. 30, 2017]. Retrieved from the Internet: URL: <https://pneumarespiratory.com/>. 3 pp.

Azzopardi, "Sauter Mean Diameter" Sep. 30, 2012, 4 pages, <https://web.archive.org/web/20120930225842/http://www.termopedia.com/content/1108>, retrieved May 26, 2014.

Ultrasonic Vibrating member catalog—Emerson. Catalog—Ultrasonic Vibrating member (2014). Available at: <https://www.emerson.com/documents/automation/catalog-ultrasonic-vibrating-member-branson-en-us-160126.pdf>. (Accessed: Aug. 9, 2023).

Lin, J. & Lin, S. "Study on a large-scale three-dimensional ultrasonic plastic welding vibration system based on a quasi-periodic phononic crystal structure," *Crystals* 2020, 10, 21, MDPI, 18 pages. Industrial resonators Available at: http://www.krell-engineering.com/fea/industr/industrial_resonators.htm, accessed Aug. 9, 2023, 7 pages.

* cited by examiner

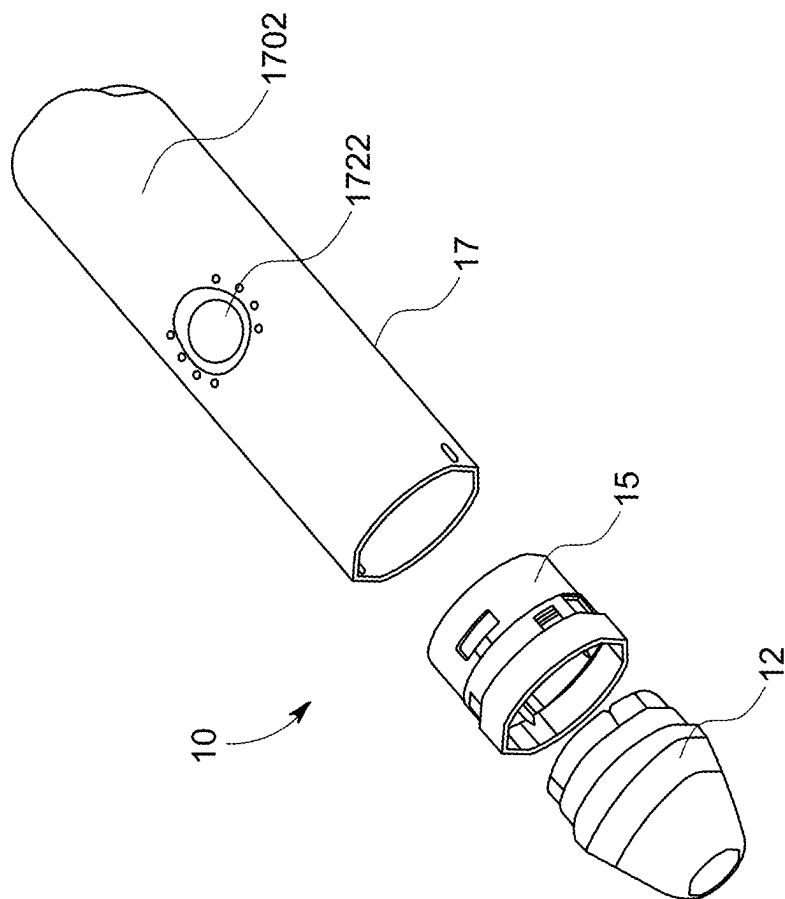


FIG. 1A

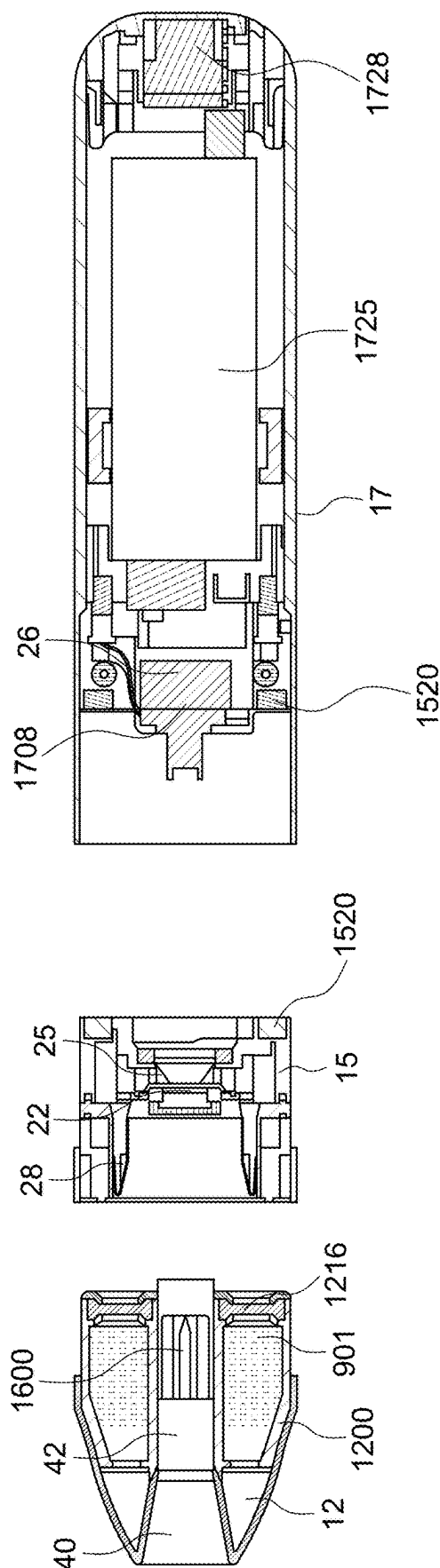


FIG. 1B

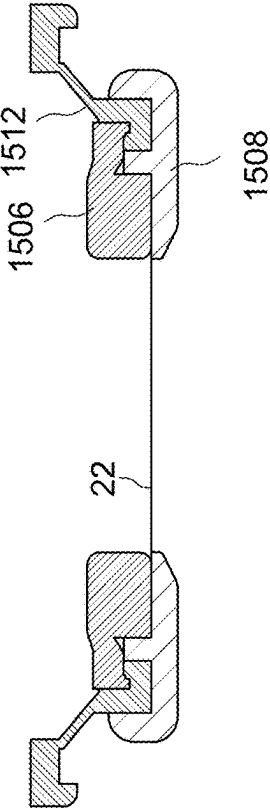


FIG. 3

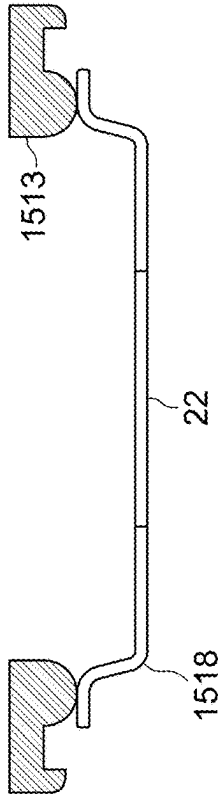


FIG. 2

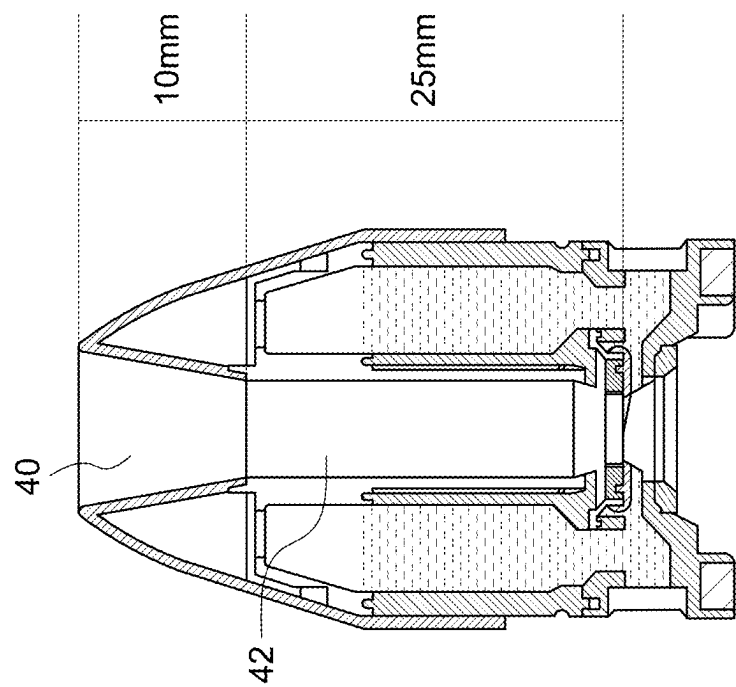


FIG. 4

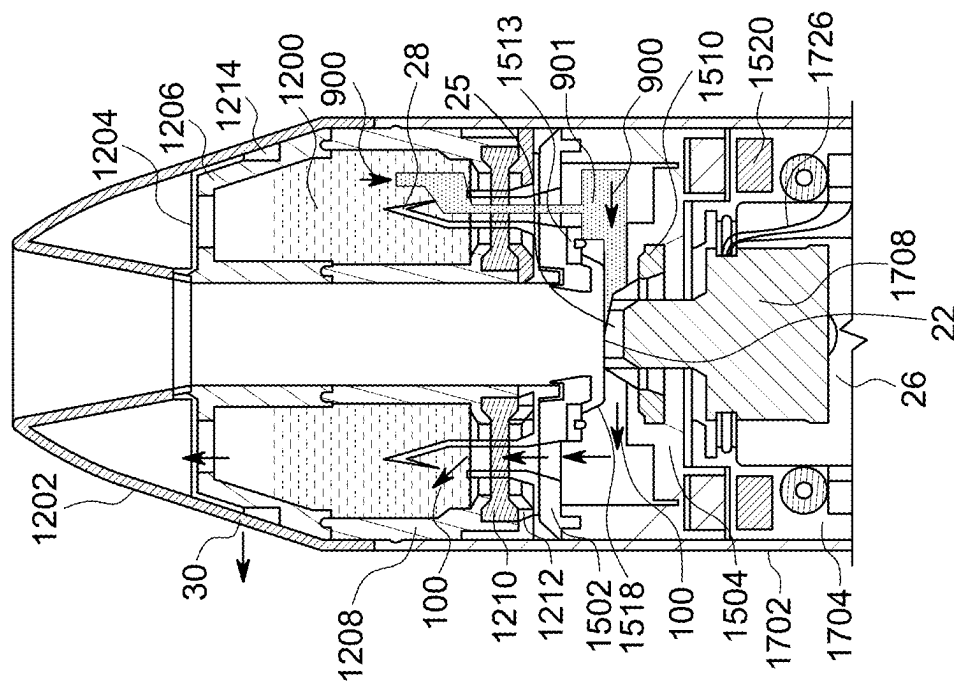


FIG. 5

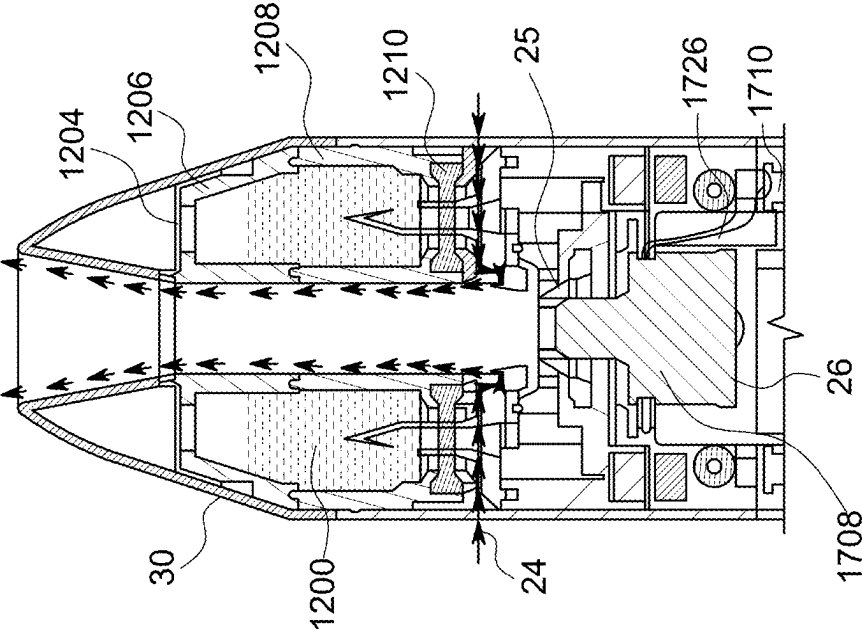


FIG. 6B

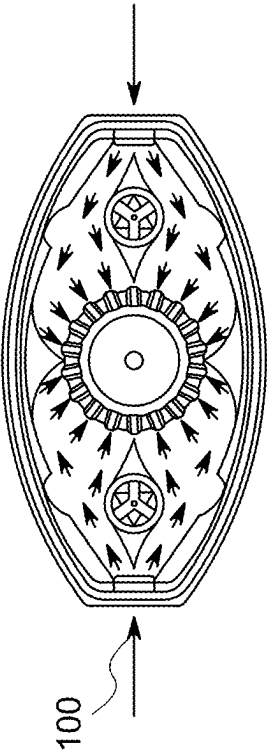


FIG. 6A

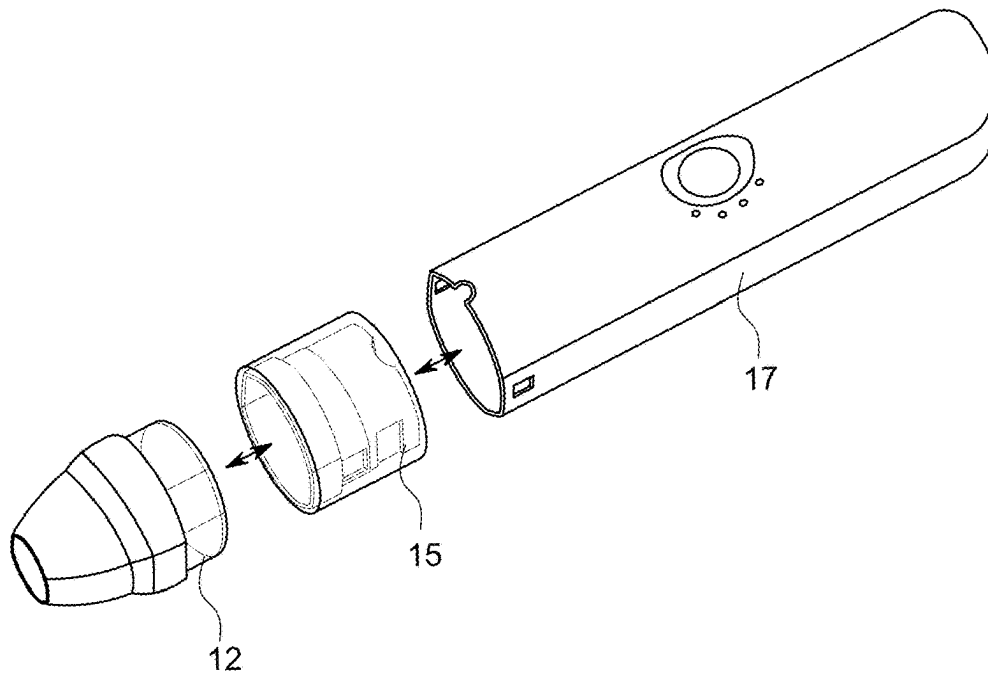


FIG. 7A

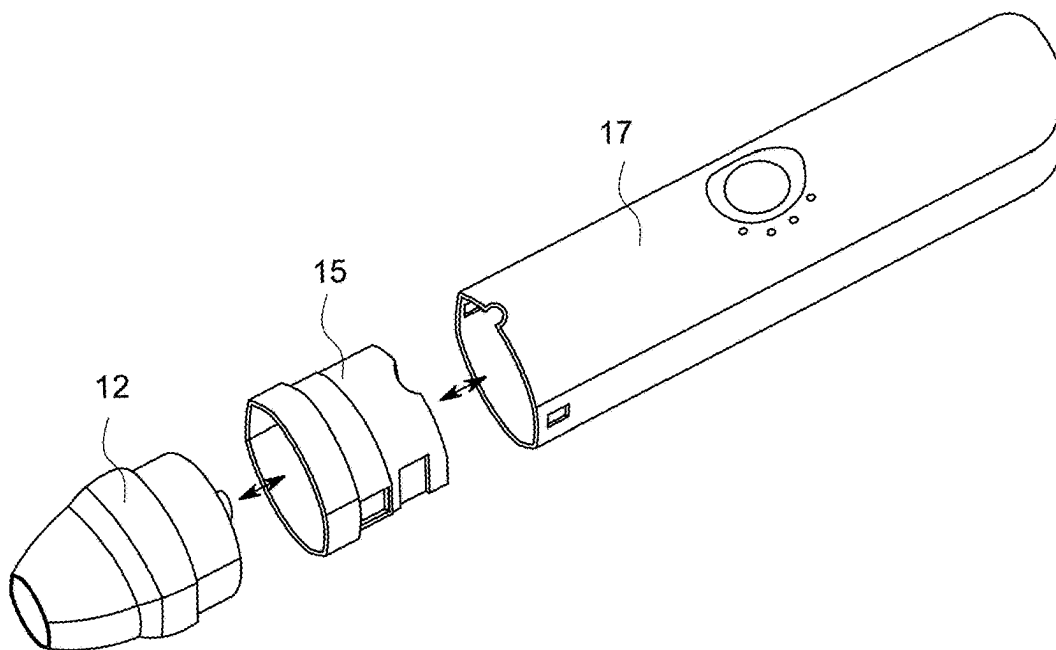


FIG. 7B

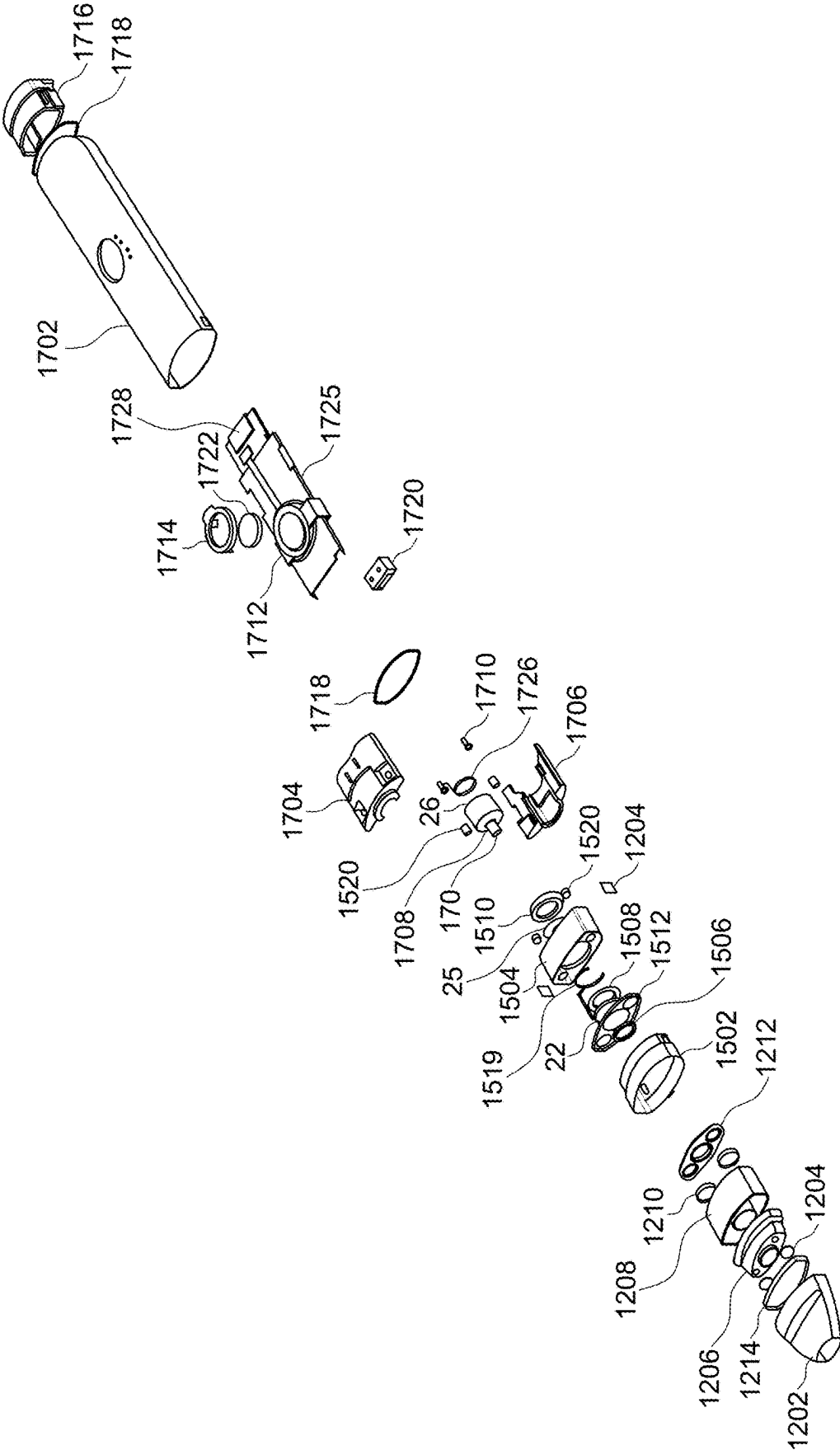


FIG. 8

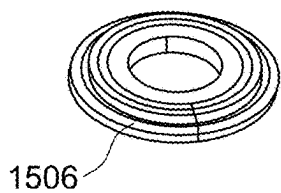


FIG. 9A

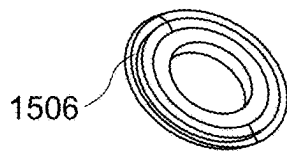


FIG. 9B

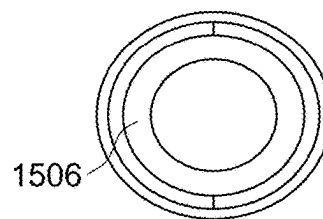


FIG. 9C



FIG. 9F

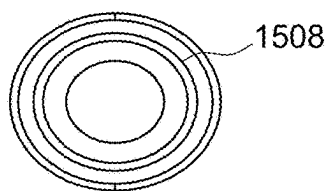


FIG. 9D

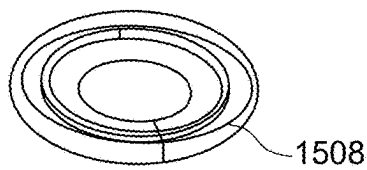


FIG. 9E

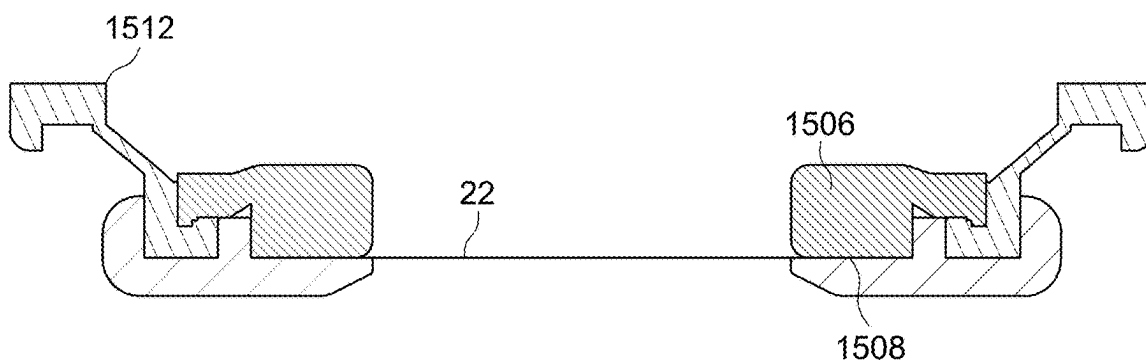


FIG. 10

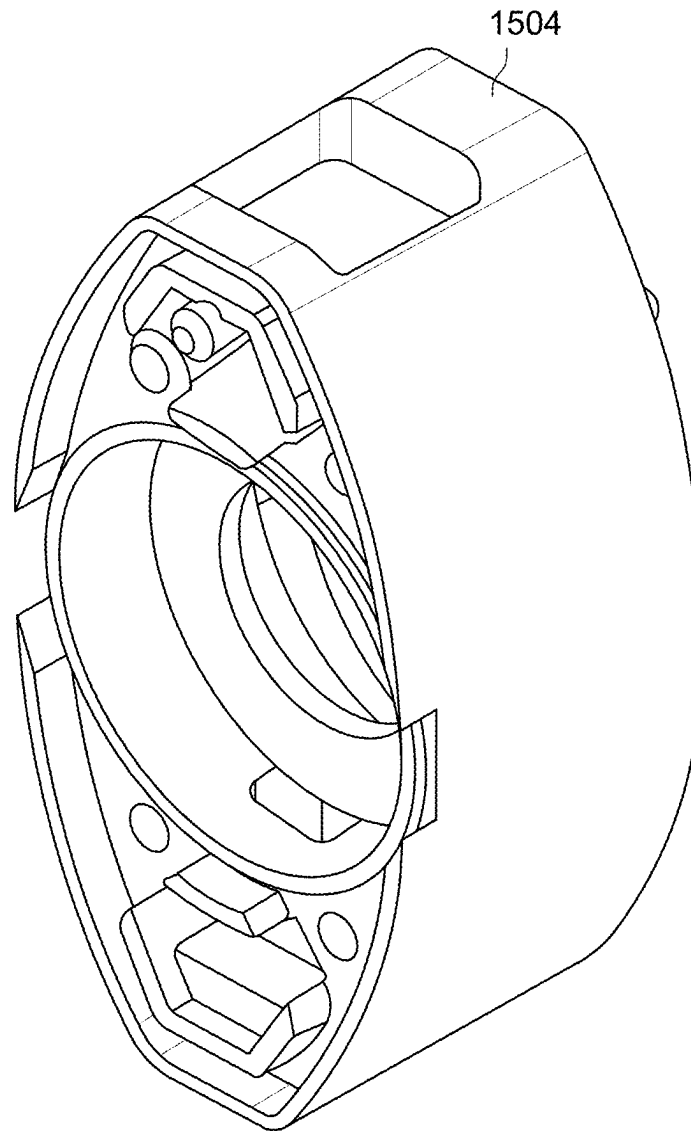


FIG. 11

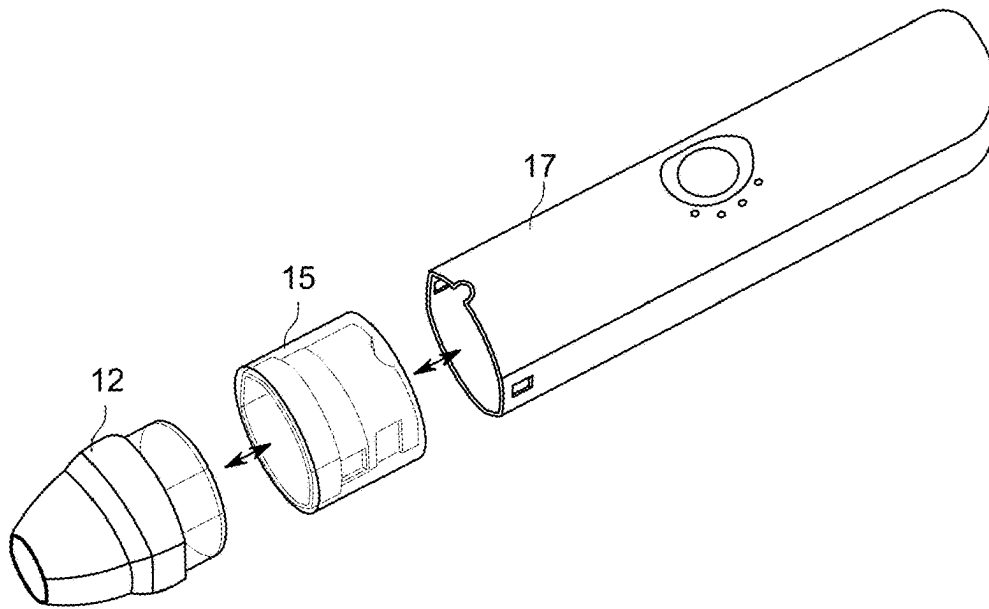


FIG. 12A

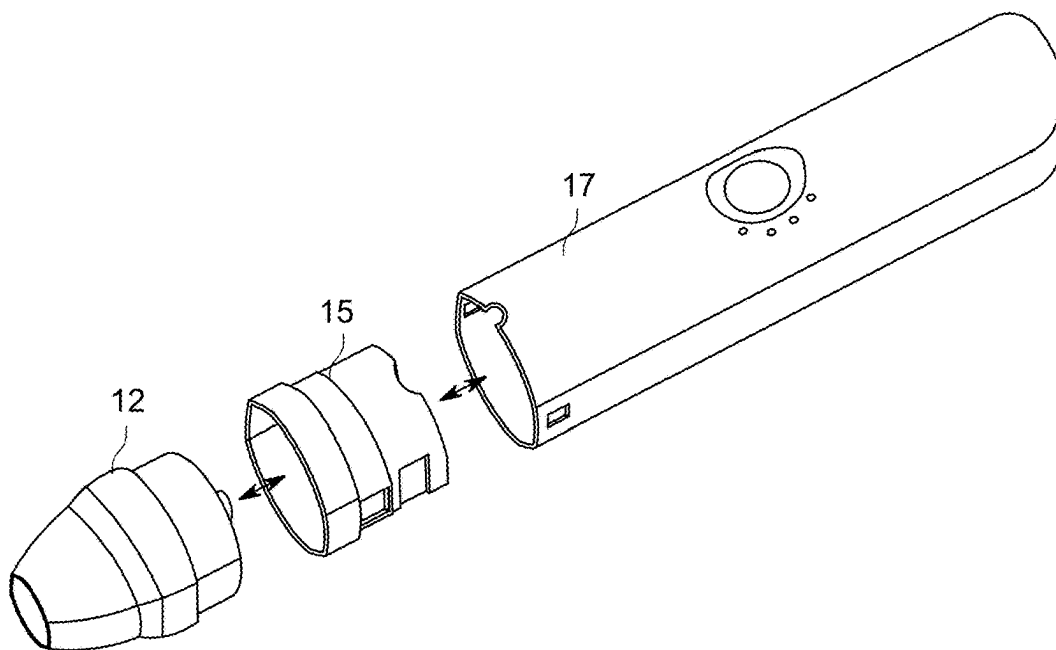


FIG. 12B

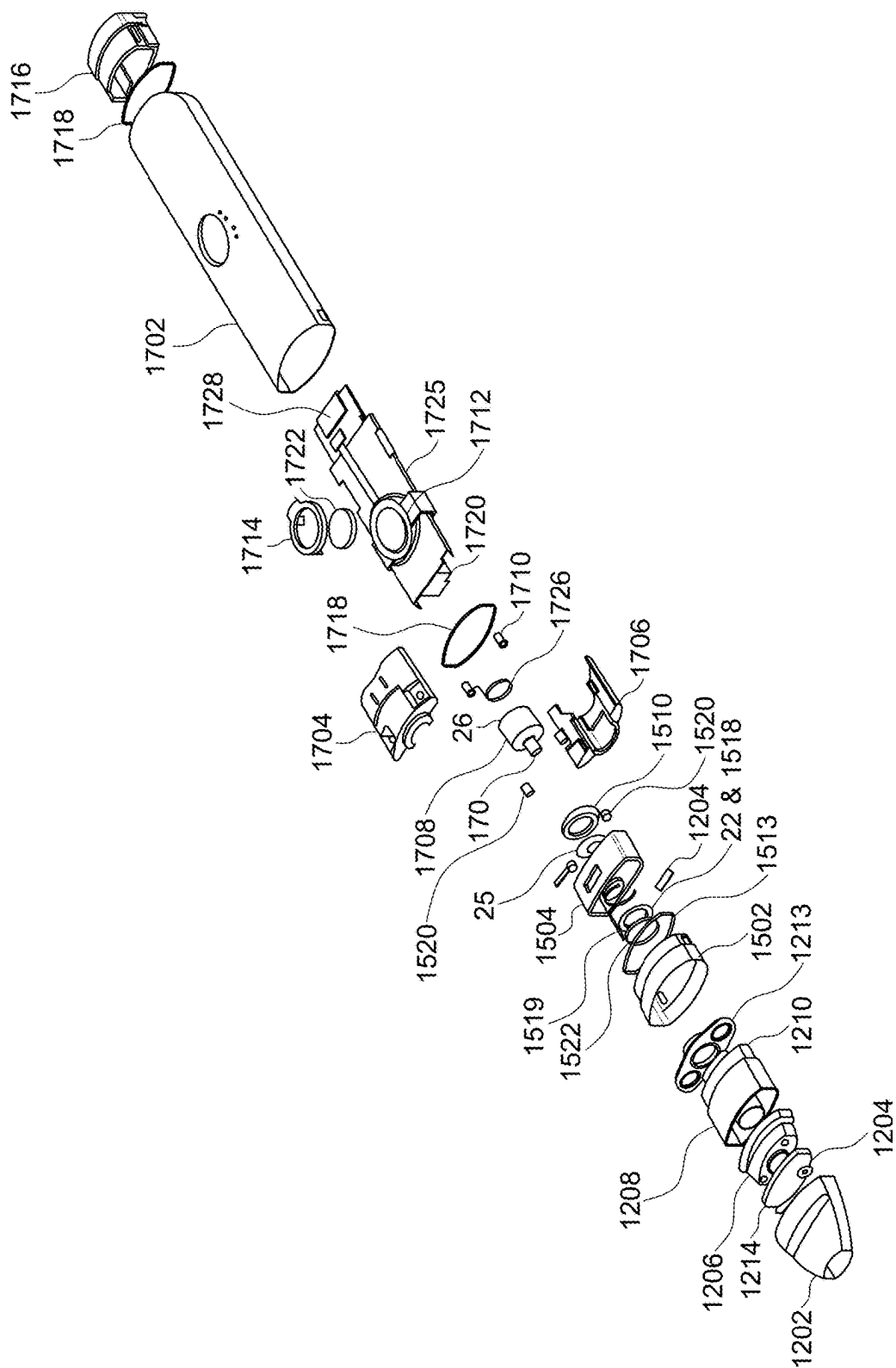


FIG. 13

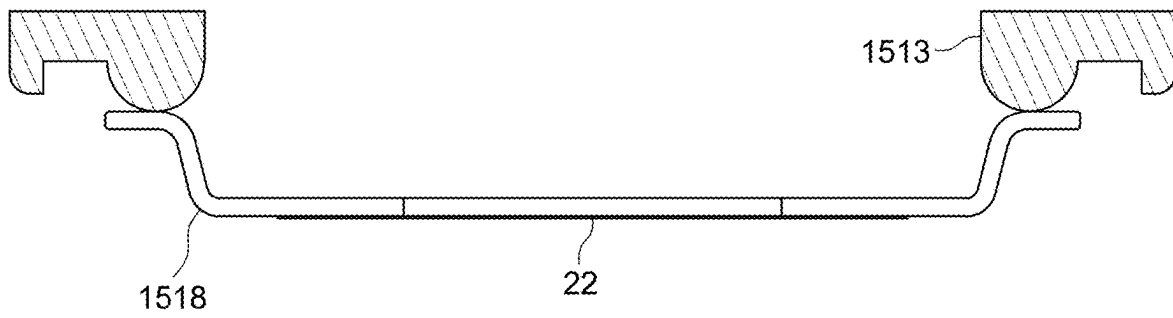


FIG. 14

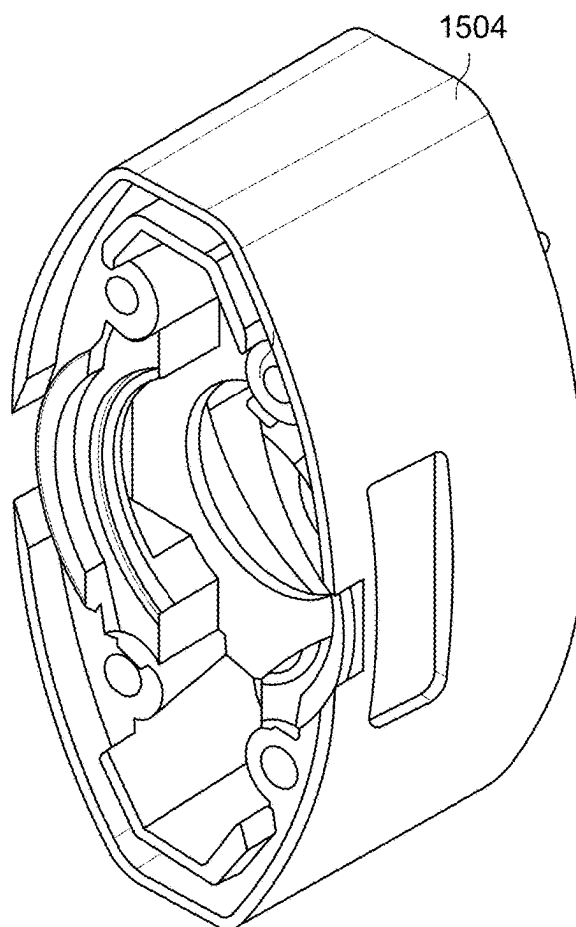


FIG. 15

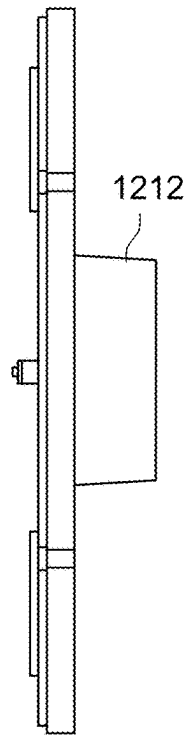


FIG. 16

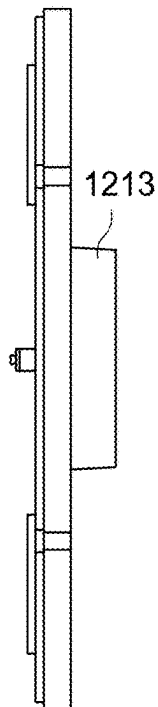


FIG. 17

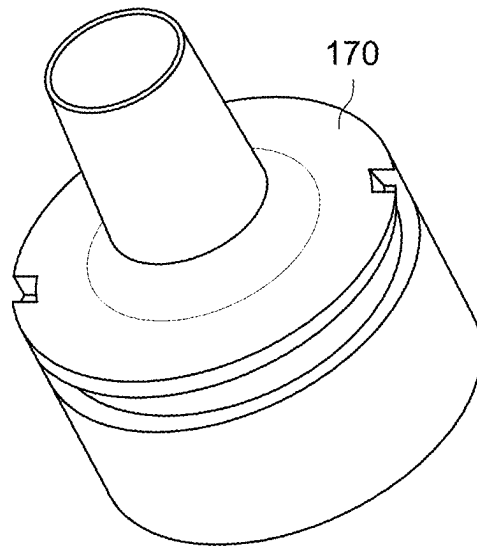


FIG. 18

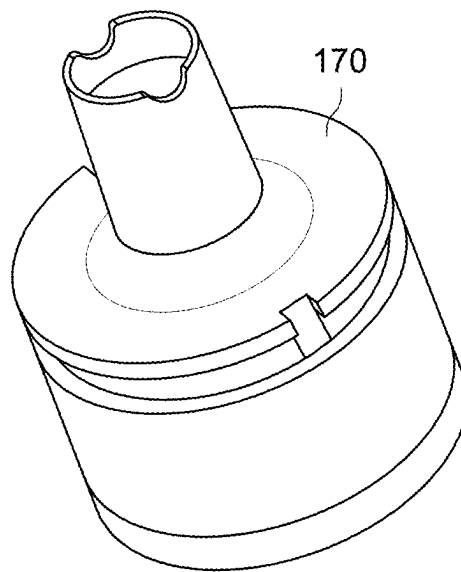


FIG. 19

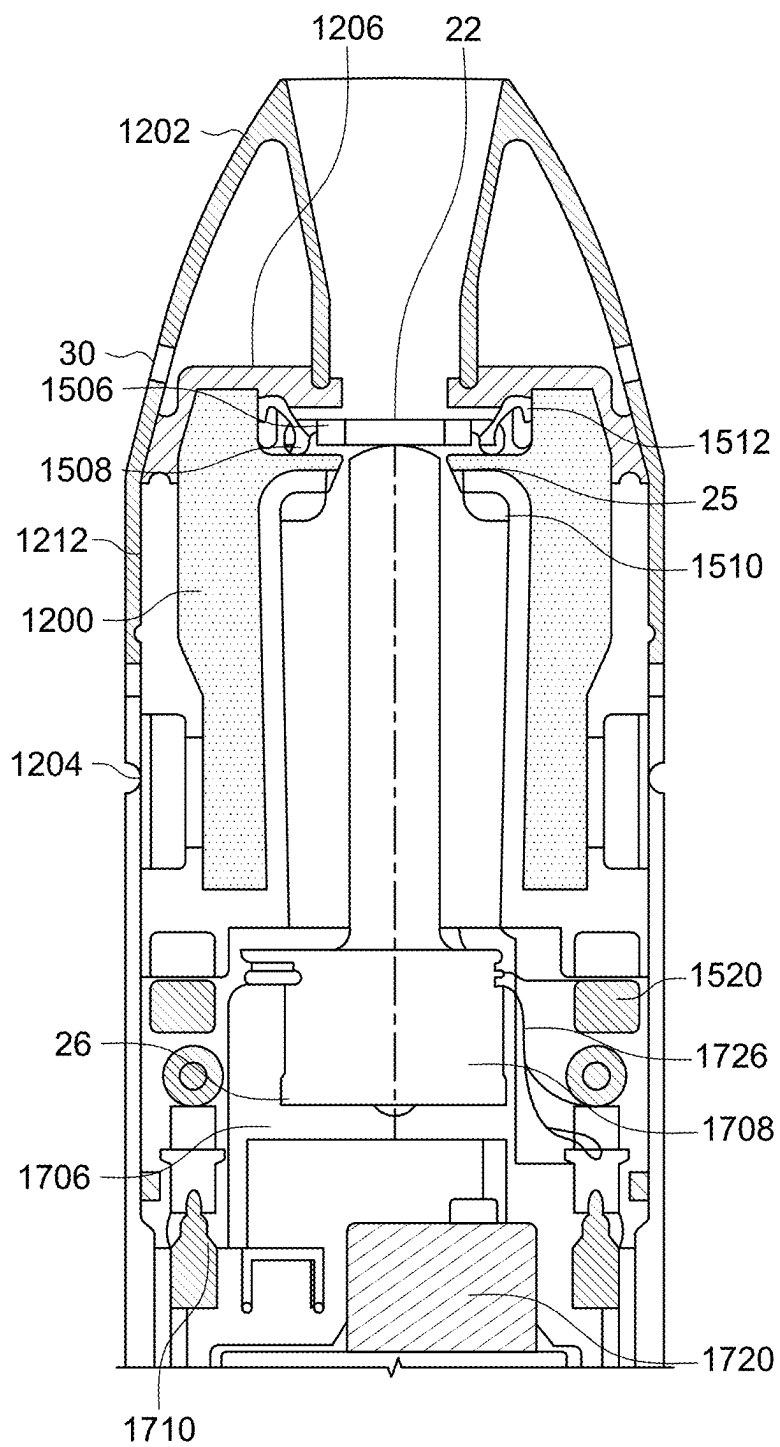


FIG. 20

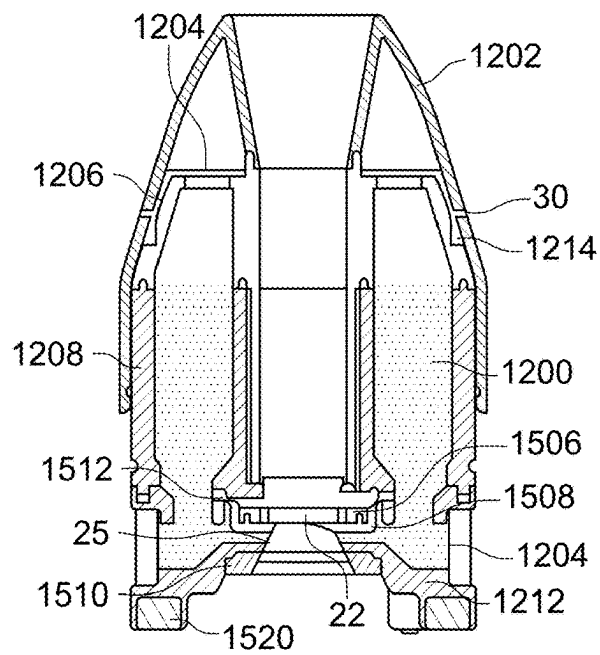


FIG. 21A

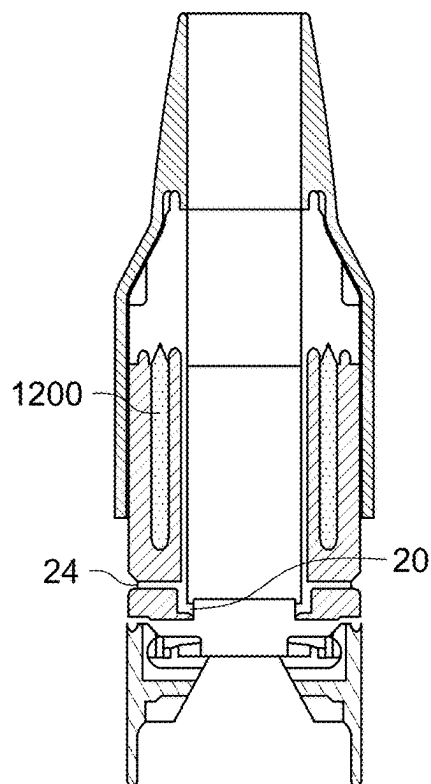


FIG. 21B

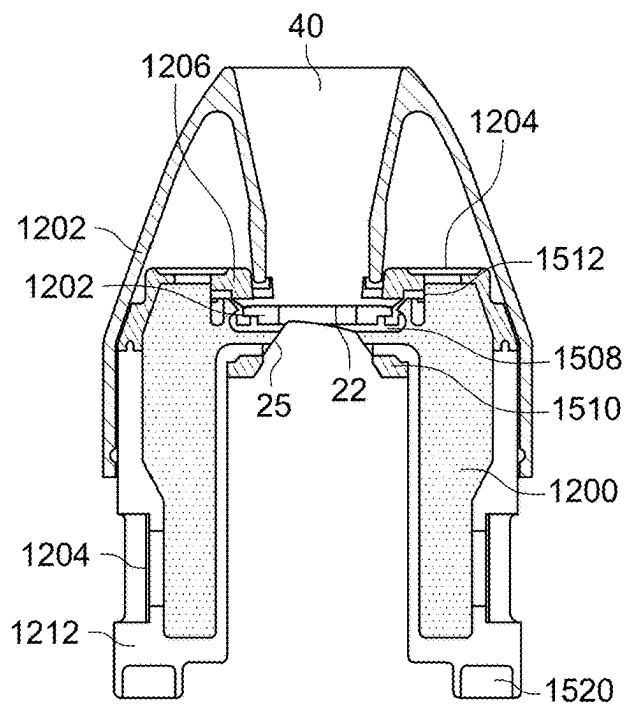


FIG. 22A

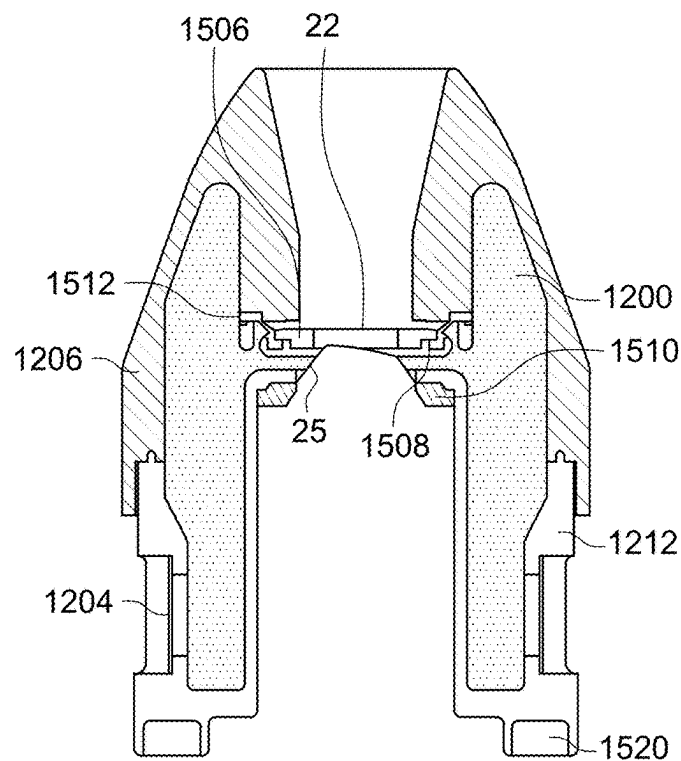


FIG. 22B

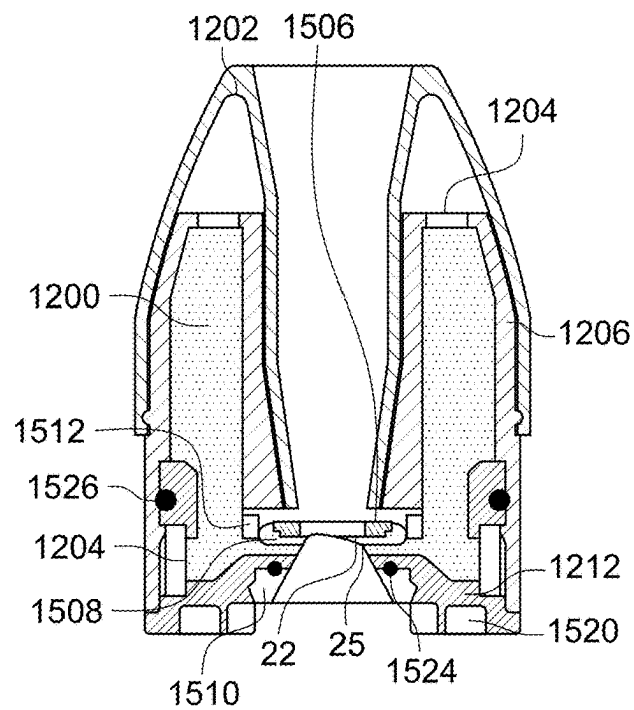


FIG. 23A

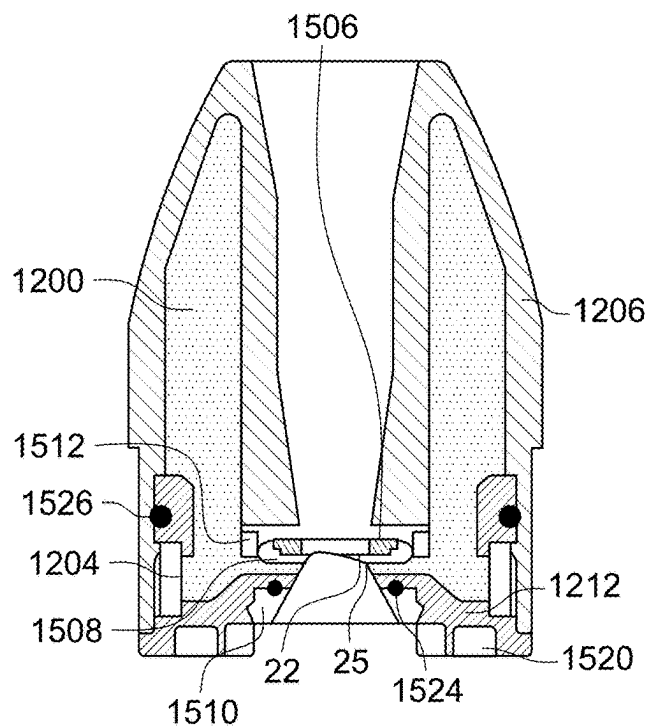


FIG. 23B

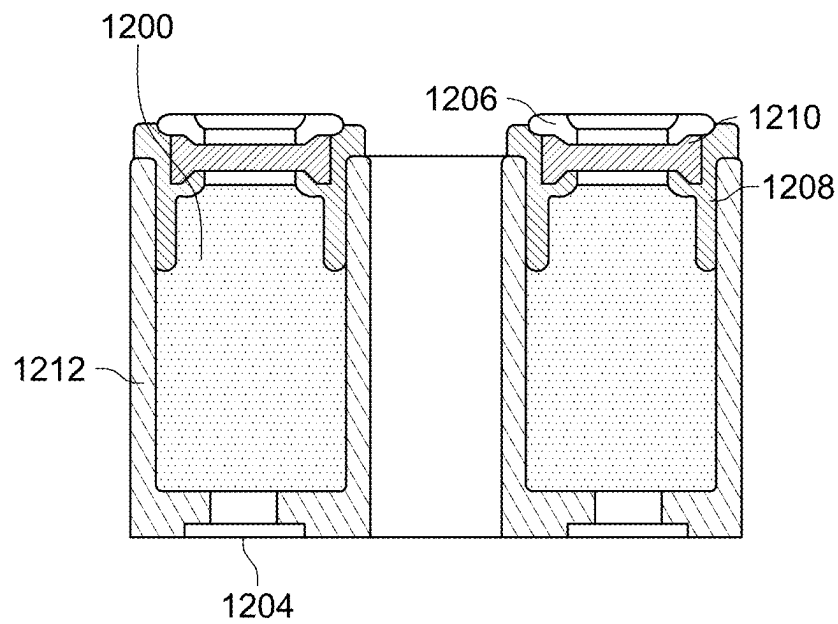
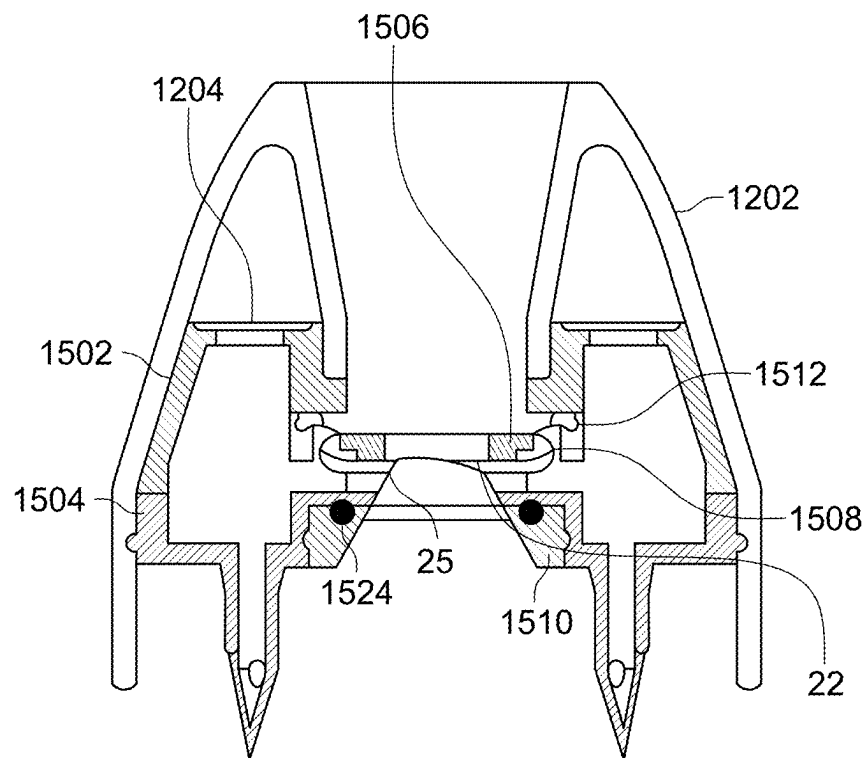


FIG. 24

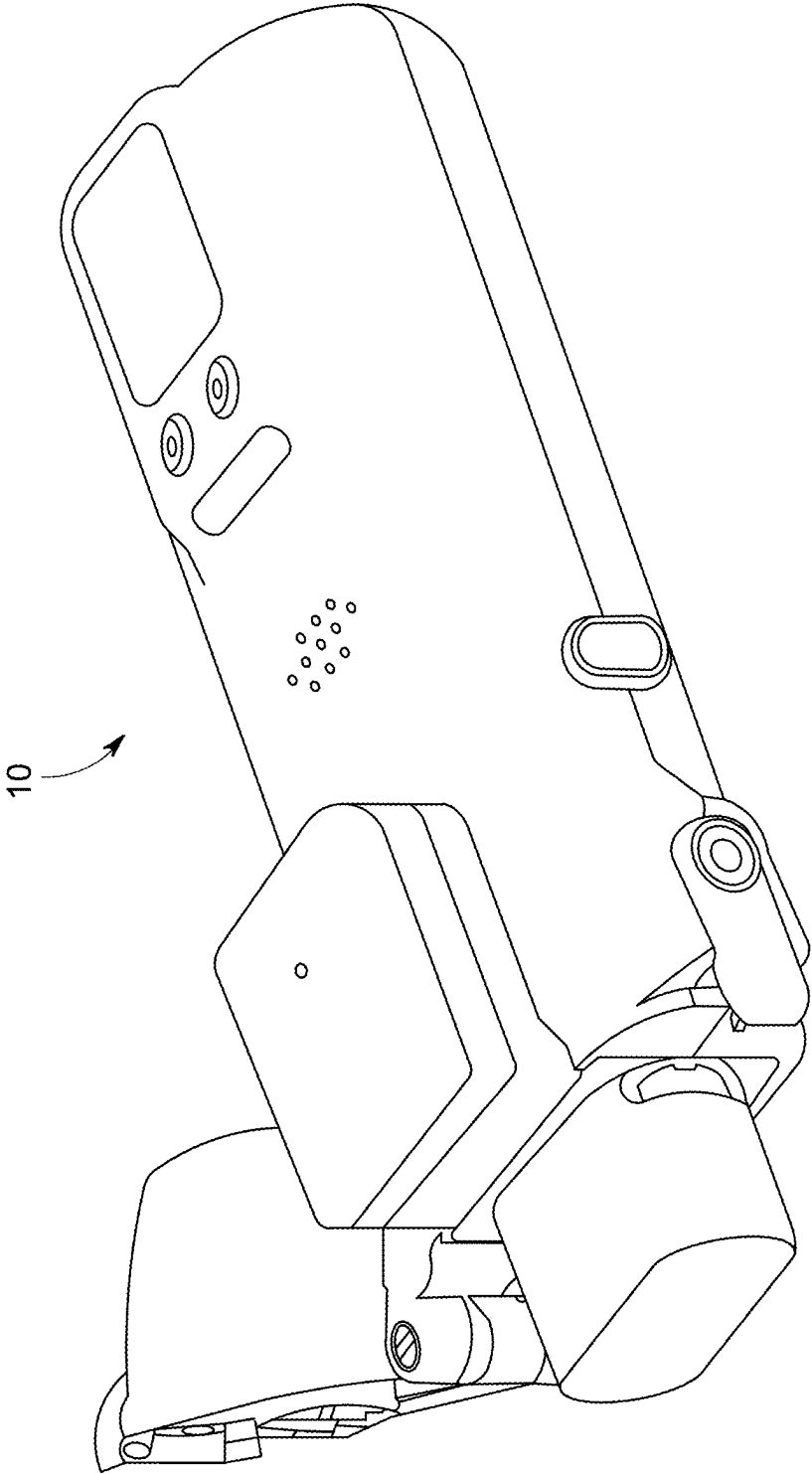


FIG. 25

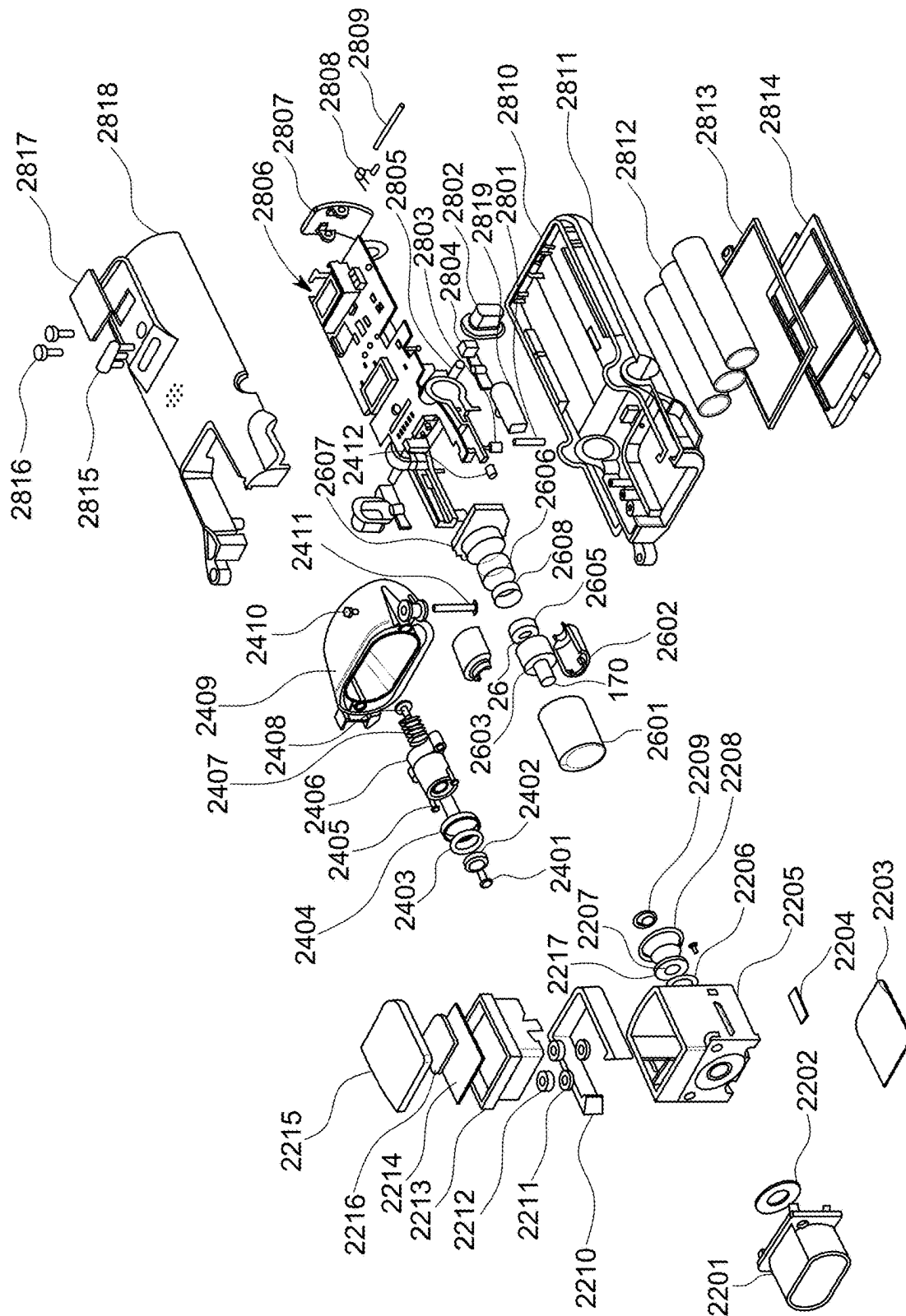


FIG. 26

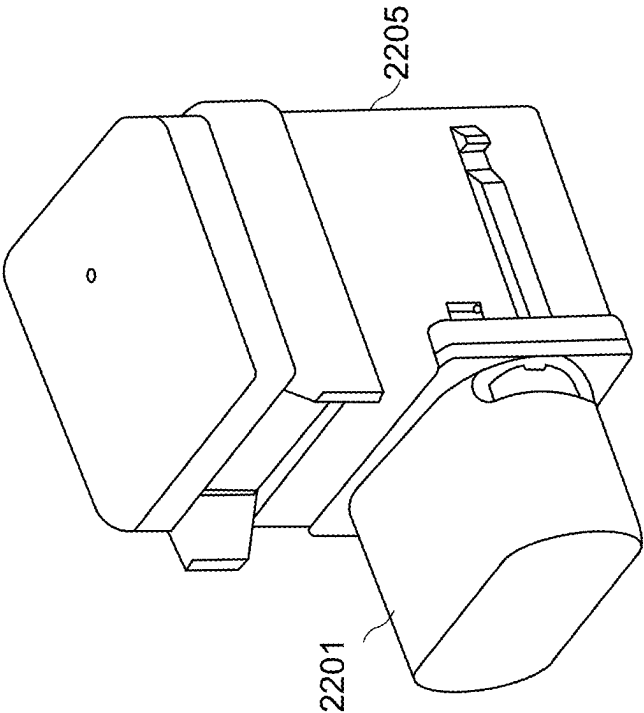


FIG. 27A

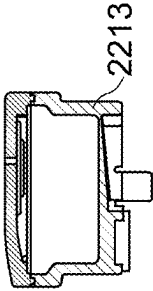


FIG. 27B



FIG. 27C

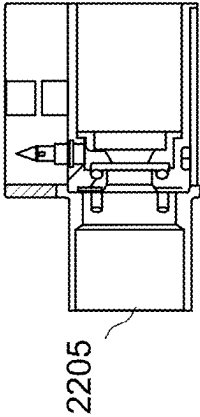


FIG. 27D

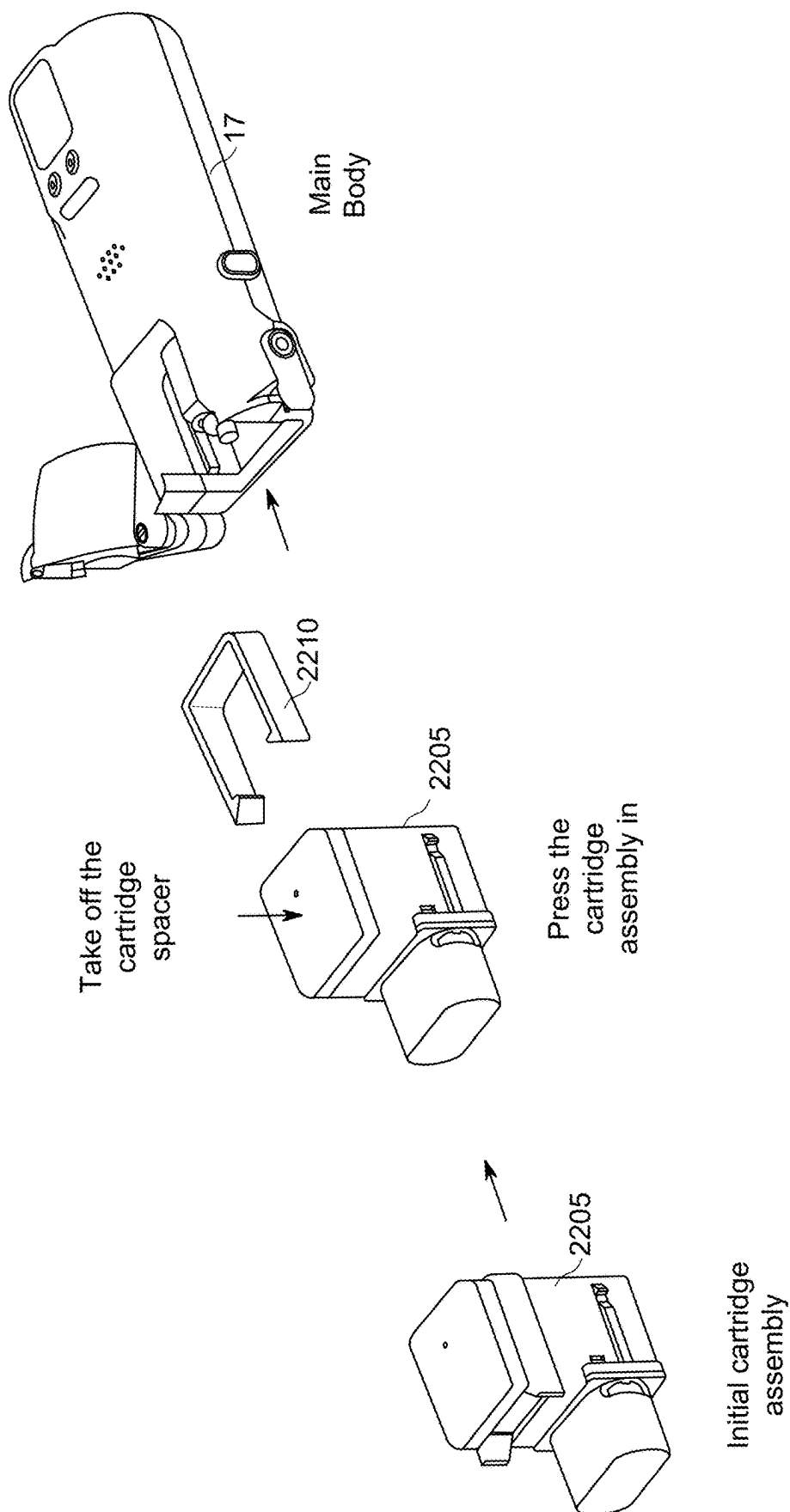


FIG. 28

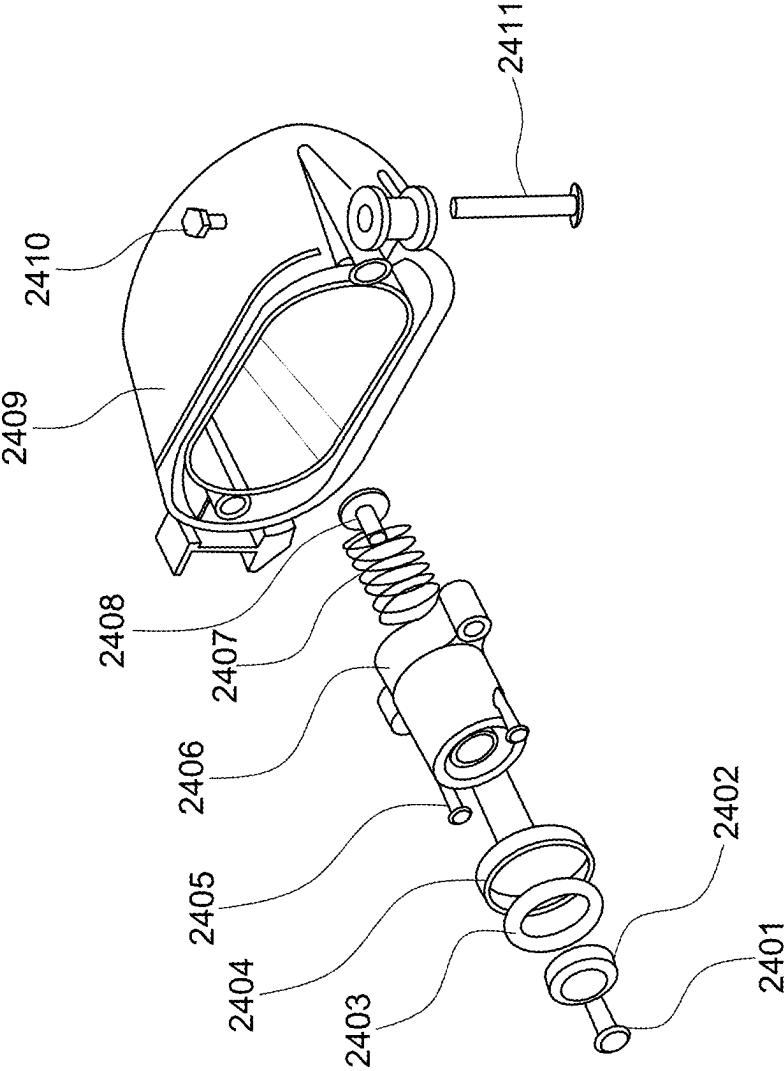


FIG. 29

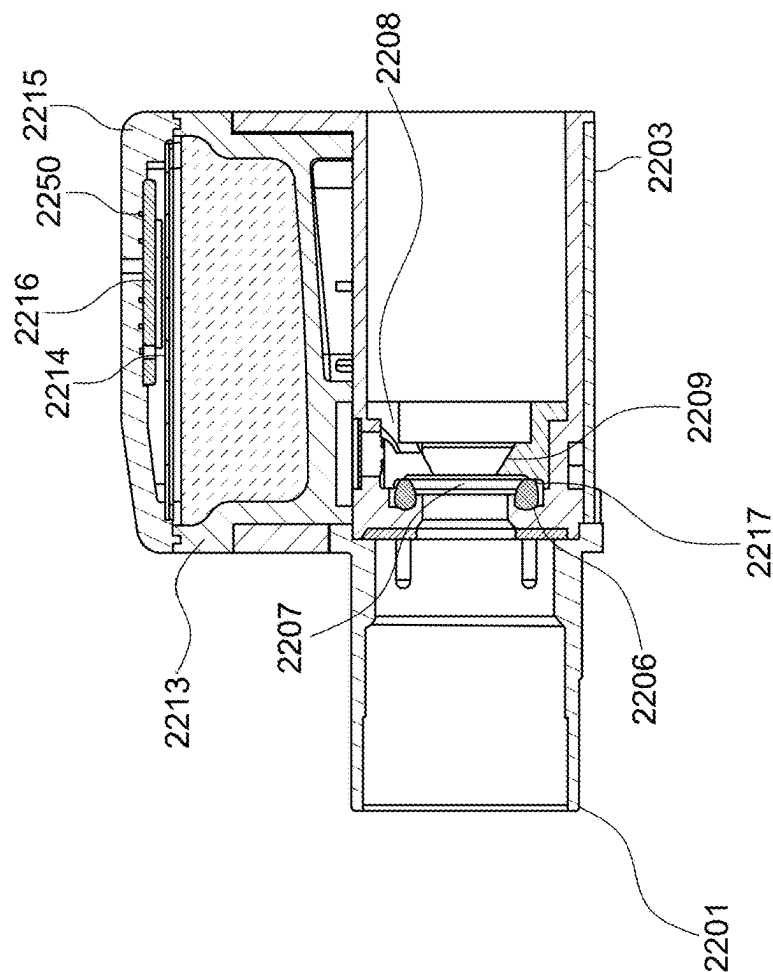


FIG. 30A

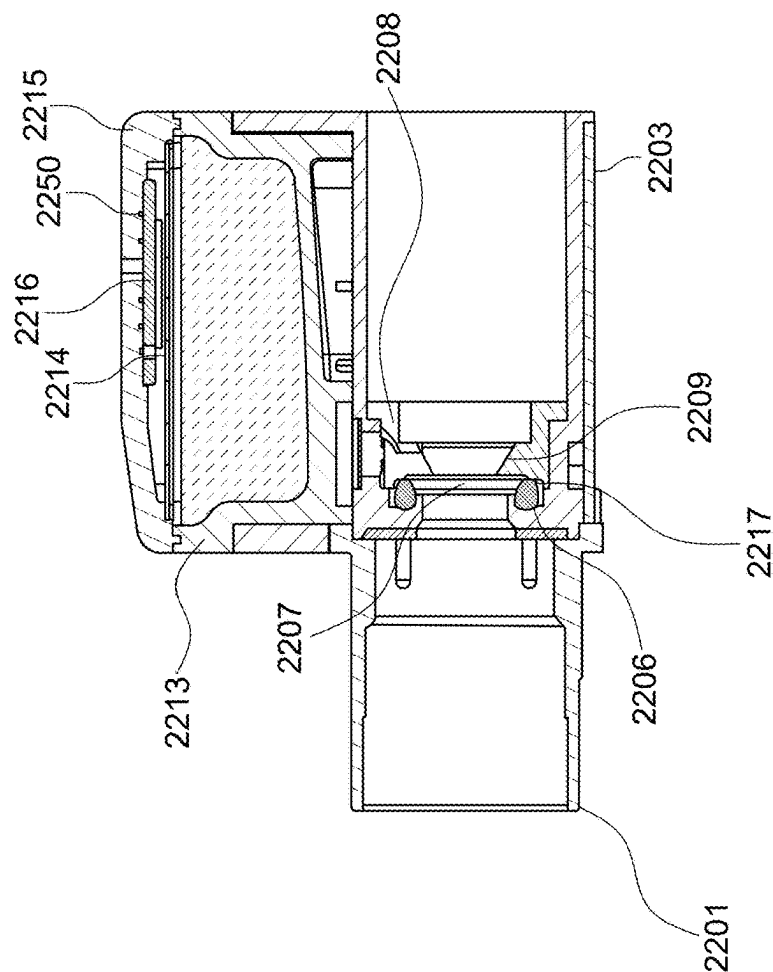


FIG. 30B

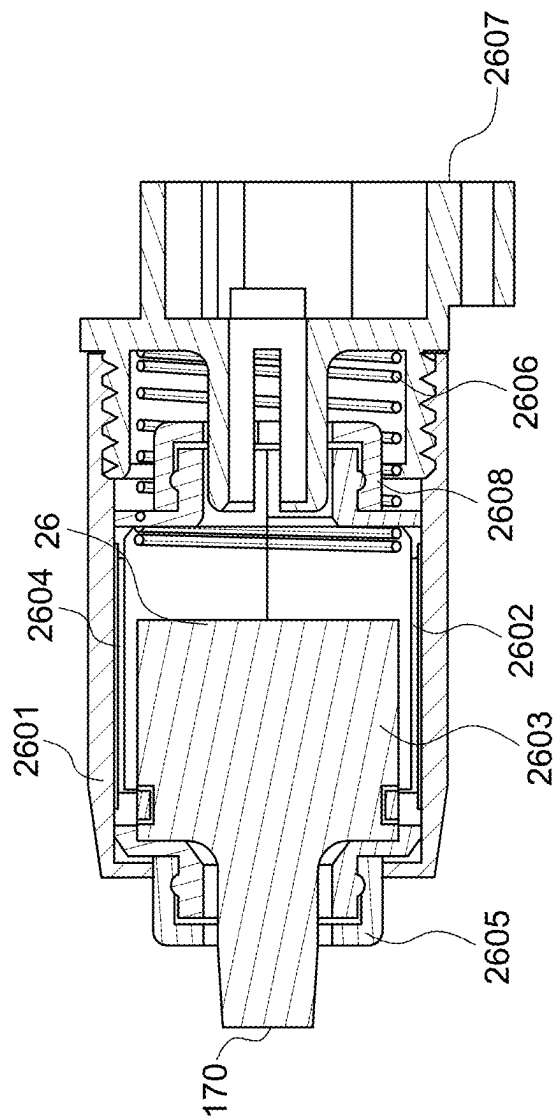


FIG. 31

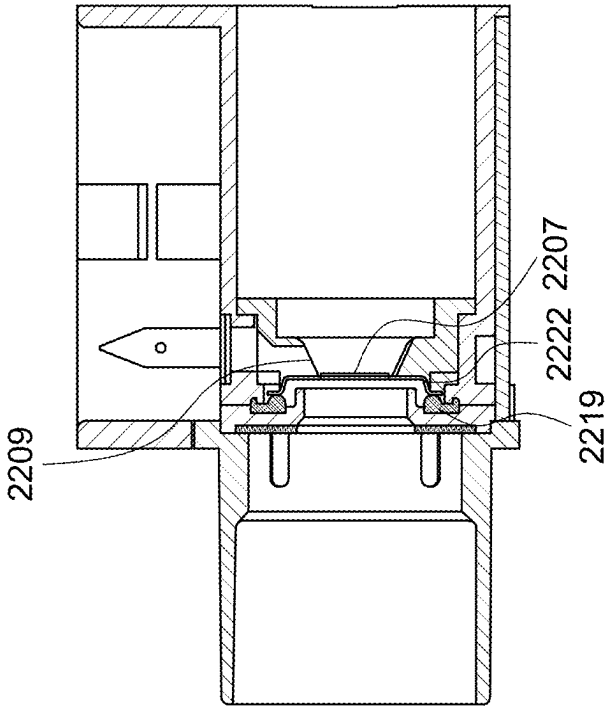


FIG. 32

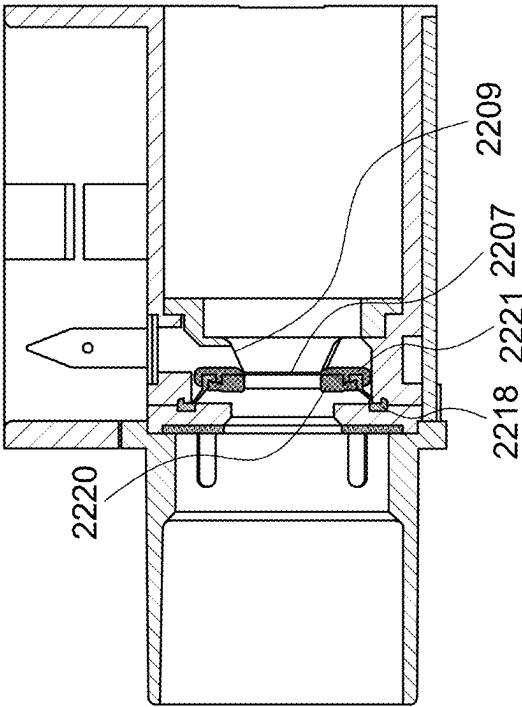


FIG. 33

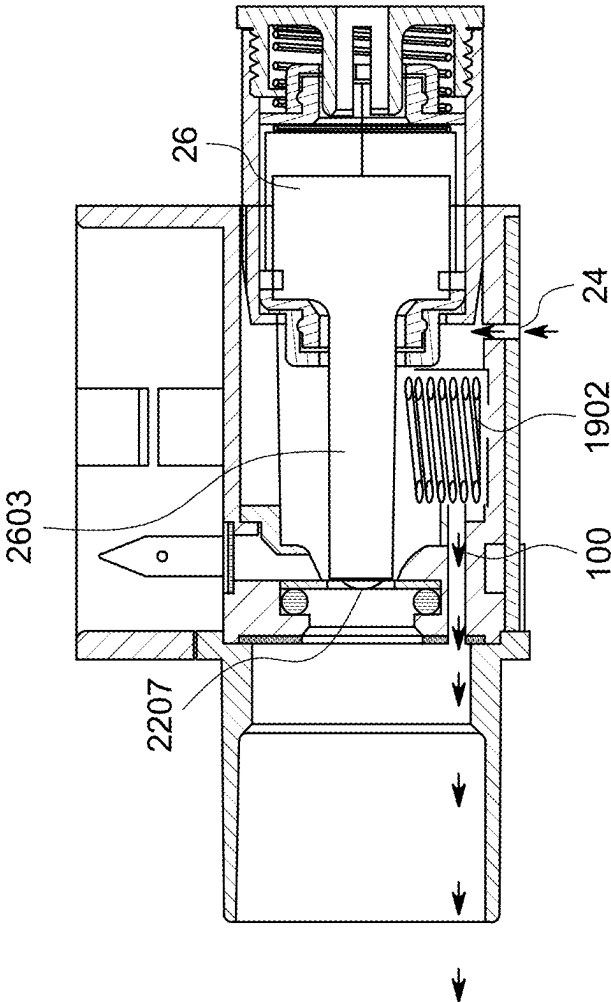


FIG. 34A

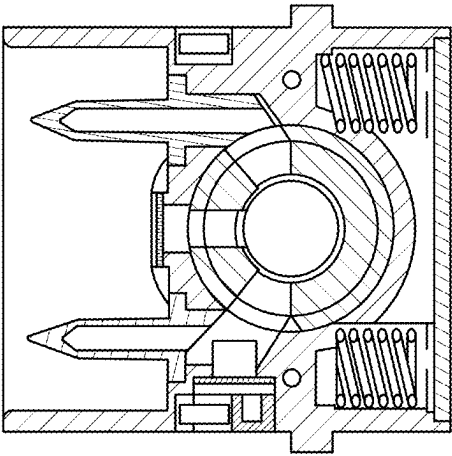


FIG. 34B

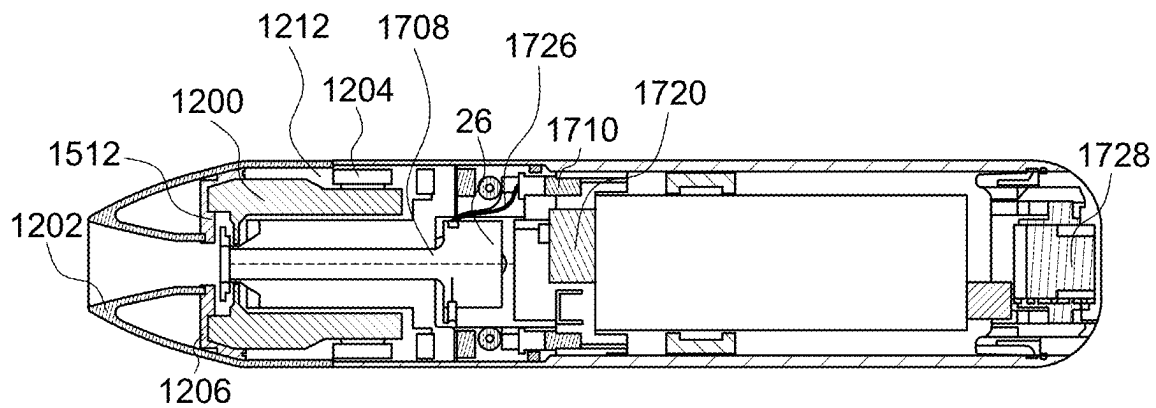


FIG. 35A

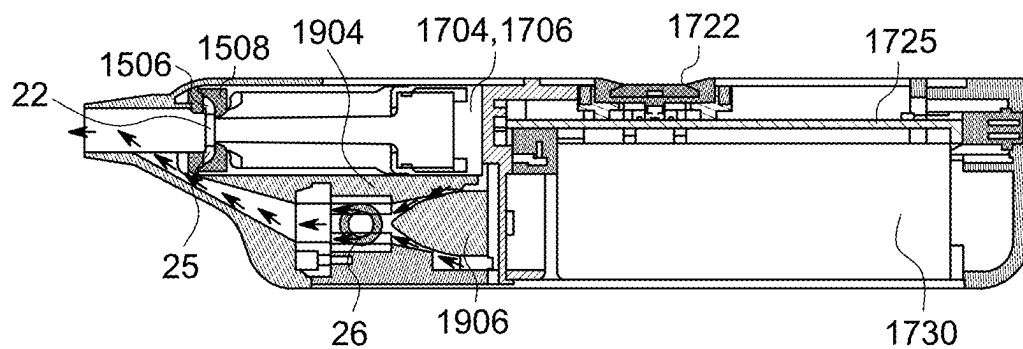


FIG. 35B

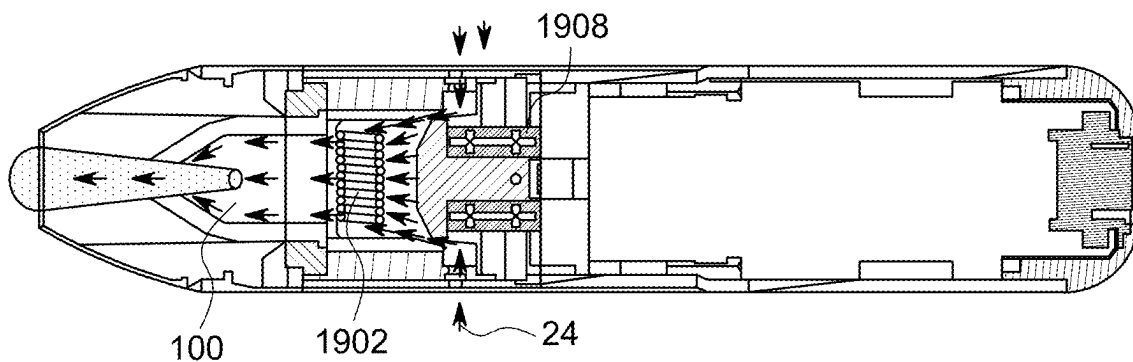


FIG. 35C

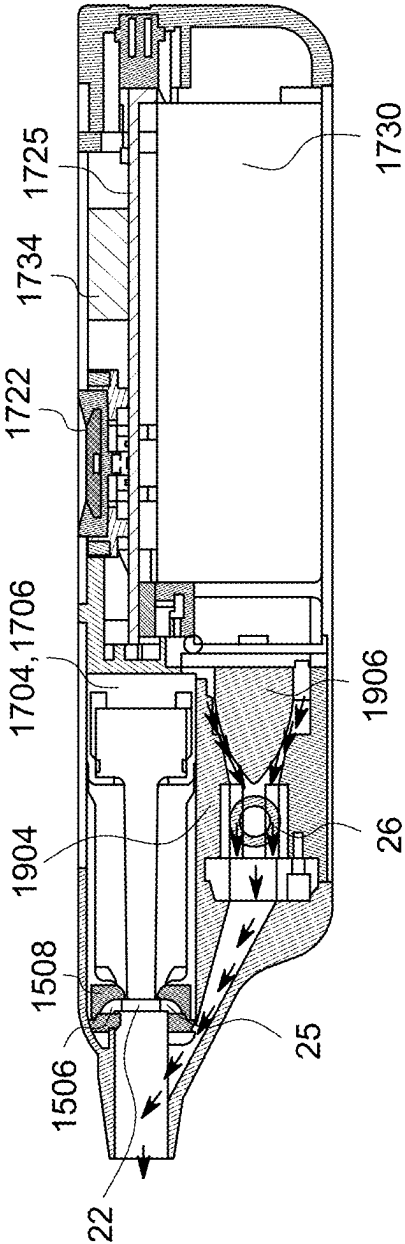


FIG. 36

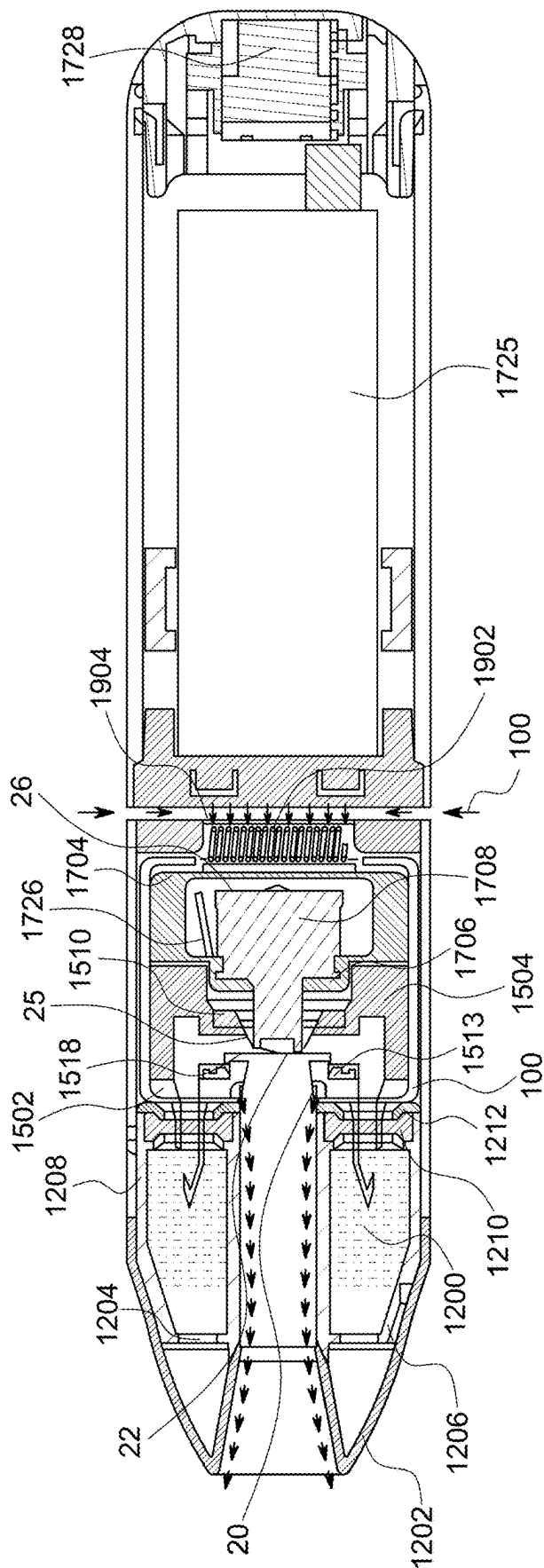


FIG. 37

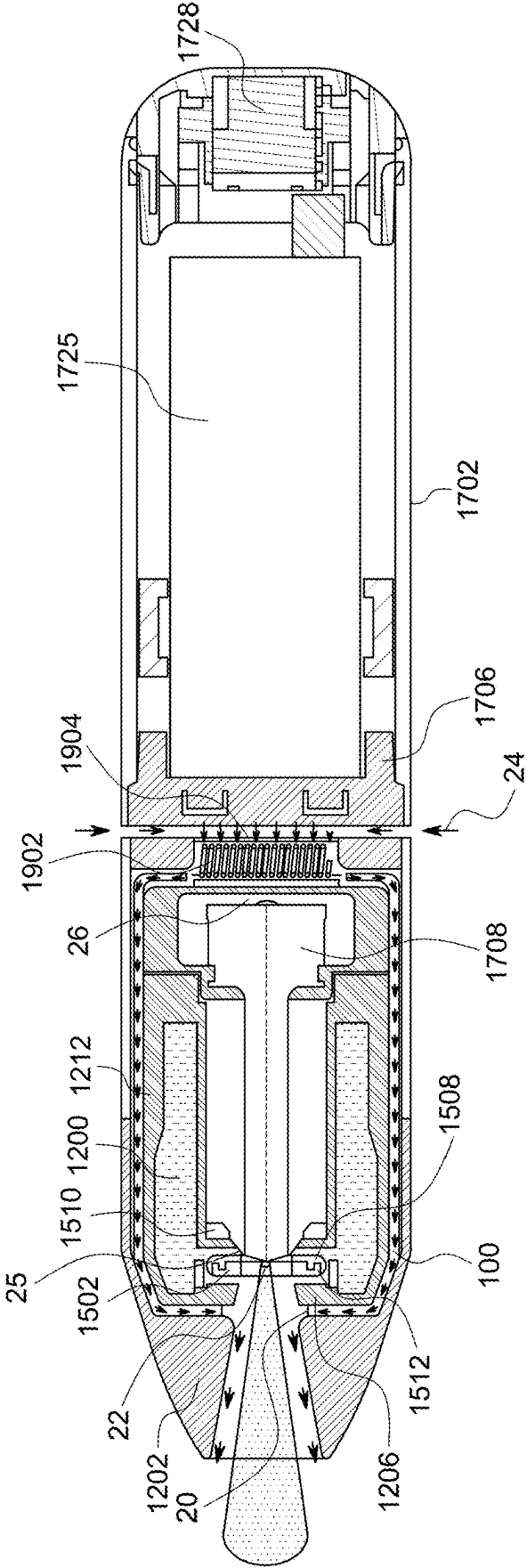


FIG. 38

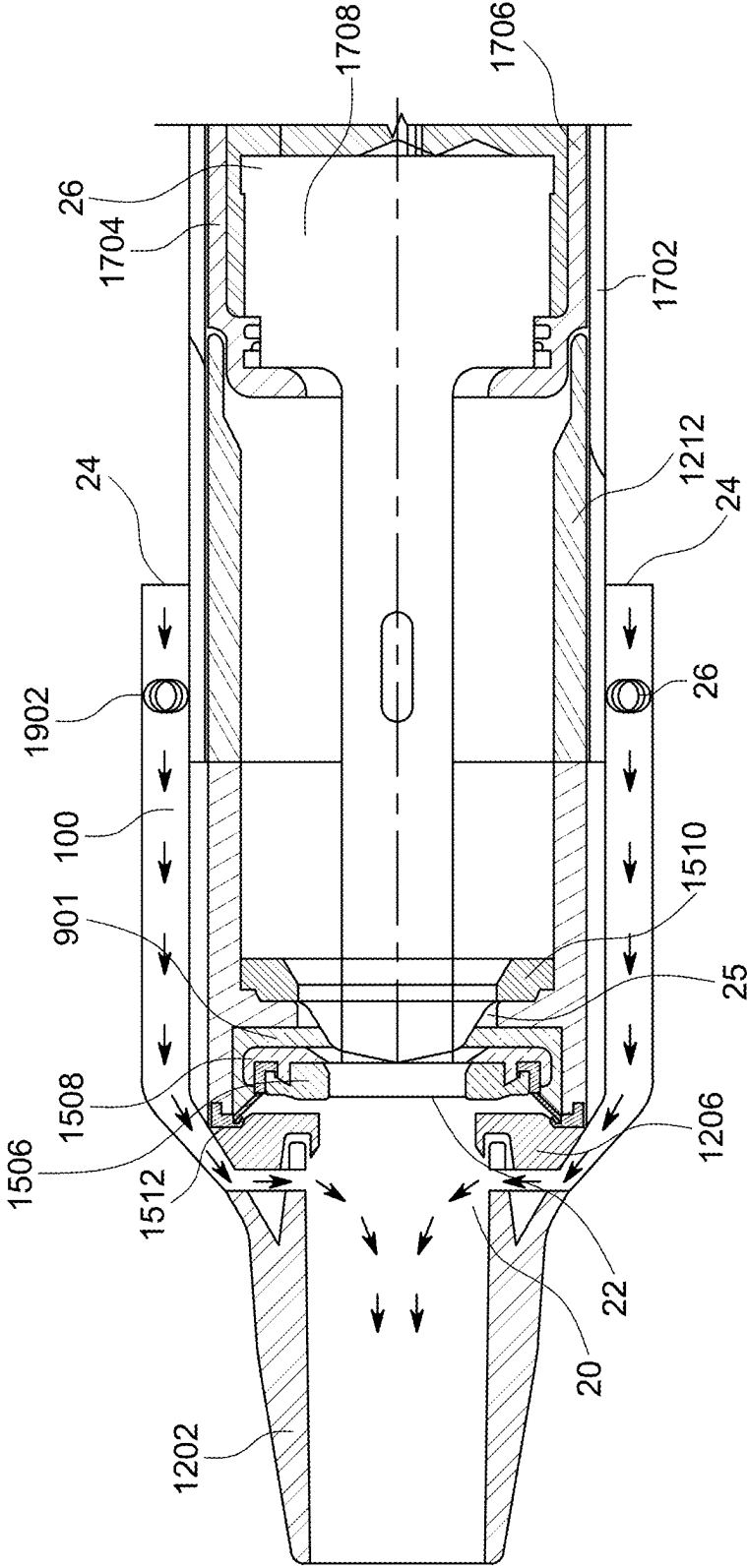


FIG. 39

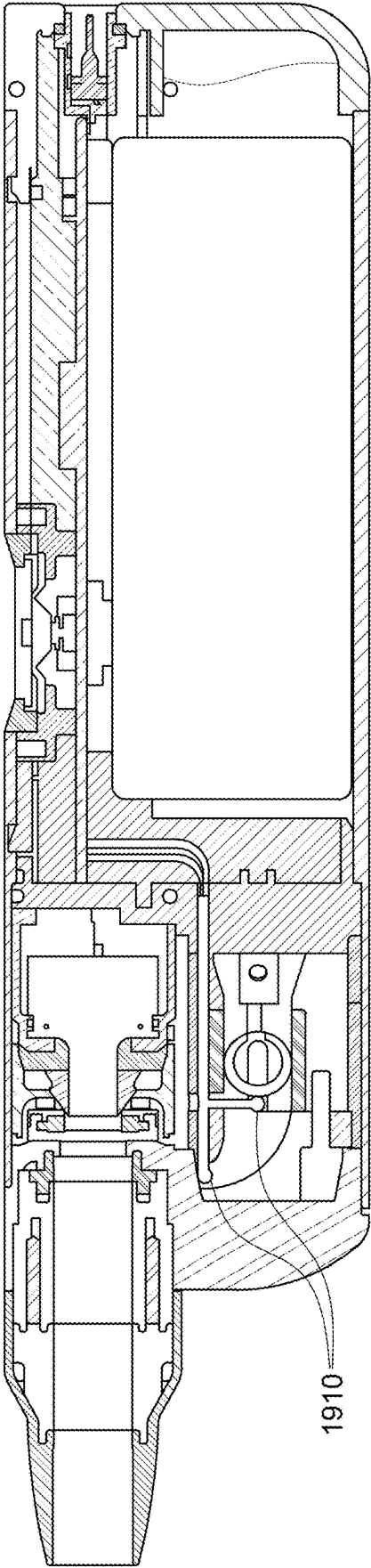


FIG. 40

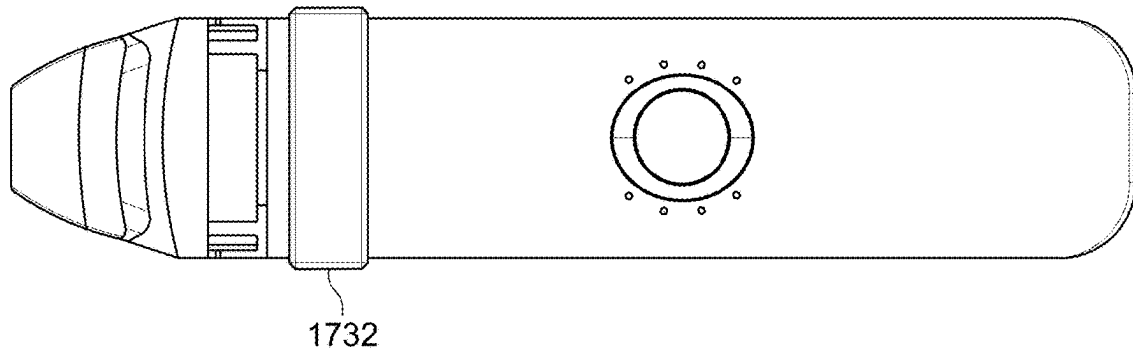


FIG. 41A

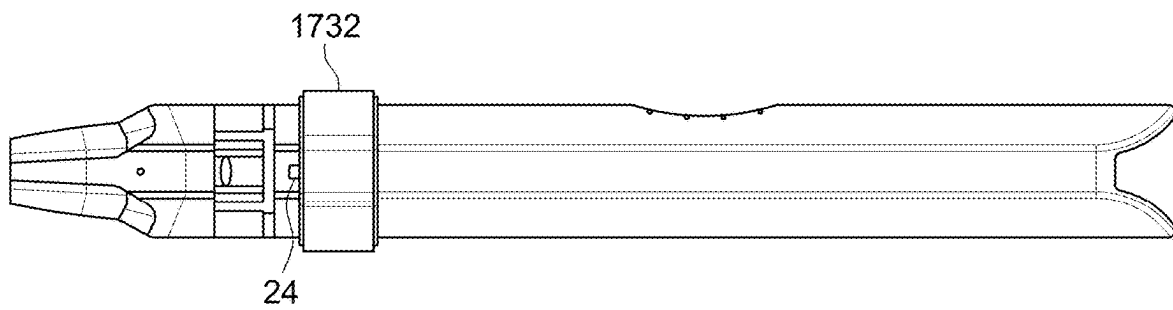


FIG. 41B

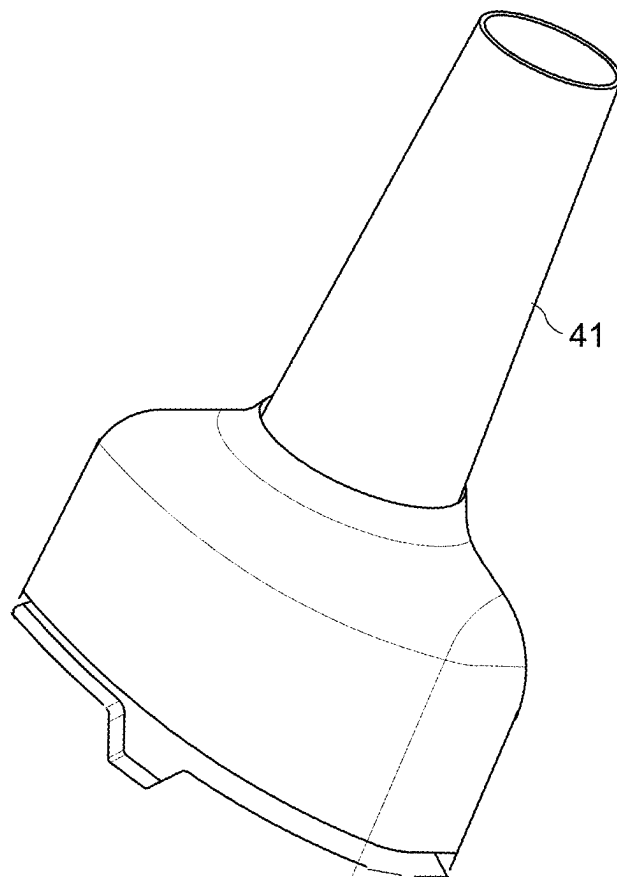


FIG. 42

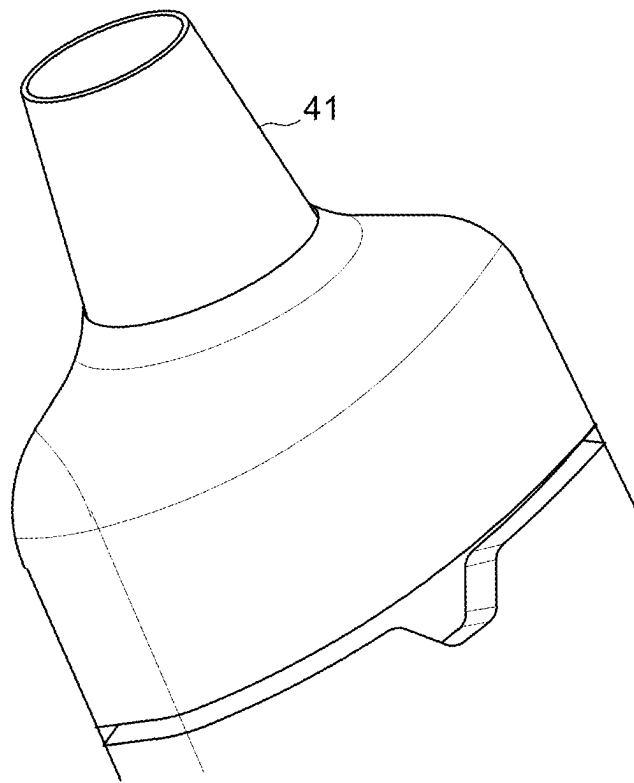


FIG. 43

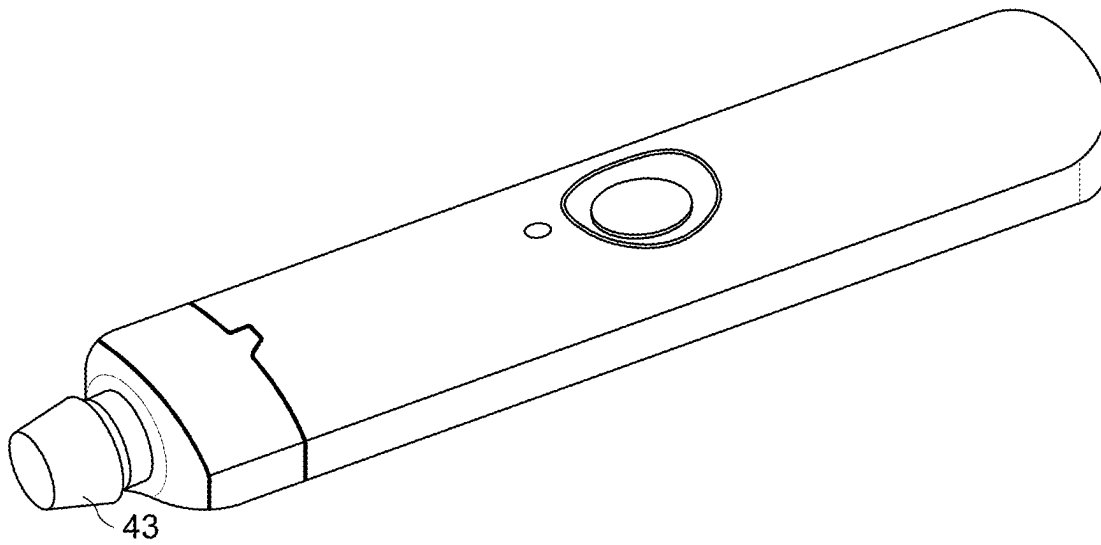
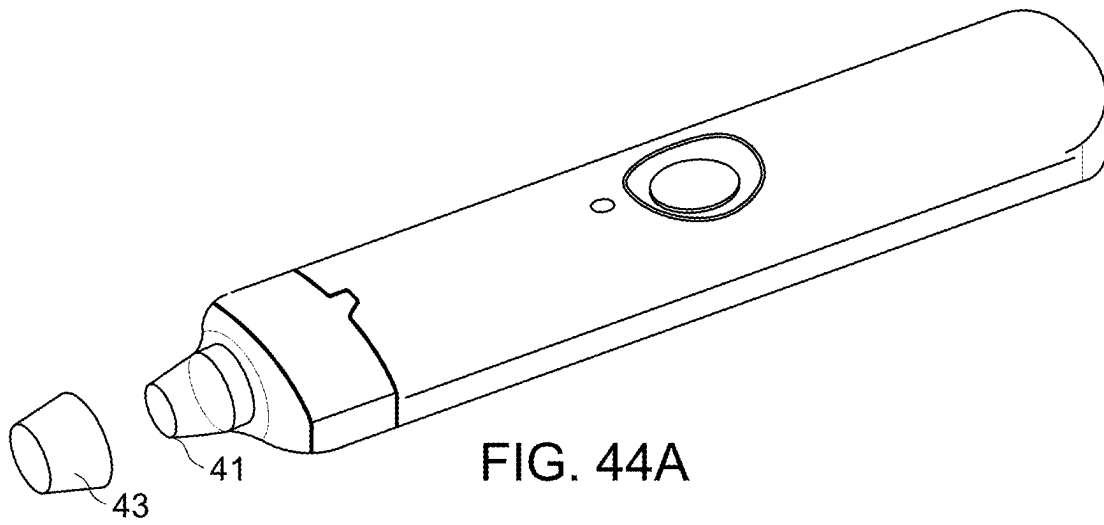


FIG. 44B

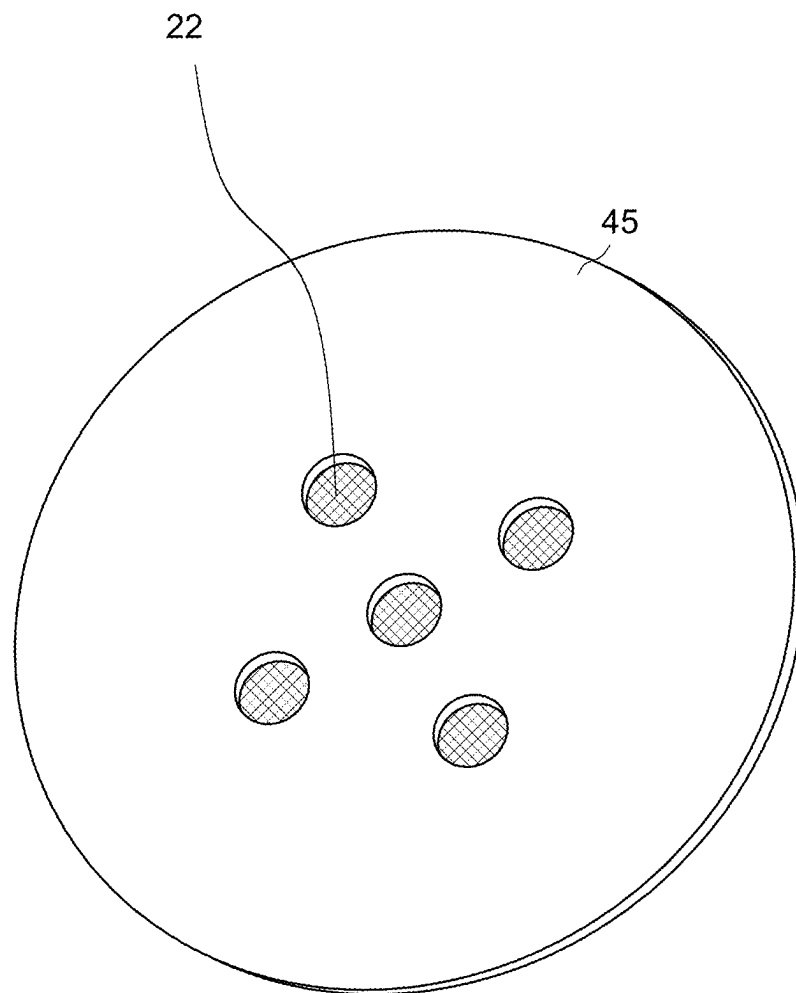


FIG. 45

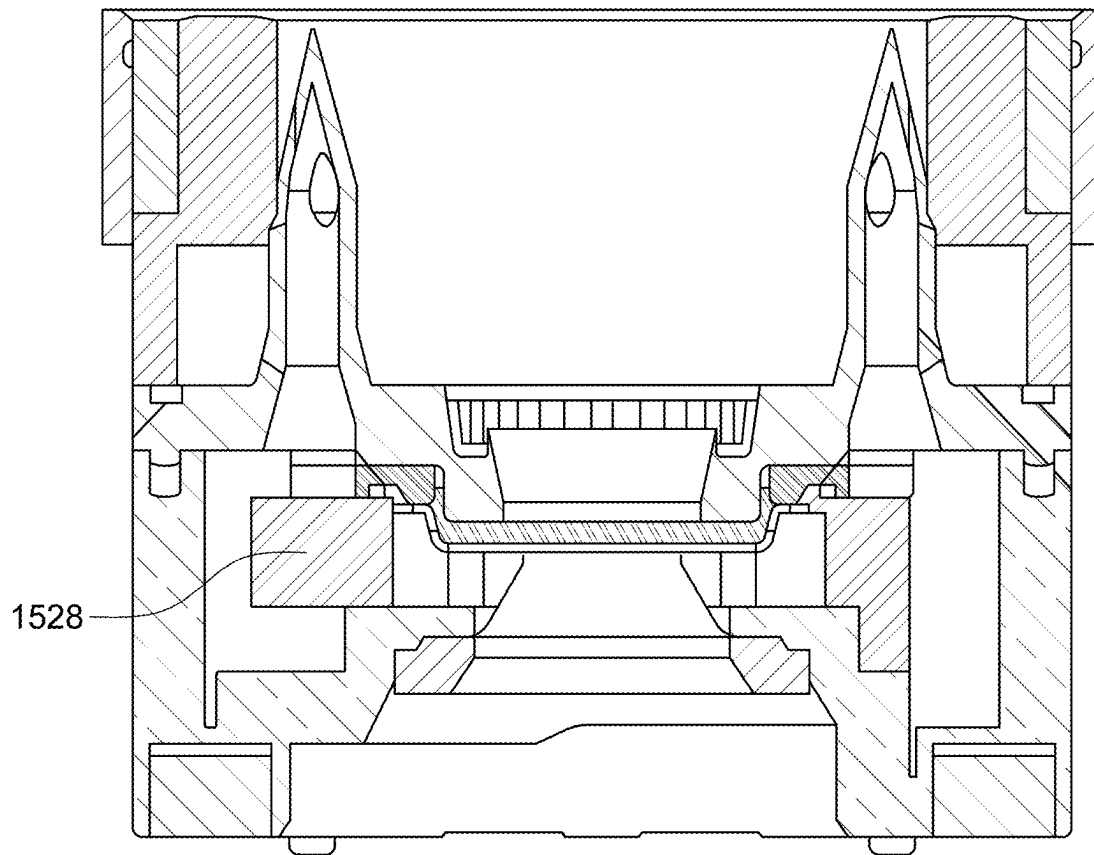
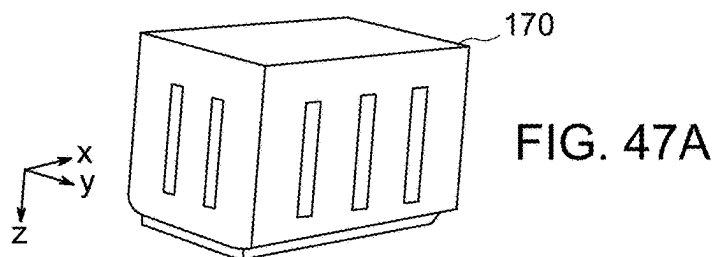
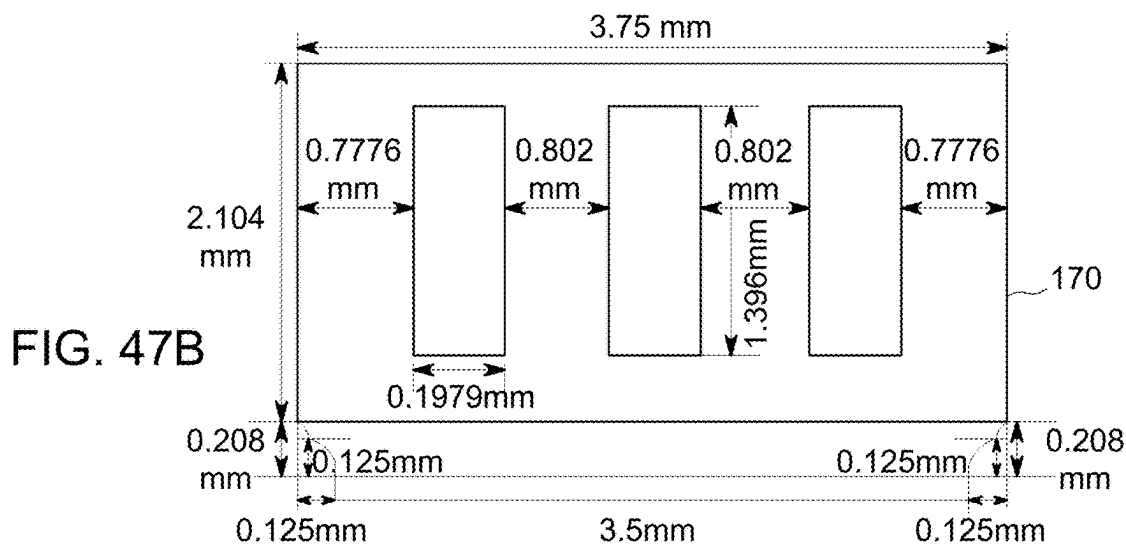


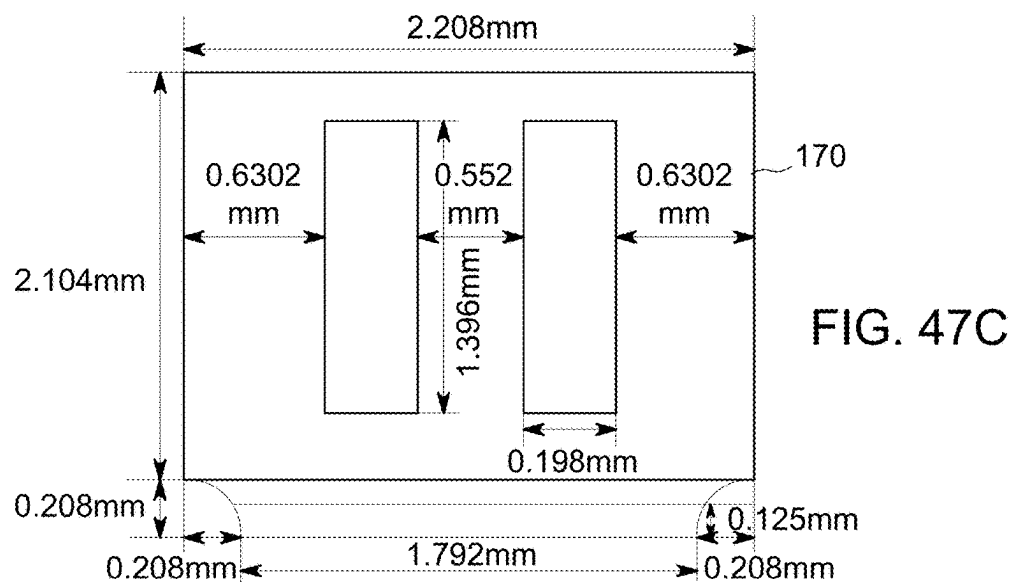
FIG. 46



(a) Structure diagram of the vibrating member tip



(b) Dimensional drawing of vibrating member tip on the XZ surface



(c) Dimensional drawing of vibrating member tip on the YZ surface

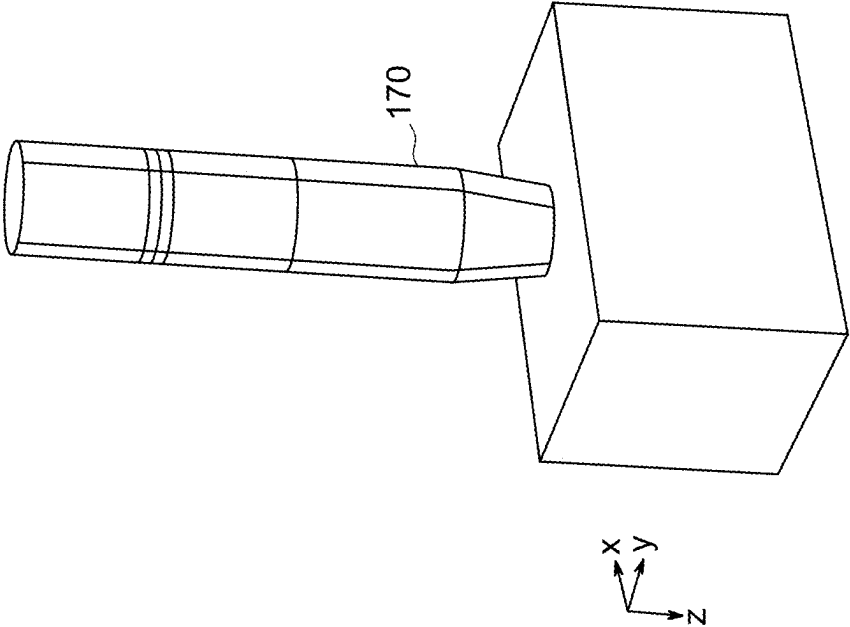


FIG. 48A

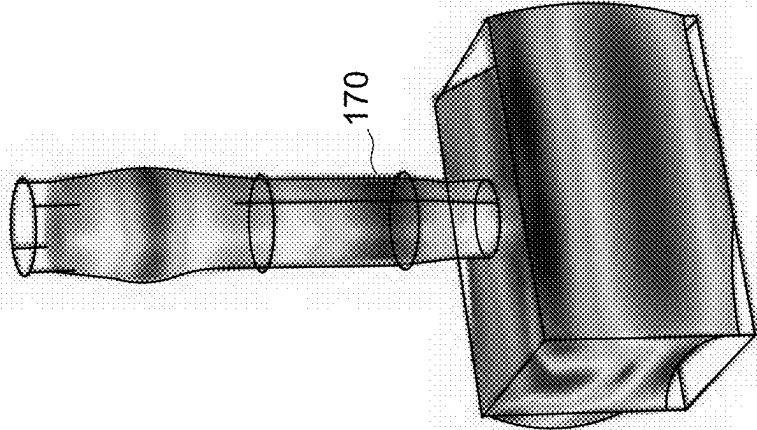


FIG. 48B

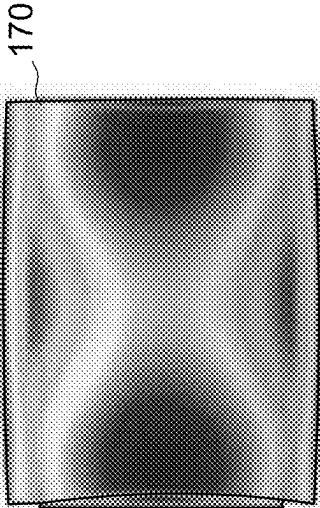


FIG. 48C

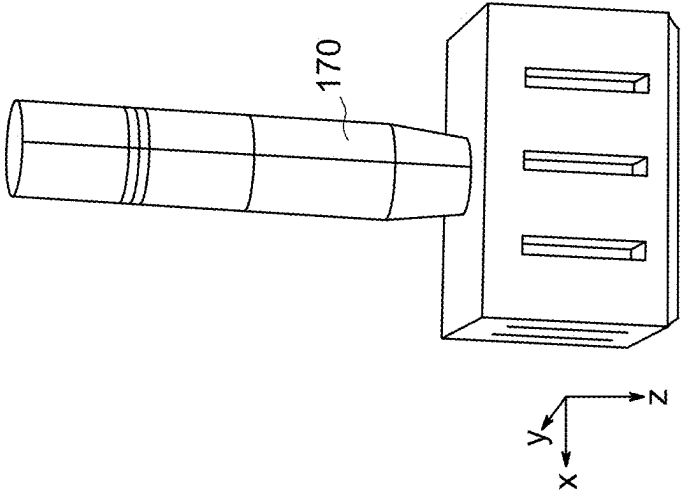


FIG. 49A

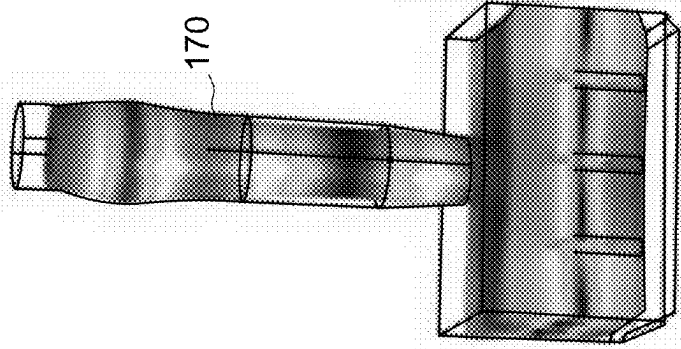


FIG. 49B

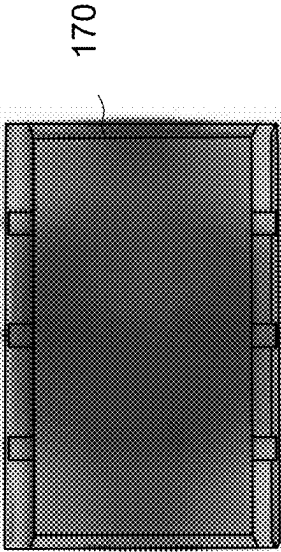


FIG. 49C

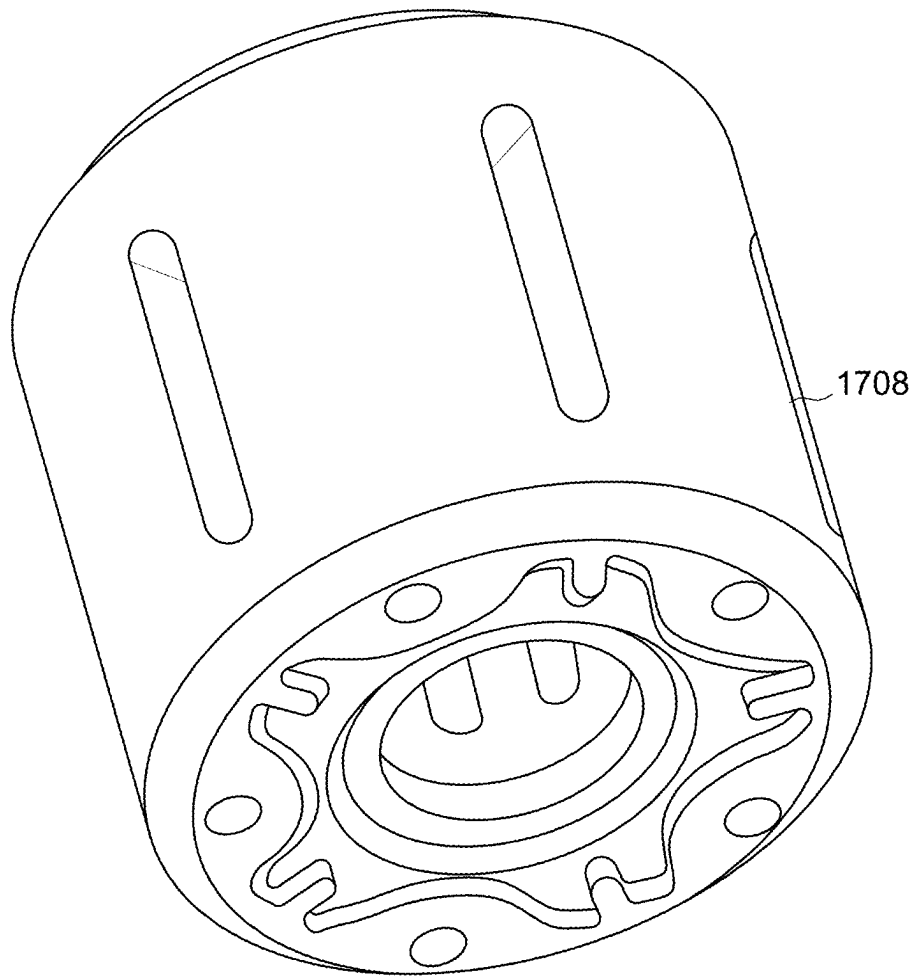


FIG. 50

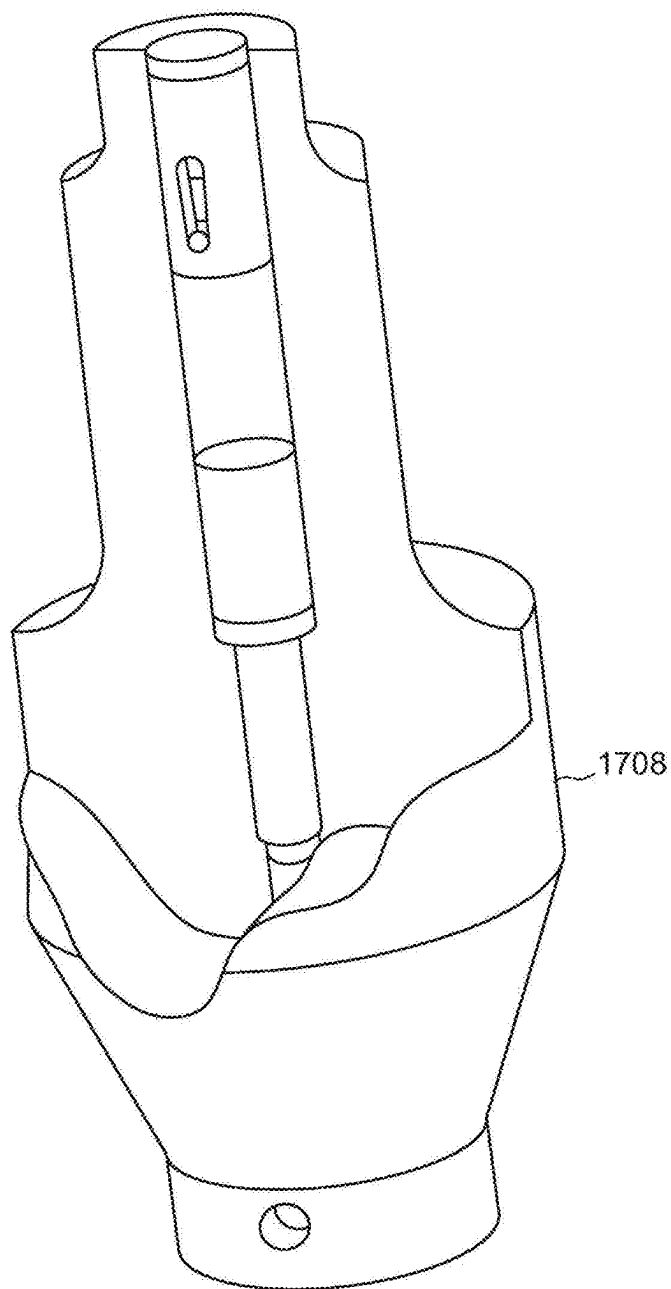


FIG. 51

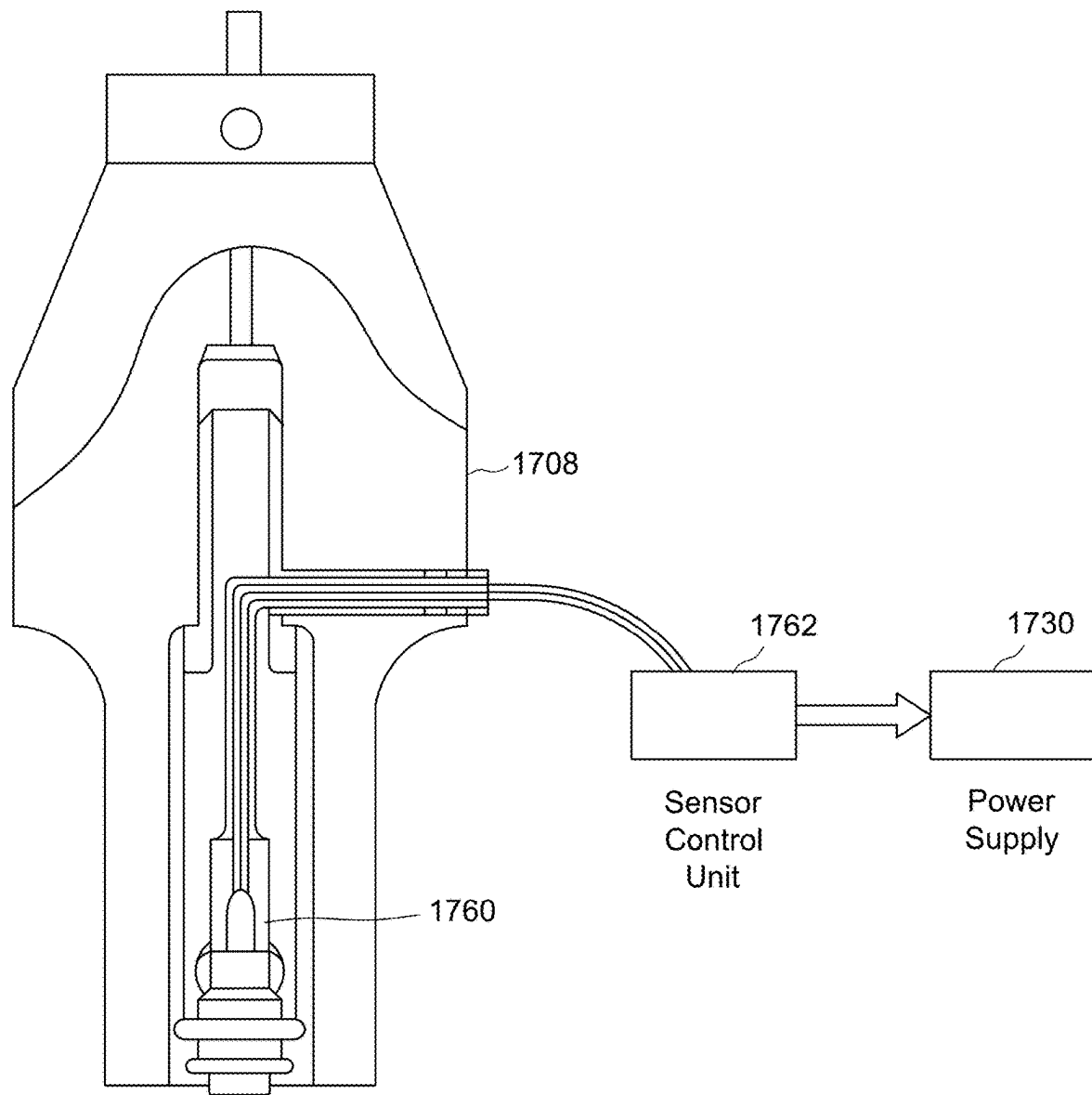


FIG. 52

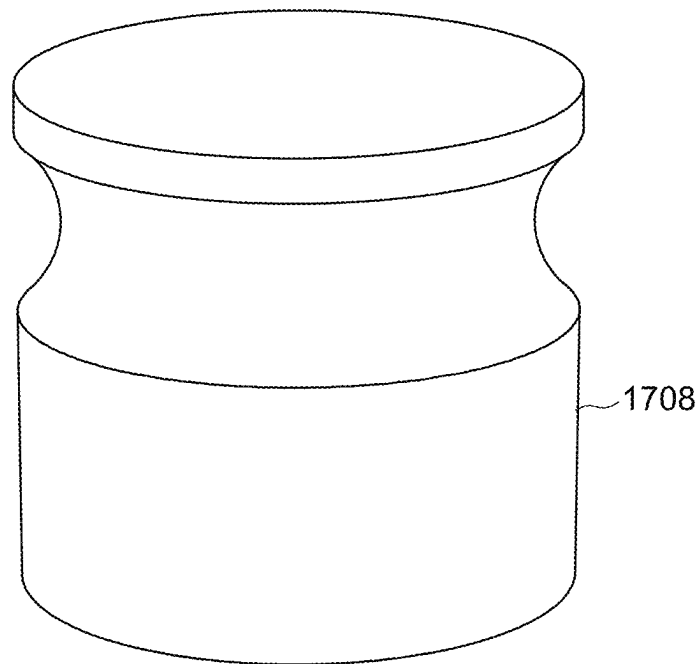


FIG. 53A

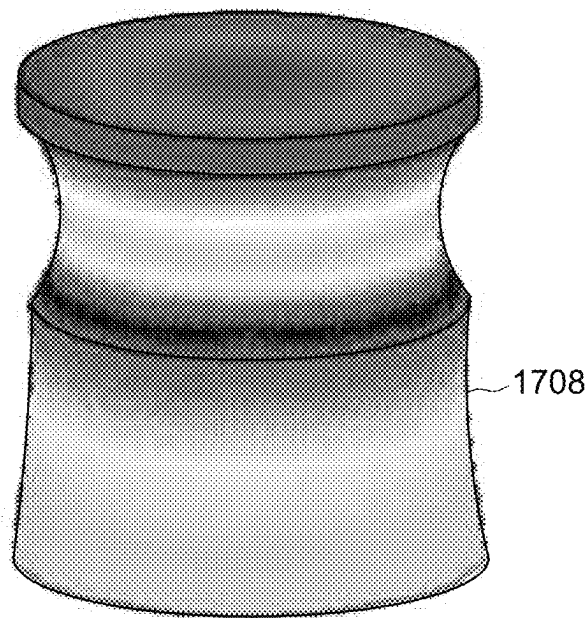


FIG. 53B

FIG. 54A

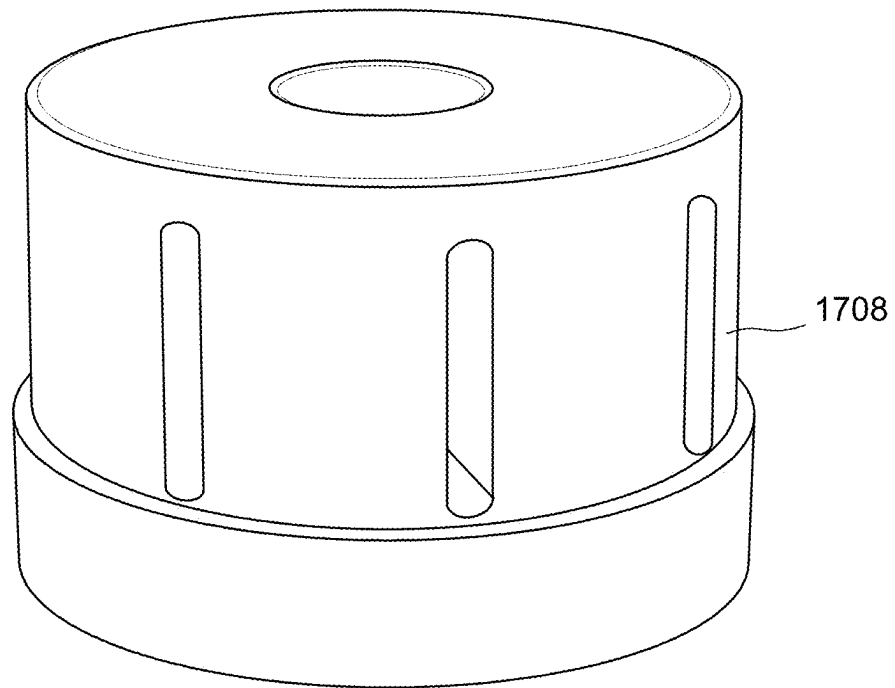


FIG. 54B

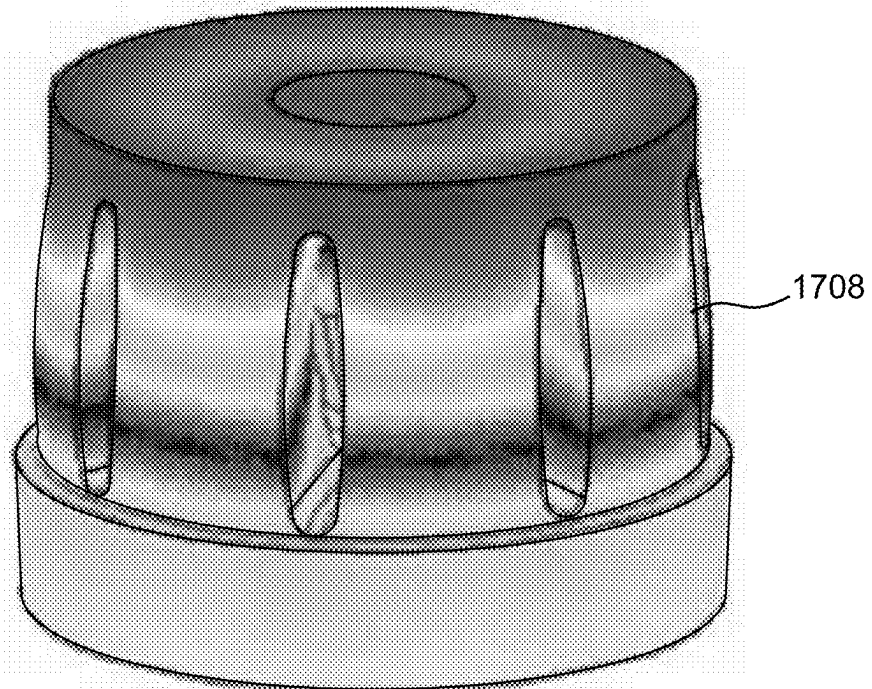


FIG. 55A

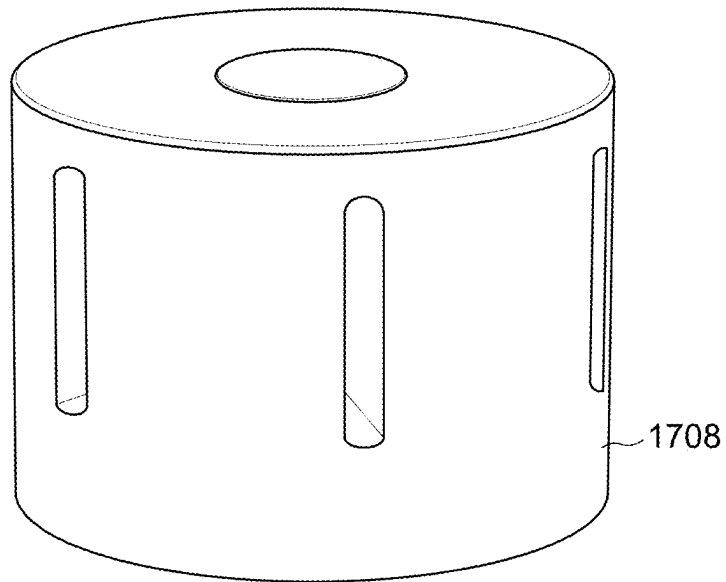
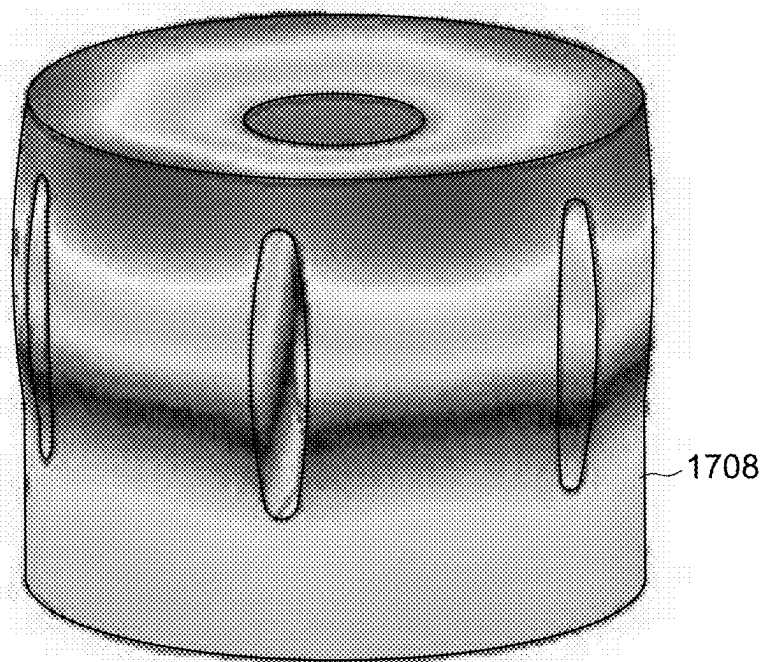


FIG. 55B



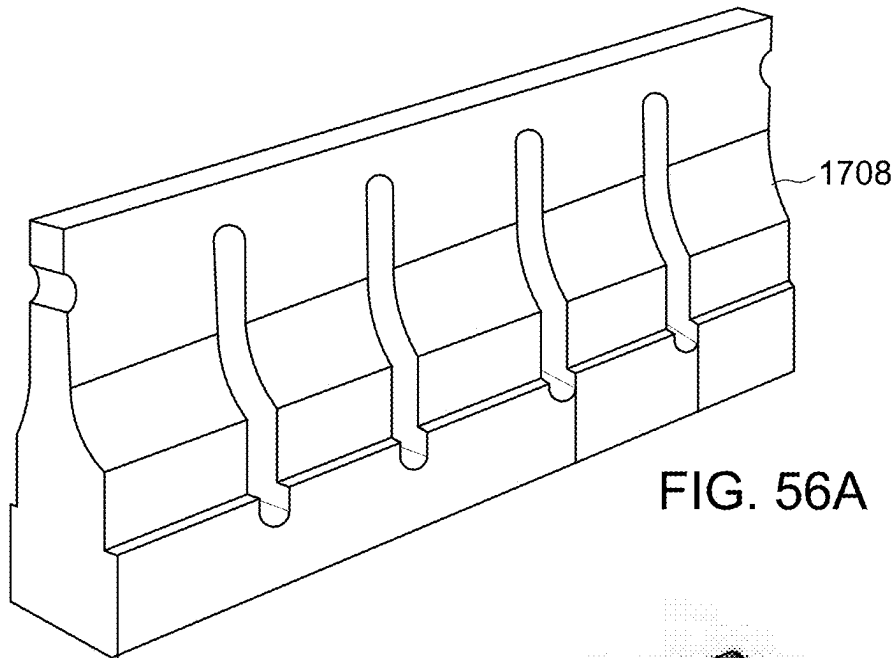


FIG. 56A

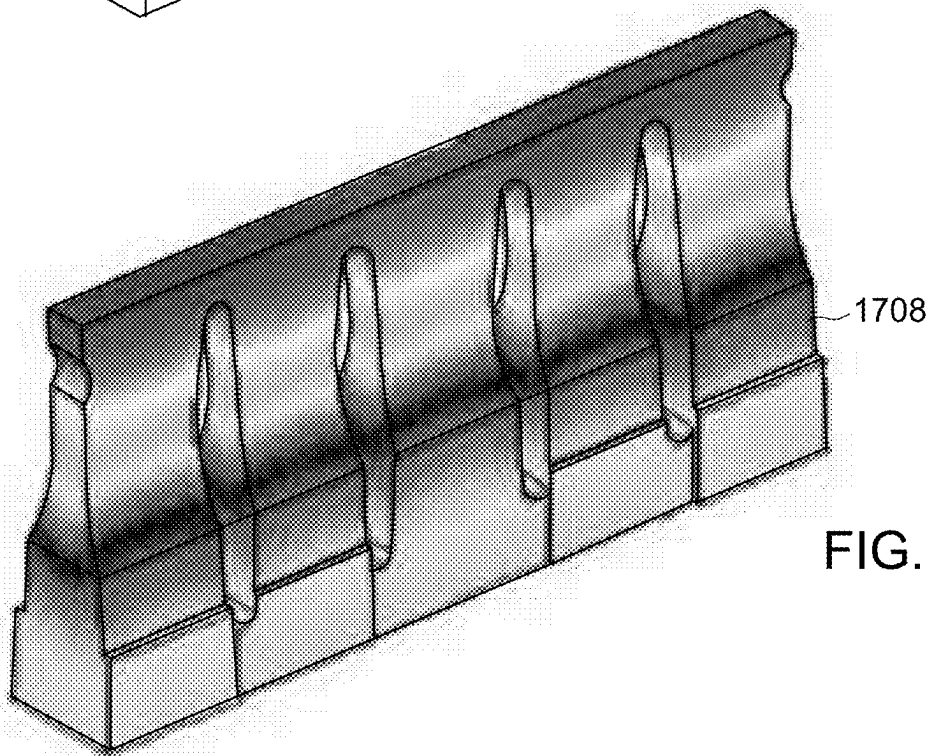


FIG. 56B

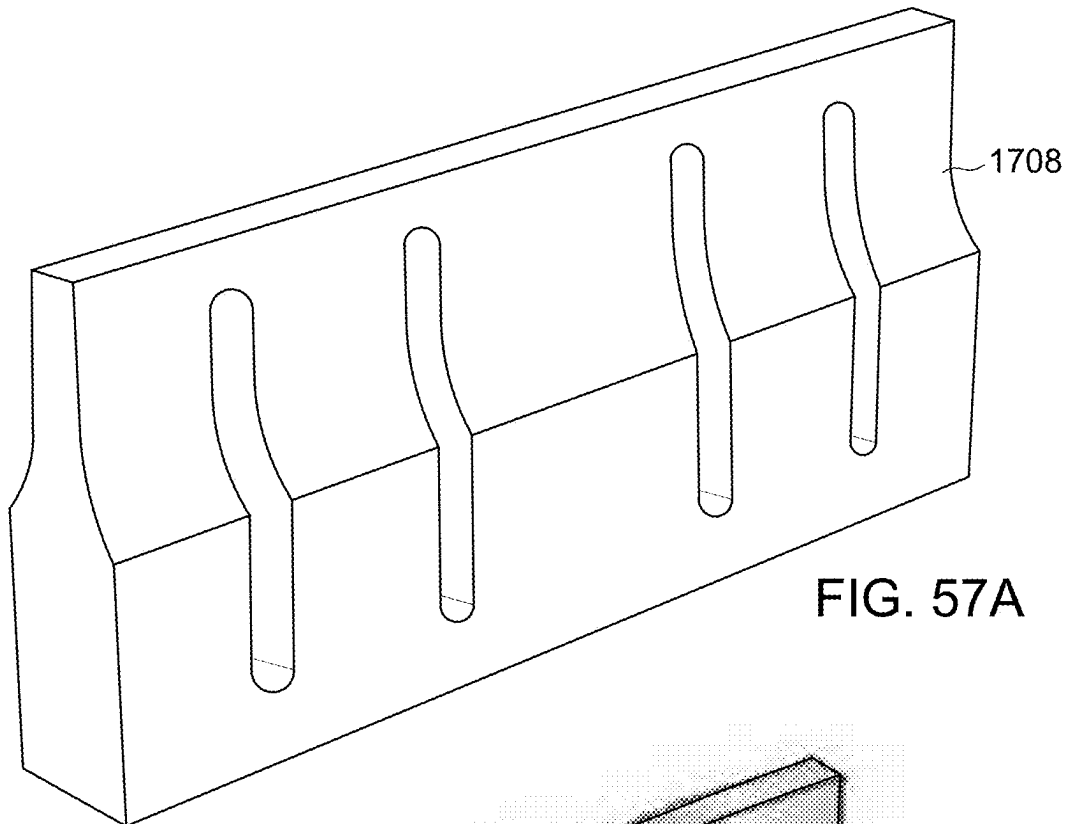


FIG. 57A

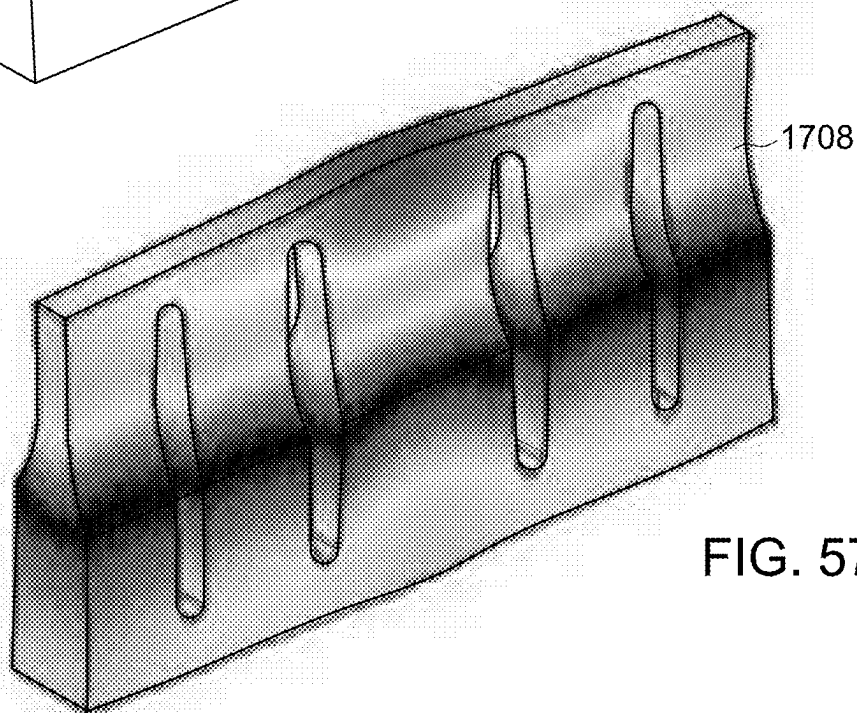


FIG. 57B

FIG. 58A

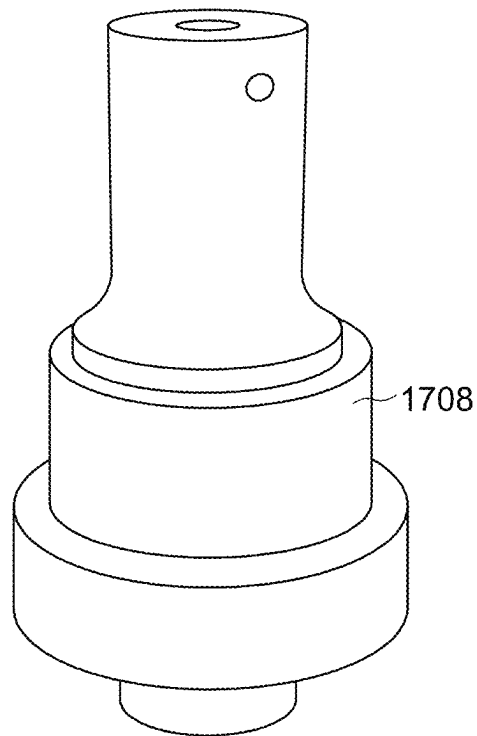
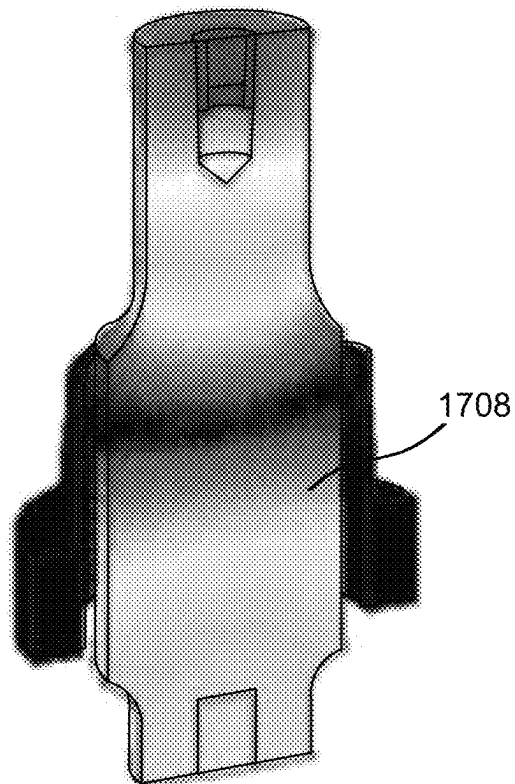


FIG. 58B



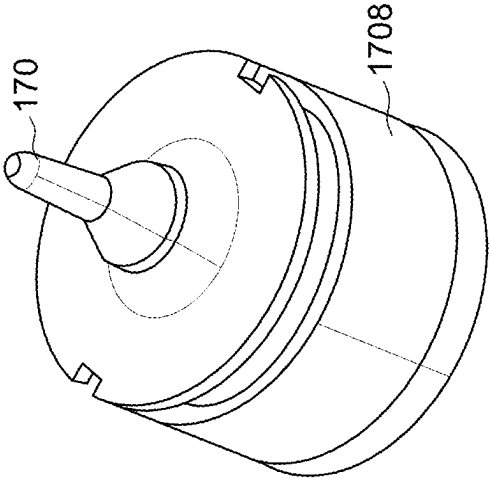


FIG. 59A

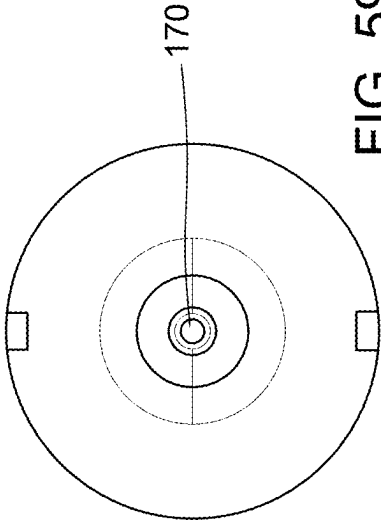


FIG. 59B

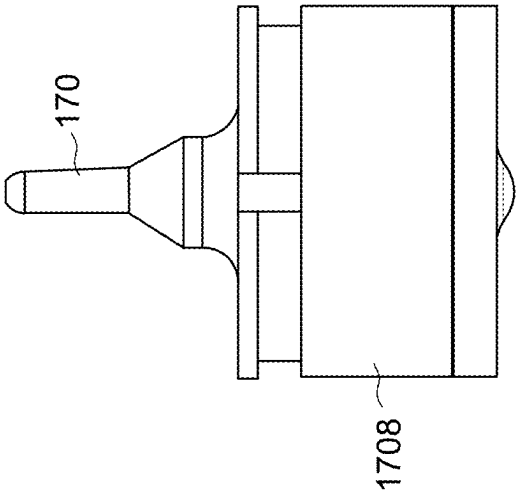


FIG. 59C

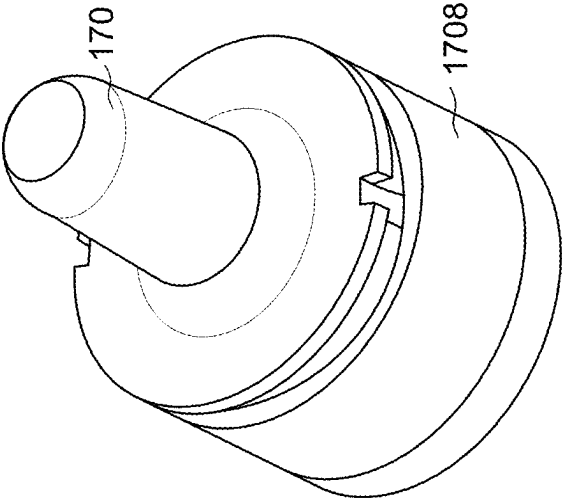


FIG. 60A

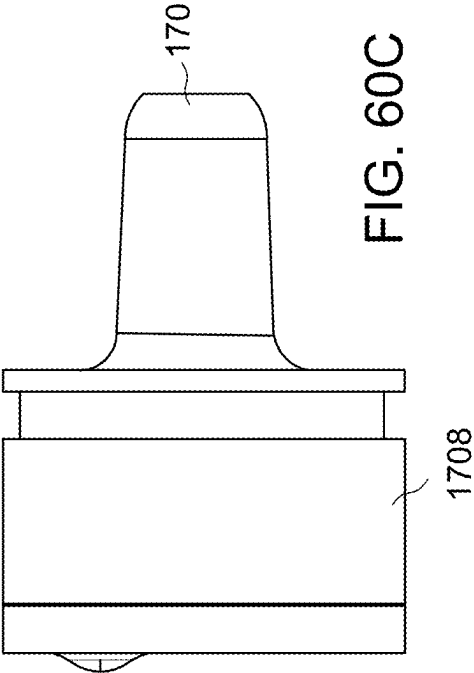


FIG. 60C

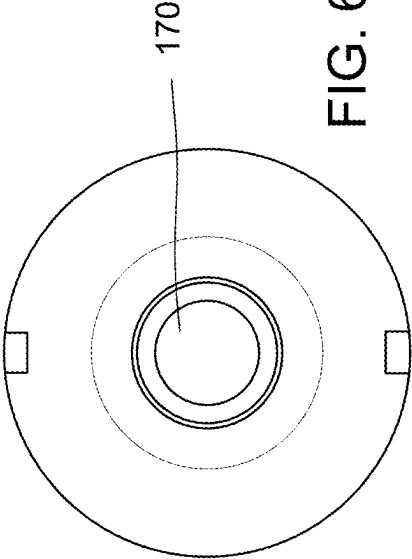


FIG. 60B

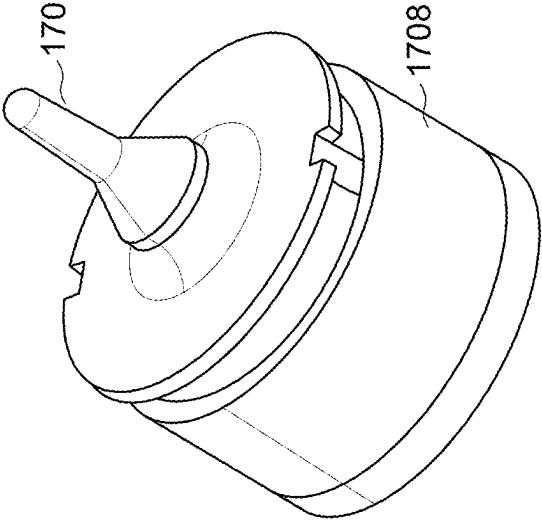


FIG. 61A

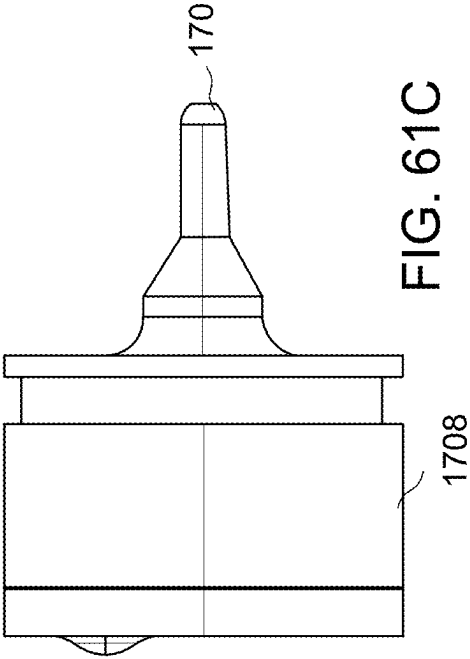


FIG. 61C

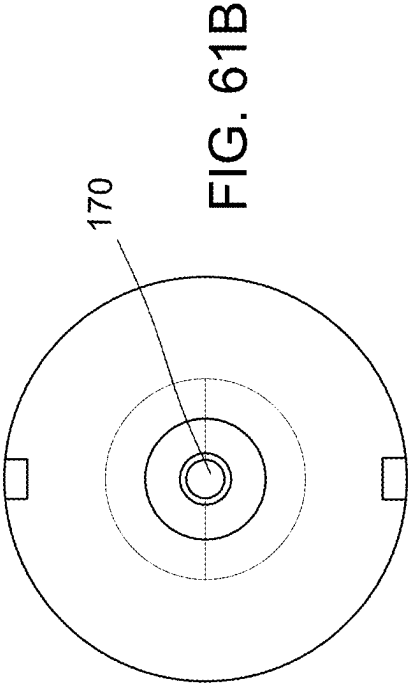


FIG. 61B

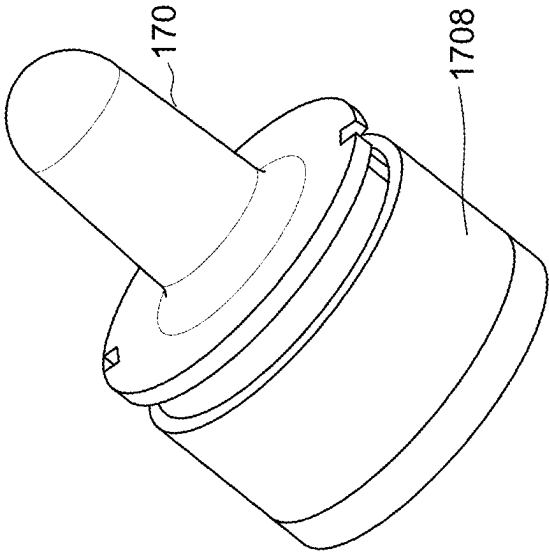


FIG. 62A

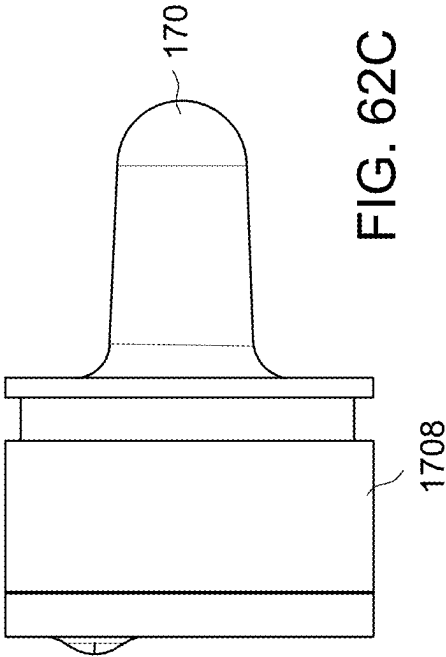


FIG. 62C

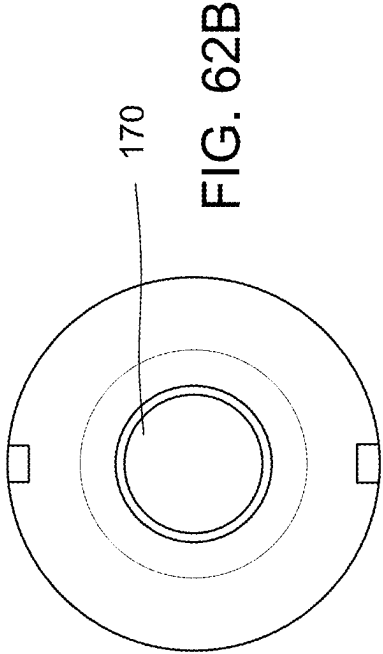
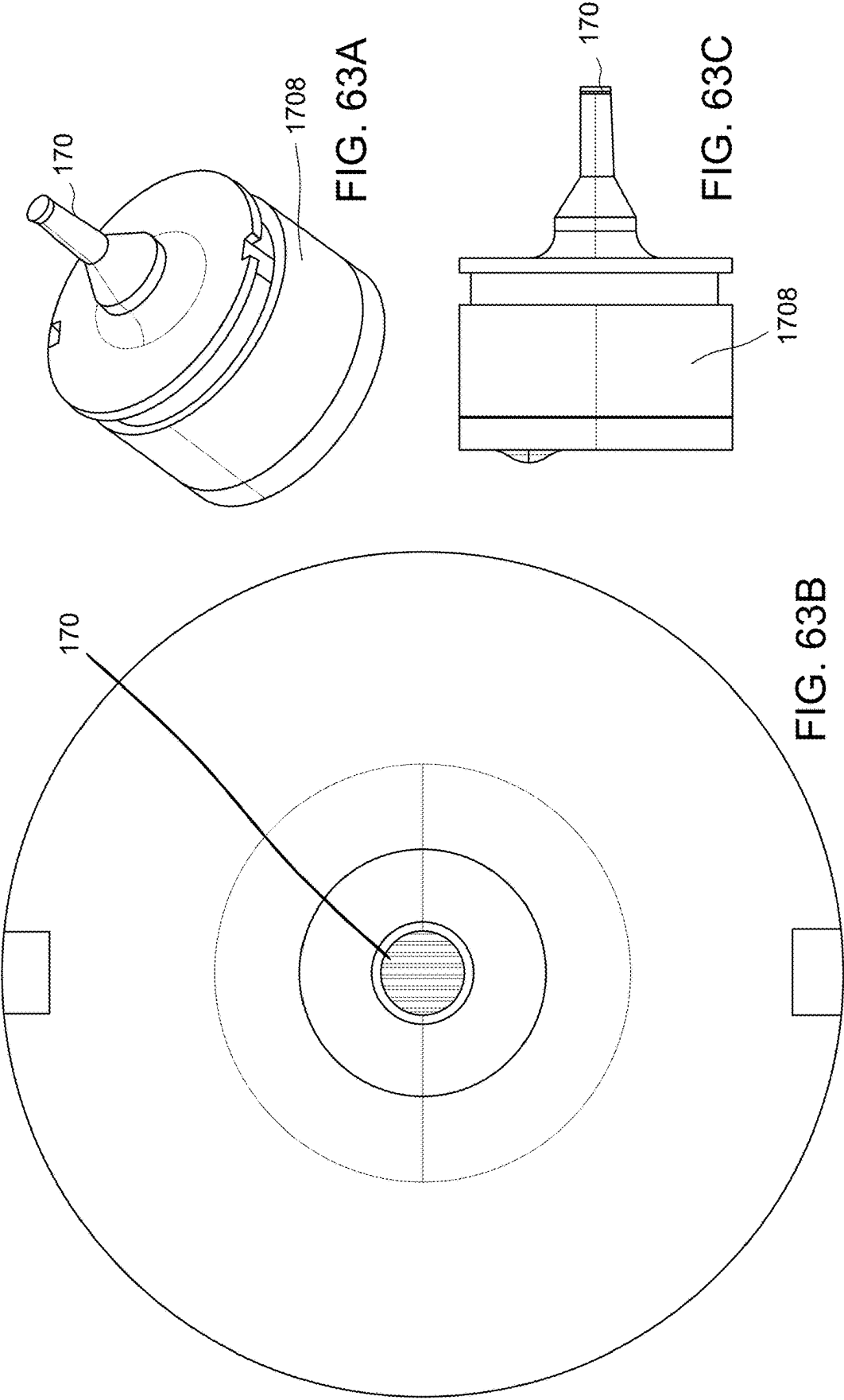


FIG. 62B



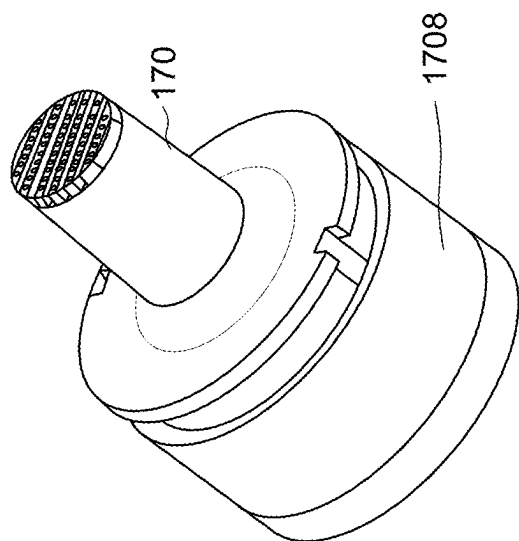


FIG. 64A

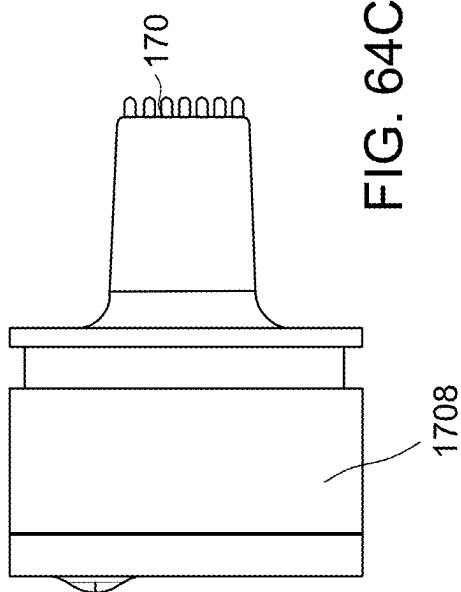


FIG. 64C

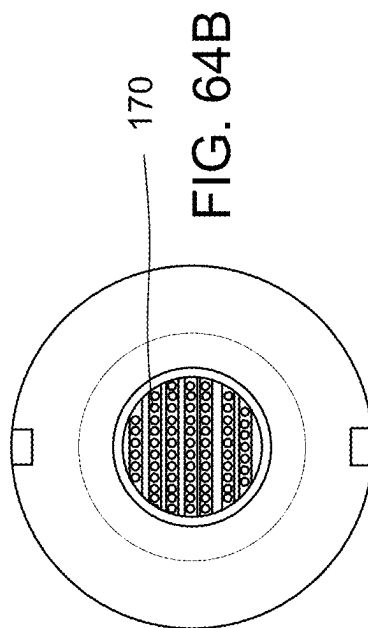


FIG. 64B

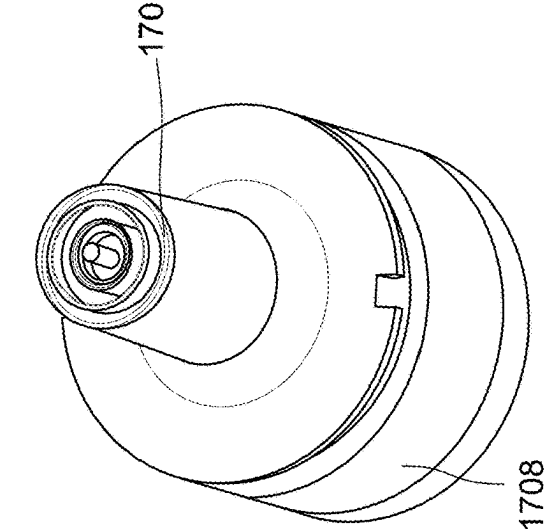
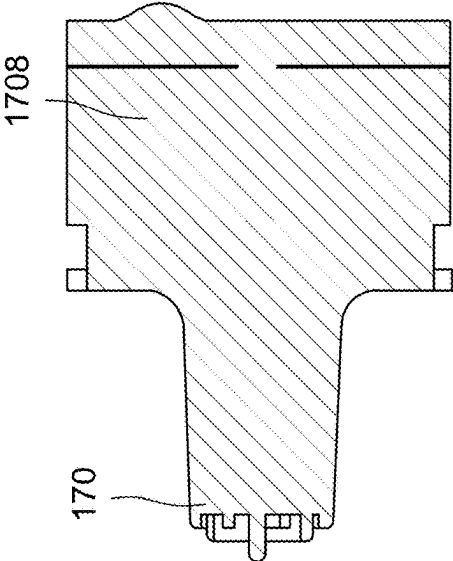


FIG. 65A



SECTION A_A

FIG. 65D

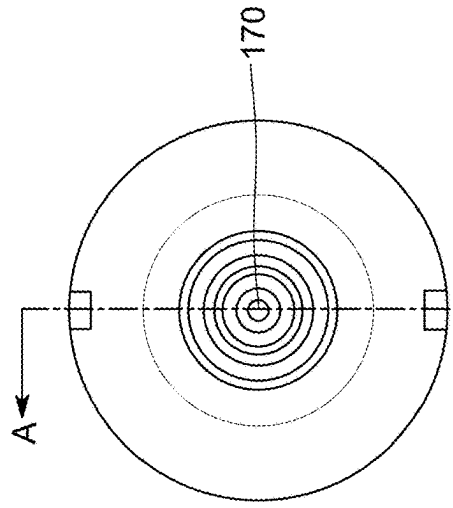


FIG. 65B

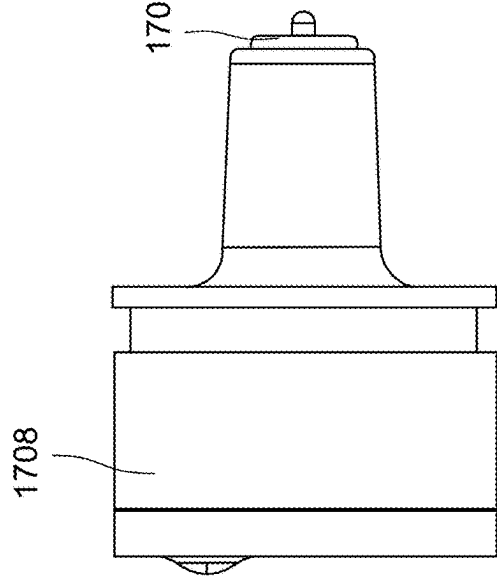


FIG. 65C

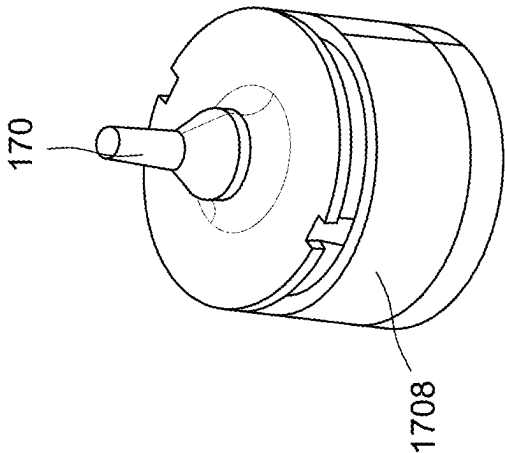


FIG. 66A

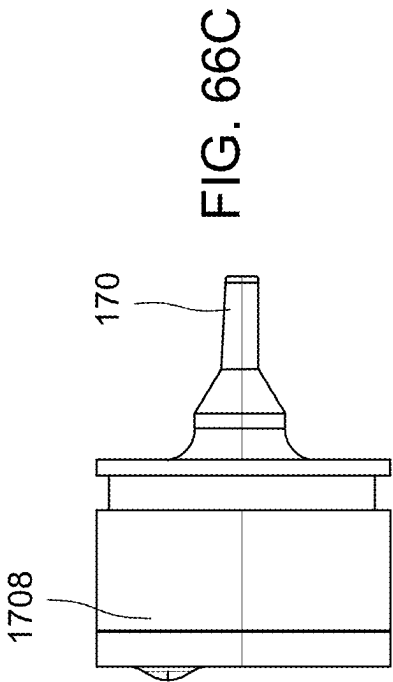


FIG. 66C

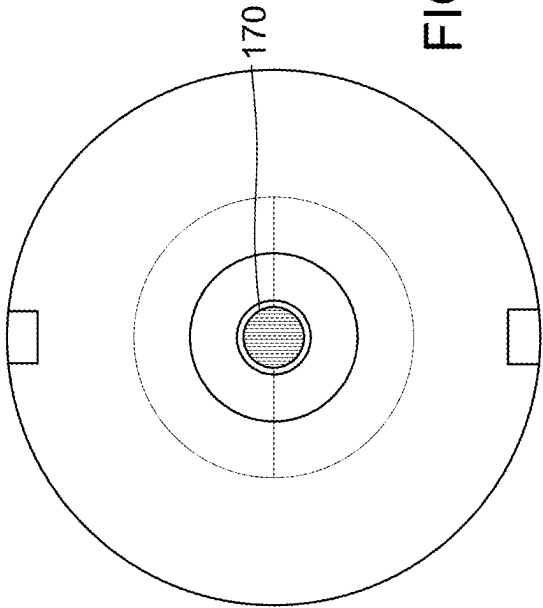


FIG. 66B

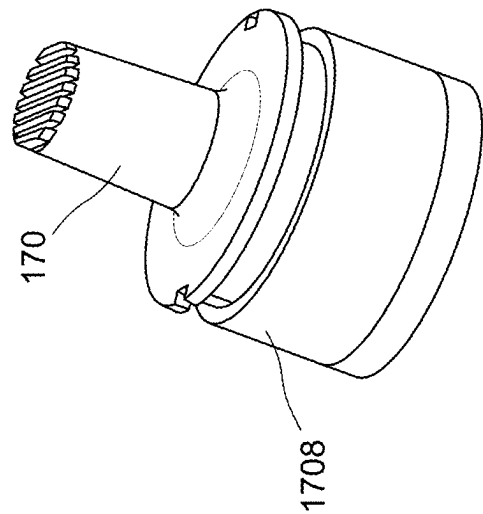


FIG. 67A

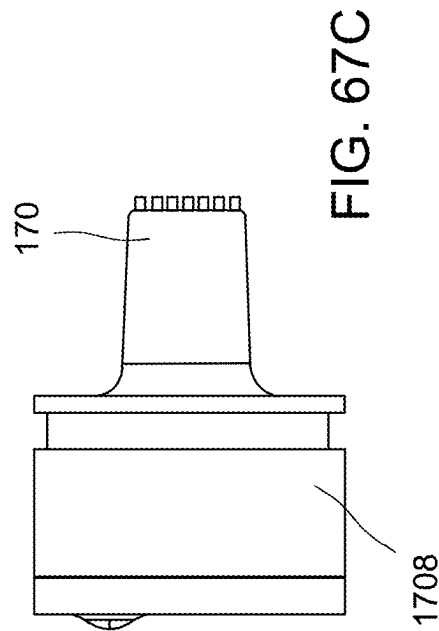


FIG. 67C

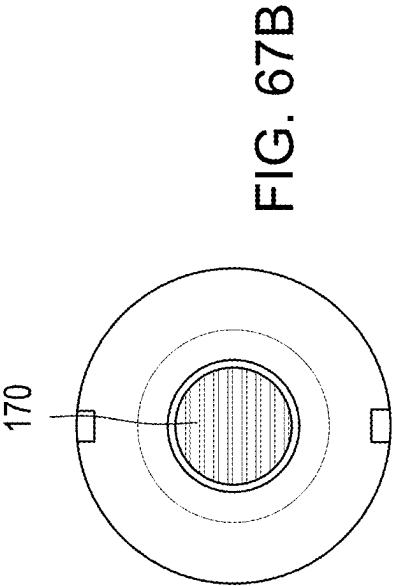


FIG. 67B

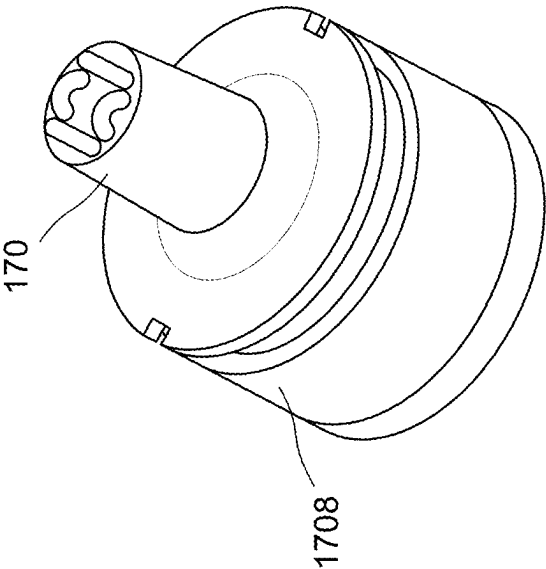


FIG. 68A

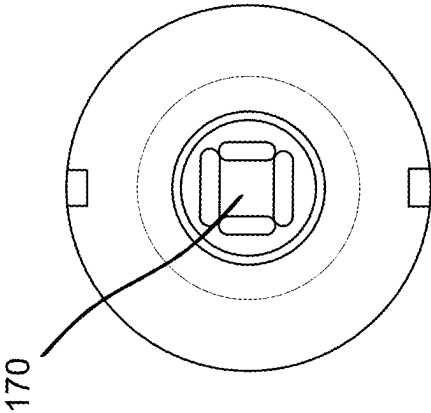


FIG. 68B

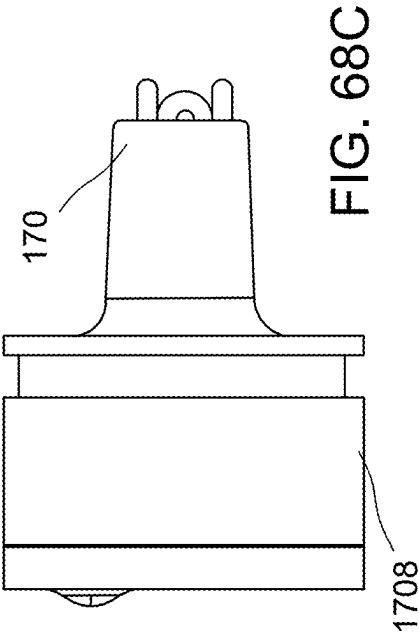


FIG. 68C

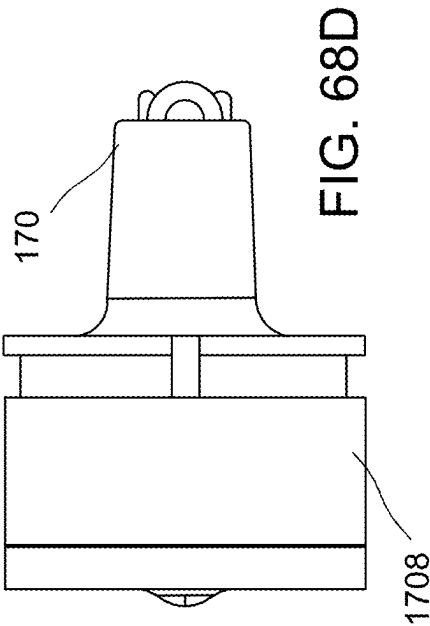


FIG. 68D

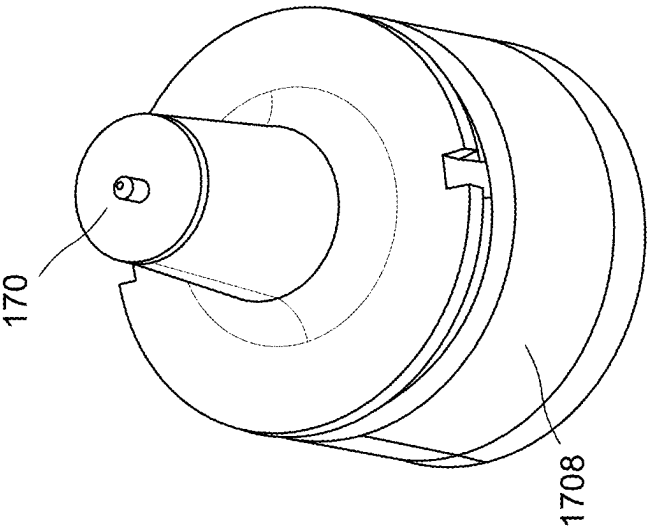


FIG. 69A

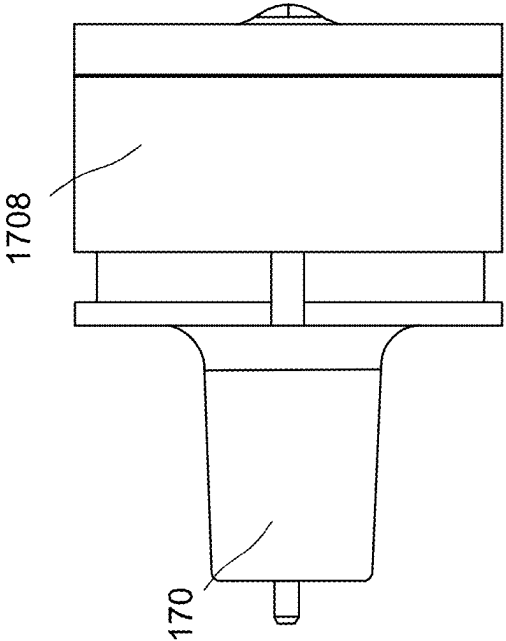


FIG. 69B

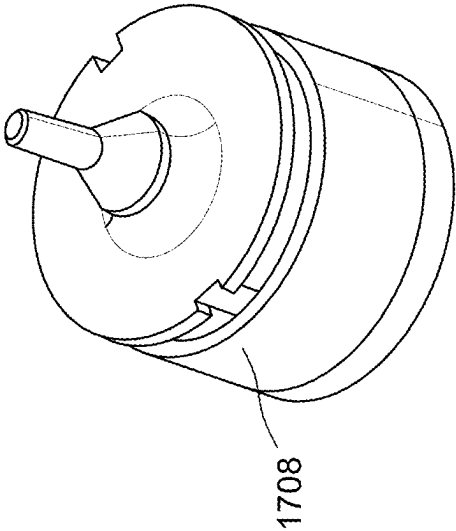


FIG. 70A

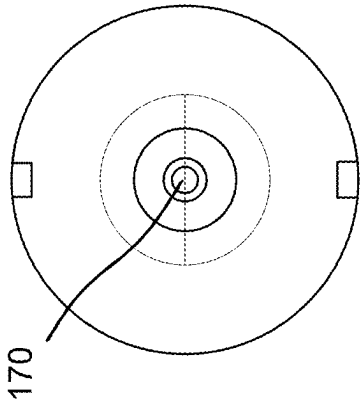


FIG. 70B

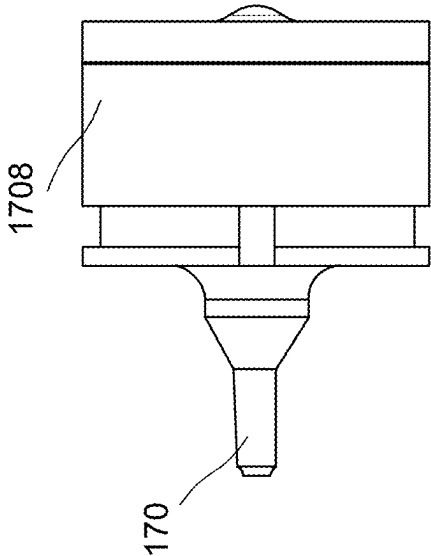


FIG. 70C

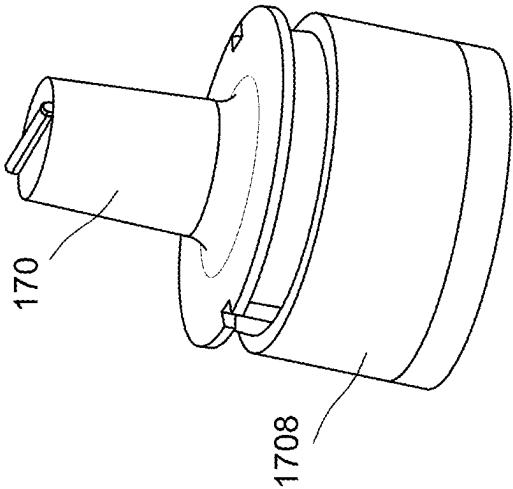


FIG. 71A

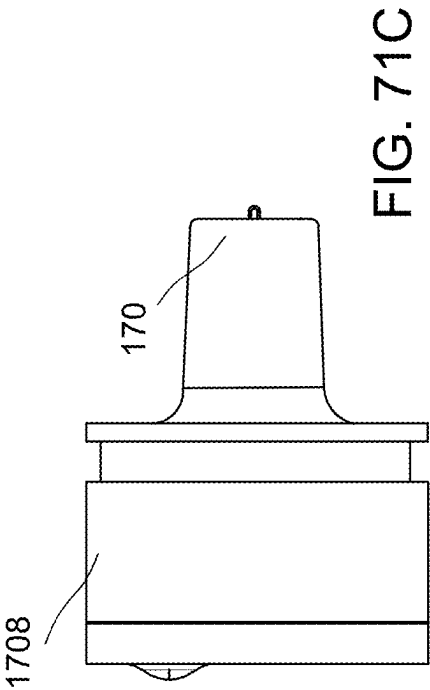


FIG. 71C

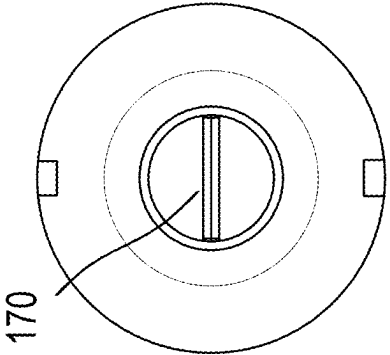


FIG. 71B

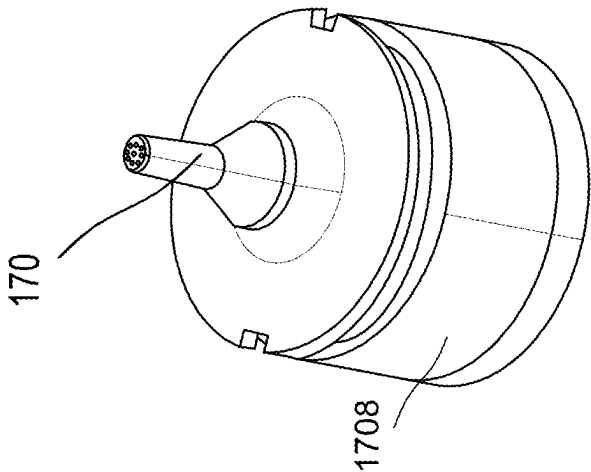


FIG. 72A

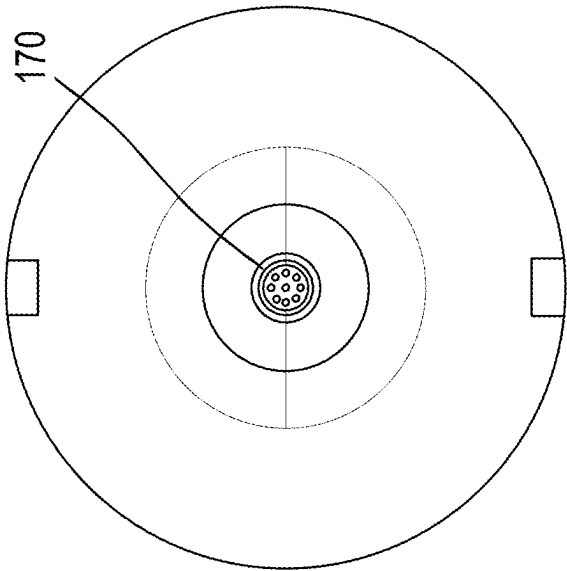


FIG. 72B

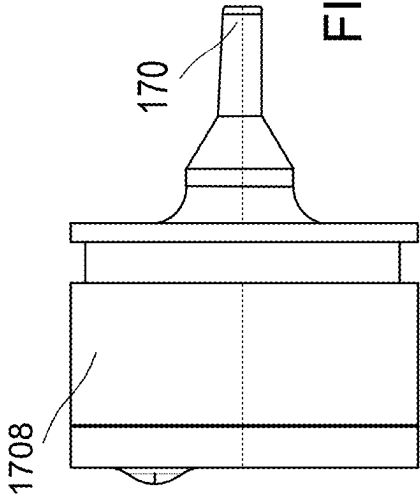


FIG. 72C

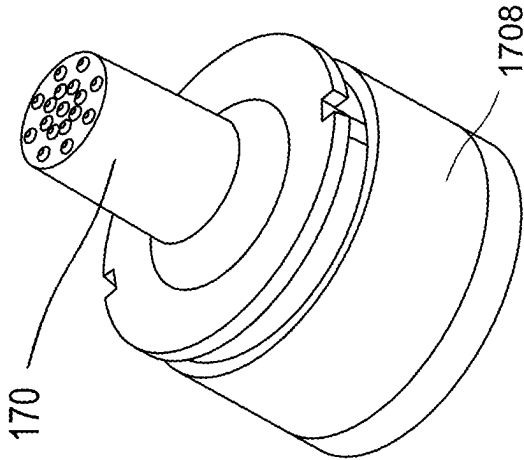


FIG. 73A

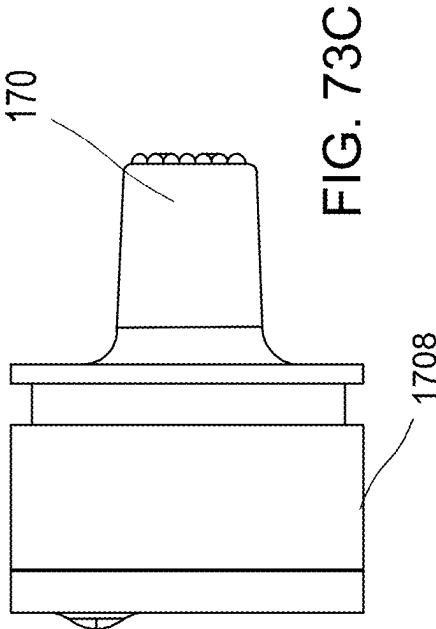


FIG. 73C

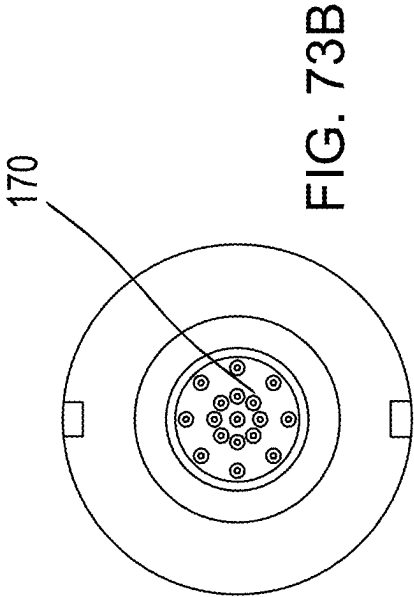


FIG. 73B

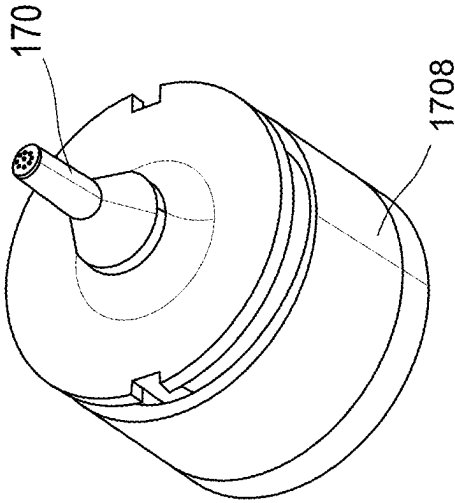


FIG. 74A

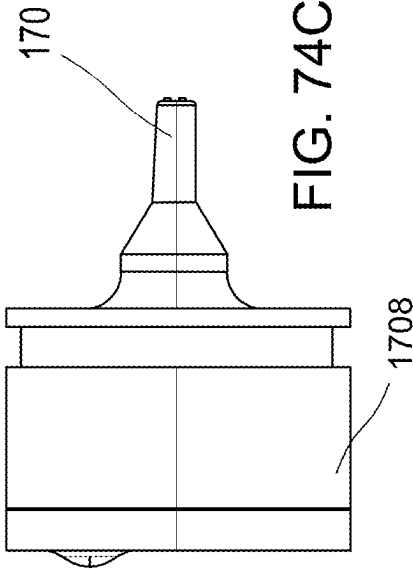


FIG. 74C

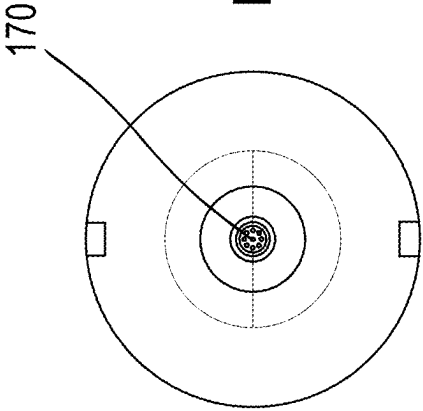


FIG. 74B

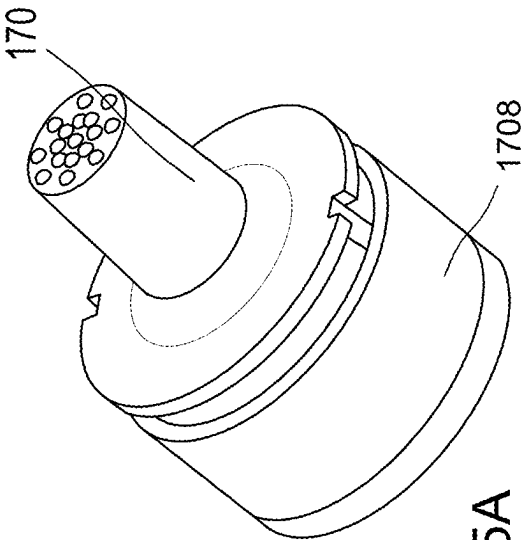


FIG. 75A

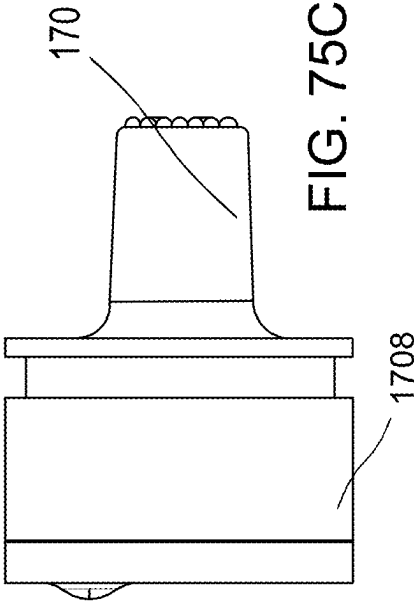


FIG. 75C

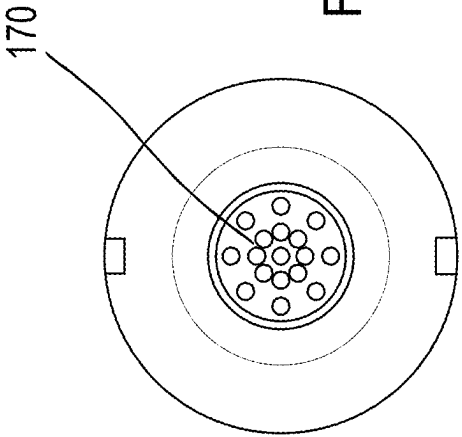


FIG. 75B

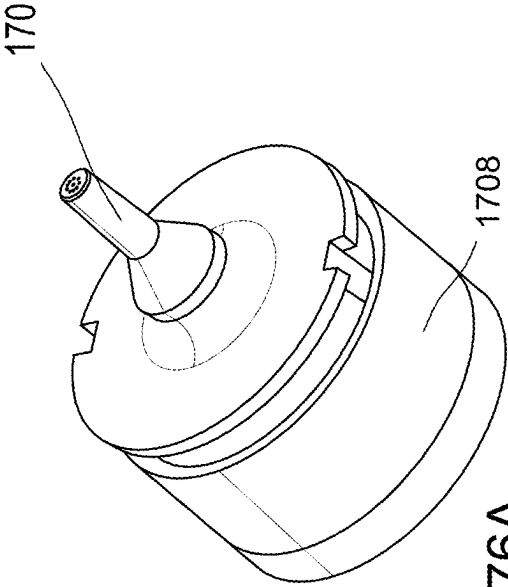


FIG. 76A

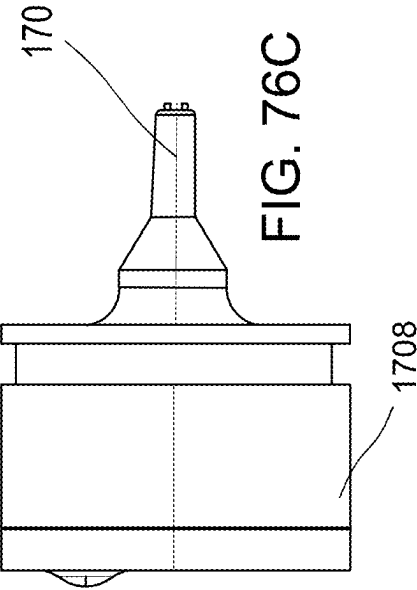


FIG. 76C

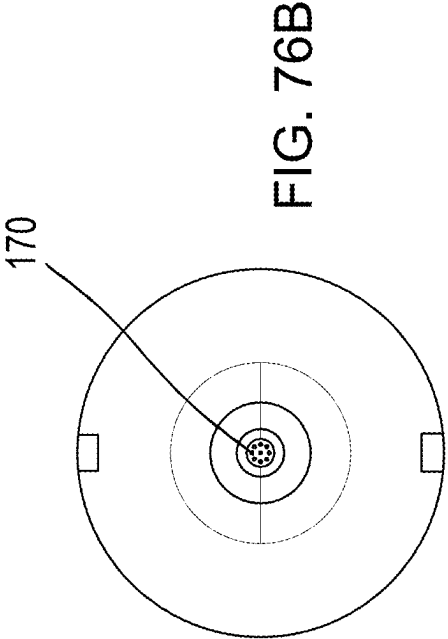


FIG. 76B

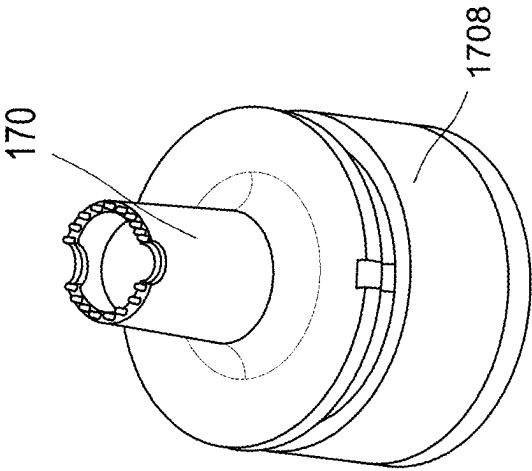


FIG. 77A

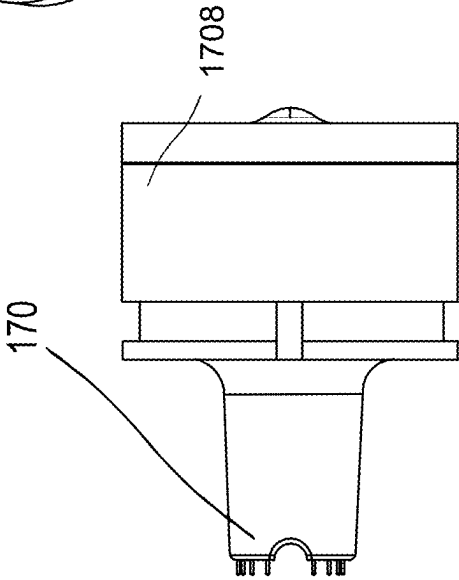


FIG. 77D

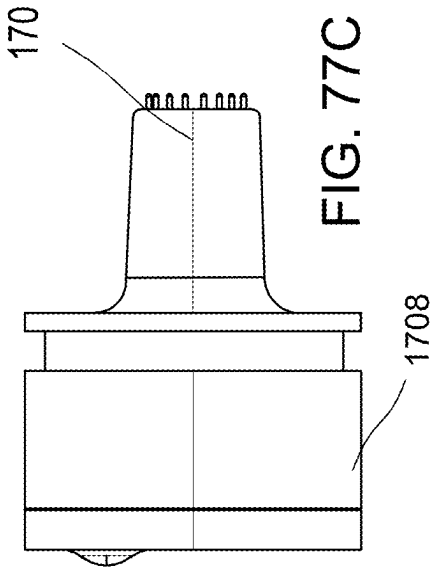


FIG. 77C

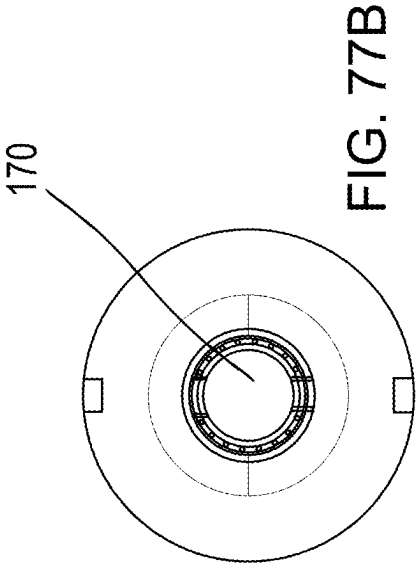


FIG. 77B

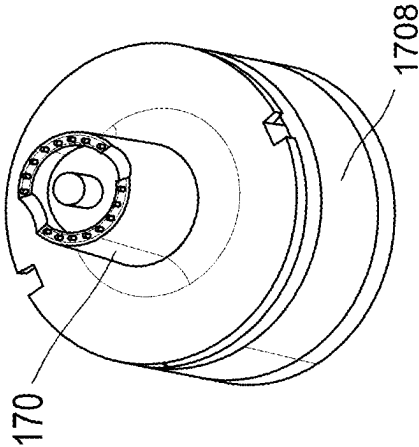


FIG. 78A

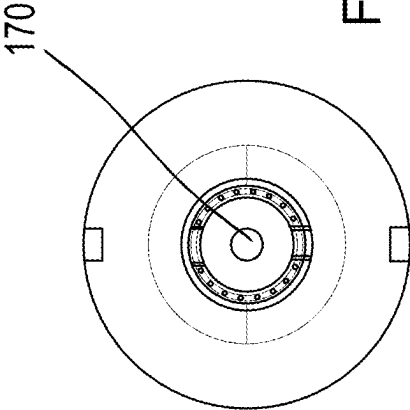


FIG. 78B

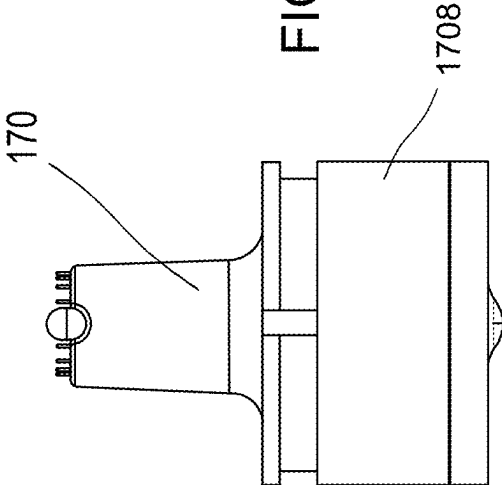


FIG. 78C

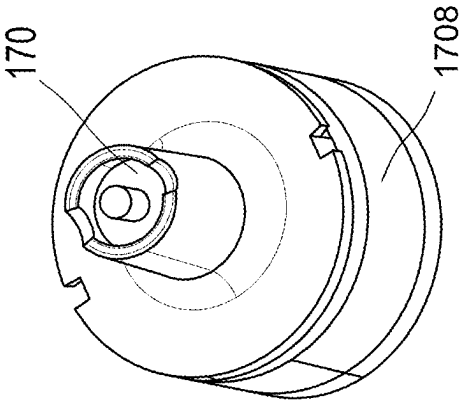


FIG. 79A

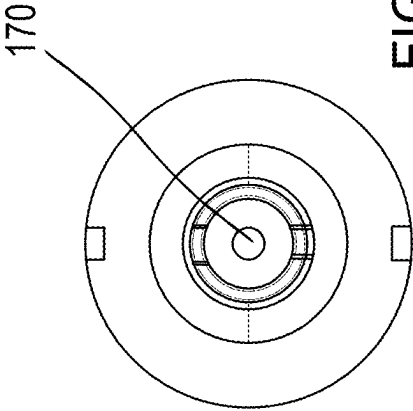


FIG. 79B

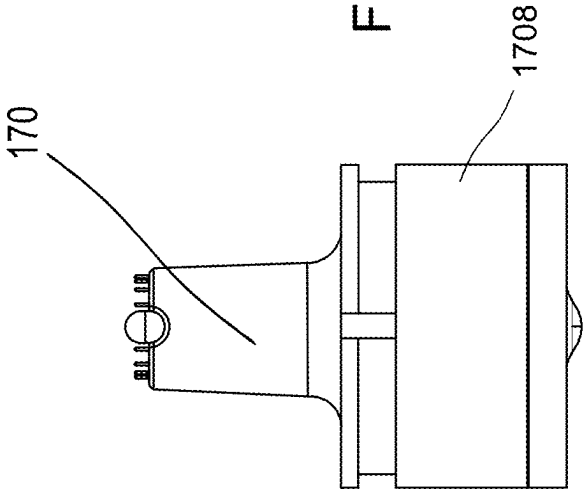
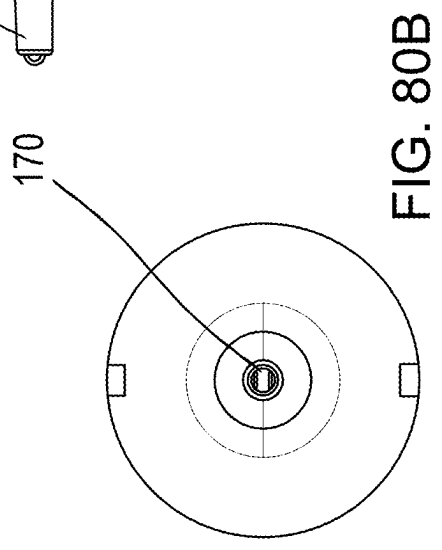
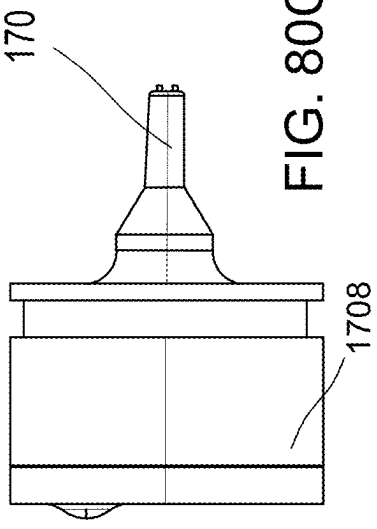
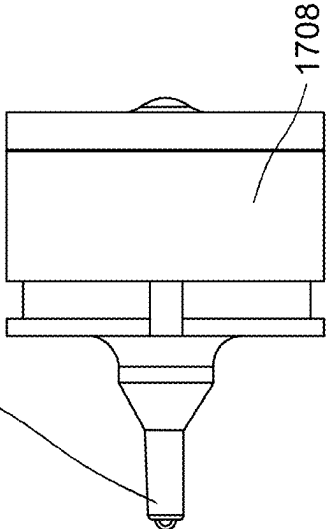
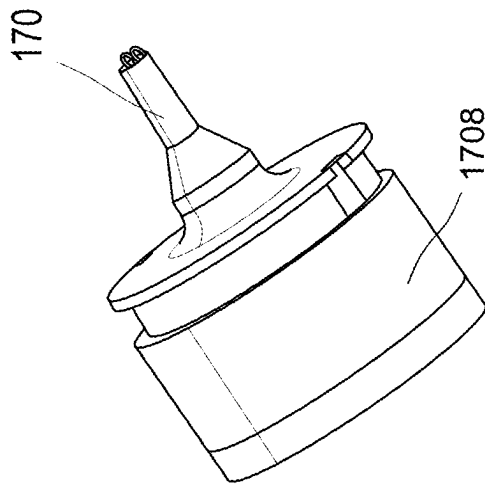


FIG. 79C



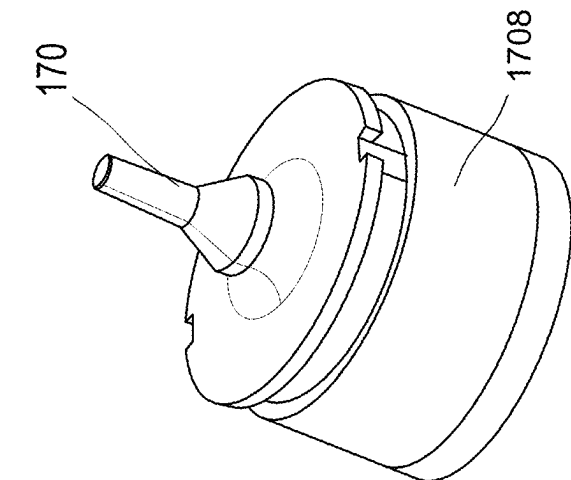


FIG. 81A

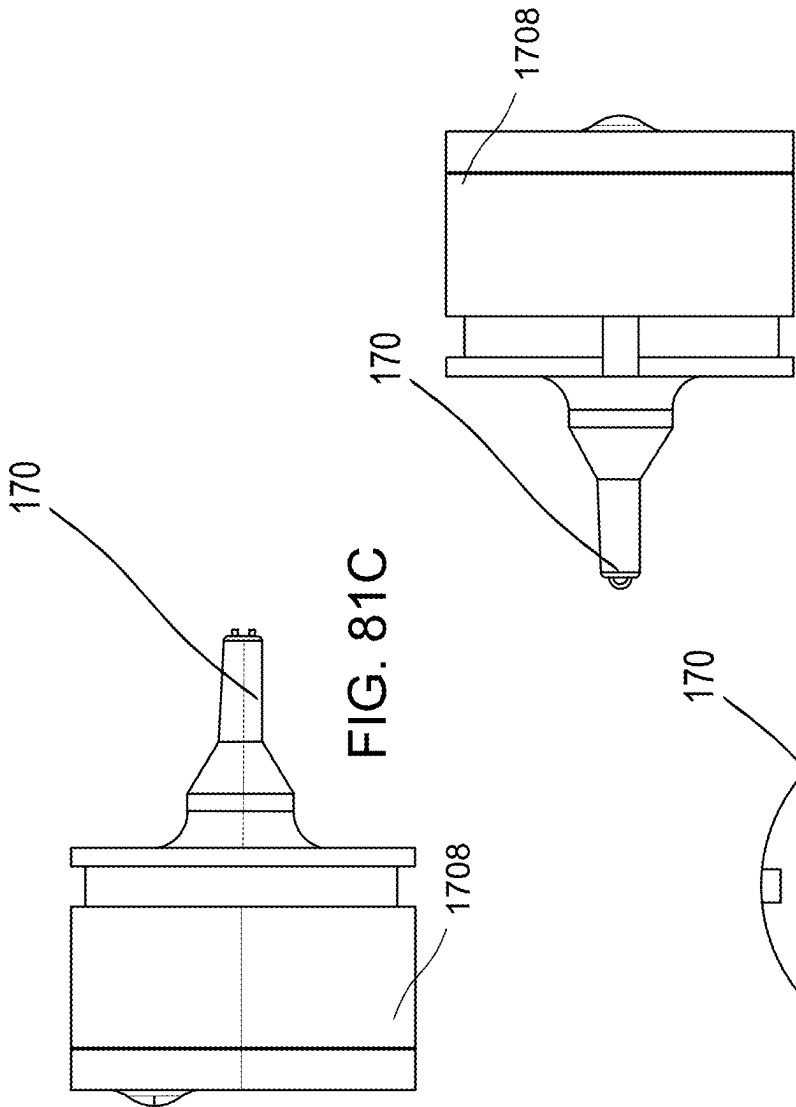
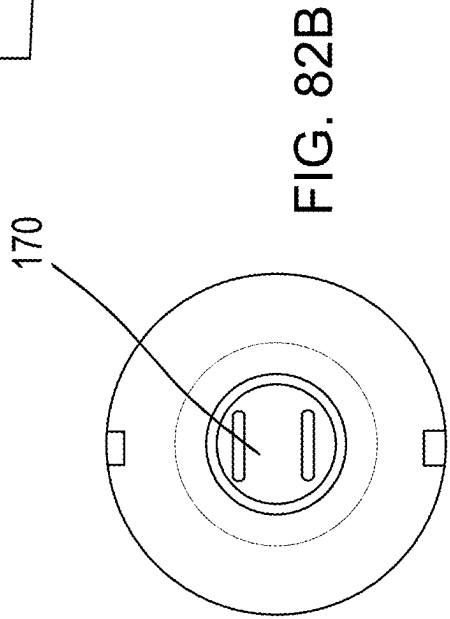
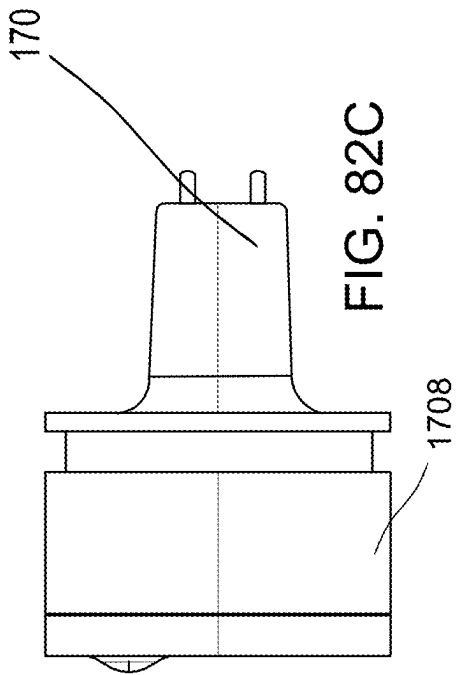
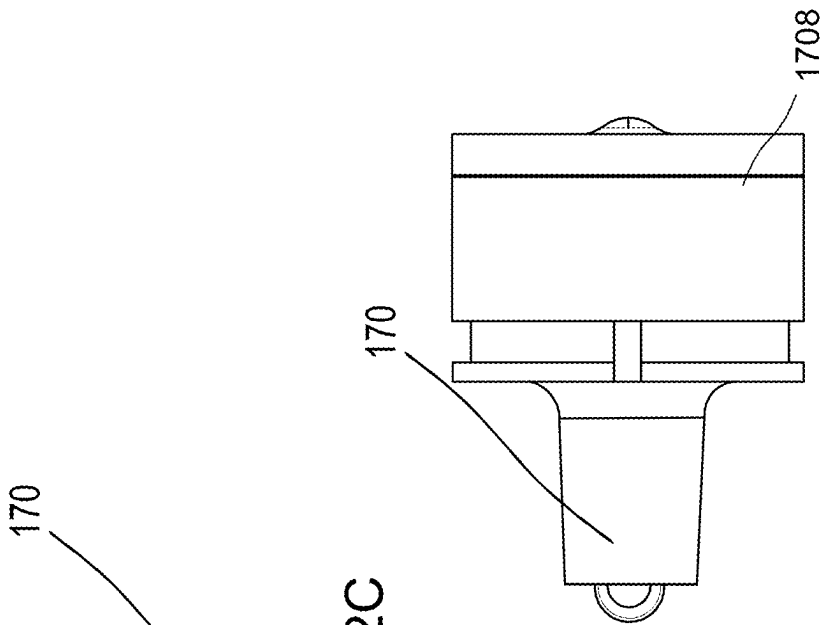
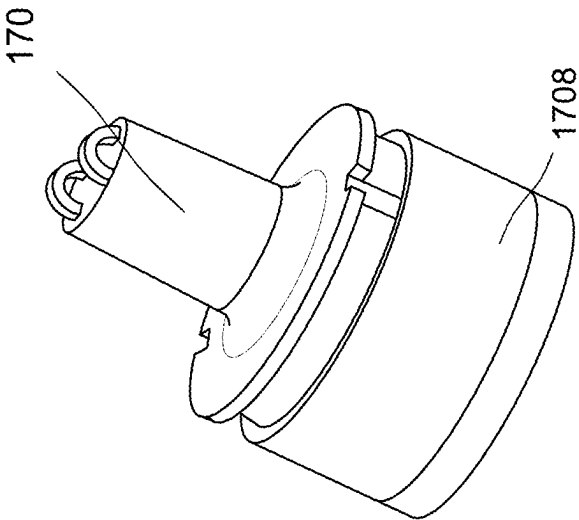


FIG. 81C

FIG. 81D

FIG. 81B



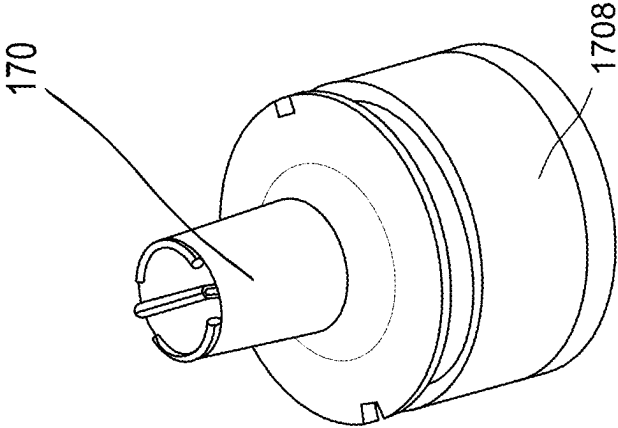


FIG. 83A

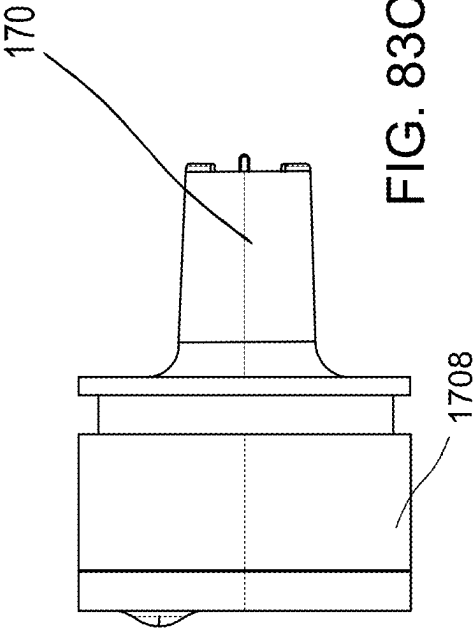


FIG. 83C

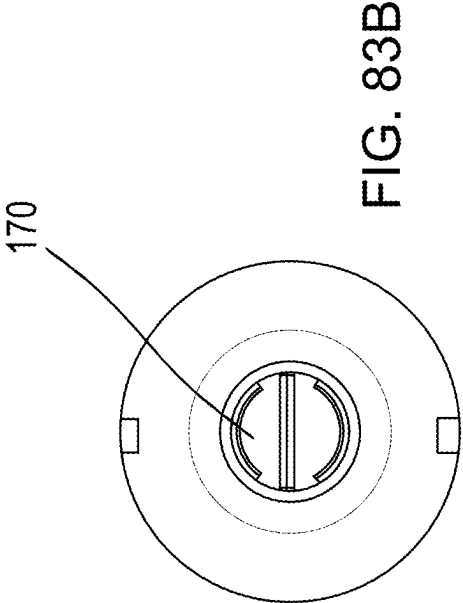


FIG. 83B

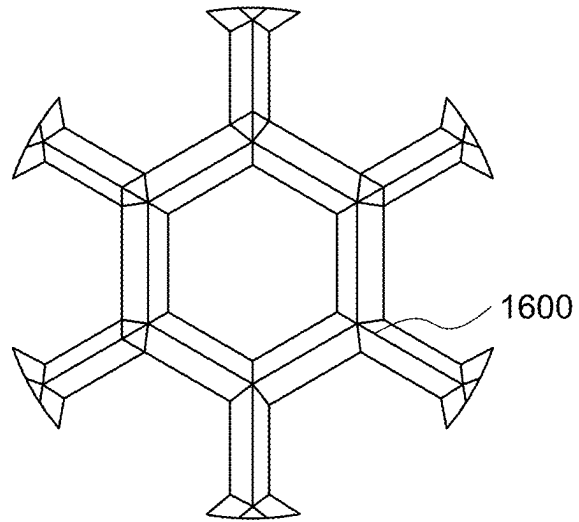


FIG. 84A

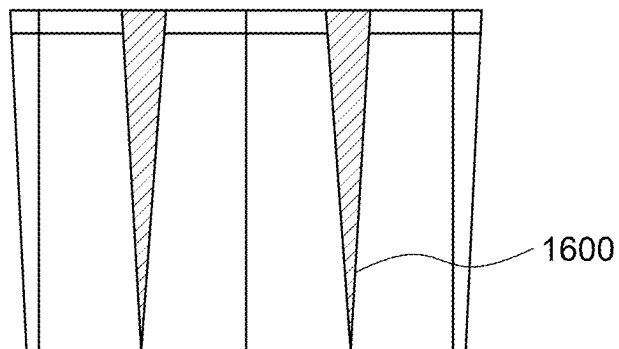


FIG. 84B

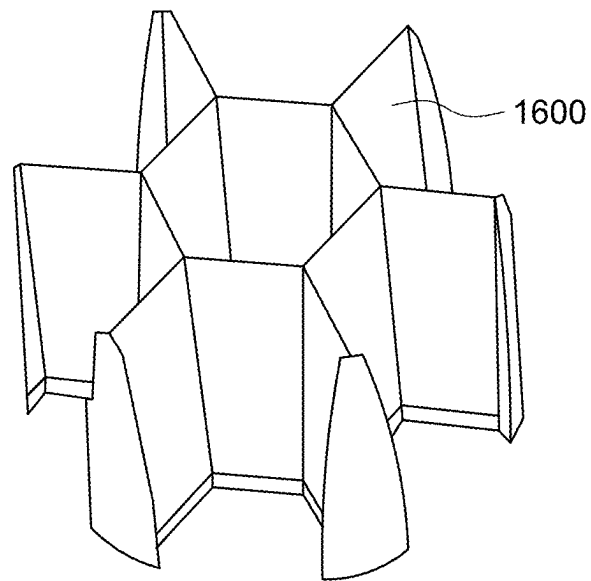


FIG. 84C

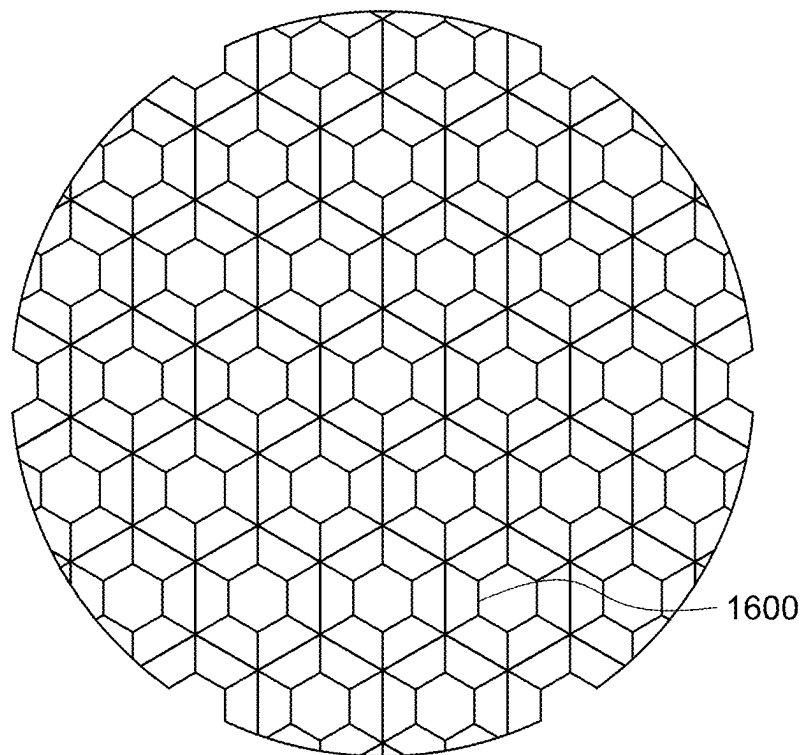


FIG. 84D

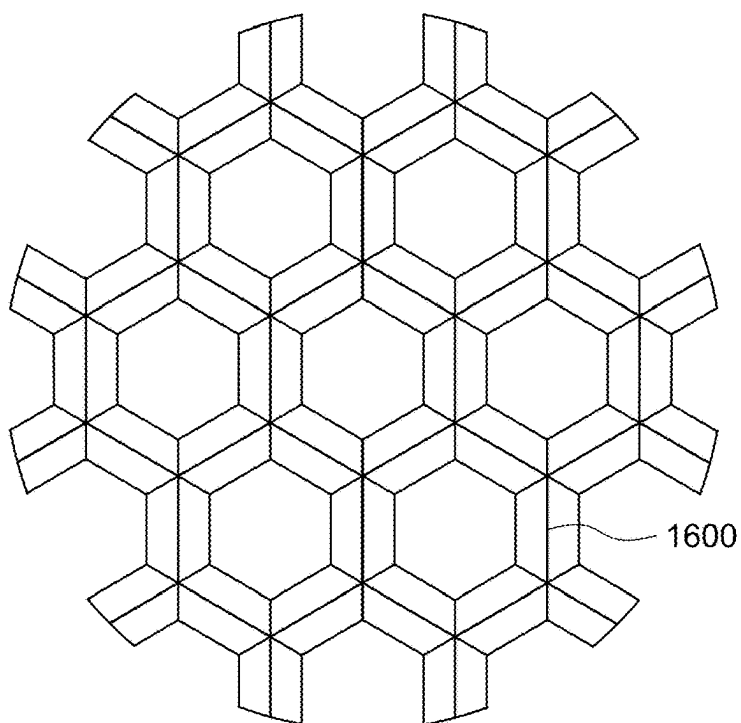


FIG. 84E

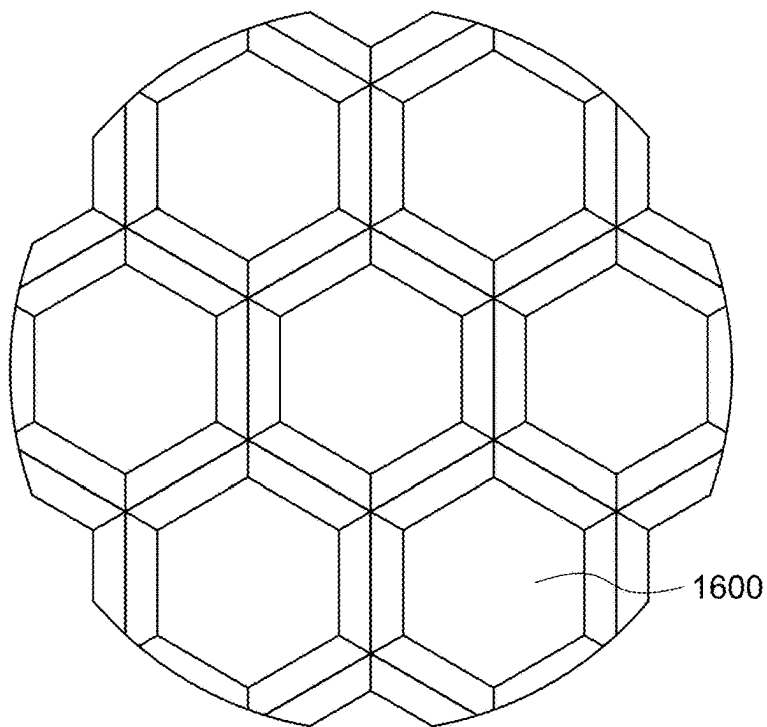


FIG. 84F

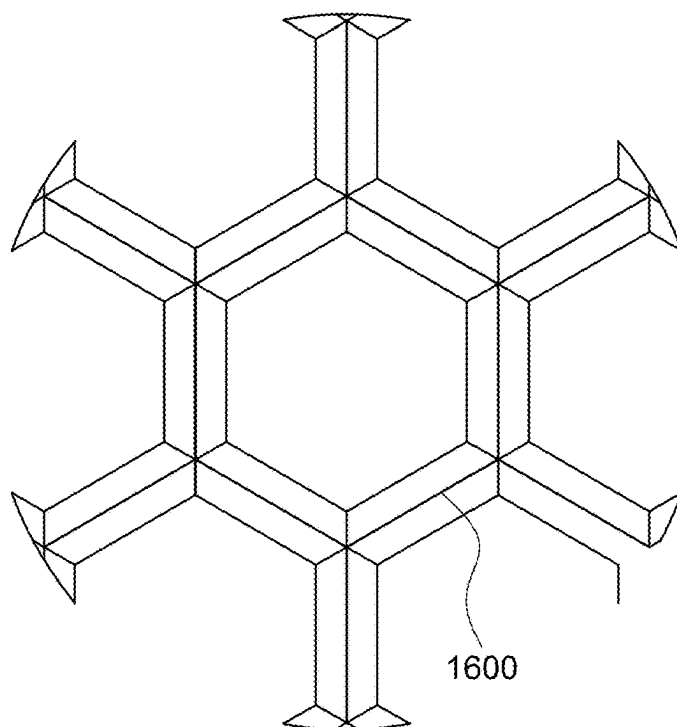


FIG. 84G

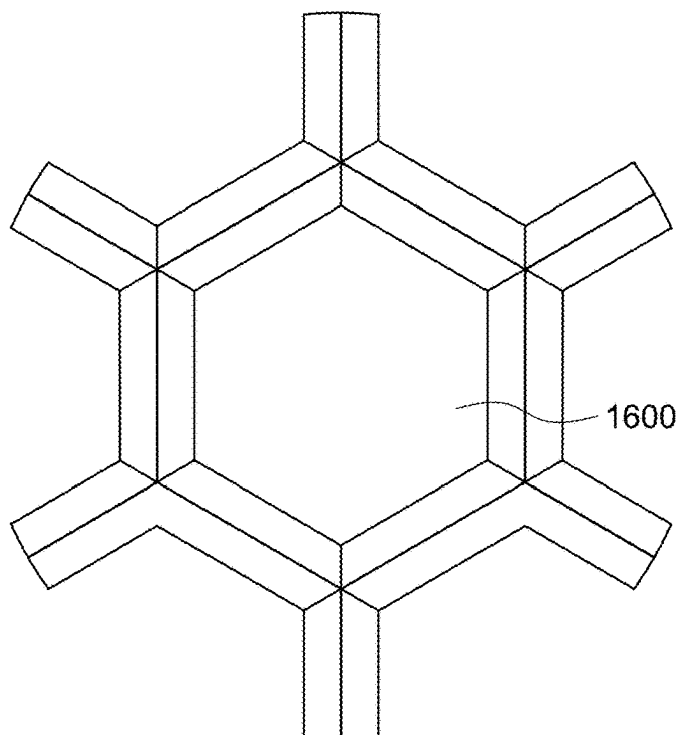


FIG. 84H

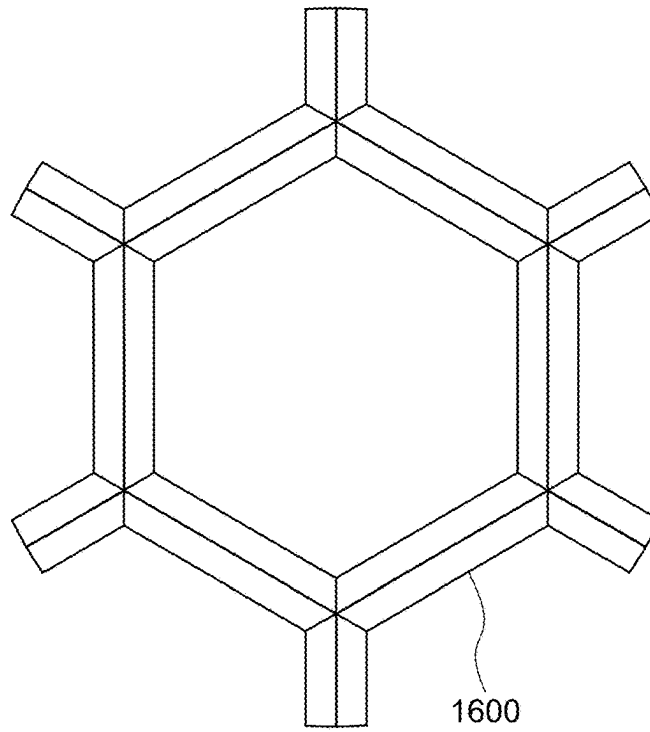


FIG. 84I

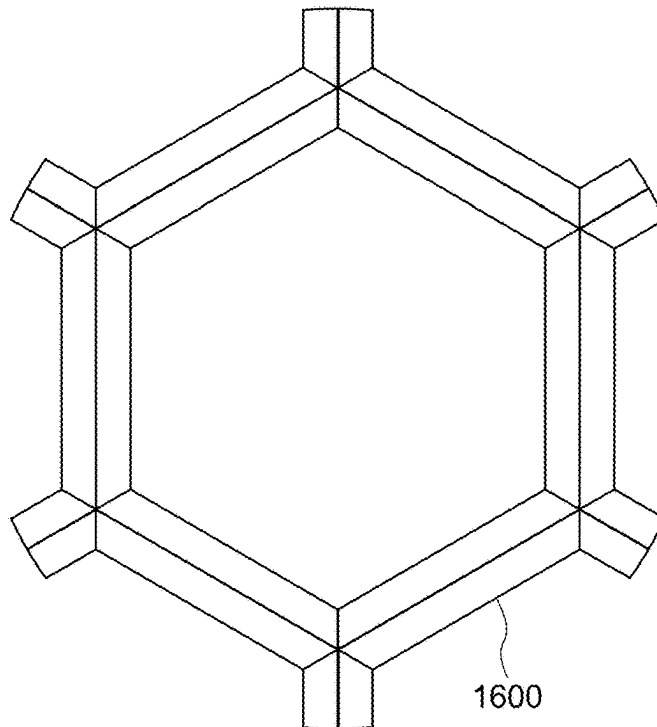


FIG. 84J

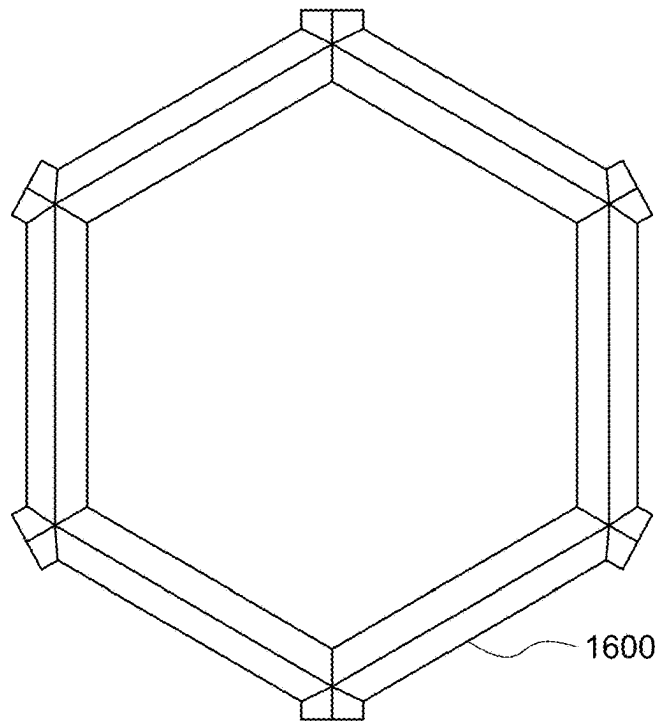


FIG. 84K

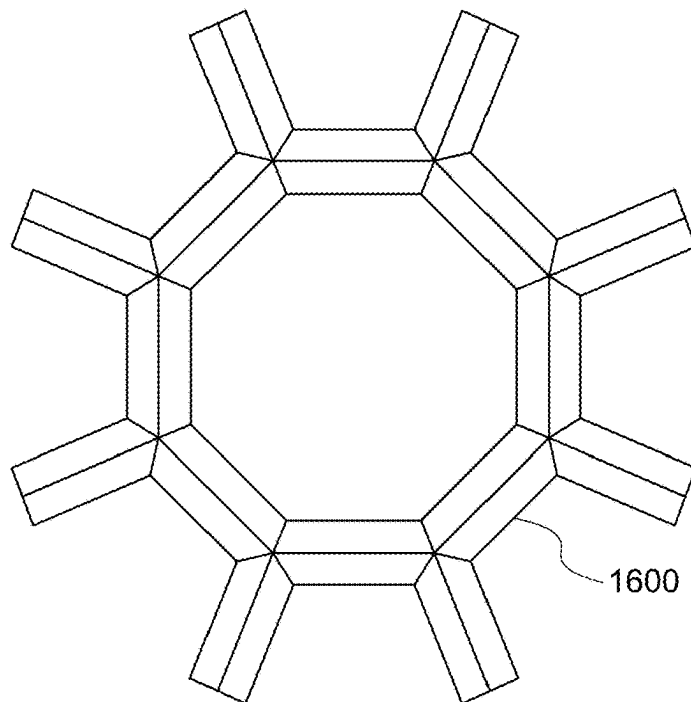


FIG. 84L

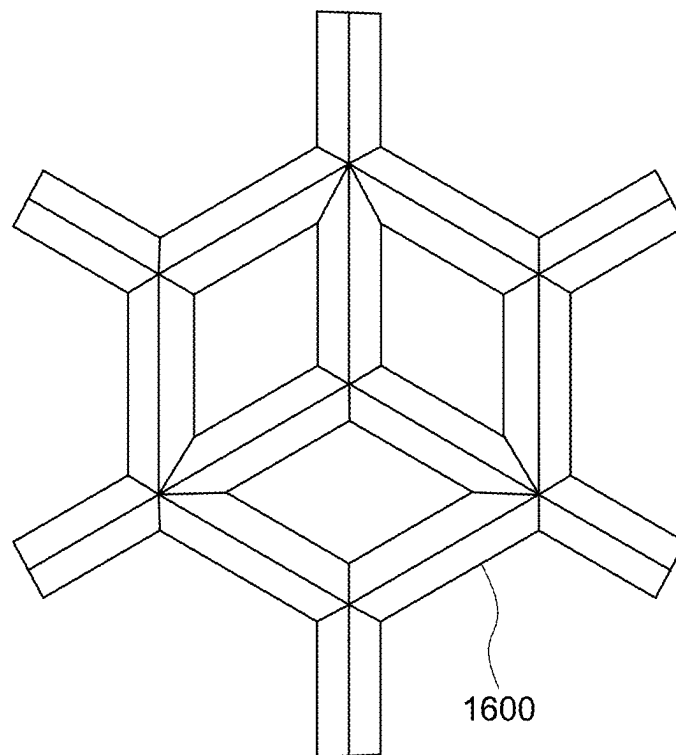


FIG. 84M

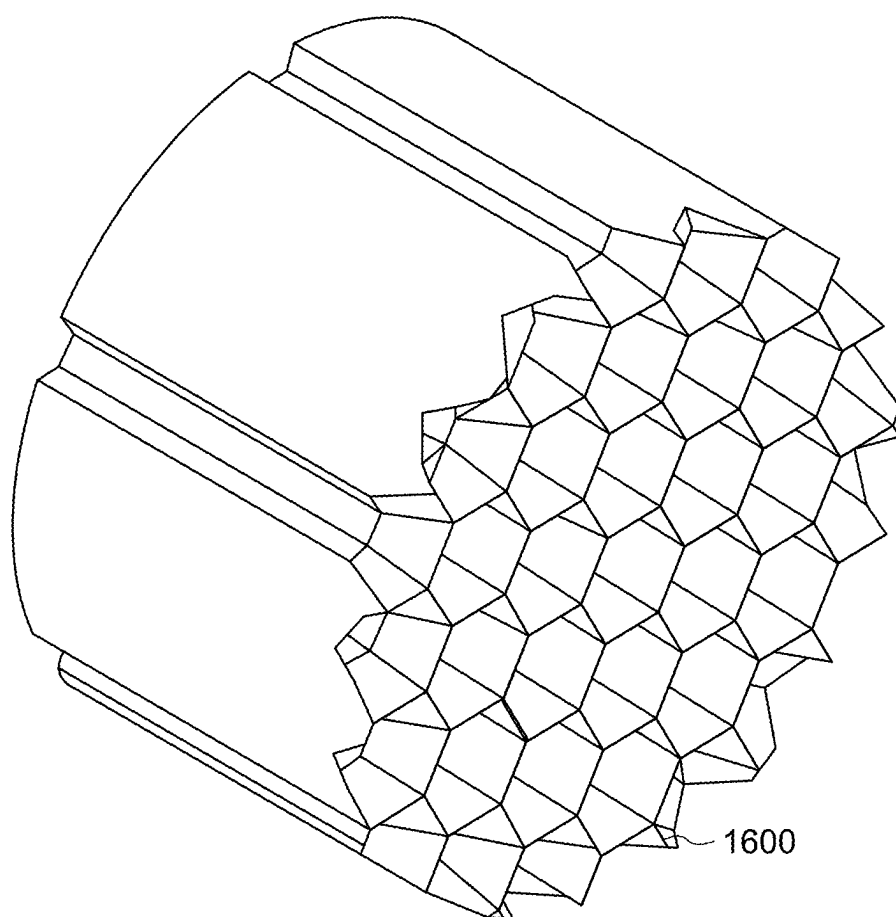


FIG. 84N

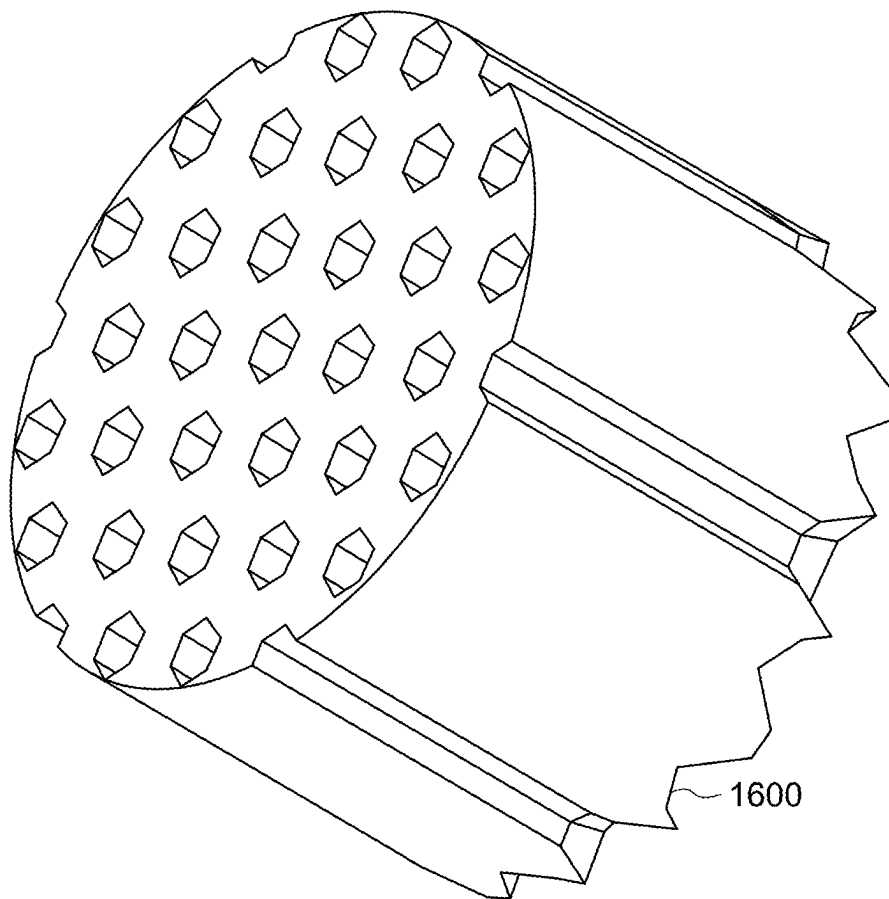


FIG. 840

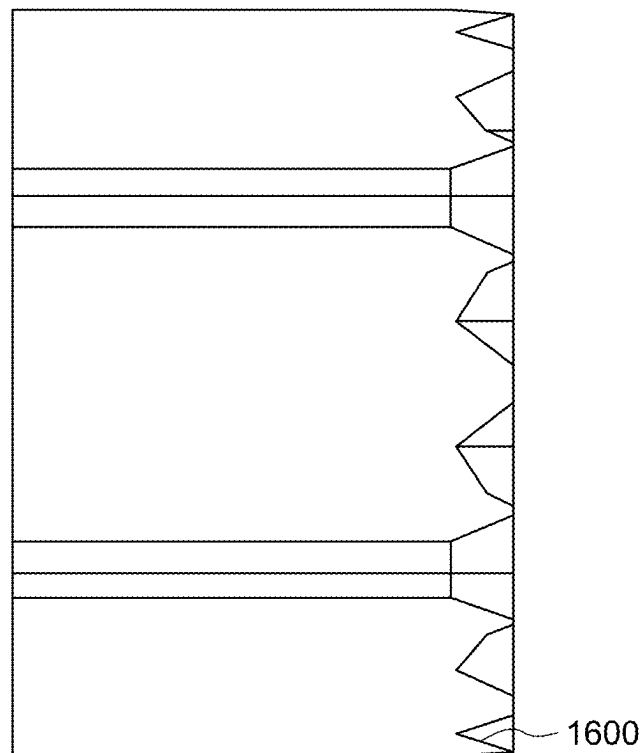


FIG. 84P

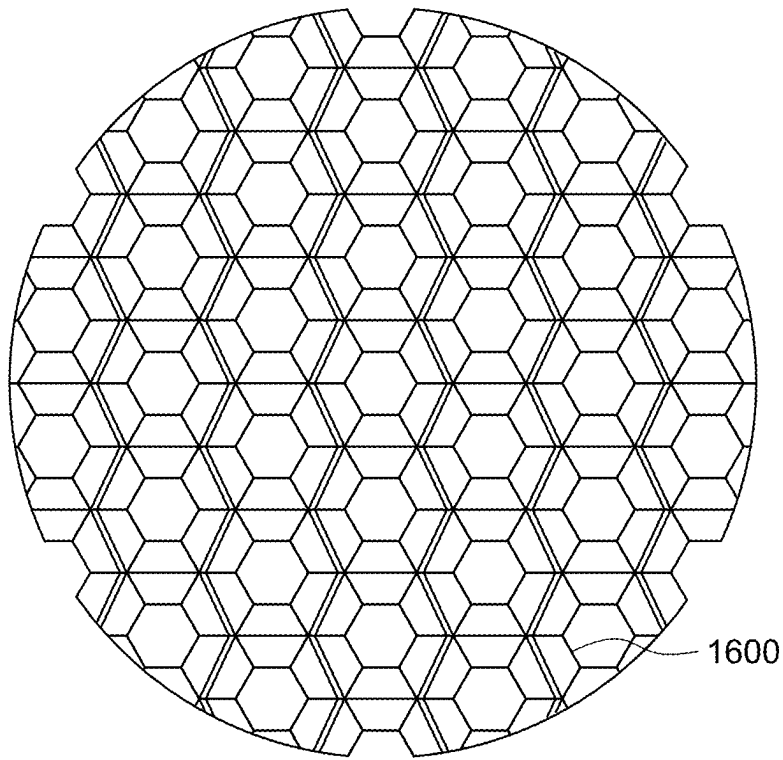


FIG. 84Q

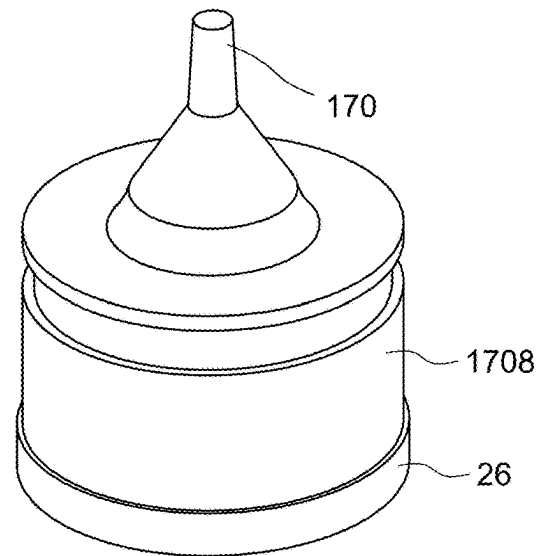


FIG. 85A

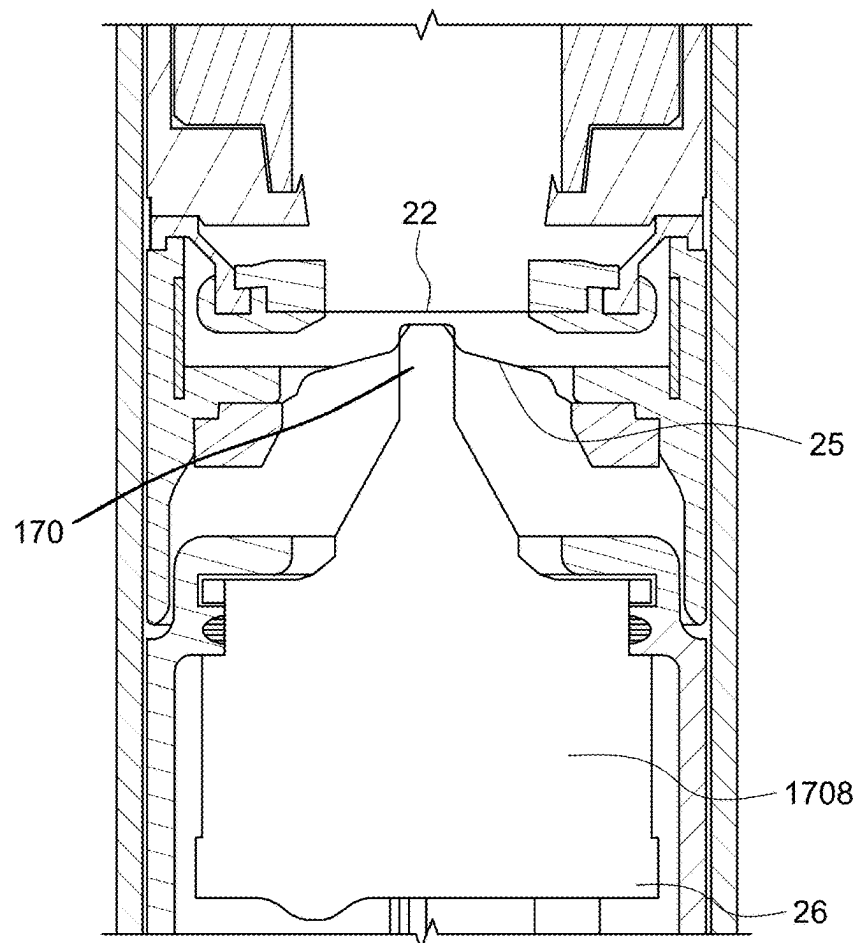


FIG. 85B

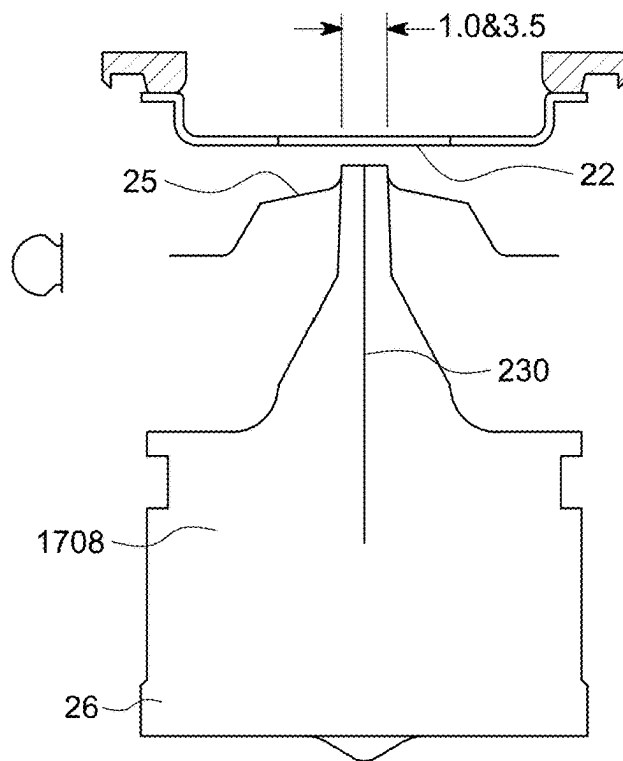


FIG. 85C

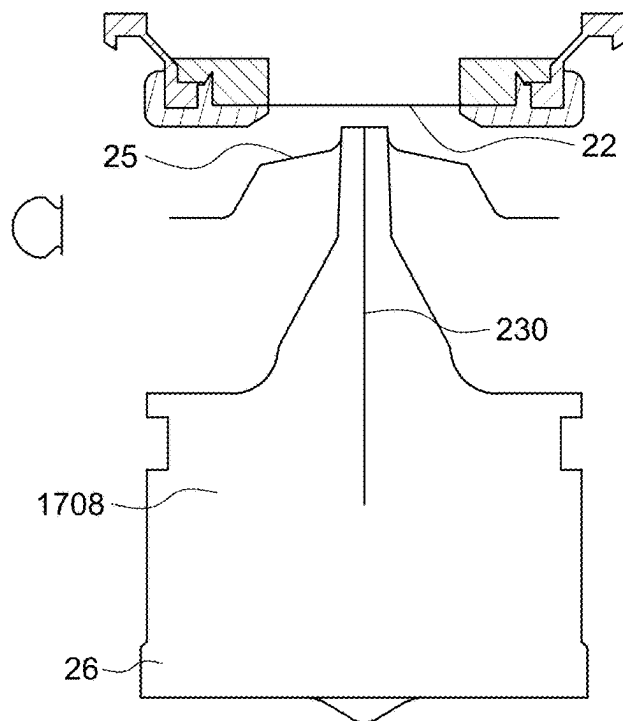


FIG. 85D

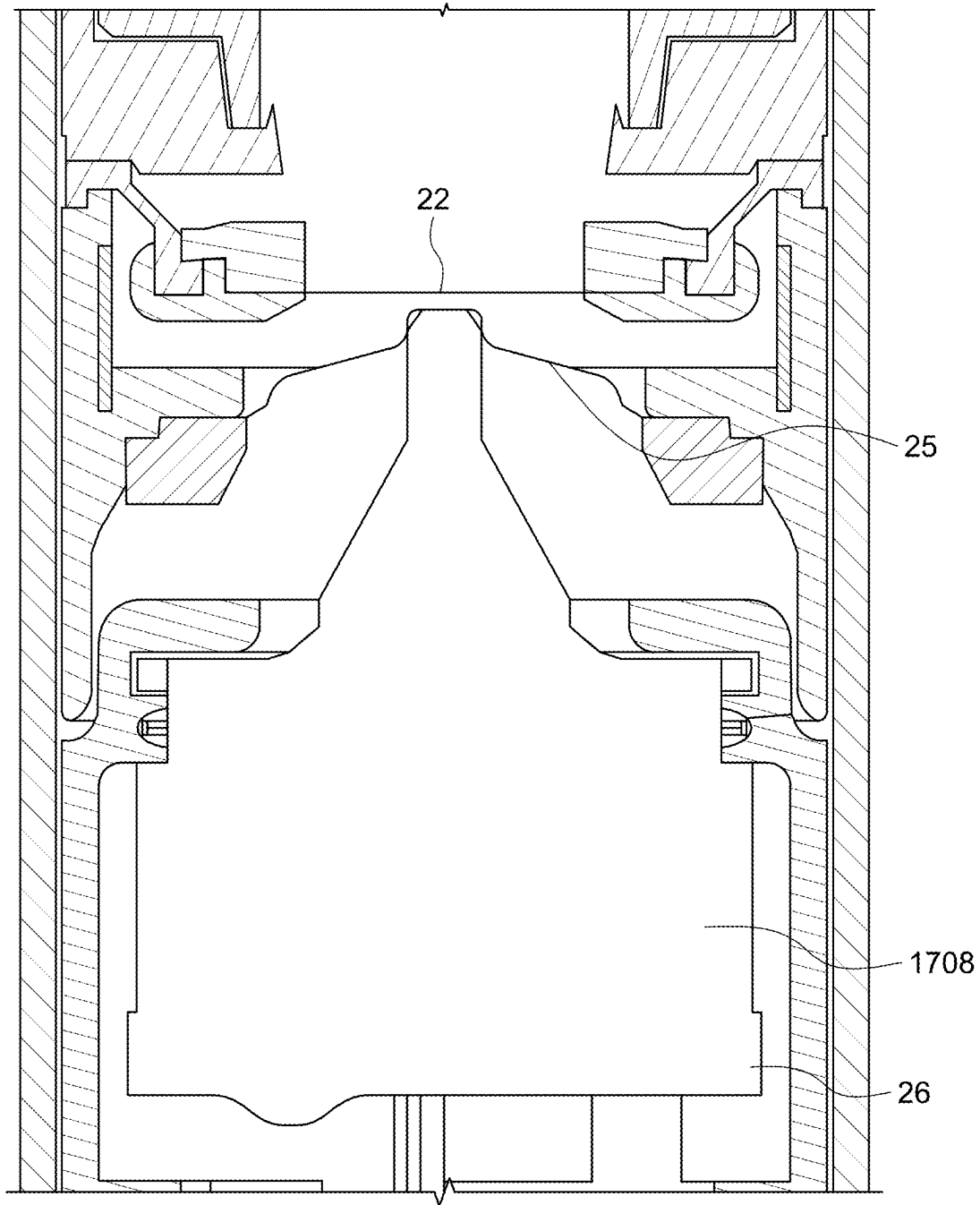


FIG. 86A

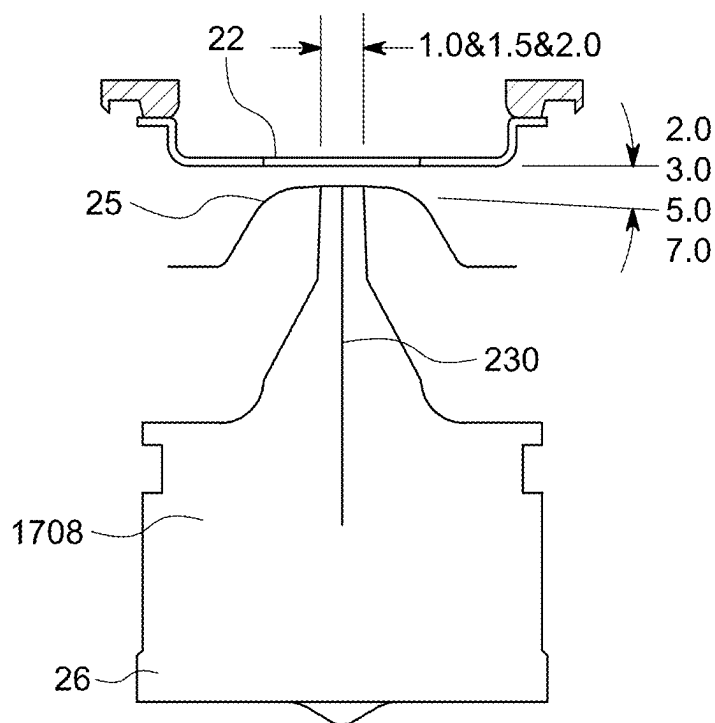


FIG. 86B

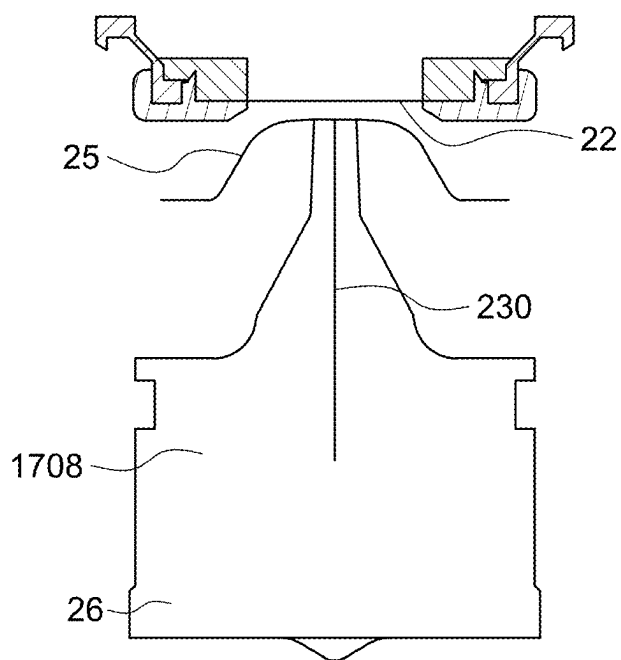


FIG. 86C

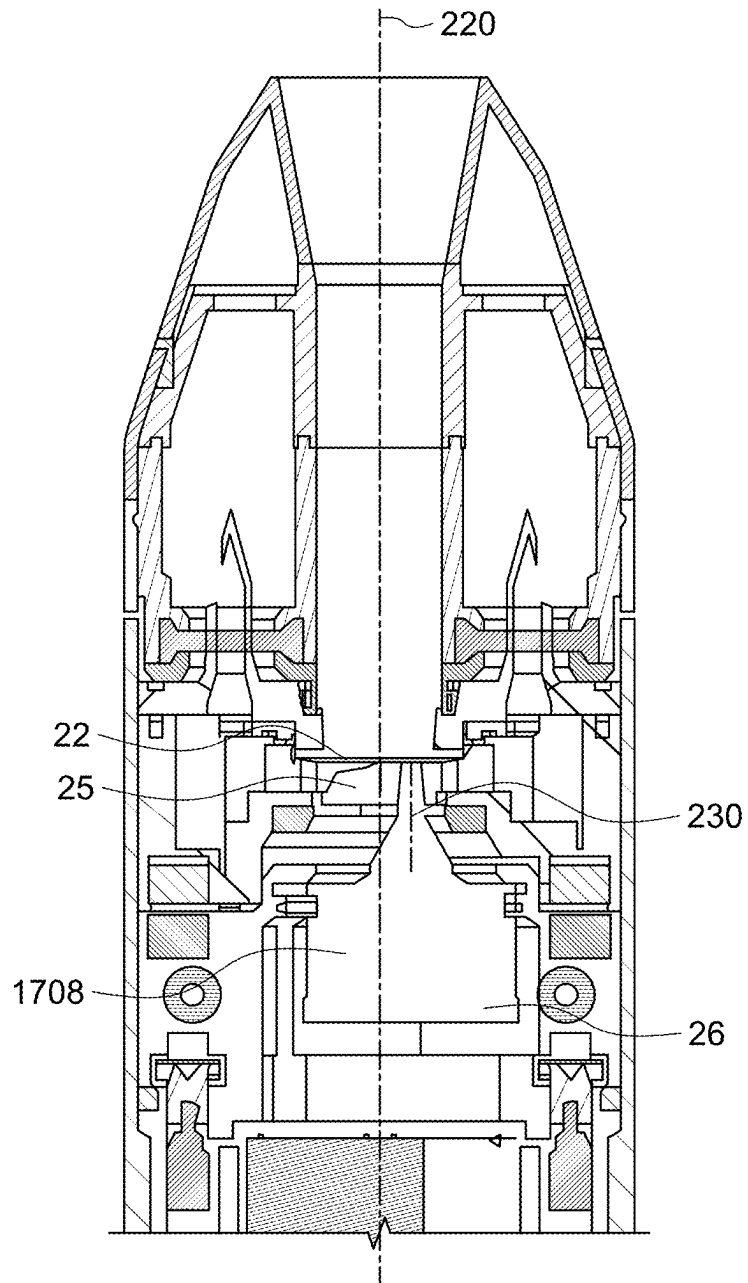


FIG. 87

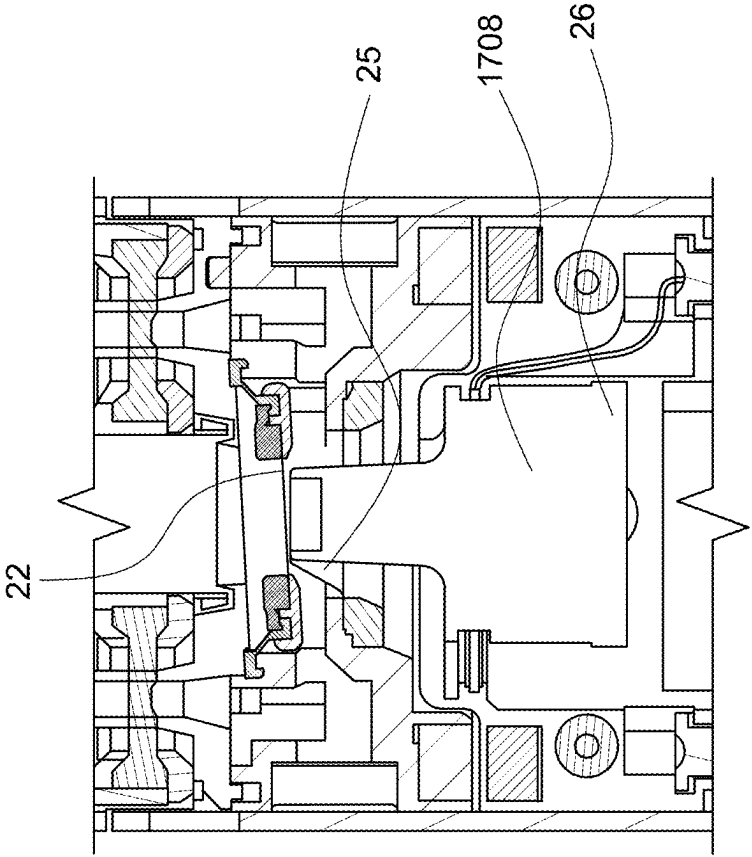


FIG. 88A

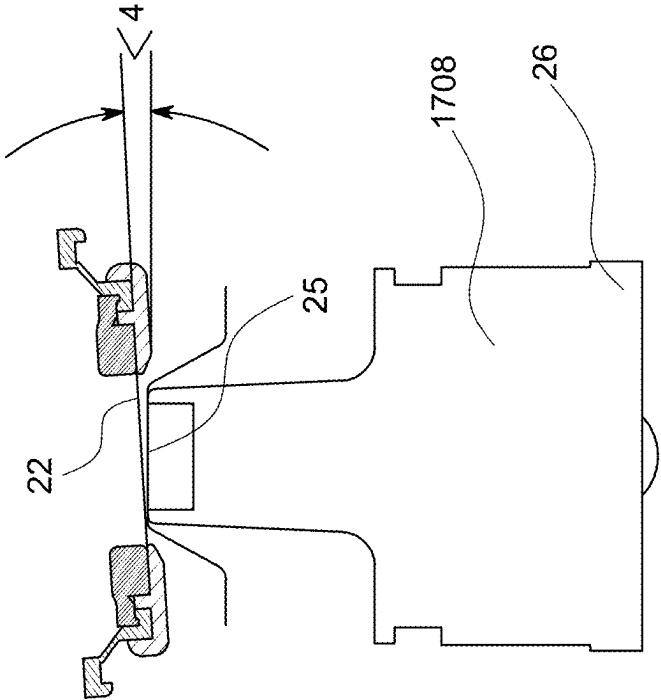


FIG. 88B

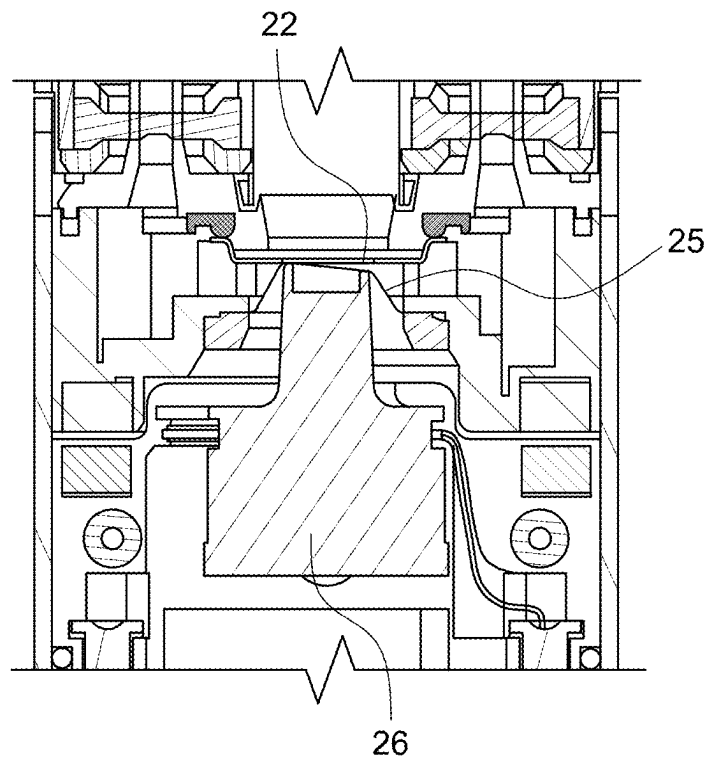


FIG. 89A

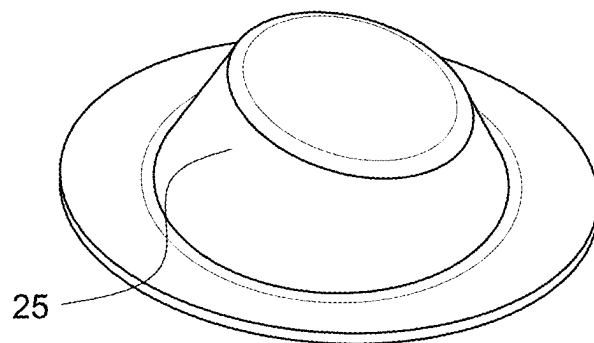


FIG. 89B

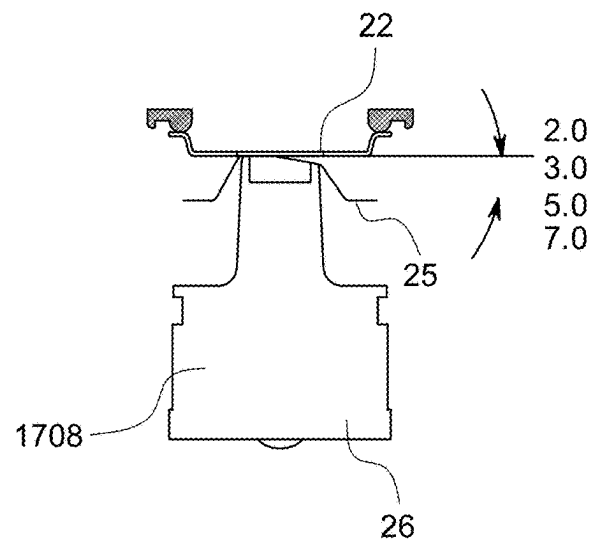


FIG. 89C

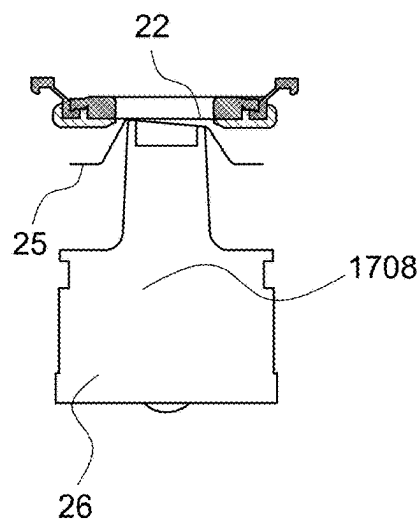


FIG. 89D

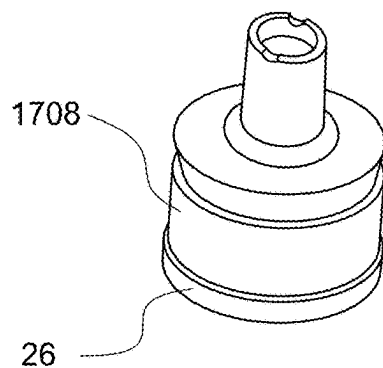


FIG. 89E

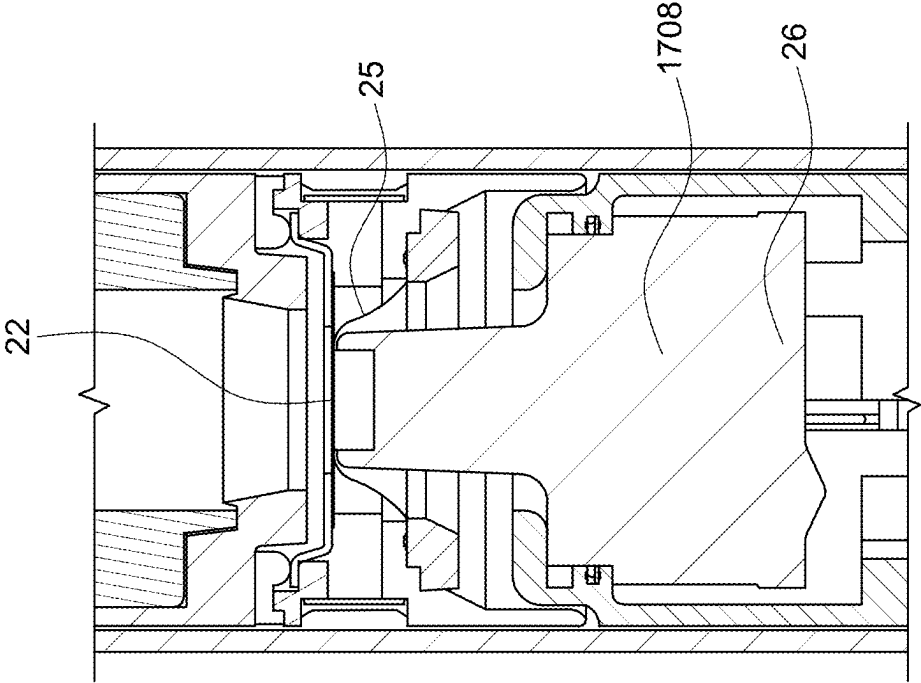


FIG. 90A

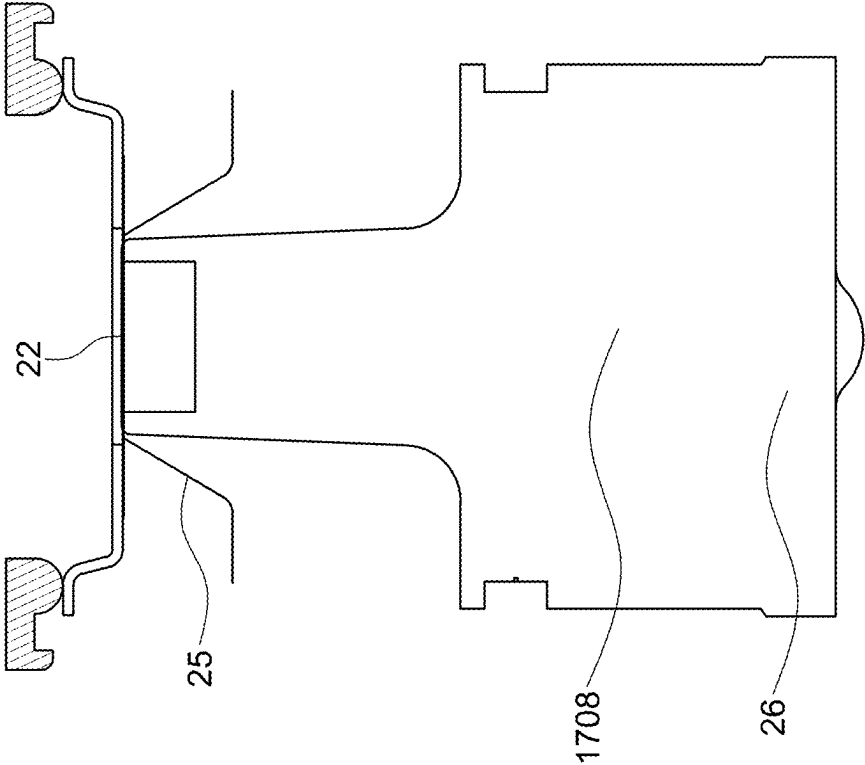


FIG. 90B

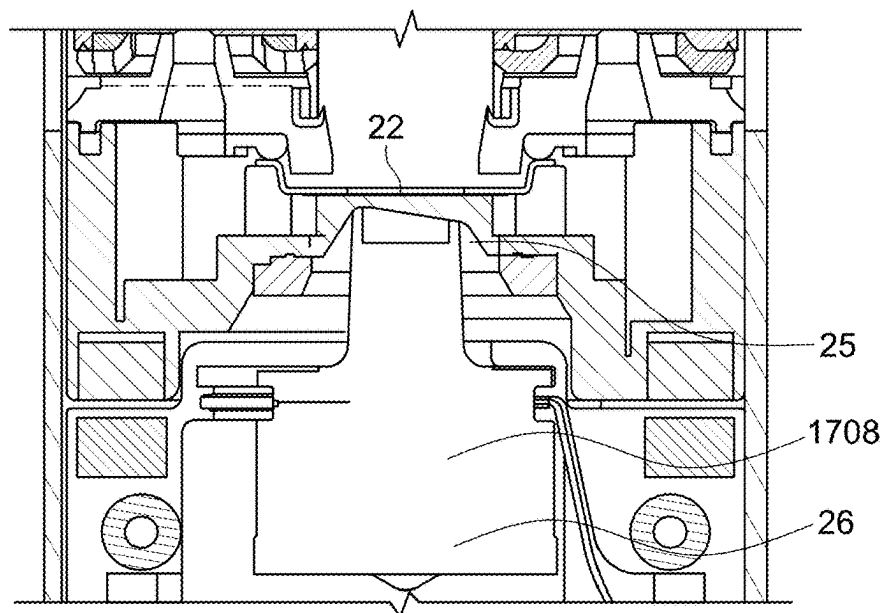


FIG. 91A

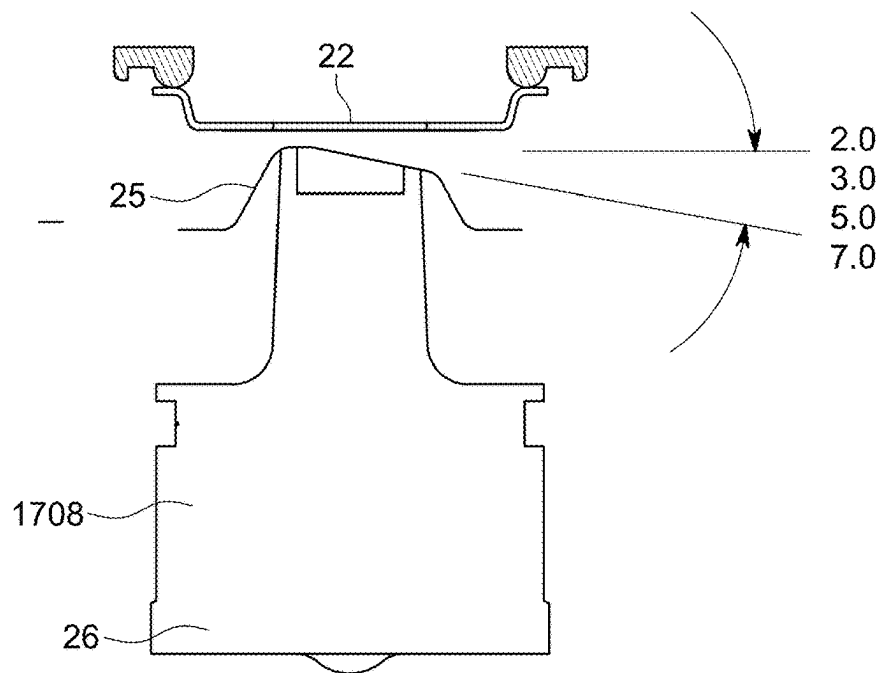


FIG. 91B

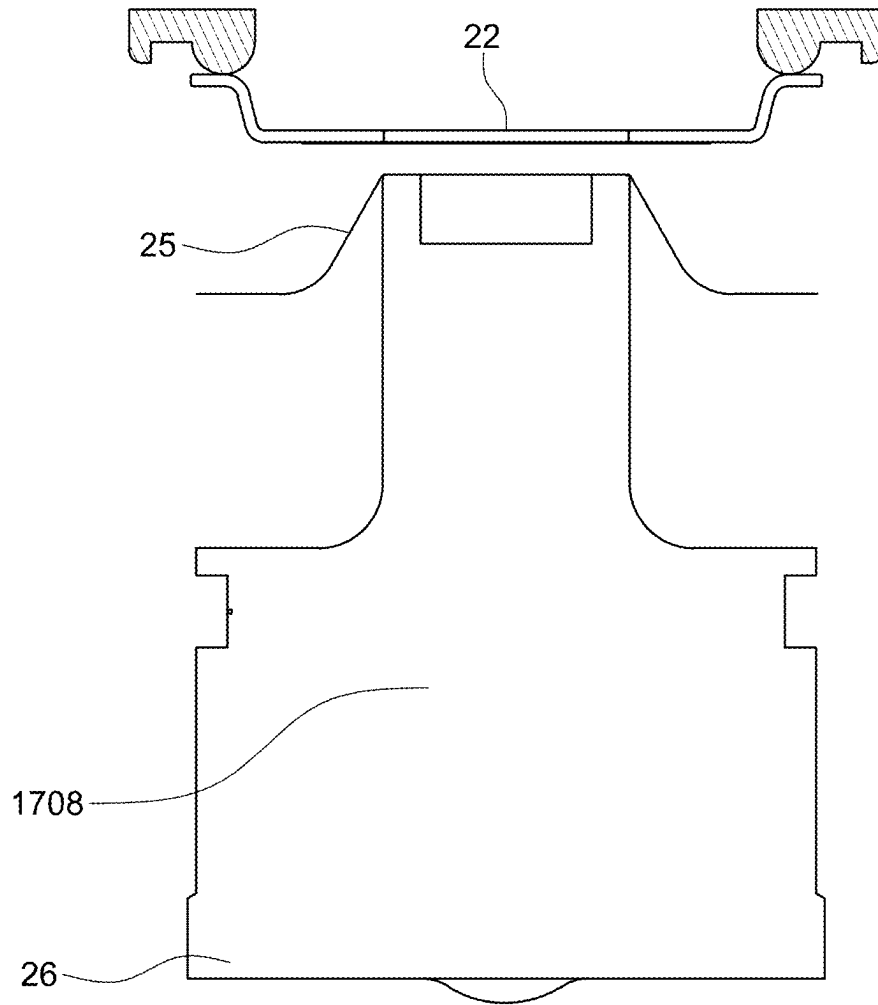


FIG. 92

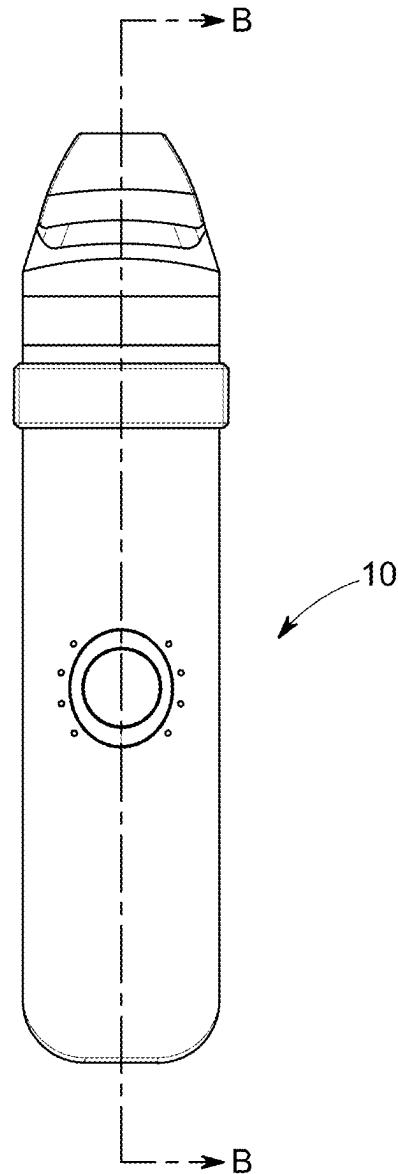


FIG. 93A

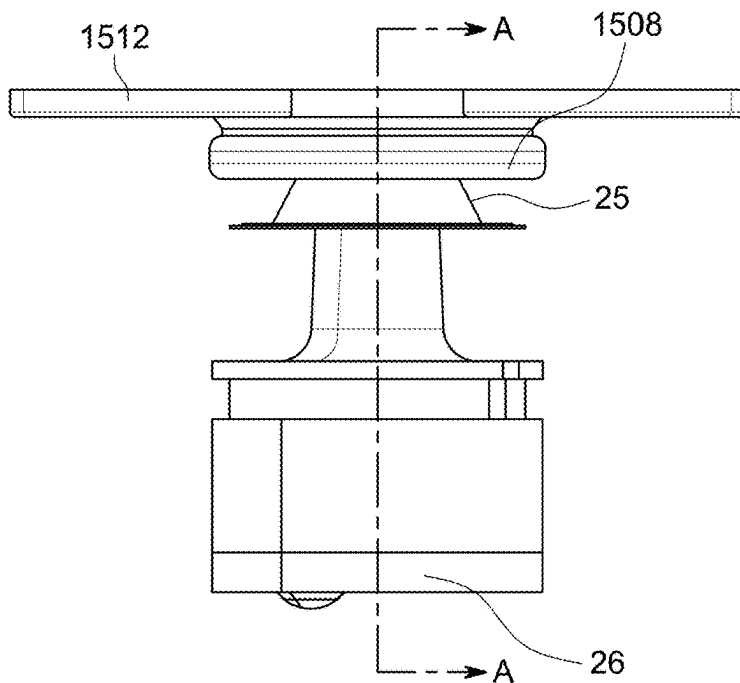
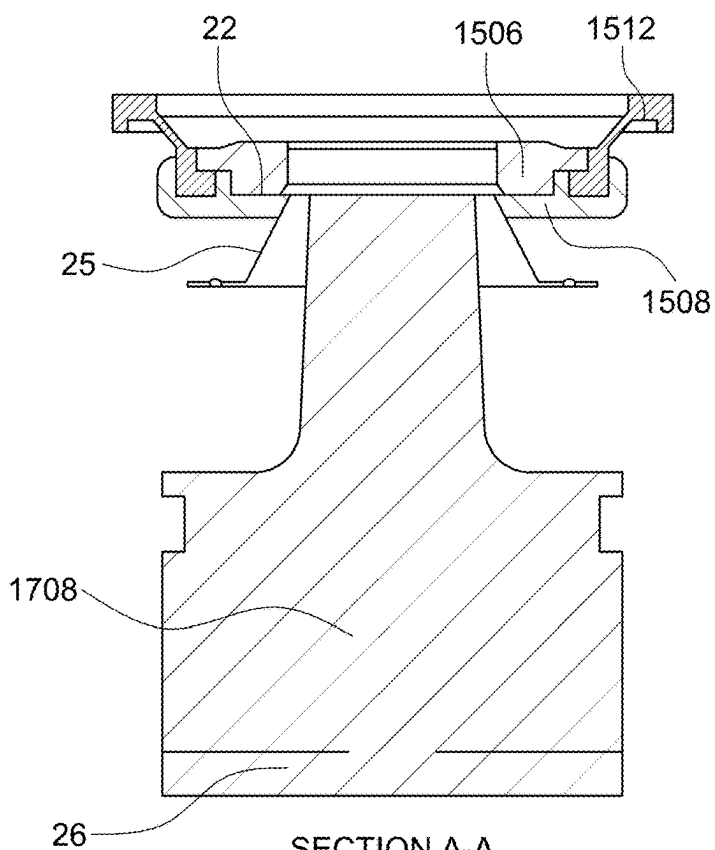


FIG. 93B



SECTION A-A
SCALE 4 : 1

FIG. 93C

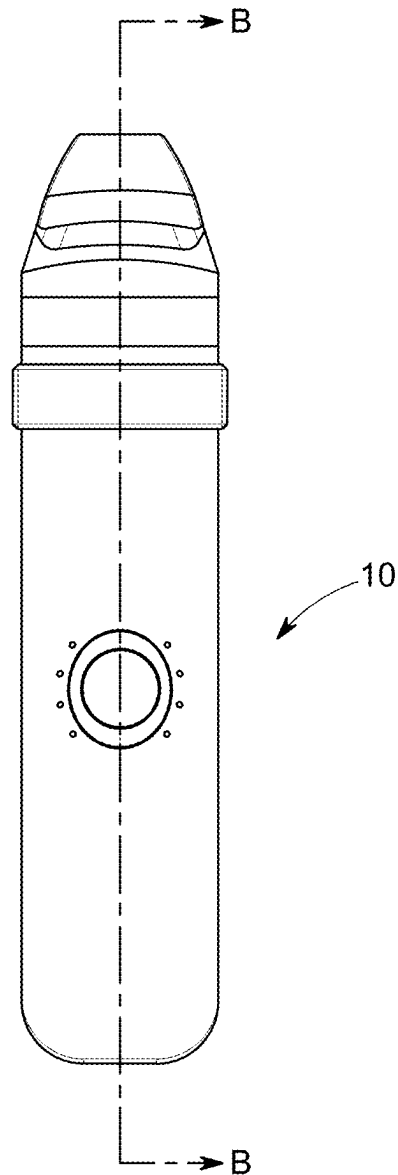
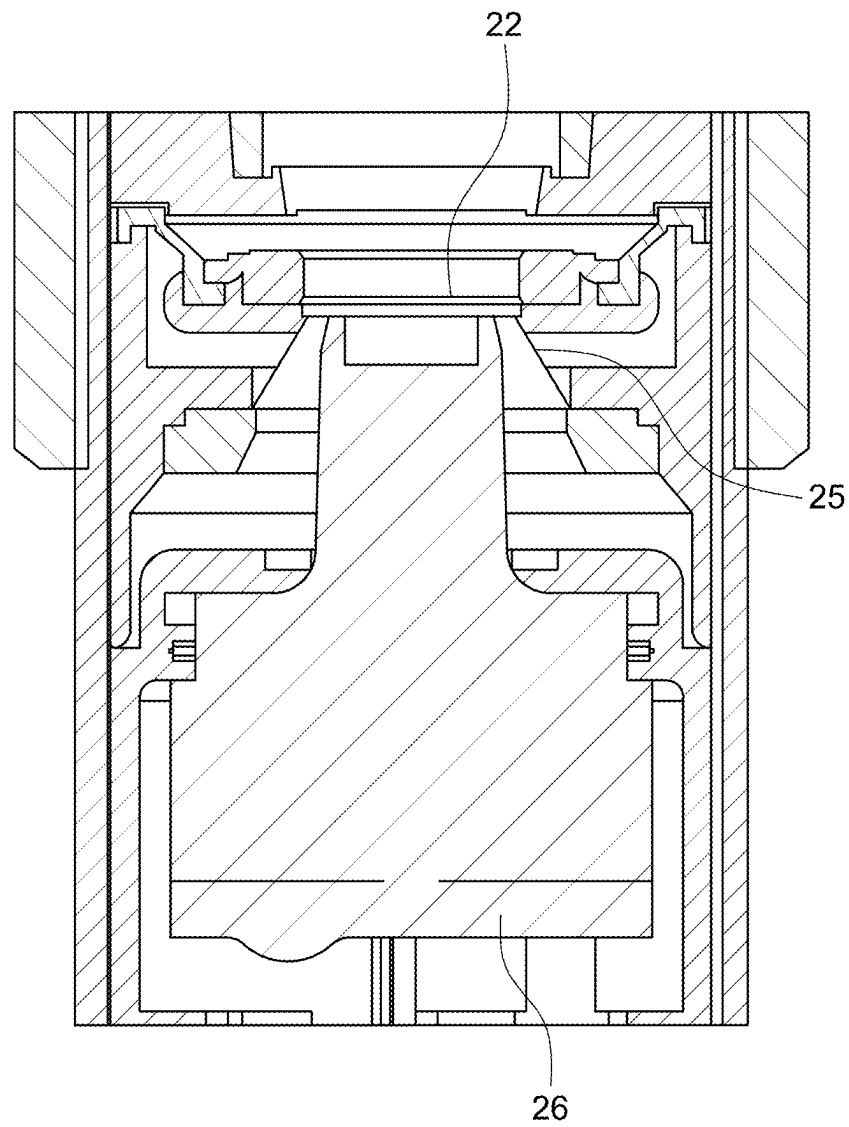


FIG. 94A



SECTION B-B
SCALE 4 : 1

FIG. 94B

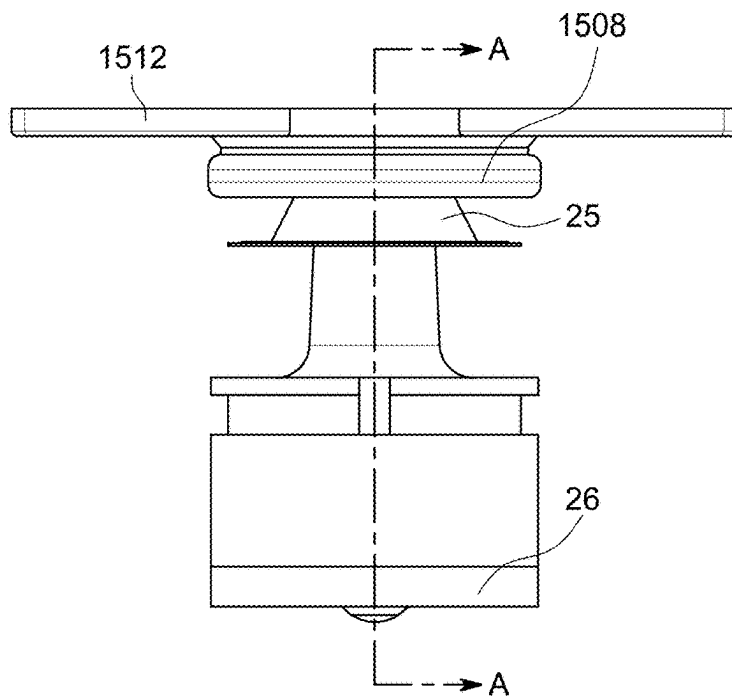
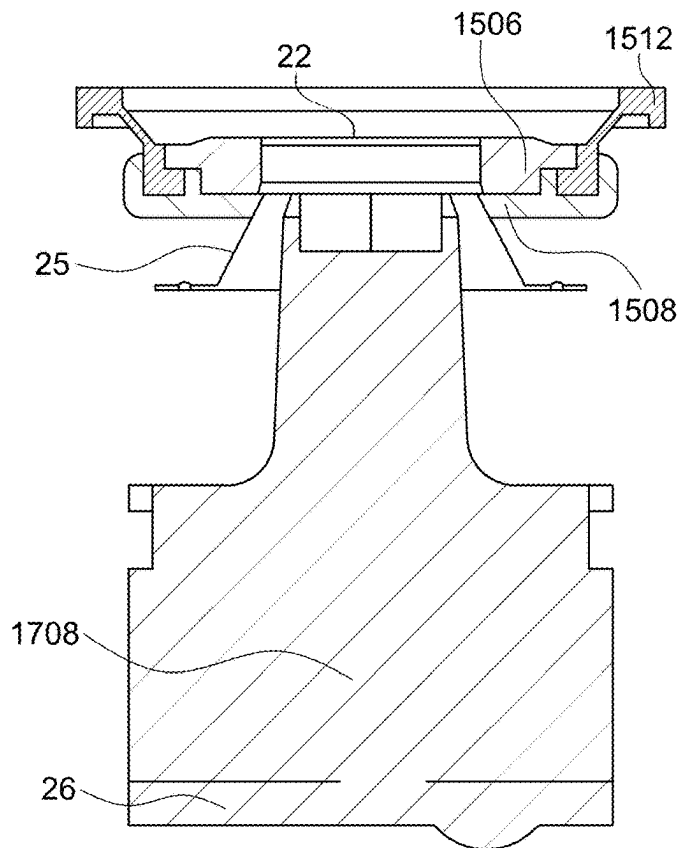


FIG. 94C



SECTION A-A
SCALE 4 : 1

FIG. 94D

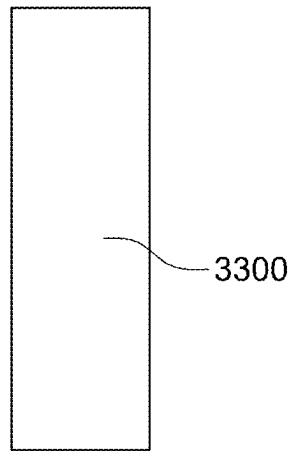


FIG. 95

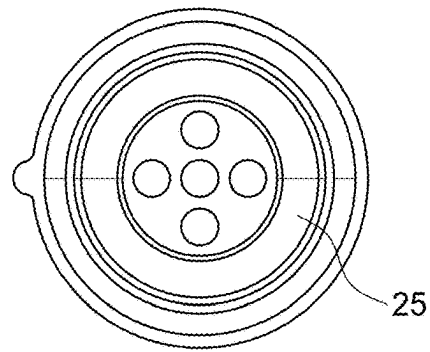


FIG. 96A

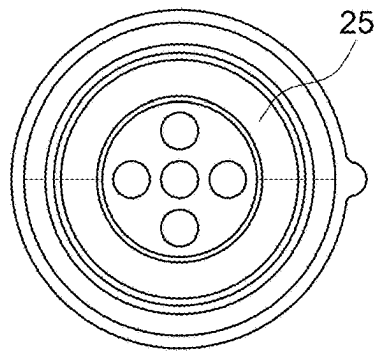


FIG. 96B

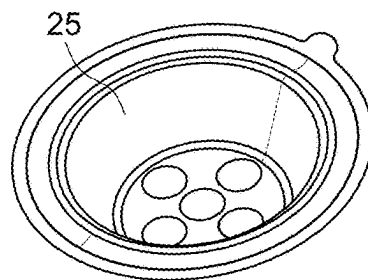


FIG. 96C

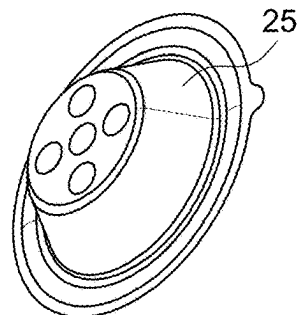


FIG. 96D

- a: SUS316 L
- b: SUS316 L
- c: Polymer mesh
- d: Plastic

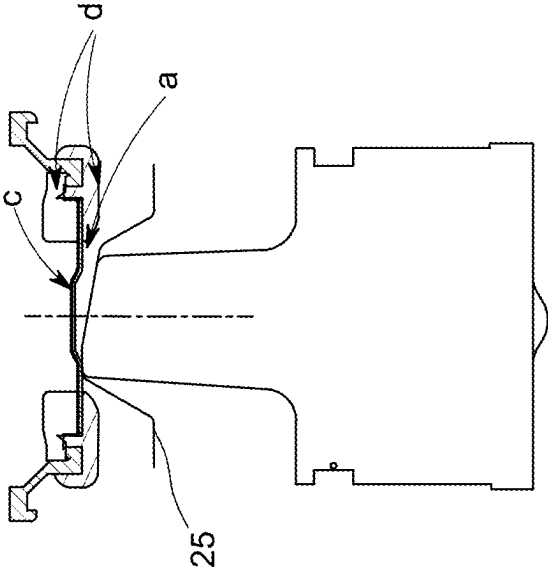


FIG. 97A

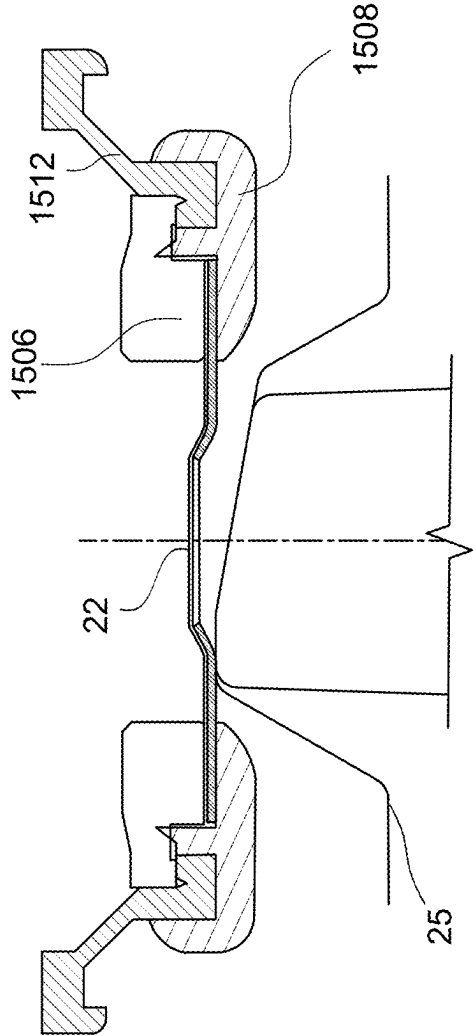


FIG. 97B

a:SUS316 L
b:SUS316 L
c:Polymer mesh
d:Plastic

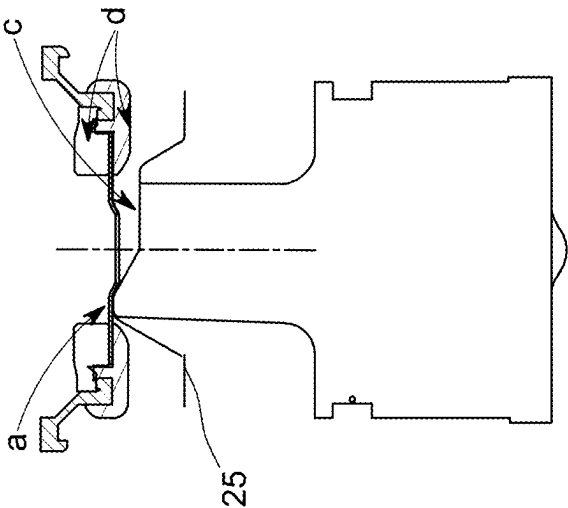


FIG. 98A

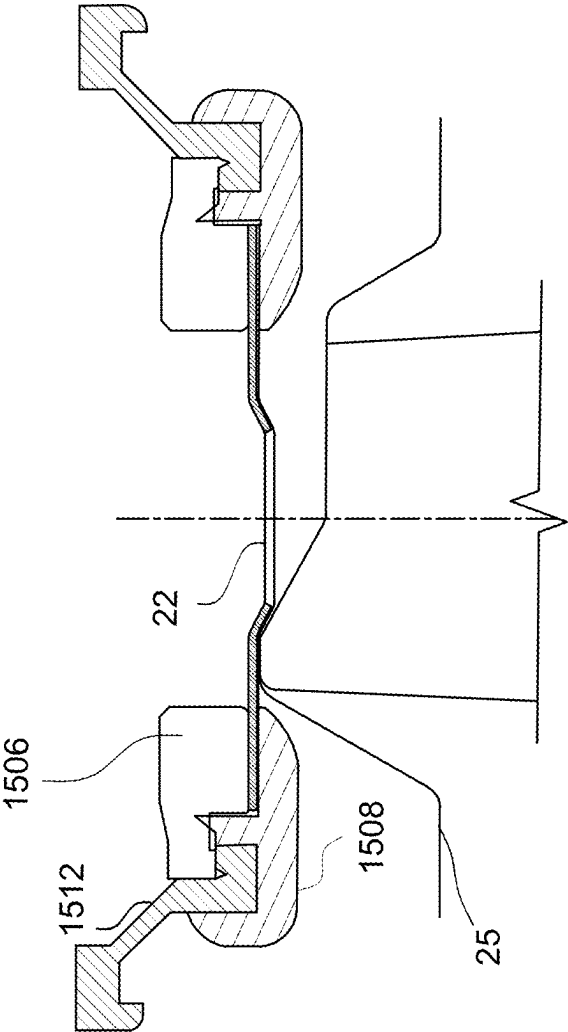


FIG. 98B

- a:SUS316 L
- b:SUS316 L
- c:Polymer mesh
- d:Plastic

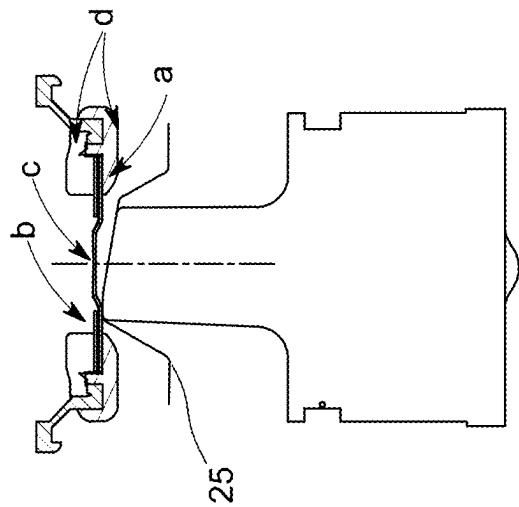


FIG. 99A

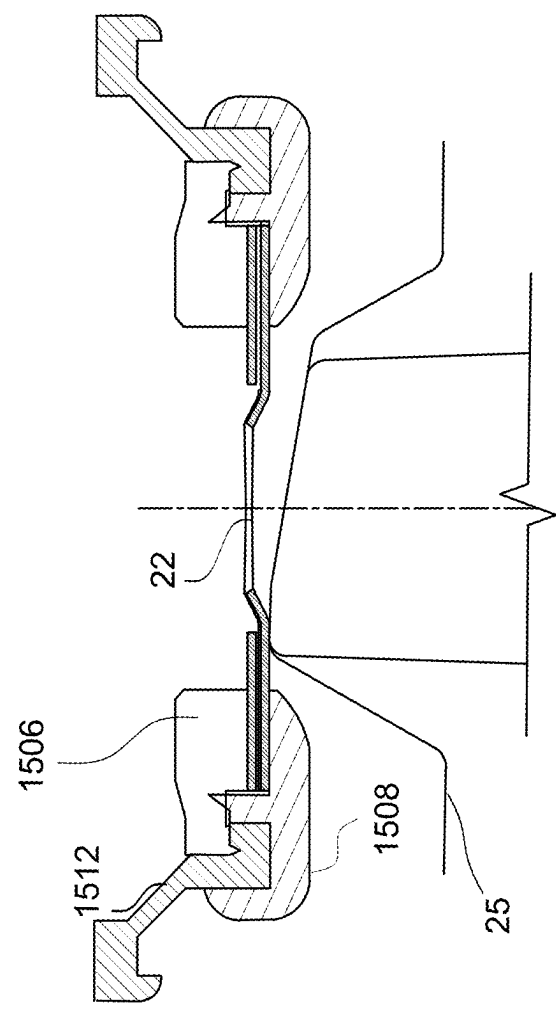


FIG. 99B

- a: SUS316 L
- b: SUS316 L
- c: Polymer mesh
- d: Plastic

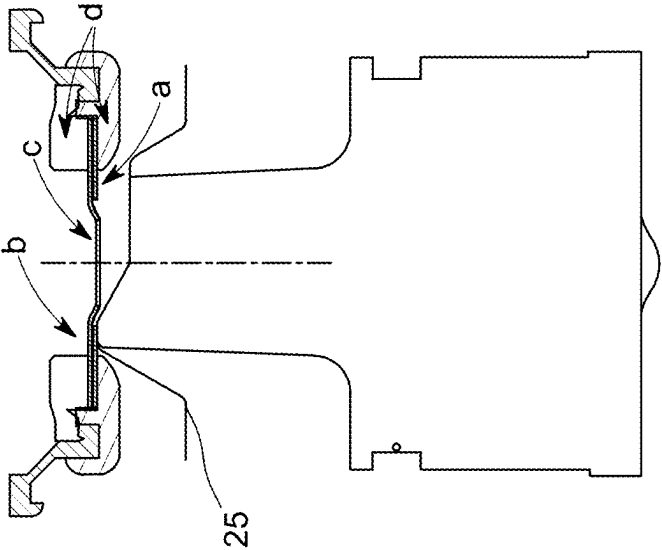


FIG. 100A

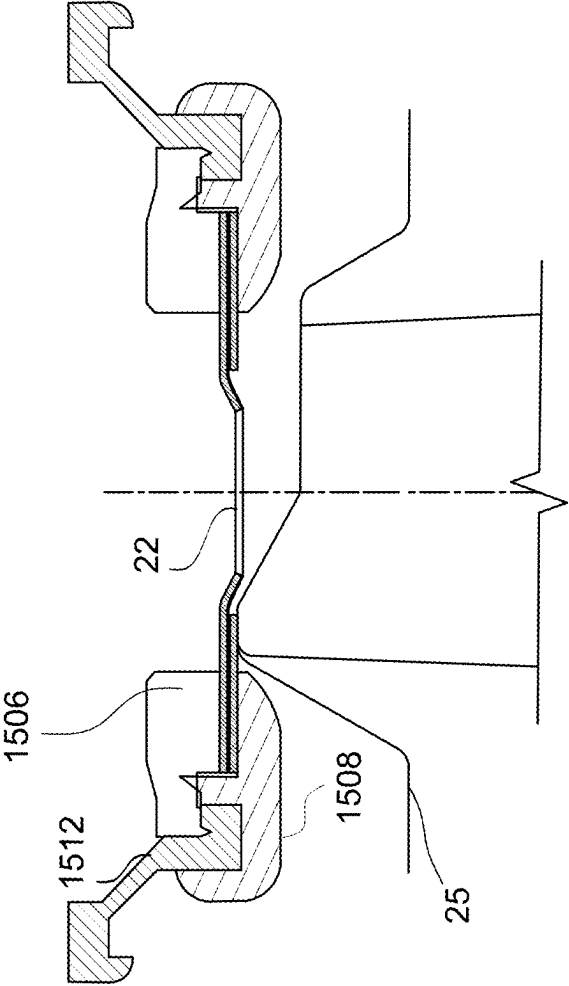


FIG. 100B

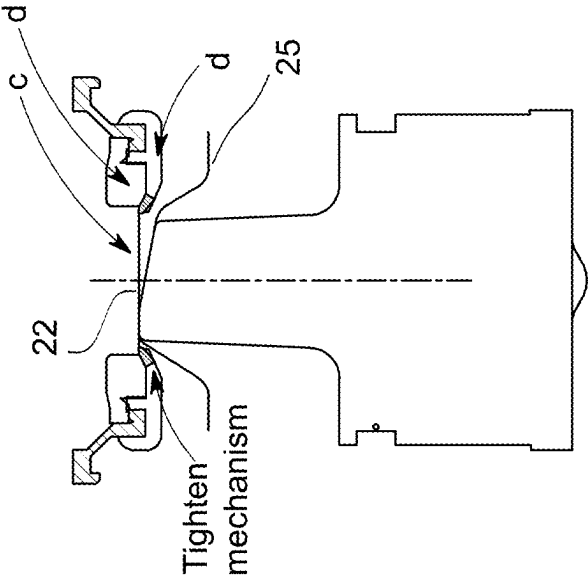


FIG. 101A

c: Polymer mesh
d: Plastic

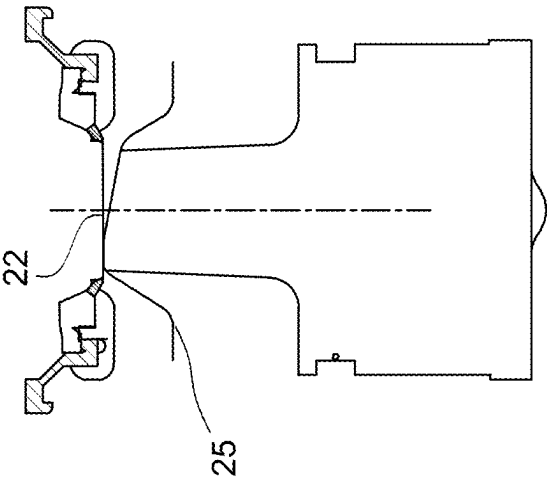


FIG. 101B

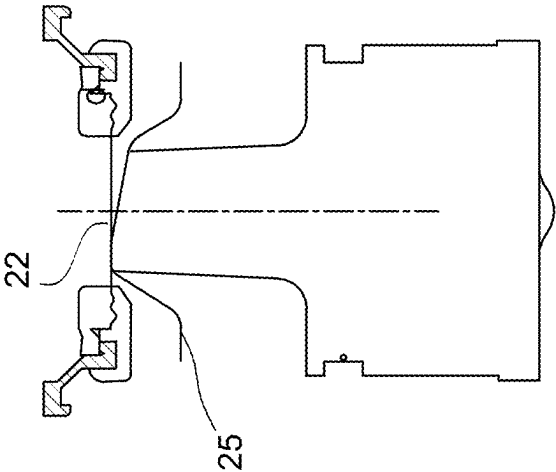


FIG. 101C

c: Polymer mesh
d: Plastic

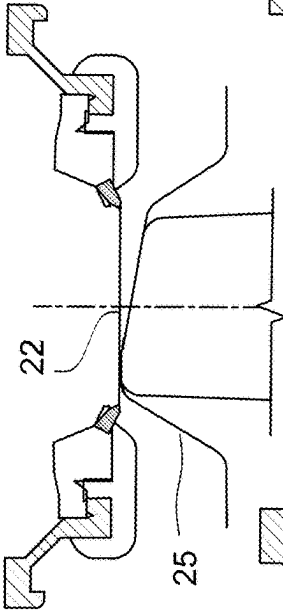


FIG. 102B

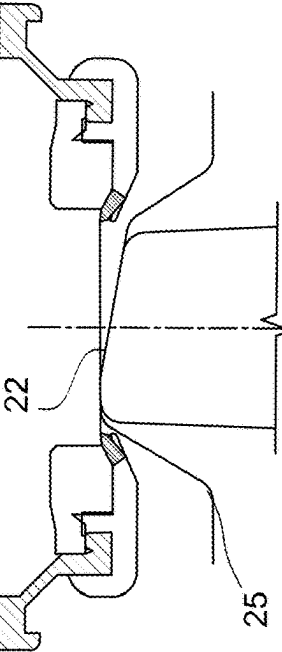
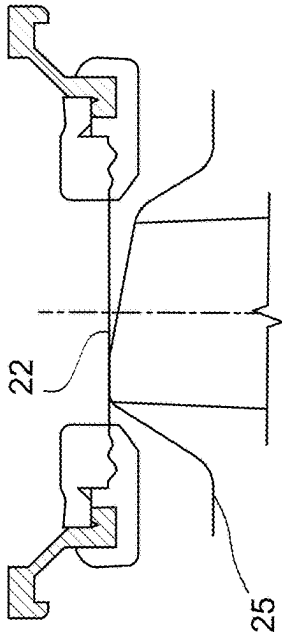
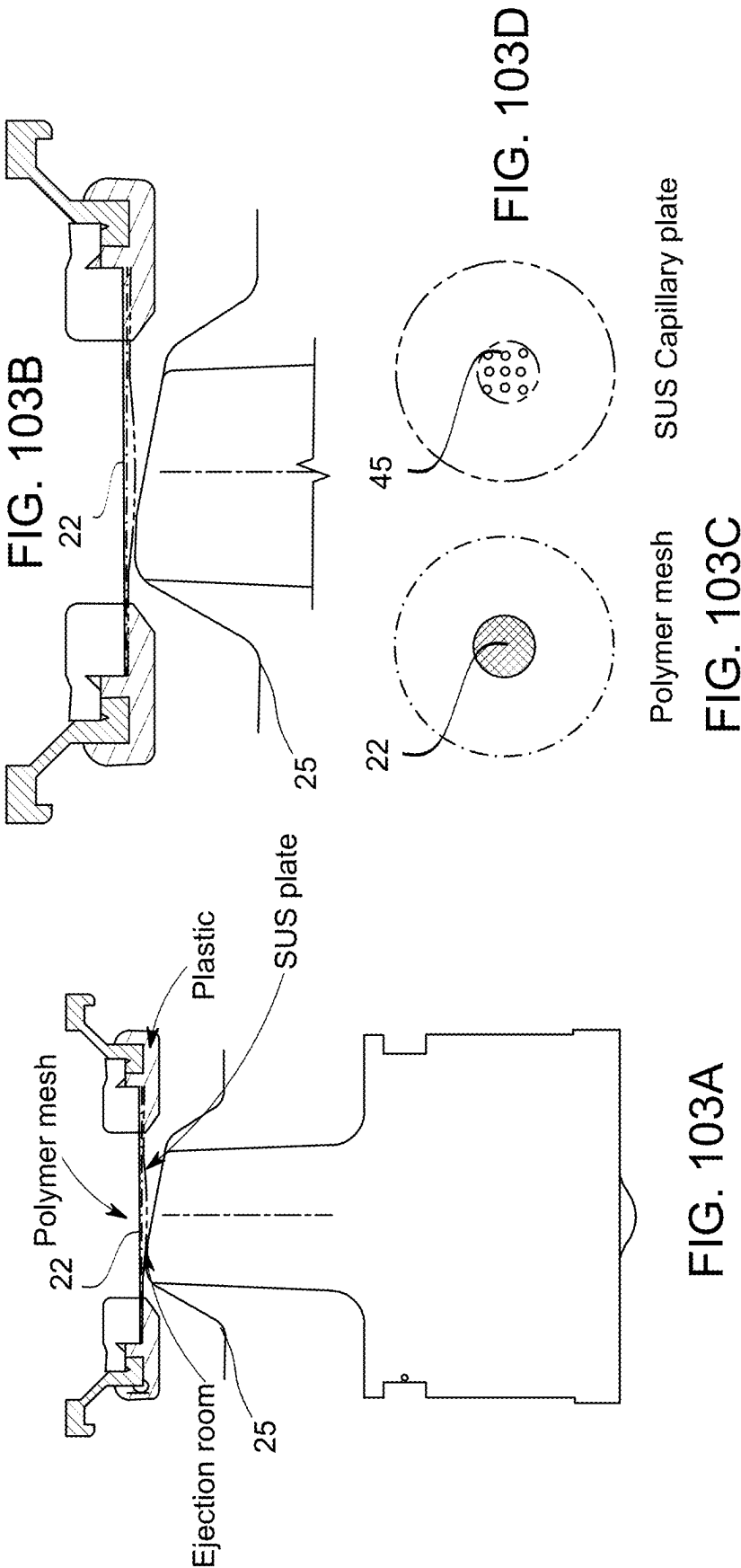


FIG. 102C





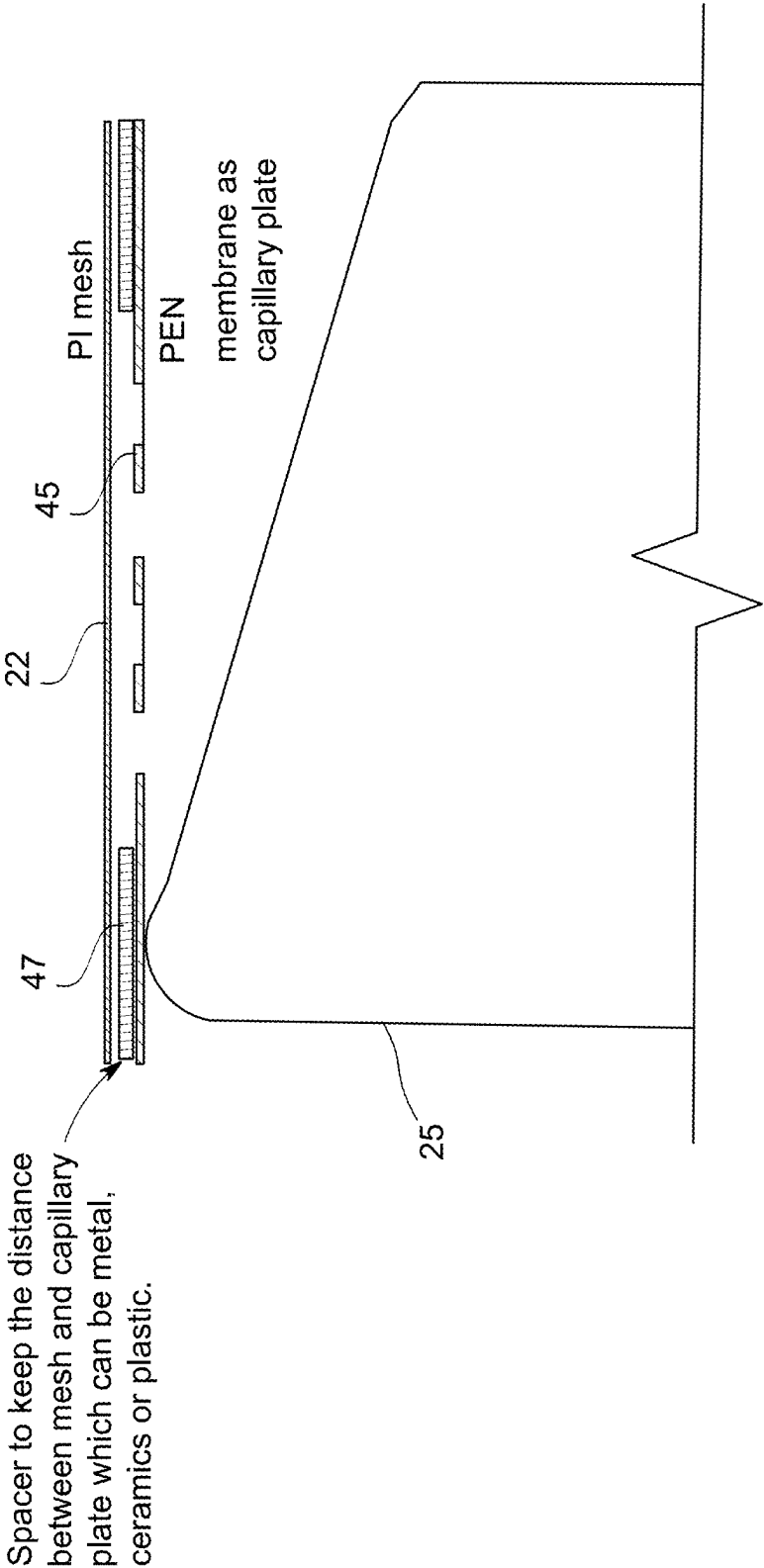


FIG. 104

a: SUS316 L
c: Polymer mesh
d: Plastic

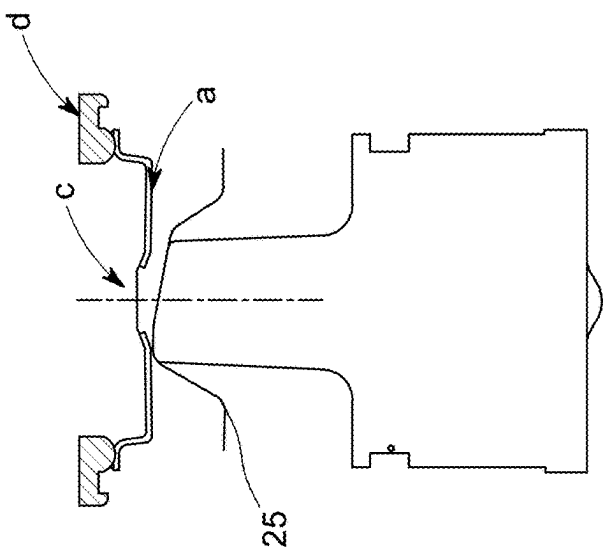
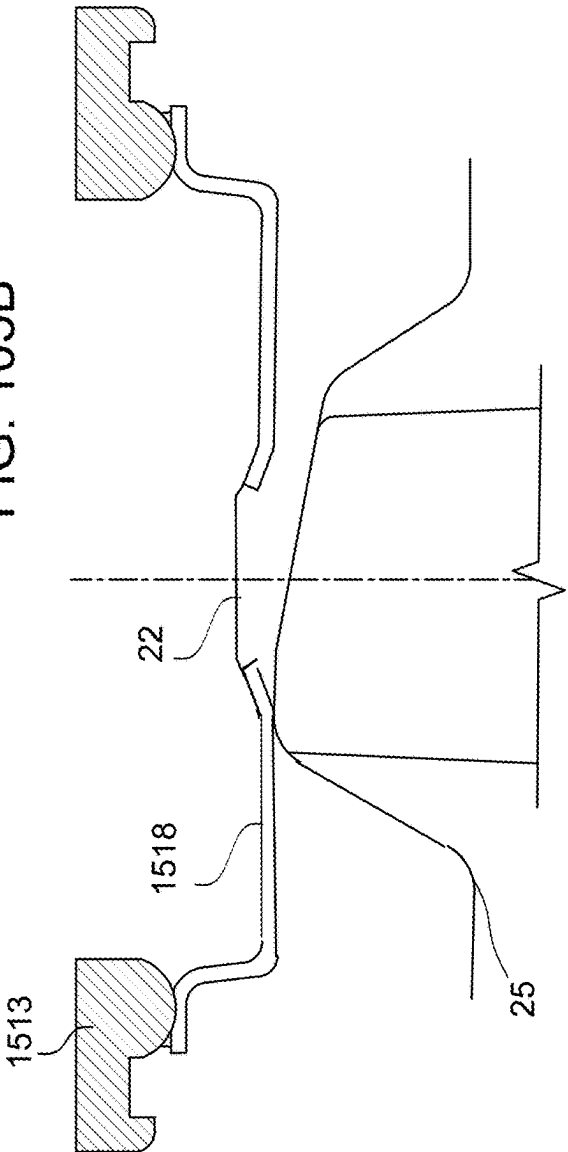


FIG. 105A

FIG. 105B



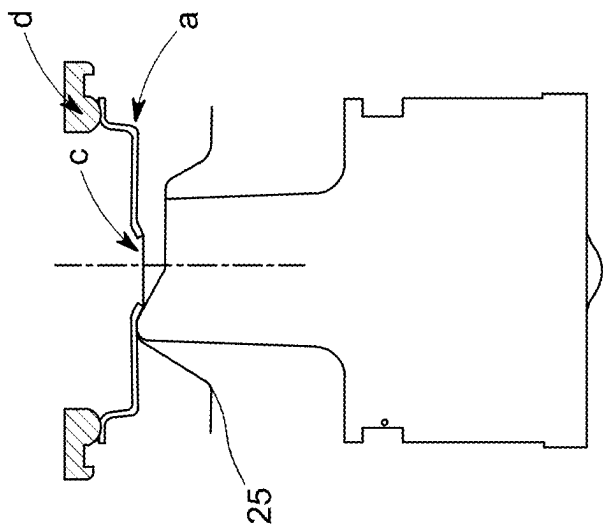


FIG. 106B

a:SUS316 L
c:Polymer mesh
d:Plastic

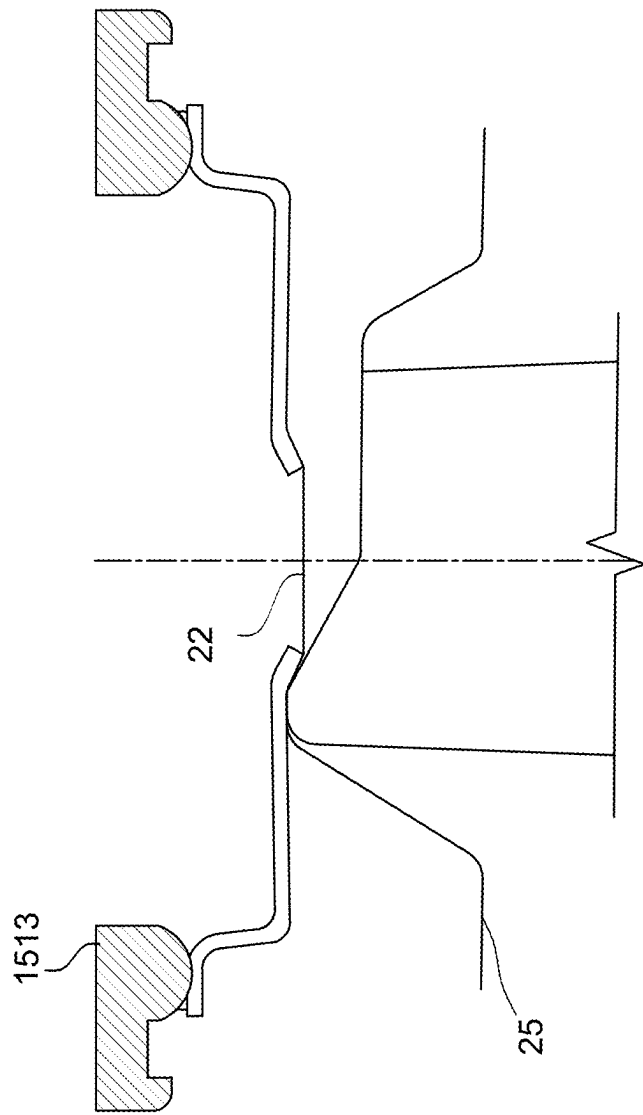


FIG. 106A

a: SUS316 L
b: SUS316 L
c: Polymer mesh
d: Plastic

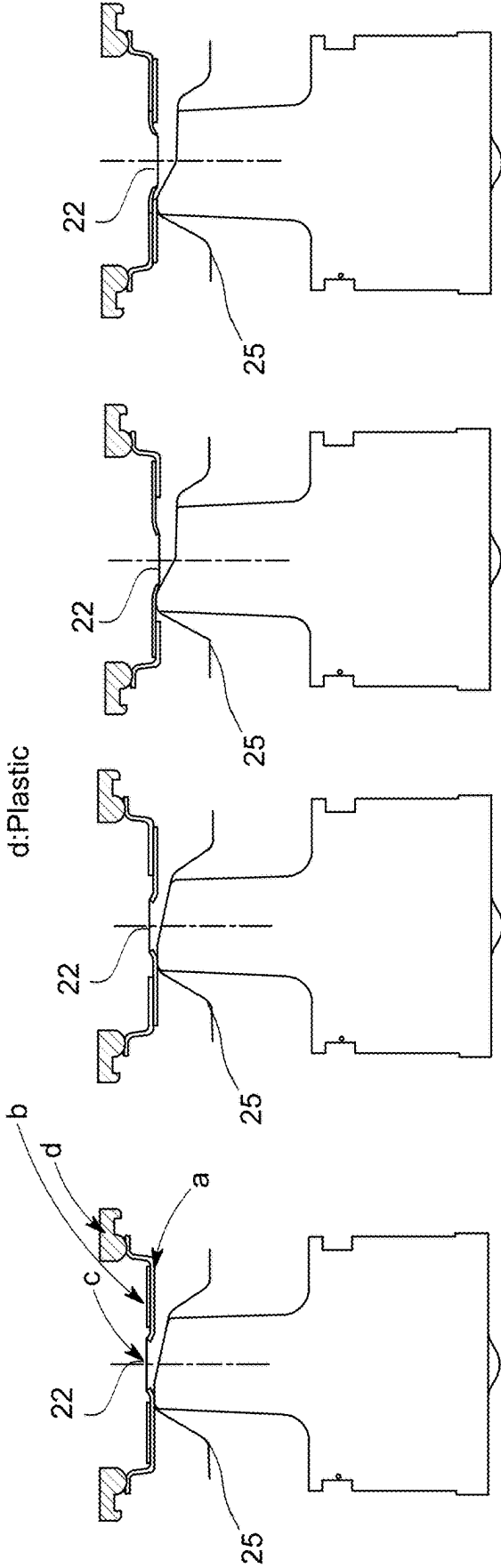
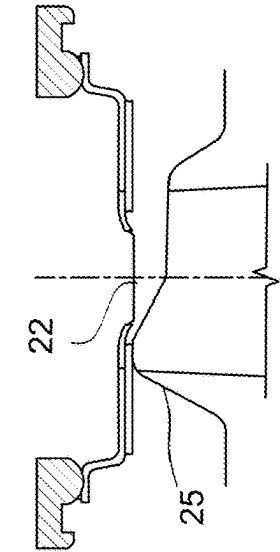
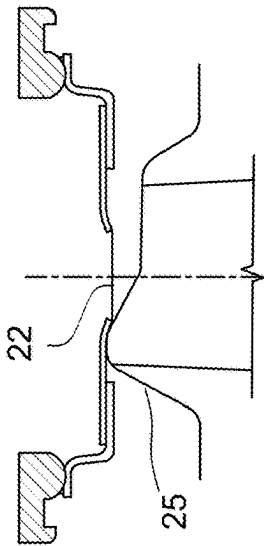
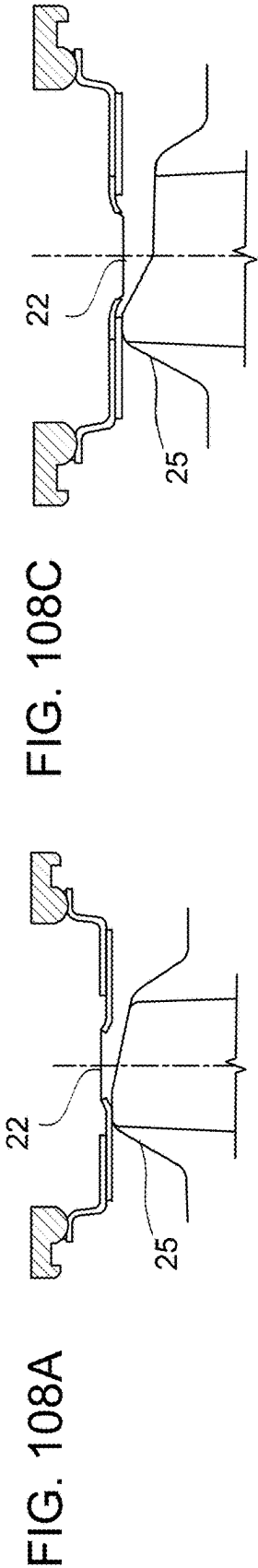
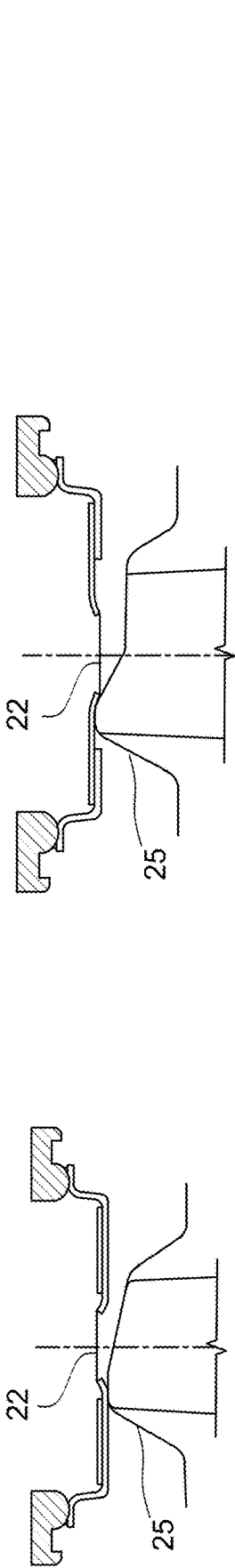


FIG. 107A

FIG. 107B

FIG. 107C

FIG. 107D



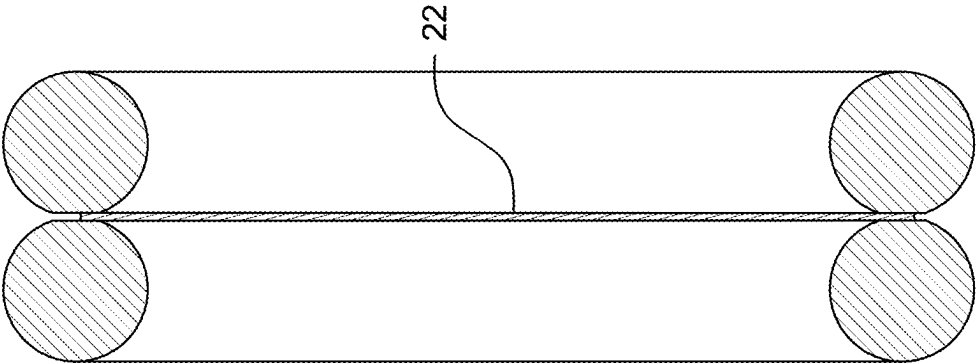


FIG. 109A

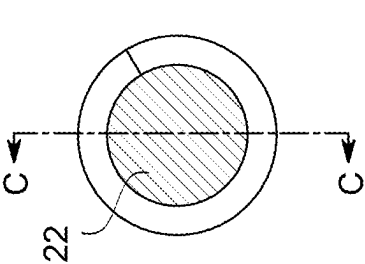


FIG. 109C

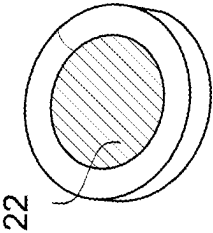
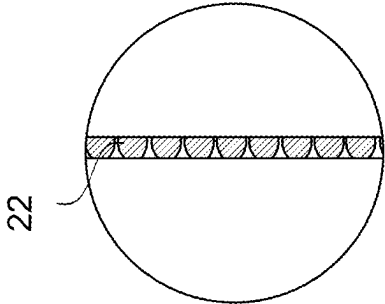


FIG. 109B



Section C-C

FIG. 109D

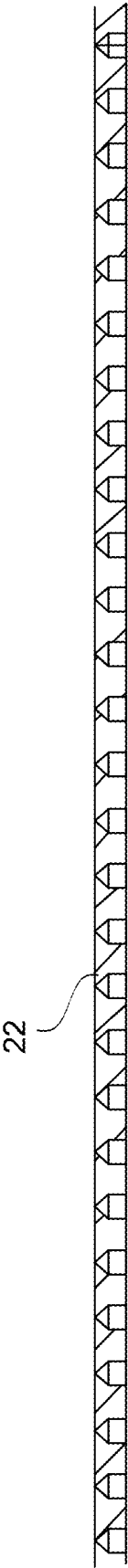


FIG. 110

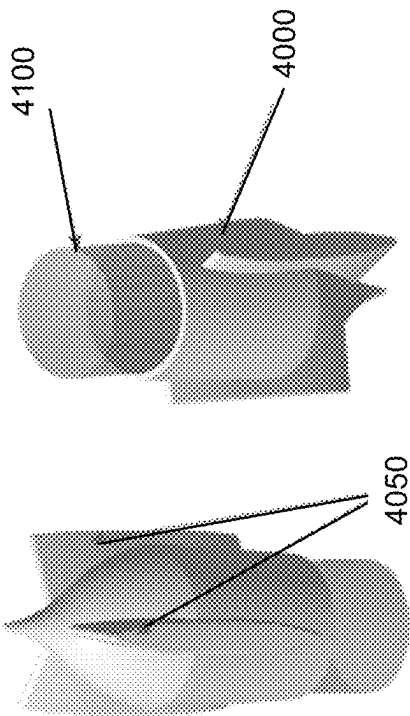


FIG. 111A

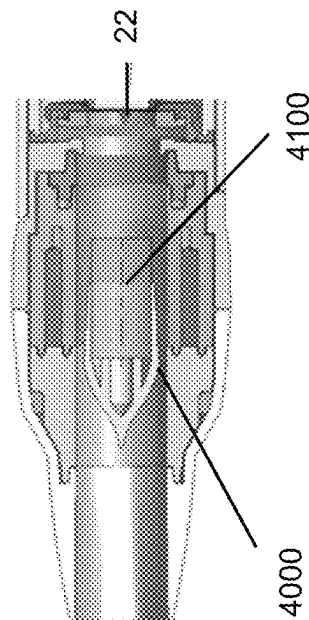


FIG. 111C

FIG. 112

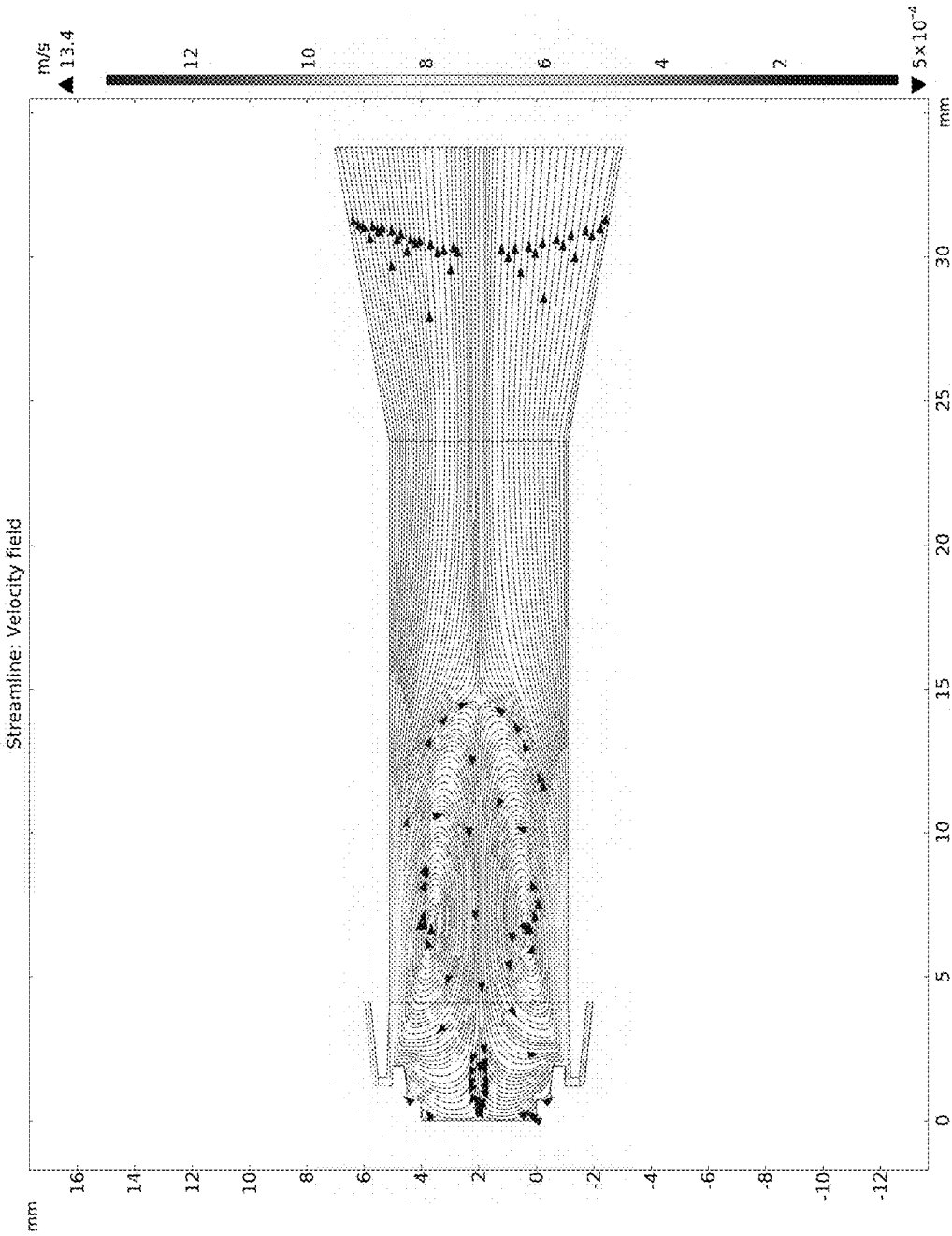


FIG. 113

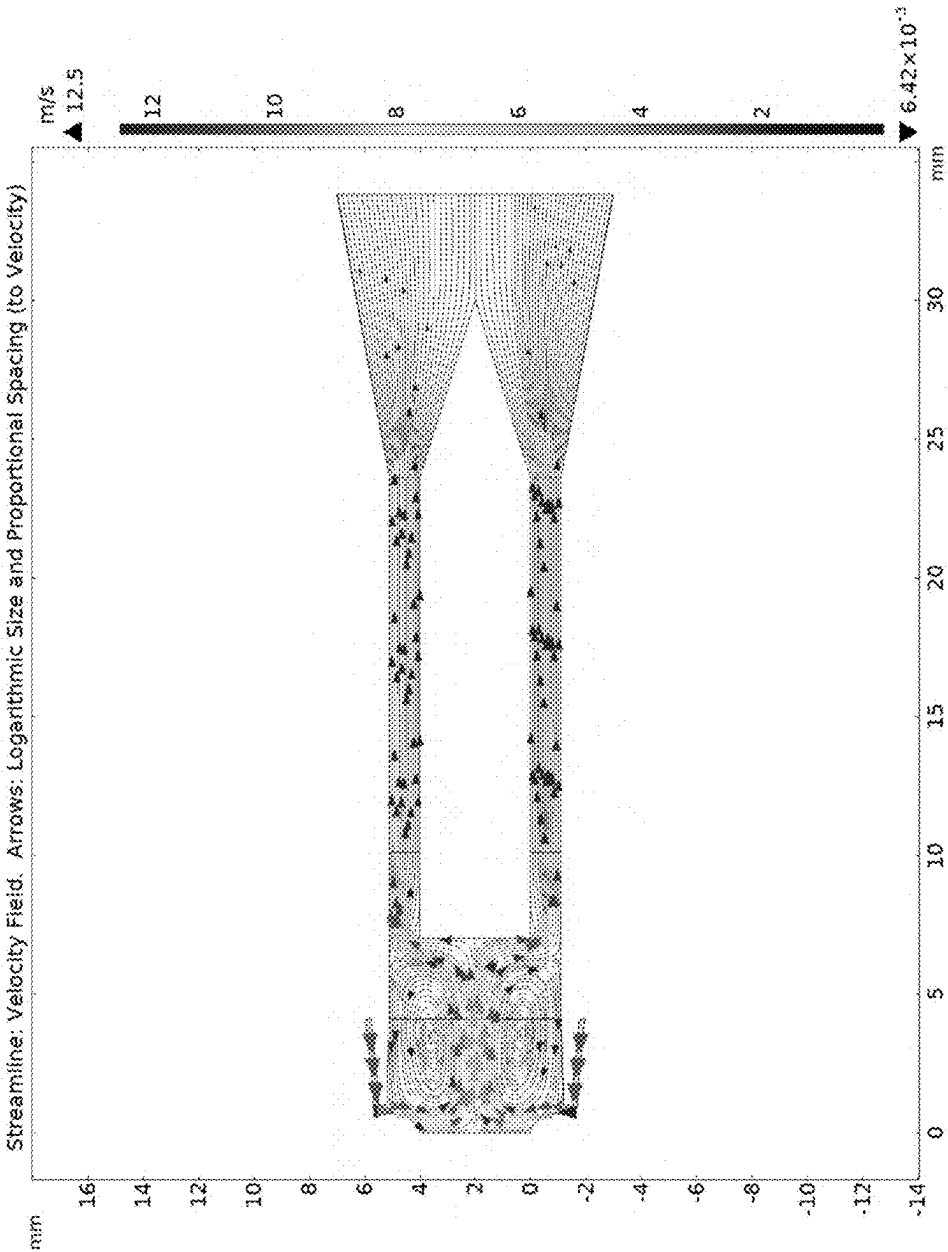
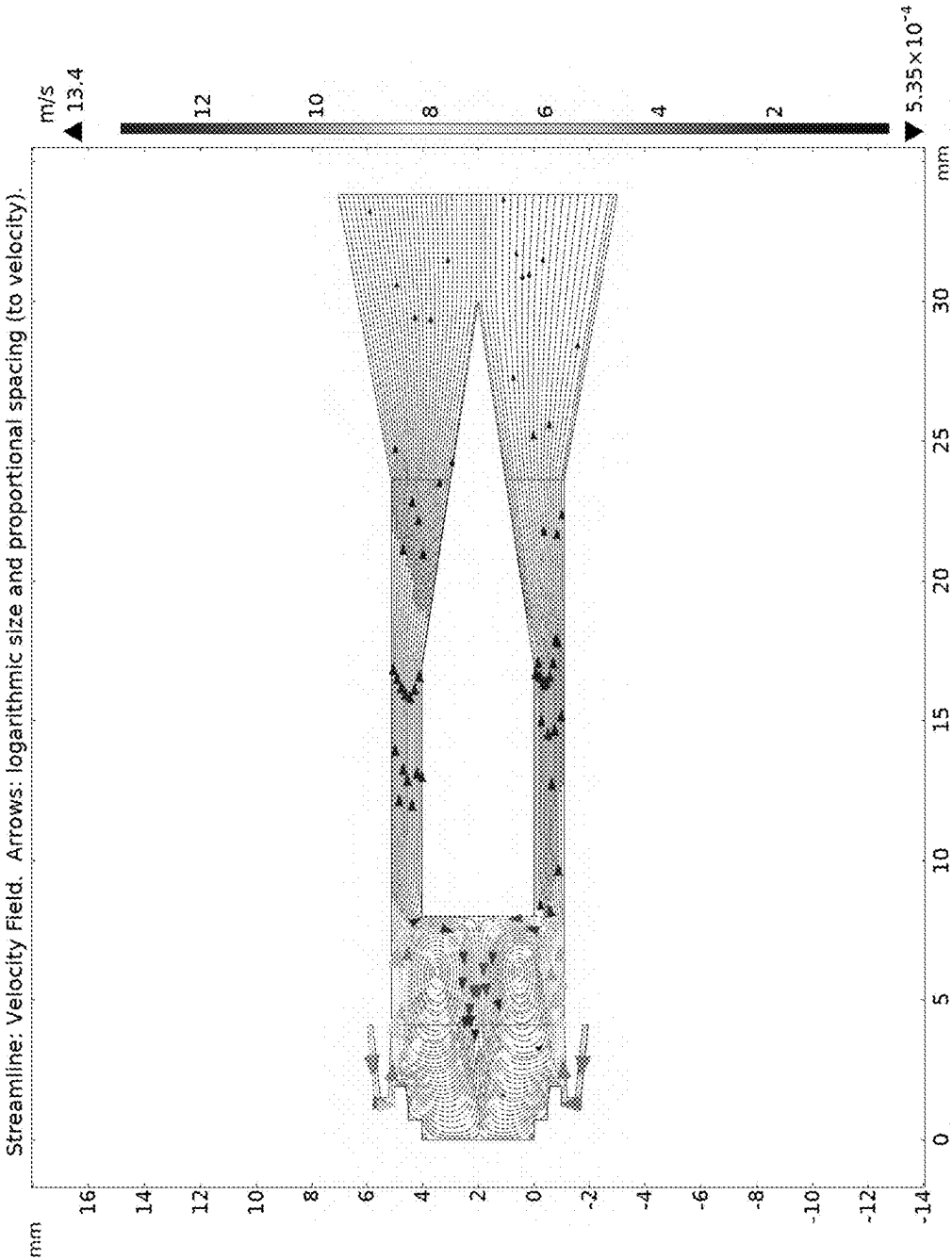


FIG. 114



1

**DROPLET DELIVERY DEVICE WITH PUSH
EJECTION****CROSS-REFERENCE TO RELATED
APPLICATIONS**

The present application is a continuation of U.S. patent application Ser. No. 17/846,902 filed Jun. 22, 2022, which claims the benefit of priority of U.S. Provisional Patent Application No. 63/280,643 filed Nov. 18, 2021, U.S. Provisional Patent Application No. 63/256,546 filed Oct. 16, 2021, Provisional Patent Application No. 63/256,245 filed Oct. 15, 2021, and Provisional Patent Application No. 63/213,634 filed Jun. 22, 2021, all of which are incorporated herein by reference in their entirety.

FIELD OF THE PUSH MODE INVENTION

This disclosure relates to droplet delivery devices with ejector mechanisms and more specifically to droplet delivery devices for the delivery of fluids that are inhaled into mouth, throat nose, and/or lungs.

**BACKGROUND OF THE PUSH MODE
INVENTION**

The use of droplet generating devices for the delivery of substances to the respiratory system is an area of large interest. A major challenge is providing a device that delivers an accurate, consistent, and verifiable amount of substance, with a droplet size that is suitable for successful delivery of substance to the targeted area of the respiratory system.

Currently most inhaler type systems, such as metered dose inhalers (MDI), pressurized metered dose inhalers (p-MDI), or pneumatic and ultrasonic-driven devices, generally produce droplets with high velocities and a wide range of droplet sizes including large droplets that have high momentum and kinetic energy. Droplet plumes with large size distributions and high momentum do not reach a targeted area in the respiratory system, but rather are deposited throughout the pulmonary passageways, mouth, and throat. Such non-targeted deposition may be undesirable for many reasons, including improper dosing and unwanted side effects.

Droplet plumes generated from current droplet delivery systems, as a result of their high ejection velocities and the rapid expansion of the substance carrying propellant, may also lead to localized cooling and subsequent condensation, deposition and crystallization of substance onto device surfaces. Blockage of device surfaces by deposited substance residue is also problematic.

Further, conventional droplet delivery devices for delivery of nicotine, including vape pens and the like, typically require fluids that are inhaled to be heated to temperatures that negatively affect the liquid being aerosolized. Specifically, such levels of heating can produce undesirable and toxic byproducts as has been documented in the news and literature.

Accordingly, there is a need for an improved droplet delivery device that delivers droplets of a suitable size range, avoids surface fluid deposition and blockage of apertures, avoids producing undesired chemical byproducts through heating, and in an amount that is consistent and reproducible.

SUMMARY OF THE PUSH MODE INVENTION

In one embodiment of the push mode invention, a “push mode” droplet delivery device does not include a heating

2

requirement that could result in undesirable byproducts and comprises: a container assembly with an mouthpiece port; a reservoir disposed within or in fluid communication with the container assembly to supply a volume of fluid, an ejector bracket in fluid communication with the reservoir, the ejector bracket including a mesh with a membrane operably coupled to an electronic transducer with the membrane between the transducer and the mesh, wherein the mesh includes a plurality of openings formed through the mesh's thickness, and wherein the transducer is coupled to a power source and is operable to oscillate the membrane and generate an ejected stream of droplets through the mesh, and an ejection channel within the container assembly configured to direct the ejected stream of droplets from the mesh to the outlet. The vibrating membrane “pushing” liquid through the mesh is referred to herein as “push mode” ejection and devices in embodiments of the push mode invention may be referred to as push mode devices.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes an ultrasonic transducer as an electronic transducer, and preferably an ultrasonic transducer that includes piezoelectric material.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes the container assembly having a fluid reservoir.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes an ejector bracket configured for releasably coupling to the container assembly and the ejector bracket further configured for releasable coupling to an enclosure system including an electronic transducer and a power source.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes magnets configured to releasably couple the ejector bracket and enclosure system.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a snap mechanism and/or magnets configured to releasably couple the ejector bracket and the container assembly.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a fluid reservoir with a self-sealing mating mechanism configured to couple to a fluid release mating mechanism of the ejector bracket.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a fluid release mating mechanism that has a fluid conduit configured for insertion into the self-sealing mating mechanism. In a preferred embodiment, a fluid release mating mechanism includes a spike-shaped structure with a hollow interior configured to provide fluid communication between the reservoir and the membrane.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh is configured so that the membrane does not contact the mesh and pushes fluid to be ejected as droplets from the droplet delivery device through openings in the mesh.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes the membrane having a slanted upper surface configured to contact fluid supplied from the reservoir.

3

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a vibrating member having a slanted tip contacting an opposite underlying surface of a slanted upper surface of the membrane.

In further embodiments of the push mode invention, an electronic transducer includes piezoelectric material that is coupled to a vibrating member with a ring-shaped beveled tip, rod-shaped beveled tip, rod-shaped tip, or a ring-shaped non-beveled tip.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a mesh with a bottom surface in a parallel configuration with an upper surface of the membrane.

In another embodiment of the push mode invention, a droplet delivery device having membrane that cooperates with a mesh further includes the mesh including a bottom surface in a non-parallel, i.e., slanted at an angle, configuration with an upper surface of the membrane.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a central axis of the droplet delivery device passing through the ejection channel and the membrane, and wherein the transducer is coupled to a vibrating member that is coupled to the membrane at a position offset from the central axis.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a fluid in the reservoir including at least one of a non-therapeutic substance, nicotine, or cannabinoid.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a fluid in the reservoir including a therapeutic substance that treats or prevents a disease or injury condition.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a laminar flow element positioned in an ejection channel of a container assembly before a mouthpiece port of the delivery device. In preferable embodiments, the laminar flow element includes a plurality of cellular apertures. In some embodiments, a laminar flow element includes blade-shaped walls defining the plurality of cellular apertures. In further embodiments, one or more of the plurality of cellular apertures include a triangular prismatic shape, quadrangular prismatic shape, pentagonal prismatic shape, hexagonal prismatic shape, heptagonal prismatic shape, or octagonal prismatic shape.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a breath-actuated sensor, such as a pressure sensor, operatively coupled to the power source, wherein the breath-actuated sensor is configured to activate the electronic transducer upon sensing a predetermined pressure change within the ejection channel or within a passageway of the droplet delivery device in fluid communication with the ejection channel.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes the mesh made of a material of at least one of palladium nickel, polytetrafluoroethylene, and polyimide.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes the mesh made of a material of

4

at least one of poly ether ketone, polyetherimide, polyvinylidene fluoride, ultra-high molecular weight polyethylene, Ni, NiCo, Pd, Pt, NiPd, and metal alloys.

In other embodiments a mesh may be made of single crystalline or poly crystalline materials such as silicon, silicon carbide, aluminum nitride or germanium with hole structures formed using semiconductor processes such as photo lithography and isotropic and anisotropic etching. With photolithography and isotropic and/or anisotropic etches different hole shapes can be formed in a single crystalline wafer with very high precision. Using sputtering, films can be deposited on the surface with different contact angles. Thin layers formed or deposited on the surface will have, in certain embodiments, much better adherence than film deposited on metal mesh formed by galvanic deposition or polymer mesh formed by laser ablation. This better adherence is because the surfaces on the single crystalline wafers "slices" are atomically smooth and can be etched to produce exact surface roughness to facilitate mechanical bonding with glue or other materials. Silicon carbide would be a preferable material because of its high strength and toughness. An important advantage of using semiconductor processes to fabricate hole structures from a single crystalline wafer "slice" in a mesh of embodiment of the push mode invention is that the holes and surface contact angles will be exact without the variation we see in conventional ejector plates using mesh made from galvanic deposition or laser ablation.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes the membrane made of a material of at least one of polyethylene naphthalate, polyethylenimine and poly ether ketone.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes the membrane made of a material of at least one of metal membranes, metalized polymers, threaded polymers, threaded nylon, threaded polymers that are coated with polymers or metal, threaded nylon coated with polymers or metal, threaded metals, threaded SiC, threaded graphite composites, metalized graphite composites, graphite composites coated with polymers, polymer sheets filled with carbon fibers, poly ether ketone filled with carbon fibers, polymer sheets filled with SiC fibers, polymer sheets filled with ceramic or metal fibers, ULPA filter media, NITTO DENKO™ Temic Grade filter media, NITTO DENKO™ polymer sheets, threaded polymers bonded to a polymer sheet, nylon weave bonded to poly ether ketone or polyimide, graphite composites bonded to polymer sheets, polymer fiber weave with metalized coating, and nylon with sputtered on Al or vapor deposited Al.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a PZT-based ultrasonic transducer coupled to a vibrating member having a tip portion made of at least one of Grade 5 titanium alloy, Grade 23 titanium alloy, and about 99% or higher purity titanium. In certain embodiments, the vibrating member's tip includes a sputtered on outer layer of and about 99% or higher purity titanium providing a smooth tip surface configured to contact an underlying bottom surface of the membrane that is opposite an exterior top surface of the membrane positioned nearest the mesh so as to help reduce wear of the membrane and increase the longevity and operation consistency of the membrane (and also possibly vibrating member's tip portion).

5

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes an exterior surface of the membrane, opposite an underlying surface of the membrane contacting the vibrating member, having a hydrophobic coating.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes an exterior surface of the membrane, opposite an underlying surface of the membrane contacting the vibrating member, having a hydrophilic coating.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a hydrophilic coating on one or more surfaces of the mesh.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a mesh including a hydrophobic coating on one or more surfaces of the mesh.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a hydrophobic coating on a first surface of the mesh and a hydrophilic coating on a second surface of the mesh.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes the membrane having an operable lifespan of over 55,000 aerosol-creating activations by the transducer.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes at least one superhydrophobic vent in fluid communication with the reservoir that is covered with a removable aluminized polymer tab during storage.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh further includes a removable aluminized polymer tab coupled to an exterior surface of the membrane adjacent the mesh during storage.

In another embodiment of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh includes a pre-assembly step of removing a sealed packaging including aluminum and/or aluminum coating that contains the reservoir with a fluid, preferably wherein the reservoir is included in the container assembly that is also packaged for storage in the sealed packaging. In some embodiments, sealed packaging may include dry nitrogen, argon or other gas that does not contain oxygen.

In various embodiments of the push mode invention, a droplet delivery device having a membrane that cooperates with a mesh may be used for mouth inhalation or nasal inhalation. The mouthpiece port may be sized, shaped and include materials that are better suited for that particular mouth or nasal inhalation use and purpose.

BRIEF DESCRIPTION OF THE DRAWINGS

The push mode invention will be more clearly understood from the following description given by way of example, in which:

FIG. 1A is an exploded view of major components of a droplet delivery device in accordance with an embodiment of the disclosure.

6

FIG. 1B is a cross-sectional view of major components of a droplet delivery device in accordance with an embodiment of the disclosure.

FIG. 2 is a schematic view of a mesh bonded to a stainless-steel ring that supports an elastic sealing ring of a droplet delivery device in an accordance with an embodiment of the disclosure referred to as push mode II.

FIG. 3 is a schematic view of a mesh supported by inner and outer tablet rings and an elastic sealing ring of a droplet delivery device in an accordance with an embodiment of the disclosure referred to as push mode I.

FIG. 4 illustrates a cross-sectional view of certain dimensions of ejection and mouthpiece ports of a droplet delivery device in accordance with one embodiment of the disclosure.

FIG. 5 illustrates a cross-sectional view of fluid flow path of a droplet delivery device with a two-part cartridge in accordance with one embodiment of the disclosure.

FIGS. 6A and 6B illustrate airflow of a droplet delivery device with a two-part cartridge in accordance with an embodiment of the disclosure.

FIGS. 7A and 7B illustrate perspective views of the disassembly of major components of a push mode I droplet delivery device (utilizing mesh support shown in FIG. 3) in an embodiment of the disclosure.

FIG. 8 illustrates an exploded view of a push mode I droplet delivery device (utilizing mesh support shown in FIG. 3) in an embodiment of the disclosure.

FIGS. 9A-9F illustrate isolated perspective views of COC (cyclic olefin copolymer) rings, including mesh (22), of a push mode I droplet delivery device (utilizing mesh support shown in FIG. 3) in an embodiment of the disclosure.

FIG. 10 illustrates a schematic view of a push mode I droplet delivery device mesh suspension system (redundant to FIG. 3) in an embodiment of the disclosure.

FIG. 11 illustrates a perspective view of lower ejector bracket including vents located on each narrow side of the bracket of a push mode I droplet delivery device (utilizing mesh support shown in FIG. 3) in an embodiment of the disclosure.

FIGS. 12A and 12B illustrate perspective views of the disassembly of major components of a push mode II droplet delivery device (utilizing mesh support shown in FIG. 2) in an embodiment of the disclosure.

FIG. 13 illustrates an exploded view of a push mode II droplet delivery device (utilizing mesh support shown in FIG. 2) in an embodiment of the disclosure.

FIG. 14 illustrates a schematic view of a push mode II droplet delivery device mesh suspension system (as also shown in FIG. 2) in an embodiment of the disclosure.

FIG. 15 illustrates a perspective view of lower ejector bracket including vents located on each wide side of the bracket of a push mode II droplet delivery device (utilizing mesh support shown in FIG. 2) in an embodiment of the disclosure.

FIG. 16 illustrates a lower container of a push mode II droplet delivery device (utilizing mesh support shown in FIG. 2) in an embodiment of the disclosure.

FIG. 17 illustrates a lower container of a push mode I droplet delivery device (utilizing mesh support shown in FIG. 3) in an embodiment of the disclosure.

FIG. 18 illustrates a perspective view of a rod tip design for a vibrating member of a droplet delivery device in accordance with one embodiment of the disclosure.

FIG. 19 illustrates a perspective view of a ring tip design for a vibrating member of a droplet delivery device in accordance with one embodiment of the disclosure.

FIG. 20 illustrates a cross-sectional view of single part cartridge design with a long vibrating member in a droplet delivery device in accordance with one embodiment of the disclosure.

FIGS. 21A and 21B illustrate cross-sectional views of single part cartridge design with a short vibrating member in a droplet delivery device in accordance with one embodiment of the disclosure.

FIGS. 22A and 22B illustrate cross-sectional views of single part cartridge alternative designs with a long vibrating member in a droplet delivery device in accordance with one embodiment of the disclosure.

FIGS. 23A and 23B illustrate cross-sectional views of single part cartridge alternative designs with a short vibrating member in a droplet delivery device in accordance with one embodiment of the disclosure.

FIG. 24 illustrates a cross-sectional and separated view of a two-part cartridge design in a droplet delivery device in accordance with one embodiment of the disclosure.

FIG. 25 illustrates a perspective view of a droplet delivery device adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 26 illustrates an exploded view of a droplet delivery device adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 27A-27D illustrate views of major components of a droplet delivery device adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 28 illustrates an assembly view of major components of a droplet delivery device adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 29 illustrates an exploded view of a cap of a droplet delivery device adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 30A and 30B illustrate respective front and side cross-sectional views of a fluid cartridge of a droplet delivery device adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 31 illustrates a cross-sectional view of a vibrating member enclosure of a droplet delivery device adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 32 illustrates a cross-sectional view of an ejector bracket adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing a mesh suspension system that follows structure and function of the mesh support shown in FIG. 14 in accordance with one embodiment of the disclosure.

FIG. 33 illustrates a cross-sectional view of an ejector bracket adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing a mesh suspension

system that follows the structure and function of the mesh support shown in FIG. 10 in accordance with one embodiment of the disclosure.

FIGS. 34A and 34B illustrate respective side and front cross-sectional views of a droplet delivery device adapted for pharmaceutical use (but may be other uses in other embodiments) and utilizing membrane-driven aerosolization (i.e. "push mode functionality") having two heating elements positioned beneath a vibrating member on either side of the ejector bracket in accordance with one embodiment of the disclosure.

FIG. 35A-35C illustrate cross-sectional views of an airflow path for a droplet delivery device having a bottom heating element with a single part cartridge design in accordance with one embodiment of the disclosure.

FIG. 36 illustrates cross-sectional view of a droplet delivery device having a bottom heating element and speaker with a single part cartridge design in accordance with one embodiment of the disclosure.

FIG. 37 illustrates a cross-sectional view of an airflow path for a droplet delivery device having an inside heating element with a two-part cartridge design in accordance with one embodiment of the disclosure.

FIG. 38 illustrates a cross-sectional view of an airflow path for a droplet delivery device having an inside heating element with a single part cartridge design in accordance with one embodiment of the disclosure.

FIG. 39 illustrates a cross-sectional view of an airflow path for a droplet delivery device having an external heating element with a single part cartridge design in accordance with one embodiment of the disclosure.

FIG. 40 illustrates a cross-sectional view of droplet delivery device with a heated airstream including a temperature sensor that is used in conjunction with a closed loop system to keep the temperature of the airstream constant, and also avoid overheating and user injury, in accordance with one embodiment of the disclosure.

FIGS. 41A and 41B illustrate views of a droplet delivery device with adjustable air resistance via sliding sleeve and associated vent in accordance with one embodiment of the disclosure.

FIG. 42 illustrates an elongated and narrow inhalation port of a droplet delivery device adapted for nasal inhalation and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 43 illustrates a shorter-version inhalation port of a droplet delivery device adapted for nasal inhalation and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 44A and 44B illustrate a removal cap of a droplet delivery device adapted for nasal inhalation and utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 45 illustrates a mesh with an attached plate having multiple openings for liquid to enter used in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 46 illustrates a cross-sectional view of a capacitance cartridge having two parallel plates placed across the liquid next to the mesh-membrane area in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 47A-47C illustrate a perspective view (FIG. 47A), front plan view (FIG. 47B) and side plan view (FIG. 47C) of a rectangular vibrating member tip in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 48A-48C illustrate a perspective view (FIG. 48A), a perspective vibration amplitude map view (FIG. 48B) and a top vibration amplitude plan view (FIG. 48C) of an eigenmode vibrating member tip without slots or tuning and the resulting vibration amplitude maps in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 49A-49C illustrate a perspective view (FIG. 49A), a perspective vibration amplitude map view (FIG. 49B) and a top vibration amplitude plan view (FIG. 49C) of an eigenmode vibrating member tip with slots and the resulting vibration amplitude maps in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 50 illustrates a contoured vibrating member in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 51 illustrates a plunger vibrating member in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIG. 52 illustrates a sensor carrier vibrating member in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 53A and 53B illustrate a spool vibrating member and the resulting vibration amplitude map in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 54A and 54B illustrate an optimized cylindrical vibrating member and the resulting vibration amplitude map in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 55A and 55B illustrate an unoptimized slotted cylindrical vibrating member and the resulting vibration amplitude map in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 56A and 56B illustrate an optimized bar vibrating member and the resulting vibration amplitude map in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 57A and 57B illustrate an unoptimized bar vibrating member and the resulting vibration amplitude map in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 58A and 58B illustrate a perspective view (FIG. 58A) and cross-sectional vibration amplitude view (58B) of a booster vibrating member and the resulting vibration amplitude map in a droplet delivery device utilizing membrane-driven aerosolization (i.e. "push mode functionality") in accordance with one embodiment of the disclosure.

FIGS. 59A-59C illustrate a perspective view (FIG. 59A), top plan view (FIG. 59B) and front plan view (FIG. 59C) of

an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 60A-60C illustrate a perspective view (FIG. 60A), top plan view (FIG. 60B) and front plan view (FIG. 60C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 61A-61C illustrate a perspective view (FIG. 61A), top plan view (FIG. 61B) and front plan view (FIG. 61C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 62A-62C illustrate a perspective view (FIG. 62A), top plan view (FIG. 62B) and front plan view (FIG. 62C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 63A-63C illustrate a perspective view (FIG. 63A), top plan view (FIG. 63B) and front plan view (FIG. 63C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 64A-64C illustrate a perspective view (FIG. 64A), top plan view (FIG. 64B) and front plan view (FIG. 64C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 65A-65D illustrate a perspective view (FIG. 65A), top plan view (FIG. 65B), front plan view (FIG. 65C) and cross-sectional view along A-A of FIG. 65B (FIG. 65D) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 66A-66C illustrate a perspective view (FIG. 66A), top plan view (FIG. 66B) and front plan view (FIG. 66C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 67A-67C illustrate a perspective view (FIG. 67A), top plan view (FIG. 67B) and front plan view (FIG. 67C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 68A-68D illustrate a perspective view (FIG. 68A), top plan view (FIG. 66B), front plan view (FIG. 66C) and side plan view (66D) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 69A and 69B illustrate a perspective view (FIG. 69A) and side plan view (FIG. 69B) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 70A-70C illustrate a perspective view (FIG. 70A), top plan view (FIG. 70B) and front plan view (FIG. 70C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 71A-71C illustrate a perspective view (FIG. 71A), top plan view (FIG. 71B) and front plan view (FIG. 71C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 72A-72C illustrate a perspective view (FIG. 72A), top plan view (FIG. 72B) and front plan view (FIG. 72C) of

11

an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 73A-73C illustrate a perspective view (FIG. 73A), top plan view (FIG. 73B) and front plan view (FIG. 73C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 74A-74C illustrate a perspective view (FIG. 74A), top plan view (FIG. 74B) and front plan view (FIG. 74C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 75A-75C illustrate a perspective view (FIG. 75A), top plan view (FIG. 75B) and front plan view (FIG. 75C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 76A-76C illustrate a perspective view (FIG. 76A), top plan view (FIG. 76B) and front plan view (FIG. 76C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 77A-77D illustrate a perspective view (FIG. 77A), top plan view (FIG. 77B), front plan view (FIG. 77C) and side plan view (FIG. 77D) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 78A-78C illustrate a perspective view (FIG. 78A), top plan view (FIG. 78B) and front plan view (FIG. 78C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 79A-79C illustrate a perspective view (FIG. 79A), top plan view (FIG. 79B) and front plan view (FIG. 79C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 80A-80D illustrate a perspective view (FIG. 80A), top plan view (FIG. 80B), front plan view (FIG. 80C) and side plan view (FIG. 80D) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 81A-81D illustrate a perspective view (FIG. 81A), top plan view (FIG. 81B), front plan view (FIG. 81C) and side plan view (FIG. 81D) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 82A-82D illustrate a perspective view (FIG. 82A), top plan view (FIG. 82B), front plan view (FIG. 82C) and side plan view (FIG. 82D) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 83A-83C illustrate a perspective view (FIG. 83A), top plan view (FIG. 83B) and front plan view (FIG. 83C) of an alternative vibrating member that couples to a transducer of droplet delivery devices in accordance with an embodiment of the disclosure.

FIGS. 84A-84Q illustrate alternative structures of laminar flow elements of a container assembly of a droplet delivery device in accordance with embodiments of the disclosure.

FIG. 85A illustrates an ultrasonic transducer, including a vibrating member tip portion, in a droplet delivery device in accordance with an embodiment of the disclosure.

12

FIG. 85B is a partial cross-sectional top view of the ultrasonic transducer of FIG. 85A coupling to a membrane in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 85C and 85D are schematic views of the ultrasonic transducer and membrane of FIG. 85B in droplet delivery devices in accordance with alternative embodiments of the disclosure wherein a mesh includes first securing mechanism in FIG. 85C and second securing mechanism in FIG. 85D.

FIG. 86A is a partial cross-sectional top view of an ultrasonic transducer coupled to a membrane in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 86B and 86C are schematic views of the ultrasonic transducer and membrane of FIG. 86A in droplet delivery devices in accordance with alternative embodiments of the disclosure wherein a mesh includes first securing mechanism in FIG. 86B and second securing mechanism in FIG. 86C.

FIG. 87 is partial cross-sectional top view of a droplet delivery device including an ultrasonic transducer with vibrating member tip portion offset from a central axis of the droplet delivery device passing through a slanted membrane and mesh in accordance with an embodiment of the disclosure.

FIG. 88A is a partial cross-sectional top view of an ultrasonic transducer with a non-beveled ring-shaped vibrating member tip portion coupled to a tilted mesh in a droplet delivery device in accordance with an embodiment of the disclosure.

FIG. 88B is a schematic view of the ultrasonic transducer and membrane of FIG. 88A in droplet delivery devices in accordance with an embodiment of the disclosure.

FIG. 89A is a partial cross-sectional top view of an ultrasonic transducer with a beveled ring-shaped vibrating member tip portion coupled to a slanted membrane in a droplet delivery device in accordance with an embodiment of the disclosure.

FIG. 89B illustrates a slanted membrane that cooperates with an ultrasonic transducer and mesh illustrated in FIG. 89A.

FIGS. 89C and 89D are schematic views of the ultrasonic transducer and membrane of FIG. 89A in droplet delivery devices in accordance with alternative embodiments of the disclosure wherein a mesh includes first securing mechanism in FIG. 89C and second securing mechanism in FIG. 89D.

FIG. 89E illustrates an ultrasonic transducer with a beveled ring-shaped vibrating member tip portion of FIG. 89A.

FIG. 90A is a partial cross-sectional top view of an ultrasonic transducer with a non-beveled ring-shaped vibrating member tip portion coupled to a membrane and touching the mesh in a droplet delivery device in accordance with an embodiment of the disclosure.

FIG. 90B is a schematic view of the ultrasonic transducer and membrane of FIG. 90A in droplet delivery devices in accordance with an embodiment of the disclosure. This embodiment can be used with a mesh carrier either of push mode I or II.

FIG. 91A is a partial cross-sectional top view of an ultrasonic transducer with a beveled ring-shaped vibrating member tip portion coupled to a slanted membrane with a space between the mesh and membrane in a droplet delivery device in accordance with an embodiment of the disclosure.

FIG. 91B is a schematic view of the ultrasonic transducer and membrane of FIG. 91A in droplet delivery devices in

13

accordance with an embodiment of the disclosure. This embodiment can be used with a mesh carrier of either push mode I or II.

FIG. 92 is schematic view of an ultrasonic transducer with a non-beveled ring-shaped vibrating member tip portion coupled to a membrane with a space between the mesh and membrane in a droplet delivery device in accordance with an embodiment of the disclosure. This embodiment can be used with a mesh carrier of either push mode I or II.

FIGS. 93A-93C are schematic views of an ultrasonic transducer of a droplet delivery device with an isolation view (FIG. 93A) with a cross-sectional view along line B-B of FIG. 93A (FIG. 93B) and a cross-sectional view along line A-A of FIG. 93B (FIG. 93C) of the ultrasonic transducer having a wide and flat vibrating member tip portion together with membrane and mesh in accordance with an embodiment of the disclosure.

FIGS. 94A-94D are schematic views of a droplet delivery device (FIG. 94A) with a cross-sectional isolation views of the ultrasonic transducer along line B-B of FIG. 94A (FIG. 94B), an isolation view (FIG. 94C) and a cross-section view along line A-A of FIG. 94C (FIG. 94D) of the ultrasonic transducer having a wide and ring-shaped tip portion together with membrane and mesh in accordance with an embodiment of the disclosure.

FIG. 95 is a schematic block illustration of an aluminized polymer tab in an embodiment of the disclosure.

FIGS. 96A-96D are perspective views of a membrane of a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 97A and 97B illustrate a cross-sectional view (FIG. 97A) and a zoomed view (FIG. 97B) of a polymer mesh supported in a raised position by a stainless-steel annulus with respect to a membrane and transducer coupled to a vibrating member having a tip portion in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 98A and 98B illustrate a cross-sectional view (FIG. 98A) and a zoomed view (FIG. 98B) of a polymer mesh supported in a lowered position by a stainless-steel annulus with respect to a membrane and transducer coupled to a vibrating member having a tip portion in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 99A and 99B illustrate a cross-sectional view (FIG. 99A) and a zoomed view (FIG. 99B) of a polymer mesh supported in a raised position by a first stainless-steel annulus and having a second stainless steel annulus as a reinforcement coupled, such as by bonding with glue or adhesive, on top of the first annulus with respect to a membrane and transducer coupled to a vibrating member having a tip portion in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 100A and 100B illustrate a cross-sectional view (FIG. 100A) and a zoomed view (FIG. 100B) of a polymer mesh supported in a lowered position by a first stainless-steel annulus and having a second stainless steel annulus as a reinforcement coupled, such as by bonding with glue or adhesive, below the first annulus with respect to a membrane and transducer coupled to a vibrating member having a tip portion in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 101A-101C illustrate cross-sectional views of a polymer mesh supported in a raised position (FIG. 101A), lowered position (FIG. 101B) and via jagged support (FIG. 101C) with plastic elements of a ring-like support (and without a metal annulus) with respect to membranes and transducer coupled to a vibrating member in droplet delivery devices in accordance with embodiments of the disclosure.

14

FIGS. 102A-C illustrate zoomed views of FIG. 101A (FIG. 102A), FIG. 101B (FIG. 102B) and FIG. 101C (FIG. 102C).

FIGS. 103A and 103B illustrate a cross-sectional view (FIG. 103A) and a zoomed view (FIG. 103B) of a polymer mesh and stainless-steel capillary plate having openings in the plate and the plate underlying the polymer mesh between a membrane covering a vibrating member tip portion and the mesh in a droplet delivery device in accordance with an embodiment of the disclosure.

FIG. 103C is a schematic top plan view of a polymer mesh illustrated in FIGS. 103A and 103B.

FIG. 103D is a schematic top plan view of a stainless-steel capillary plate illustrated in FIGS. 103A and 103B.

FIG. 104 illustrates a schematic view of a polymer mesh and capillary plate wherein the capillary plate is made of PEN material like the membrane covering the vibrating member (also made of PEN material) and further including a spacer (such as metal, ceramics or plastic) between the capillary plate and the mesh in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 105A and 105B illustrate a cross-sectional view (FIG. 105A) and zoomed view (FIG. 105B) of a polymer mesh with a plastic or silicone ring-shaped type bracket (d) coupled to a stainless steel annulus shaped downward and then up toward a center portion of the annulus that couples to a polymer mesh with respect to a membrane and transducer coupled to a vibrating member having a tip portion in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 106A and 106B illustrates a cross-sectional view (FIG. 106A) and zoomed view (FIG. 106B) of a polymer mesh with a plastic or silicone ring-shaped type bracket center portion of the annulus that couples to a polymer mesh with respect to a membrane and transducer coupled to a vibrating member having a tip portion in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 107A-107D illustrate cross-sectional views of a polymer mesh with a plastic or silicone ring-shaped type bracket coupled to double-reinforced stainless-steel annuli (similar to FIGS. 99 and 100) wherein the polymer mesh is raised with further extending top reinforcement (FIG. 107A), the polymer mesh is raised with extending top reinforcement (FIG. 107B), the polymer mesh is lowered with extending underlying reinforcement (FIG. 107C), the polymer mesh is lowered with further extending underlying reinforcement (FIG. 107D) with respect to membrane and transducer coupled to a vibrating member having a tip portion in droplet delivery devices in accordance with embodiments of the disclosure.

FIGS. 108A-108D illustrates zoomed views of FIG. 107A (FIG. 108A), FIG. 107B (FIG. 108B), FIG. 107C (FIG. 108C) and FIG. 107D (FIG. 108D).

FIGS. 109A-109D illustrate a cross-sectional view (FIG. 109A), a perspective view (FIG. 109B), a top plan view (FIG. 109C) and a cross-sectional zoomed view along line C-C of FIG. 109C (FIG. 109D) of a crystalline silicon or silicon carbide "wafer"-type mesh between ring-structured supports and processed with semiconductor technology to provide exact fabrication of smooth openings, such as pseudospherical, (zoomed cross-sectional view of FIG. 109D intended to show openings fully through mesh), in the mesh in a droplet delivery device in accordance with an embodiment of the disclosure.

FIG. 110 illustrates a cross-sectional and zoomed view of a crystalline silicon or silicon carbide "wafer"-type mesh with well-type openings that begin larger though the thick-

15

ness of the mesh and then terminate or are finished with smaller apertures in the openings (and which also may be angled with semiconductor technology processing) in the mesh in a droplet delivery device in accordance with an embodiment of the disclosure.

FIGS. 111A-111C illustrate a perspective view of a first end of absorber and baffle with fins (FIG. 111A), a perspective view of a second opposite end of a baffle with fins (FIG. 111B) and a cross-sectional, partial schematic view a droplet delivery device airway and ejector plate with mesh including a baffle with fins in accordance with an embodiment of the disclosure.

FIG. 112 is a streamline velocity field graphical map of an airflow path of a droplet delivery device including airflow directors without a baffle in accordance with an embodiment of the disclosure.

FIG. 113 is a streamline velocity field graphical map of an airflow path of a droplet delivery device including a baffle with wicking material and no airflow directors in accordance with an embodiment of the disclosure.

FIG. 114 is a streamline velocity field graphical map of an airflow path of a droplet delivery device including a baffle with wicking material and also including airflow directors in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION

Push Mode Overview

Push mode has been developed as a reduced-risk product to deliver (i) nicotine, cannabinoids, and other non-therapeutic substances (devices described herein as “BlueSky” are preferable for use with such substances), as well as (ii) therapeutic and prescriptive drug products (devices described herein as “Norway” are preferable for use with such products). The push mode device is designed to deliver the user a safe and controlled dose. The push mode droplet delivery device **10** is capable of delivering aqueous and nonaqueous solutions and suspensions at room temperature. Large molecule formulations, whether water soluble or not, can also be delivered with this technology. Harmful chemical by-products commonly found with heated nicotine, and other substances, are eliminated in the push mode device making it a safer option for aerosol delivery.

Push mode utilizes a vibrating member **1708** and transducer **26** that work in conjunction with a membrane **25** and mesh **2** to aerosolize fluid **901**, which is held in a reservoir **1200** and supplied to the mesh **22** using various methods (e.g., wick material, hydrophilic coatings, capillary action, etc.). Preferably the vibrating member is coupled to the transducer, such as by bonding (e.g. adhesives and the like), welding, gluing, physical connections (e.g. brackets and other mechanical connectors), and the like. The transducer and vibrating member interact with the membrane to push fluid through the mesh. As illustrated and described in various embodiments, the membrane may in some cases contact the mesh while also “pushing” fluid through holes in the mesh, and may in other cases be separated without contacting the mesh to push liquid through holes in the mesh. The transducer may comprise one or more of a variety of materials (e.g., PZT, etc.). In certain embodiments the transducer is made of lead-free piezoelectric materials to avoid creation of unwanted or toxic materials in a droplet delivery device intended for human inhalation. The vibrating member may be made of one or more of a variety of different materials (e.g., titanium, etc.). The mesh may be one or more of a variety of materials (e.g., palladium nickel,

16

polyimide, etc.). After the fluid is pushed through the mesh, a droplet spray is formed and ejected through a mouthpiece port, carried by entrained air.

The device is tunable and precise. The device can be optimized for individual user preferences or needs. The aerosol mass ejection and mass median aerodynamic diameter (MMAD) can be tuned to desired parameters via the mesh hole size, mesh treatment, membrane design, vibrating member design, airflow, manipulation of power to the transducer, etc. The design produces an aerosol comprised of droplets with a high respirable fraction, such that the lungs can absorb the aerosol most efficiently.

The vibrating member and transducer are both separate from the cartridge, isolated by the membrane. Not only does this create a safer product, but it eases manufacturability. The vibrating member and transducer are both typically expensive components. Keeping these components in the enclosure system rather than the cartridge reduces the cost of goods sold (COGS).

Element Number Tables

Substance, feature, and part numbers are provided for convenient reference with respect to the descriptions and figures provided herein in Table 1:

TABLE 1

Element Numbers			
		Substance Number	Substance Name
		100	Airflow
		800	Air
		900	Fluid flow
		901	Fluid
		Feature Number	Feature Name
		10	Droplet delivery device
		12	Container assembly
		15	Ejector bracket
		17	Enclosure system
		20	Airflow outlet
		24	Airflow inlet
		26	Heat exchange area
		28	Spike
		30	Air exchange outlet
		40	Mouthpiece Port
		41	Nasal Inhalation Port
		42	Ejection Port
		43	Nasal Inhalation Cap
		45	Mesh plate with holes
		47	Mesh and mesh plate spacer
		170	Vibrating member tip
		220	Central Axis
		230	Vibrating member central axis
		1200	Fluid reservoir
		2200	Fluid reservoir
		2250	Spiral
Part Number	Part Name	Material	Embodiment Push Mode I and/or II
22	Mesh	Palladium nickel	I and II
25	Membrane	Various Material	I and II
26	Transducer	PZT, lead-free material	I and II
1200's	BlueSky Container		
1202	Mouthpiece	COC	I and II
1204	Vent material	Sintered PTFE—PMA20	I and II
1206	Upper container	COC	I and II
1208	Middle container	COC	I and II
1210	Septum	Butyl rubber	I and II

17

TABLE 1-continued

Element Numbers			
1212	Lower container	COC	I
1213	Extended lower container	COC	II
1214	Container ring	COC	I and II
1500's	BlueSky Ejector Bracket		
1502	Upper ejector bracket	COC	I and II
1504	Lower ejector bracket	COC	I and II
1506	Upper mesh carrier	COC	I
1508	Lower mesh carrier	COC	I
1510	Membrane holder	COC	I and II
1512	Suspension gasket	Silicone	I
1513	Carrier gasket	Silicone	II
1518	Stainless steel mesh carrier	SUS316L	II
1519	Wick		I and II
1520	Magnet	N54 Ni coating	I and II
1522	Carrier O-Ring	Silicone	II
1524	Small O-Ring	Silicone	Simplified cartridge
1526	Large O-Ring	Silicone	Simplified cartridge
1528	Parallel plate capacitor		
1600	Laminar flow element	COC	I and II
1700's	BlueSky Enclosure System		
1702	Enclosure	Aluminum 6063 alloy	I and II
1704	Vibrating member front cover	PC/ABS	I and II
1706	Vibrating member rear cover	PC/ABS	I and II
1708	Vibrating member	Titanium alloy	I and II
1710	Transducer contact pin	Brass 3604 alloy, gold-plated	I and II
1712	Fingerprint/button/sealing bracket	PC/ABS	I and II
1714	Fingerprint/button cover	PC/ABS	I and II
1716	Enclosure rear cover	PC/ABS	I and II
1718	Enclosure sealing ring	Silicone	I and II
1720	Sensor director	Silicone	I and II
1722	Fingerprint/button	PC/ABS	I and II
1725	PCB	Many materials	All
1726	Spring electrode	SUS304	All
1728	USB-C port	PC/ABS	All
1730	Battery/Power supply	Li-Ion	Heater-Beluga
1732	Airflow Sleeve	PC/ABS	Air resistance
1734	Speaker		Speaker-Beluga
1760	Nodal-mounted sensing device		
1762	Sensor control unit		
1900's	Heating Components		
1902	Heating element	Nichrome 80	Heater
1904	Heat insulation		Heater
1906	Airflow accelerator		
1908	Electrode		All
1910	Temperature Sensor		Heater
2200's	Norway Ejector Bracket and Container		
2201	Mouthpiece	COC	Norway D
2202	Faceplate	SUS316L	Norway D
2203	Ejector Bracket Bottom Cover	COC	Norway D
2204	Ejector Bracket ID Chip	Many Materials	Norway D
2205	Ejector Bracket	COC	Norway D
2206	O-Ring	Silicon	All
2207	Mesh		Norway D
2208	Membrane Carrier	COC	Norway D
2209	Membrane		Norway D
2210	Cartridge Spacer	COC	Norway D
2211	Septum Cap	COC	Norway D
2212	Septum	Butyl Rubber	Norway D
2213	Lower Container	COC	Norway D
2214	Vent Material		Norway D
2215	Upper Container	COC	Norway D
2216	Vent Spacer		Norway D
2217	Mesh Carrier		
2218	Suspension gasket		
2219	Carrier gasket		
2220	Upper mesh carrier	COC	
2221	Lower mesh carrier	COC	

18

TABLE 1-continued

Element Numbers			
2222	Stainless steel mesh carrier	SUS316L	
5 2400's	Norway Face Seal		
2401	Screw		Norway D
2402	Cartridge Sealer Top Piece		Norway D
2403	O-Ring	Silicon	Norway D
2404	Cartridge Sealer Middle Piece		Norway D
10 2405	Screw		Norway D
2406	Cartridge Sealer Bottom Piece		Norway D
2407	Cap Spring	Steel Alloy	Norway D
2408	Screw		Norway D
2409	Device Cap	COC	Norway D
15 2410	Screw		Norway D
2411	Pin Screw	COC	Norway D
2412	Magnet		All
2600's	Norway Vibrating Member Components		
2601	Vibrating Member Enclosure	Aluminum Alloy	Norway D
20 2602	Vibrating Member Front Cover	COC	Norway D
2603	Vibrating Member and Transducer Assembly	Titanium Alloy and PZT4	Norway D
2604	Vibrating Member Rear Cover	COC	Norway D
25 2605	Vibrating Member Cover Front Holder	Silicon	Norway D
2606	Vibrating member assembly Spring	Steel Alloy	Norway D
2607	Vibrating Member Device Bracket	COC	Norway D
30 2608	Vibrating Member Cover Rear Holder		
2609	Screw		
2800's	Norway Enclosure System		
2801	Device Cap Release Axel		Norway D
2802	Cartridge Release Button Cover		Norway D
35 2803	Cartridge Release Button Actuator		Norway D
2804	Spacer		Norway D
2805	Cartridge Release Spring		Norway D
2806	7-Seg Display		Norway D
2807	Device Battery Cover Release		Norway D
40 2808	Device Battery Cover Release Spring		Norway D
2809	Device Battery Cover Axel		Norway D
2810	Device Enclosure Gasket		Norway D
2811	Device Bottom Enclosure		Norway D
45 2812	AAA Batteries		Norway D
2813	Device Battery Cover Gasket		Norway D
2814	Device Battery Cover		Norway D
2815	LED Display Cover		Norway D
50 2816	Device Front Cover Buttons		Norway D
2817	7-Seg Display Cover		Norway D
2818	Device Top Enclosure		Norway D
2819	Device Cap Release		Norway D
3300	Aluminized polymer tab		
4000	Baffle		
55 4050	Baffle fin		
4100	Absorbent Plug		

“BlueSky” Embodiments

Referring to FIGS. 1A and 1B, a BlueSky push mode device **10** includes main components of container assembly **12**, ejector bracket **15** and enclosure system **17**. Currently, two embodiments of BlueSky push mode, I and II, have been prototyped and tested. Referring to FIG. 2, inclusion of a mesh supported by a stainless-steel ring and elastic sealing ring in a droplet delivery device **10** is referred to as “push mode II” herein. Referring to FIG. 3 inclusion of a mesh

supported by upper and lower mesh carrier and an elastic sealing ring in a droplet delivery device **10** is referred to as “push mode I” herein.

The push mode I and II embodiments have a transducer consisting of a lead zirconate titanate (PZT) disc bonded to the bottom of a vibrating member made of titanium alloy. The vibrating member and transducer are encased by a plastic cover in an enclosure system **17**. A membrane made of polyethylene naphthalate (PEN) in the ejector bracket **15** isolates the transducer and vibrating member from the fluid that is supplied from a reservoir in the container assembly **12**. The membrane can be thermoformed to the shape of the vibrating member tip. The embedded system on the device consists of the transducer, pressure sensor, and lithium-ion battery all connected on a single board microcontroller. The aluminum enclosure that houses the embedded system contains a button that can double as a fingerprint sensor for use with controlled substances. The device is charged through a USB-C charging port. Magnets are used to hold the cartridge in the enclosure.

Embodiments use a two-component cartridge system to keep the fluid from contacting the mesh in storage. This design involves two spikes, one of which contains wicking material, on one part of the cartridge, the ejector bracket. The other part of the cartridge, the container, houses a fluid reservoir and two septa. The user pushes the ejector bracket and container together, and the spikes puncture the septa, creating a path for fluid to flow to the mesh. The wicking material in one spike aids in the supply of fluid to the mesh. The other spike, which does not include wicking material, allows air to enter the container for pressure equalization. Vents covered with vent material are located at the top of each side of the fluid reservoir and are connected to the open atmosphere via airflow outlets, allowing for equalization of pressure.

Referring to FIG. 4, there is an ejection port **42** with a length of 25 mm and a mouthpiece port with a length of 10 mm. The preferred length of the ejection port is 0 mm-50 mm. The preferred mouthpiece port length is 0 mm-50 mm. FIG. 5 shows the fluid **900** and ventilation **100** flow paths through the spikes **28** in prototyped embodiments. FIGS. 6A and 6B show the entrained air path of prototyped embodiments.

BlueSky I Push Mode

FIGS. 7A and 7B show a rendering and a CAD overview, respectively, of the push mode I embodiment. The overviews in FIGS. 7A and 7B show the container assembly **12**, ejector bracket **15**, and the enclosure system **17**, from left to right.

FIG. 8 provides an exploded view of the components from the push mode I embodiment.

Referring to FIG. 9, the push mode I embodiment includes a mesh carrier that includes two COC rings **1506**, **1508** that are ultrasonically welded holding the mesh **22** and suspension gasket **1512**. The COC rings sandwich the mesh and suspension gasket as shown in FIG. 10. The gasket is placed between the upper and lower ejector brackets.

Referring to FIG. 11, two vents are located on the narrow sides of the lower ejector bracket **1504** in the push mode I embodiment. The spikes are located on the upper ejector bracket **1502**. The container, which houses the fluid reservoir **1200**, includes three COC pieces. The two septa **1210** are held between the middle and lower container pieces. A container ring is bonded onto the upper **1206** and middle **1208** container pieces and the mouthpiece **1202** snaps onto the upper container piece **1206**.

BlueSky II Push Mode

FIGS. 12A and 12B show a rendering and a diagrammatic overview of the push mode II embodiment, respectively. The overviews in FIGS. 12A and 12B show the container assembly **12**, ejector bracket **15**, and the enclosure assembly **17**, from left to right.

FIG. 13 illustrates an exploded view of the components of the push mode II embodiment.

In the push mode II embodiment, a stainless-steel annulus carrier **1518** is bonded to the mesh **22**. A gasket **1513** is placed above the mesh and mesh carrier between the upper **1502** and lower **1504** ejector brackets. FIG. 14 illustrates the push mode II embodiment mesh carrier **1518** and gasket **1513**.

Two vents are located on the wide sides of the lower ejector bracket **1504** as shown in FIG. 15. The spikes are located on the upper ejector bracket **1502**.

As in push mode I, the container, which houses the fluid reservoir, includes three COC pieces. The lower container for the push mode II embodiment extends further than in push mode I, with the tubular portion extending into the upper ejector bracket.

FIG. 16 (push mode II) and FIG. 17 (push mode I) illustrate a comparison of the lower containers of each embodiment. The extension is necessary because the mesh sits lower, compared to I, due to the stainless-steel mesh carrier being thinner than the COC carrier of I. The two septa are held between the middle and lower containers. A container ring is bonded onto the upper and middle container pieces and the mouthpiece snaps onto the upper container piece.

BlueSky Vibrating Member and Membranes

Push mode has multiple vibrating member and membrane designs. Table 2 and Table 3 contain descriptions of the vibrating member and membrane designs, respectively, that have been prototyped and tested. Referring to FIGS. 18 and 19, there are currently two different tips for the vibrating member rod tip and ring tip, respectively.

TABLE 2

Description of vibrating members		
Vibrating Member	Description	
H1	1.0 mm diameter rod tip	
H2	1.5 mm diameter rod tip	
H3	2.0 mm diameter rod tip	
H4	3.5 mm diameter rod tip	
H5	3.5 mm diameter ring tip with a 2-degree tilt	
H6	3.5 mm diameter ring tip with a 5-degree tilt	
H7	3.5 mm diameter ring tip with an 8-degree tilt	
H8	3.5 mm diameter ring tip with no tilt	
H9	3.5 mm diameter rod tip with 3-degree tilt	

TABLE 3

Description of membranes		
Membrane Style	Description	
M1	Thermoformed to a 1.0 mm diameter circle with a round plateau, used with H1	
M2	Thermoformed to a 1.0 mm diameter circle with a 2-degree tilt, used with H1	
M3	Thermoformed to a 1.0 mm diameter circle with a 5-degree tilt, used with H1	
M4	Thermoformed to a 1.0 mm diameter circle with an 8-degree tilt, used with H1	

21

TABLE 3-continued

Description of membranes	
Membrane Style	Description
M5	Thermoformed to a 1.5 mm diameter circle with a 2-degree tilt, used with H2
M6	Thermoformed to a 1.5 mm diameter circle with a 5-degree tilt, used with H2
M7	Thermoformed to a 1.5 mm diameter circle with an 8-degree tilt, used with H2
M8	Thermoformed to a 2.0 mm diameter circle with a 2-degree tilt, used with H3
M9	Thermoformed to a 2.0 mm diameter circle with a 5-degree tilt, used with H3
M10	Thermoformed to a 2.0 mm diameter circle with an 8-degree tilt, used with H3
M11	Thermoformed to a 3.5 mm diameter circle with a round plateau, used with H4
M12	Thermoformed to a 3.5 mm diameter circle with a 2-degree tilt, used with H5
M13	Thermoformed to a 3.5 mm diameter circle with a 5-degree tilt, used with H6
M14	Thermoformed to a 3.5 mm diameter circle with an 8-degree tilt, used with H7
M15	Thermoformed to a 3.5 mm diameter circle with no tilt, used with H4 or H8

The transducer requires a large amount of power during the actuation of the device. As the power usage increases, the heat generated by the printed circuit board assembly (PCBA) increases. The effect from the heat is mitigated through several design features in the PCBA. A four-layer PCBA increases anti-interference and heat dissipation capabilities. The PCBA also contains a large amount of copper foil, making it conducive to heat dissipation. The MOSFET driving the transducer adopts a high-current package to avoid damage caused by heating in long-term continuous operation. The automatic transformer, to increase the voltage output, it is suspended to insulate it from the rest of PCBA. These features allow the device to operate for days without concern of overheating or being subjected to electrical noise.

BlueSky Life Testing

The prototype BlueSky push mode embodiments, I and II, have gone through life testing. The life test consisted of repeated three-second dosing with one-second resting intervals over the course of several days. Mass ejection was done before and after the life test. Mass ejection is defined as the mass the device aerosolizes over one three-second dose. Mass ejection data before the life test is listed in Table 3 and the data for after life testing is listed in Table 4. The mass ejection of one embodiment remained consistent before and after 55,000 doses and can likely go beyond. This embodiment, II push mode with H4 and M11, has a stainless-steel mesh carrier. There is a second embodiment, I push mode, which has a COC plastic mesh carrier. Due to heat from the extreme dosing cycling, the plastic mesh carrier warped during testing. This led to a decrease in mass ejection after the life test. However, the stainless-steel carrier in II push mode did not warp from the heat, which allowed it to remain consistent after testing. In both I and II, thermal management is improved through a four-layer PCBA, a larger than standard amount of copper foil, and a high current MOSFET driver. The conditions of the testing are not representative of normal consumer use. During normal daily use, where extreme heating does not occur, both embodiments, I and II, show consistent mass ejection. Tables 2 and 3 provide details of the referenced Vibrating Member and Membrane, respectively.

22

TABLE 4

Mass ejection data for push mode devices before life testing						
Type	Vibrating Member	Membrane	Ejector	Average Scale Ejection (mg)	Average Nicotine (μg)	
II	H4	M11	R52	2.45	73.5	
I	H5	M12	W11	3.30	99	
I	H8	M15	W11	3.60	108	

TABLE 5

Mass ejection data for push mode devices after life testing						
Type	Vibrating Member	Membrane	Ejector	Average Scale Ejection (mg)	Average Nicotine (μg)	
II	H4	M11	R52	3.26	97.8	
I	H5	M12	W11	1.16	34.8	
I	H8	M15	W11	1.54	46.2	

Comparison of Push Mode and Prior Art Ring Mode

As set forth in Example 1 described subsequently, prototypes of BlueSky I and II push mode were tested and compared to prior technology, referred to as BlueSky ring mode (such as described and shown with respective test data for that technology in WO 2020/264501), is provided as follows:

Example 1

Ejectors with a hole size of 2.0 μm were tested in each device. Half of the ejectors tested had a hydrophilic entrance and hydrophobic exit (R). The other half had a hydrophobic entrance and exit (W). The testing was performed with a TSI Mini-MOUDI Model 135 and a THERMO FISHER™ Vanquish UHPLC. Eight different design combinations (vibrating members, membranes, ejector treatments) were tested with BlueSky I and II. Based on the results of the testing, push mode I appears to be the preferred embodiment for push mode. The push mode I design resulted in more consistent mass ejection and MMAD values versus II. Seven of the eight design combinations resulted in comparable mass ejections and MMADs. One outlier, H5 with M12 and R-treated ejector, had a significantly higher mass ejection than the others. Upon comparison of I push mode to BlueSky ring mode, I delivered higher and more consistent mass ejection and lower MMADs. Table 6, Table 7 and Table 8 provide the data obtained from ring mode, I push mode, and II push mode, respectively. The data in the tables include micrograms of nicotine ejected, MMAD, geometric standard deviation (GSD), and the percentage of ejected solution in stage 1 and stage 2 of the mini-MOUDI. All the vibrating member and membrane combinations tested with I push mode, found in Table 7, performed well with both ejector treatments. As seen in Table 8, the best performing combinations with II push mode were H4 with M11 and H5 with M12, both using W-treated ejectors.

TABLE 6

Ring Mode Mini-MOUDI results:						
Treatment	Nicotine (μg)	MMAD (μm)	GSD	Stage 1 (%)	Stage 2 (%)	Stage 1 and 2 (%)
R	47.040	1.38	1.76	0.299	1.322	1.62
R	59.367	1.61	1.68	0.406	4.340	4.75

23

TABLE 6-continued

Ring Mode Mini-MOUDI results:						
Treatment	Nicotine (μ g)	MMAD (μ m)	GSD	Stage 1 (%)	Stage 2 (%)	Stage 1 and 2 (%)
R	23.830	1.12	1.76	0.340	1.073	1.41
W	38.057	1.50	1.77	0.288	4.159	4.45
W	69.387	1.77	1.69	1.057	10.316	11.37
W	39.653	1.22	1.86	0.207	1.443	1.65

The results obtained from Push Mode I device are shown in Table 7. Tables 2 and 3 provide details of the referenced Vibrating Member and Membrane, respectively.

TABLE 7

Push Mode I Mini-MOUDI results:								
Treatment	Vibrating member Style	Membrane Style	Nicotine (μ g)	MMAD (μ m)	GSD	Stage 1 (%)	Stage 2 (%)	Stage 1 and 2 (%)
R	H4	M11	85.880	1.40	1.95	6.20	3.07	9.27
R	H4	M15	76.250	1.14	1.56	1.06	0.45	1.51
R	H5	M12	141.957	1.30	1.81	2.00	1.66	3.66
R	H8	M15	90.124	1.26	1.81	1.61	1.27	2.88
W	H4	M11	99.705	1.40	1.78	0.72	2.78	3.50
W	H4	M15	108.750	1.27	1.82	1.07	1.71	2.78
W	H5	M12	102.493	1.37	1.79	0.87	2.41	3.28
W	H8	M15	102.177	1.22	1.69	0.55	1.14	1.69

The results obtained from Push Mode II device are shown in Table 8. Tables 2 and 3 provide details of the referenced Vibrating Member and Membrane, respectively.

TABLE 8

Push Mode II Mini-MOUDI results:								
Treatment	Vibrating member Style	Membrane Style	Nicotine (μ g)	MMAD (μ m)	GSD	Stage 1 (%)	Stage 2 (%)	Stage 1 and 2 (%)
R	H3	M10	32.64	1.02	1.59	0.51	0.53	1.04
R	H4	M11	96.29	1.85	1.59	0.84	9.37	10.21
R	H5	M12	50.58	1.21	1.84	1.69	2.57	4.26
R	H7	M14	45.22	1.17	1.66	0.69	1.32	2.01
W	H3	M10	13.75	0.96	1.63	1.35	0.27	1.62
W	H4	M11	75.77	1.4	1.73	0.64	1.47	2.11
W	H5	M12	88.26	1.28	1.87	4.45	1.24	5.69
W	H7	M14	229.53	1.45	4.37	20.5	7.84	28.34

Based on the results of the testing, I push mode is the preferred embodiment when compared to II. BlueSky Single Piece Cartridge and Low Cost of Goods Sold Designs

Another embodiment of push mode incorporates the two-part cartridge system into a singular component. Having the cartridge in one piece simplifies setup for the user and increases manufacturability while reducing cost. FIGS. 20, 21A and 21B show two single piece cartridge embodiments. The embodiment shown in FIG. 20 includes a long vibrating member with the fluid reservoir residing under the mesh. In this design, the container is two pieces that are assembled during manufacturing.

In another embodiment, there is a short vibrating member with the fluid reservoir above the mesh (see FIGS. 21A and 21B). In this design, the container is comprised of three pieces that are assembled during manufacturing. After the fluid reservoir is filled, the mouthpiece snaps onto the container with the container ring between.

24

The vibrating member and transducer work in conjunction with a membrane and mesh, as previously described embodiments of BlueSky push mode. The membrane also serves to isolate the vibrating member and transducer from the fluid. A mesh carrier is used in both designs. Magnets on the bottom of the containers hold the cartridge in the enclosure.

Further embodiments, shown in FIGS. 22A and 22B, of a single piece cartridge include a simpler design, reducing the COGS in manufacturing by decreasing the number of injection molded parts and bonds. FIG. 22A illustrates a simplified version of the design in FIG. 21A but with a long vibrating member. The design in FIG. 22A reduces the

number of ultrasonic welds and injection molded parts. FIG. 22B further simplifies the design from FIG. 21A with fewer ultrasonic welds and injection molded parts.

The low COGS designs shown in FIGS. 23A and 23B are a simplification of the design shown in FIG. 21B. This design is a single part cartridge that can be inserted into the enclosure. Air exchanges between the seal of the mouthpiece and the upper container. The cartridges shown in FIGS. 22A-22B and FIG. 24 have removed the ejection port leaving the 10 mm mouthpiece port. The preferable ejection port and mouthpiece port lengths are the same as previously set forth, 0 mm-50 mm.

BlueSky Two-Piece Cartridge

FIG. 24 illustrates a two-piece cartridge design for a long vibrating member. The container and ejector bracket are swapped where the ejector bracket is connected to the mouthpiece and the container is below. The spikes on the ejector bracket face downward onto the septa on the container.

Pharmaceutical/Therapeutic (Norway) Embodiments

Another embodiment of push mode, Norway, is similar to its BlueSky counterpart in most aspects, except that is

25

tailored for prescriptive and medical use. Much like BlueSky, Norway features a releasable cartridge which contains a fluid reservoir and ejector bracket. The device can also be used to assess lung health using spirometry. FIG. 25 shows one embodiment of Norway push mode.

Patients diagnosed with lung diseases can use the Norway device to track their medication dosages and take lung function tests so their treatment progression can be assessed. The patient can perform lung function tests and view dosage history via a phone app which pairs to the Norway device with Bluetooth. The device saves pressure sensor measurements from each dosage of medication. Inspiratory flow measurements can be derived from the pressure sensor measurements to ensure the user is inhaling their medication at a flow rate which delivers the solution most efficiently. The device can also perform lung function tests to measure a patient's forced expiratory volume over 1 second, forced vital capacity, peak expiratory flow, and other spirometry measurements. The data from dosage tracking and lung function tests are uploaded to the cloud so that the patient and doctor may view the patient's progression.

The ejector bracket has been designed to accept many different sizes of containers, where the fluid reservoir volume changes. This makes the device capable of being used with biologics, or one time use ejections. Possible fluid reservoir volumes range from 1 μ L to 20 mL.

The mouthpiece for the Norway embodiment has a preferred length of 15 mm. There are two slits on the sides of the mouthpiece with a dimension of 9 mm by 3 mm for an area of 27 mm². The length of the mouthpiece could be anywhere from 5 mm to 30 mm. The area for the mouthpiece could be from 1 mm² to 100 mm². The mouthpiece opening has dimensions of 14 mm $\times$ 24 mm for an area of 336 mm². The area of the mouthpiece opening could be anywhere from 10 mm² to 500 mm².

The cartridge can be inserted into the main body of the device. The front of the cartridge can be sealed by an O-Ring attached to the cap that presses around the mesh on a stainless-steel annulus when closed to prevent any evaporation through the mesh, this is the face seal. The device features voice coaching and LED lights to guide the user through the ejection inhalation. There is an LCD screen to display dose count, and other necessary information. FIG. 26 shows an exploded view of one embodiment of Norway push mode.

Referring to FIGS. 27A-D, the cartridge assembly (FIG. 27A) is composed of three parts: the container (FIG. 27B), cartridge spacer (FIG. 27C), and the ejector bracket (FIG. 27D). The cartridge spacer keeps the ejector bracket separated from the container to prevent the fluid from contacting the mesh during storage prior to the push mode Initial use.

The cartridge spacer can be removed so the container can be pushed down onto the ejector bracket such that the spikes pierce the septa making the cartridge one piece. Then, the cartridge can be pushed into the main body of the device to complete the device. This process is illustrated in FIG. 28.

The cap of the Norway embodiment is designed to create a firm seal around the cartridge after each use. An O-Ring is seated on a spring-loaded plastic piece which lightly compresses onto the cartridge assembly when the cap is closed, generating a seal between the cartridge and open atmosphere. The components of the cap are shown isolated in as illustrated in FIG. 29.

The critical components to generate precise aerosol of the ejector bracket include the mesh, gasket, membrane, vent material, and mouthpiece. The membrane is positioned such that the membrane face is held parallel to the mesh face, or

26

at a small precise angle. The ejector bracket also has two spikes protruding out of the top that pierce the container. One is for fluid supply, and the other provides a ventilation path for air generated by ejection. On the side of the ejector bracket with the air ventilation spike there is an opening covered by vent material to help relieve pressure and build-up of air. The mouthpiece is positioned following the face of the mesh.

The critical components of the container to maintain consistent aerosol are vent material, a spiral, septa, and septa caps. The vent material is positioned between the fluid and the spiral. The spiral is created by the upper container and vent spacer which minimizes evaporation of the fluid through the vent material. The vent spacer is bonded onto the top of the upper container to create the sealed spiral with an opening to the push mode. Inside of the container assembly and another opening to atmosphere. The septa are at the bottom of the container. The septa are placed into a cavity in the lower container and held in place with septa caps that are bonded onto the lower container. The critical components of both the ejector bracket and container can be seen in FIGS. 30A and 30B.

The main body of the Norway contains the vibrating member and transducer assembly. In one embodiment, as shown in FIG. 31, the vibrating member and transducer assembly is encased by a vibrating member front cover and vibrating member rear cover. The covers are held together by circular caps called the front and rear vibrating member cover holders. The encased vibrating member is then put into the vibrating member enclosure, followed by the vibrating member assembly spring, and finally seated into the vibrating member device bracket. The vibrating member enclosure allows the spring to press the vibrating member and transducer assembly to the membrane.

Additional embodiments of Norway push mode include different suspension systems to hold the mesh in the cartridge, similar to those in BlueSky push mode. With the suspension systems seen in FIGS. 32 and 33, the vibrating member and transducer assembly no longer has a spring; therefore, it no longer needs to be in the vibrating member enclosure, nor does it need the vibrating member device bracket.

An additional embodiment of the Norway push mode device includes a heating element that increases the push mode Inhaled air temperature to roughly 50° C. to make the dose more comfortable. As with the BlueSky designs that include heating elements, the heated air temperature is kept below thermal degradation levels, so the push mode Integrity of the formulation is maintained, and no harmful by-products are produced. This can be accomplished because, as with BlueSky, the device does not depend on heat to aerosolize. FIGS. 34A and 34B illustrate one design that includes two heating elements positioned beneath the vibrating member on either side of the ejector bracket. As seen in FIGS. 34A and 34B, air enters through openings in the bottom of the ejector bracket, passes through the heating elements, and exits into the mouthpiece. Additionally, the warmer air will cause minimal evaporation of the aerosolized fluid resulting in a decrease in MMAD.

Biocompatibility

In the push mode design, the vibrating member and transducer are completely isolated from the push mode Inhaled solution by a membrane. The transducer, which typically contains heavy metals, is located behind a vibrating member, such that it is completely removed from the ejection area and fluid reservoir. The membrane separates the fluid reservoir from the vibrating member, presenting a

chemically inert barrier that permits little or no diffusion, and subsequent evaporation. In one embodiment, a palladium nickel alloy mesh is used to atomize the fluid. A polyimide mesh has also been tested and was shown to be a viable option. Using a polymer mesh would significantly reduce manufacturing cost and potentially improve the extractable/leachable profile of the device. The non-metallic components in prototyped embodiments are primarily comprised of cyclic olefin copolymer (COC) and silicone, both widely accepted materials used in the medical device industry.

Heated Air Design

FIGS. 35A-35C through FIG. 38 show embodiments which include a heating element to increase the push mode I inhaled air temperature to roughly 50° C., making the dose more comfortable. Air passes perpendicularly through the heating element to be most efficiently heated. Since the heated air temperature is kept below thermal degradation levels, the push mode Integrity of the formulation is maintained, and no harmful by-products are produced. Also, the specific heat of the fluid is much greater than air; therefore, the temperature of the aerosolized fluid will heat minimally. This can be accomplished because the device does not depend on heat to aerosolize. Here, the heat is only used to optimize the user experience. Additionally, the warmer air will cause minimal evaporation of the aerosolized fluid resulting in a decrease in MMAD. Finally, the heating element will be surrounded by insulation material to keep all the components of the device insulated from heat.

The heating element is breath actuated such that the element only heats air as the user inhales. This allows the battery to have a much longer life. It also creates a much safer device in that the heating element is not always on. This can be accomplished due to the push mode Incorporation of small gauge wire. This wire heats up very quickly, so the heating element responds as soon as the user inhales.

In the embodiment shown in FIGS. 35A-35C, after air enters the device, the air pathway is narrowed by the airflow accelerator to increase velocity. Then, the air is passed through the heating element, which is positioned in the heat exchange area. Finally, the heated air flows into the mouthpiece. FIGS. 35A-35C features three views of this embodiment. This design allows for a larger battery to be installed in the device which supplements the heating element.

Referring to FIG. 36, a speaker can also be incorporated into any of the heated air BlueSky embodiments. This will allow for an additional sensory experience for the user (i.e., crackling/heating sound upon inhalation).

In the embodiments shown in FIG. 37 and FIG. 38, the heating element is positioned below the vibrating member in a separate chamber inside the enclosure. The air enters through the airflow inlet, is passed through the heating element, and exits above the ejector. This design can be used in the two-part cartridge design (FIG. 37) or the single piece cartridge design (FIG. 98). These embodiments offer the advantage of a more compact device, compared to the embodiment shown in FIGS. 35A-35C, at the cost of battery life.

Another embodiment features external heating elements seated on the outside of the enclosure (FIG. 39). Air passes through the heating elements, enters the mouthpiece above the mesh, and exits through the end of the mouthpiece. This design may in some embodiments provide a removable heating element.

In another embodiment of a heated air push mode device, closed loop control is used to regulate the power delivered to the heating element. The power is adjusted to keep the

airstream temperature constant and at safe levels. Referring to FIG. 40, the airstream temperature is measured by a temperature sensor such as an RTD. The power delivered to the heating element changes as a result of the temperature sensor readings.

In another embodiment of the heated air push mode device, open loop control is used to regulate the power delivered to the heating element. The power is adjusted to keep the airstream temperature constant. The pressure drop from inhalation is sensed. The amount of power needed to supply the heating element to keep the air stream temperature constant due to changes in pressure drop is known. A look-up table is created to determine the amount of power needed to supply the heating element to keep the air stream temperature constant based upon the pressure sensor value.

In another embodiment of the heated air push mode device, one or more of the push mode Internal device components that are in contact with heated air is preferably made of metal (i.e., aluminum, Inconel, etc.). This will insulate the heating element and enhance biocompatibility of the device.

In another embodiment of the heated air push mode device, any component that could be compromised by the heated air is preferably made of metal (i.e., titanium, aluminum, Inconel, etc.). These components include, but are not limited to the mouthpiece, the heating chamber, and like components that heated air could negatively affect.

In one embodiment of the heated air push mode device, the metal components that are in contact with the heated air are preferably made of a material with a low thermal conductivity, such as Inconel.

In one embodiment of the heated air push mode device, ceramic is used to insulate the heating element.

Adjustable Air Resistance Design

Another embodiment of push mode incorporates a mechanism to adjust the size of the airflow inlets. The airflow inlets can be opened and closed using a sleeve or an adjustable aperture. In this way, the resistance experienced by the user can be adjusted to individual preferences. FIGS. 41A and 41B show a BlueSky device with a sliding sleeve 1732 around the enclosure. The sleeve can be adjusted to partially or completely cover the airflow inlets, increasing the resistance felt by the user. Additionally, the airflow in the mouthpiece will change as the position of the sleeve is changed. This will also change the MMAD of the dose due to changes in the airflow current.

Nasal Device Embodiments

BlueSky push mode has also been adapted for nasal inhalation. FIGS. 42-44 show several embodiments of a nasal BlueSky push mode device. As seen in FIGS. 42-44, there are multiple variations of the push mode Inhalation port. However, preferable embodiments of the nasal device have longer and narrower inhalation ports (see FIG. 42) than in other designs with shorter inhalation ports (see FIG. 43) for optimal nostril use. As seen in FIG. 44, a cap may be added to protect the push mode Inhalation port and keep it clean. The preferred droplet sizes are between 1-110 micron range, but 2-23 microns is preferred.

Additional Features

Hydrophilic/Hydrophobic Tubes

Another embodiment of push mode incorporates a tube with a hydrophilic interior that supplies fluid from the fluid reservoir to the mesh. A hydrophilic tube eliminates the need for wicking material and allows for a wider variety of suspensions and solutions to be delivered from the device. An example of one of these tubes is the spike on BlueSky I and II.

Another embodiment of push mode incorporates a tube with a hydrophilic interior that supplies fluid from the fluid reservoir to the mesh without a wick material, allowing for a wider variety of suspensions and solutions to be delivered from the device; and an opposite hydrophobic tube that encourages gas migration from the fluid supply area between the membrane and mesh.

Polymer Mesh Holes

In another embodiment, as shown in FIG. 45, a polymer mesh 22 is used with a plate 45 attached to it. It has been found that a 2 mm hole on a plate works best for ejection. Therefore, another embodiment is where the plate has multiple 2 mm openings for the liquid to enter. The holes on the plate can range from 0.1 mm-20 mm.

Tidal Breathing

Another embodiment of push mode uses a tidal breathing system that can be used for pediatric therapy. The push mode technology supplies aerosol a mask similar to the AERO CHAMBER PLUS™ Z-Stat Pediatric Mask (Monaghan Medical). This allows for long use therapy. When a user inhales, the device will start ejection and when the user exhales the device will stop ejection. Due to the robustness of push mode, this can be a very effective device for extended therapies.

Capacitance Cartridge

In another embodiment, two parallel plates 1528 surround the fluid next to the mesh and membrane area. These two parallel plates will measure the capacitance of the fluid. The capacitance of the supplied fluid is known. If the capacitance measured is different than the known capacitance, the device will not work. This will prevent tampering of the cartridge, and it will prevent unauthorized fluids to be inserted into the cartridge. One of the parallel plates is shown in FIG. 46.

Microfluidic Pump

Another embodiment of push mode utilizes vibrating member and membrane geometries at their coupled interface to act as both an atomizer and microfluidic pump in applications where wicking materials are not incorporated into the preferred embodiment for certain suspensions, solutions, and other medical, therapeutic, and consumer applications. The tip of the vibrating member is coupled to a membrane matching the desired geometry allowing fluid to enter between the mesh and membrane while also encouraging any gas to exit freely. These membranes may be treated by technologies mentioned previously to be hydrophilic or hydrophobic.

Another embodiment utilizes a separate microfluidic pump to direct the proper amount of fluid and pressure between the mesh and membrane when powered on, at breath actuation, at set intervals, etc. to ensure proper dosing.

Vibrating Member Geometry Optimization

Vibrating members of the embodiments are to be made of materials featuring proper acoustical and mechanical properties. Thin film sputtering of various nonreactive metals such as titanium, palladium, gold, silver, etc. can be performed on the vibrating member tip section to further enhance biocompatibility. According to industry leaders, titanium has the best acoustical properties of the high strength alloys, has a high fatigue strength enabling it to withstand high cycle rates at high amplitudes, and has a higher hardness than aluminum, making it more robust. Correct material must be selected, vibrating members must be balanced, designed for the required amplitude, and be accurately tuned to a specific frequency. One aspect of tuning is making the vibrating member have the correct elongated length. Another aspect of tuning is matching the

vibrating member to the mesh and having the correct gain ratio. Incorrectly tuned vibrating members may cause damage to the power supply and won't be resonating at the device's optimized frequency, decreasing mass ejection and longevity. (see also *Ultrasonic Vibrating member catalog*—Emerson. Catalog—Ultrasonic Vibrating member (2014). Available at: www.emerson.com/documents/automation/catalog-ultrasonic-vibrating-member-branson-en-us-160126.pdf. (Accessed: 2 Nov. 2021)—incorporated herein by reference.)

For example, Titanium 7-4 material has far more uniform wave propagation in one direction (axial) than Titanium 6-4.

Embodiments must have vibrating members with proper moduli of elasticity, acoustical properties, sound speeds, mechanical properties, molecular structure, etc. such as Ti Grade 23, Ti Grade 5, Ti Pure >99.9%, TITETAL® 7-4, 302 Stainless Steel, 303 Stainless Steel, 304 Stainless Steel, 304L Stainless Steel, 316 Stainless Steel, 347 Stainless Steel, Al 6061, Al 6063, Al 3003, etc.

Other embodiments have crystalline vibrating members with proper moduli of elasticity, acoustic properties, sound speeds, mechanical properties, molecular structure, etc. such as: Sapphire (Al₂O₃Aluminum oxide), monocrystalline silicon, etc.

In one embodiment, vibrating member design is based on industrial ultrasonic vibrating member design such as disclosed by the push mode Indicated reference subsequently noted, but optimized to be used for the purposes of aerosol generation in the delivery of fluids to the lungs, nose, ear, eye, etc.

Referring to FIG. 47, the vibrating member is rectangular at the membrane interface. This rectangular tip features three periodic slots along the X directions and two periodic slots along the Y directions of the member tip based on a quasi-periodic phononic crystal structure.

Referring to FIGS. 48 and 49, the rectangular vibrating member tip combined with a conical section and a cylindrical section can effectively improve the output amplitude gain and utilizes the band gap property of the structure to effectively suppress lateral vibration of the vibrating member tip, improving the amplitude distribution uniformity at the membrane interface (see also Lin, J. & Lin, S. *Study on a large-scale three-dimensional ultrasonic plastic welding vibration system based on a quasi-periodic phononic crystal structure*. MDPI (2020). Available at: www.mdpi.com/2073-4352/10/1/21/htm. (Accessed: 2 Nov. 2021)—incorporated herein by reference).

In other embodiments, shown in FIGS. 50-58, the vibrating member 1708 is tuned and machined similarly to industrial ultrasonic vibrating member designs (such drawings being disclosed in the noted reference) but optimized for aerosol generation in the delivery of fluids to the lungs, nose, ear, eye, etc. such as contoured vibrating member (FIG. 50), plunger vibrating member (FIG. 51), product authenticity sensor vibrating member (FIG. 52), spool vibrating member (FIG. 53), slotted cylindrical vibrating member (FIGS. 54 and 55), bar vibrating member (FIGS. 56 and 57), and booster vibrating member (FIG. 58). See also Industrial resonators Available at: www.krell-engineering.com/fea/industry/industrial_resonators.htm. (Accessed: 2 Nov. 2021)—incorporated herein by reference.

Referring to FIG. 50, vibrating members can be contoured to make intimate contact with the membrane geometry.

Referring to FIG. 51, plunger members have nodally-mounted plungers that can be used to exert pressure on a given surface of the membrane contacted by the vibrating member.

31

Referring to FIG. 52, sensor carrier vibrating members feature an internal cavity partially or fully encapsulating a nodal-mounted sensing device. The sensing device is coupled with a sensor control unit which outputs a signal to the PCBA. This signal can be used to disable aerosol generation when non-compliant, incorrect, unlicensed, etc. cartridges are attempted to be used.

Referring to FIG. 53, spool vibrating members are unslotted cylindrical members featuring undercut sides behind the face to form a spool shape. This spool shape improves the face amplitude uniformity. Because a spool vibrating member does not have slots, its stresses are much lower than comparable slotted cylindrical vibrating members making machining costs much lower. Using cavities, slots, and back extension to optimize axial resonance creates a very uniform amplitude across the members face. The member is one half-wavelength long at axial resonance, as indicated by the single node that is generally transverse to the principal direction of vibration. Spool vibrating members generally have about 1:1 gain, although somewhat higher gain is possible.

Referring to FIGS. 54 (optimized) and 55 (unoptimized), slotted cylindrical vibrating members feature longitudinal slots used to reduce the transverse coupling due to the Poisson effect. Such slots are usually radial, although other configurations are sometimes useful. Without such slots, the vibrating member will either have very uneven amplitude across the face or may even resonate in a nonaxial manner. They also have a face cavity that extends deep within the member to increase its gain. The vibrating member is one half-wavelength long at axial resonance, as indicated by the single node that is generally transverse to the principal direction of vibration. Slotted cylindrical vibrating members generally have low-to-moderate gain (1:1 to 2:1).

Referring to FIGS. 56 (optimized) and 57 (unoptimized), bar vibrating members are rectangular and either unslotted or slotted only through the thickness. Special design techniques give optimum face amplitude uniformity. The vibrating member's thickness has been reduced in the blade section in order to provide reasonable gain. The vibrating member is one half-wavelength long at the axial resonance, as indicated by the single node that is generally transverse to the principal direction of vibration. Bar vibrating members generally have low-to-moderate gain (1:1 to 4:1).

Referring to FIG. 58, A booster is a coupling resonator that is placed between a transducer and vibrating member in order to change the member's amplitude and or as a means of supporting the resonator stack. The booster body is rigidly supported by a collar that is bonded to the booster's node. Because the rigid booster is constructed only of metal (no compliant elastomers), it has excellent axial and lateral stiffness. For additional stiffness a second collar can be incorporated into a full-wave design. The collar is tuned to isolate the motion of the booster body from the support structure. This is shown in the following image of a displaced booster, where the coolest colors indicate the lowest amplitudes. Each booster has a fixed gain (ratio of output amplitude to input amplitude), generally between 0.5:1 and 3.0:1.

With further reference to FIGS. 59-83, further alternative embodiments of vibrating members 1708 with vibrating member tip 170 that couple to transducers 26 of droplet delivery devices 10 in accordance with various embodiments of the disclosure are shown.

32

Other Vibrating Member and Membrane Alignments and Designs

In other embodiments, the vibrating member 1708 may include other shapes and the membrane 25 may also include alternative shapes. For example, FIG. 85A illustrates an ultrasonic transducer coupled to a rod-shaped vibrating member tip portion 170. FIG. 85 shows the vibrating member of FIG. 85A coupled to a centrally peaked or pointed membrane 25 in a droplet delivery device 10. FIGS. 85C and 85D show ultrasonic transducer 26 and membrane 25 of FIG. 85B in alternative embodiments wherein a mesh 22 includes first securing mechanism in FIG. 85C (see FIG. 2 and accompanying description) and second securing mechanism in FIG. 85D (see FIG. 3 and accompanying description).

FIG. 86A further illustrates in another embodiment an ultrasonic transducer 26 with a rod-shaped tip portion 170 coupled to a membrane 25 with a wide or dome/rounded exterior surface in a droplet delivery device 10. FIGS. 86B and 86C show ultrasonic transducer 26 and membrane 25 of FIG. 86A in alternative embodiments wherein a mesh 22 includes first securing mechanism in FIG. 86B (see FIG. 2 and accompanying description) and second securing mechanism in FIG. 86C (see FIG. 3 and accompanying description).

FIG. 87 shows an alternative embodiment of a droplet delivery service including an ultrasonic transducer 26 with rod-shaped vibrating member tip portion 170 offset from a central axis 220 of the droplet delivery device passing through the ejection channel 23, a slanted/sloped membrane 25 and mesh 22 and wherein the central axis of the vibrating member 230 is not aligned with central axis 220 of the device 10.

In another embodiment, FIGS. 88A and 88B illustrate an ultrasonic transducer 26 with a non-beveled ring-shaped vibrating member tip portion 170 coupled to a tilted mesh 22 in contact with a membrane 25 having a generally flat exterior top surface (nearest the mesh 22) in a droplet delivery device 10.

In further embodiment shown in FIG. 89A an ultrasonic transducer 26 with a beveled ring-shaped vibrating member tip portion 170 may be coupled to a slanted/sloped membrane 25 in contact with a membrane 25 in a droplet delivery device 10. FIG. 89B illustrates the slanted membrane 25 of FIG. 89A and FIG. 89E illustrate an ultrasonic transducer with a beveled ring-shaped vibrating member tip portion 170 also shown in FIG. 89A. FIGS. 89C and 89D show the ultrasonic transducer 26 and membrane 25 of FIG. 89A in droplet delivery devices in accordance with alternative embodiments of the disclosure wherein a mesh 22 includes first securing mechanism in FIG. 89C (see FIG. 2 and accompanying description) and second securing mechanism in FIG. 89D (see FIG. 3 and accompanying description).

FIGS. 90A and 90B show an ultrasound transducer 26 with a non-beveled ring-shaped vibrating member tip portion 170 coupled to a membrane with a generally flat exterior surface in contact and in a parallel plane to the plane of the fluid-entry underlying surface of mesh 22.

FIGS. 91A and 91B show ultrasonic transducer 26 with a beveled ring-shaped vibrating member tip portion 170 coupled to a slanted/sloped membrane 25 with a space between the membrane 25 and the mesh 22.

FIGS. 90A and 92B illustrate an ultrasonic transducer 26 with a non-beveled ring-shaped vibrating member tip portion 170 coupled to a membrane 25 having a generally flat

and parallel exterior surface relative to and not in contact with the underlying fluid-facing flat surface of the mesh **22** in a further embodiment.

FIGS. **93A-93D** show an alternative embodiment of a droplet delivery device **10** with an ultrasonic transducer **26** having a wide and flat vibrating member tip portion **170** together with membrane **25** having a generally flat surface and mesh **22** being generally flat. A preferable suspension system for mesh **22** is further illustrated by FIGS. **30C** and **30D**.

FIGS. **94A-94D** shown another embodiment with an ultrasonic transducer **26** having a wide and ring-shaped tip portion **170** together with membrane **25** having a generally flat surface and mesh **22** being generally flat. A preferable suspension system for mesh **22** is further illustrated by FIGS. **94C** and **94D**.

Membranes

The membranes **25** of the embodiments are made of materials featuring robust and proper acoustical and mechanical properties such as polyethylene naphthalate, polyethylenimine, poly ether ketone, polyamide, polymethyl methacrylate, polyetherimide, polyvinylidene fluoride, ultra-high molecular weight polyethylene, and the like.

The membranes of the embodiments may have a hydrophobic coating, hydrophobic etching, hydrophilic etching, hydrophilic coating, roughening etch, etc.

In some embodiments, such as shown in FIGS. **96A-96D**, membranes may include various shapes and surface textures, including "bumps" in one embodiment.

Meshes

Meshes **22** of the embodiments are to be made of materials featuring robust and proper acoustical and mechanical properties such as poly-methyl methacrylate, poly ether ketone, polyetherimide, polyvinylidene fluoride, ultra-high molecular weight polyethylene, polytetrafluoroethylene (PTFE), Ni, NiCo, Pd, Pt, NiPd, and metal alloys.

In one embodiment, the mesh is made from single crystalline or poly crystalline materials such as silicon, silicon carbide, aluminum nitride, boron nitride, silicon nitride, or aluminum oxide. Different hole shapes can be formed in a single crystalline wafer via high precision photolithography with and without using greyscale masks, and isotropic and/or anisotropic etches. Sputtered films can be deposited on the mesh to modify the wettability of the surface. Thin layers formed or deposited on the surface will have, in certain embodiments, much better adherence than films deposited on metal mesh formed by galvanic deposition or polymer mesh formed by laser ablation. The surfaces on the single crystalline wafers "slices" are atomically smooth and can be etched to produce exact surface roughnesses. Exact surface roughnesses can be used for better adherence of mechanical bonding with glue or other materials. Silicon carbide would be a preferable material because of its high strength and toughness. An important advantage of using semiconductor processes to fabricate hole structures from a single crystalline wafer "slice" in a mesh of embodiment of the push mode invention is that the holes and surface contact angles will be exact without the variation seen in conventional ejector plates using mesh made from galvanic deposition or laser ablation. This mesh, as noted in Table 9 may be fixed as in II, or suspended as in I, and the membrane is coupled with an optimized vibrating member with a thin film sputtering of nonreactive metals such as palladium or gold member tip section to further enhance biocompatibility.

The hole structures of other embodiments are formed using semiconductor processes such as photo lithography and isotropic and anisotropic etching, laser ablation, femto-

second laser ablation, electron beam drilling, EDM (Electrical discharge machining) drilling, diamond slurry grinding, etc. See also FIGS. **109** and **110**.

TABLE 9

Mesh Design	Embodiment	Brief Description
Single Crystalline Wafer	II	Fixed mesh coupled to optimized vibrating member
Single Crystalline Wafer	I	Suspended mesh coupled to optimized vibrating member

The meshes of the embodiments may have a hydrophobic coating, hydrophobic etching, hydrophilic etching, hydrophilic coating, roughening etch, etc. or a combination thereof.

In other embodiments, FIGS. **97-108** illustrate various implementations of polymer meshes utilized in push mode I and II devices.

Laminar Flow Element

In embodiments of the push mode invention, a laminar flow element **1600**, such as shown in FIG. **1B**, is preferably secured in the ejection port before the mouthpiece port of a droplet delivery device. In preferable embodiments, laminar flow element includes a plurality of cellular apertures. In some embodiments a laminar flow element includes blade-shaped walls defining the plurality of cellular apertures. In further embodiments, one or more of the plurality of cellular apertures include a triangular prismatic shape, quadrangular prismatic shape, pentagonal prismatic shape, hexagonal prismatic shape, heptagonal prismatic shape or octagonal prismatic shape. FIGS. **84A-84Q** show various embodiments of a laminar flow element.

Preventing Oxygen Diffusion

Referring to FIG. **95**, a droplet delivery device in an embodiment where an ejector bracket and container assembly are integrated as a single assembly includes a membrane cooperating with a mesh further preferably includes at least one superhydrophobic vent in such single assembly in fluid communication with the reservoir and is covered in storage with a removable aluminized polymer tab **3300** to help prevent oxygen diffusion into the fluid in the reservoir during such storage. In another embodiment of the push mode invention, a droplet delivery device in an embodiment where an ejector bracket and container assembly are integrated as single assembly that includes a membrane cooperating with a mesh further preferably further includes a removable aluminized polymer tab **3300** coupled to an exterior surface of the membrane adjacent the mesh during storage to help prevent oxygen diffusion into the fluid in the reservoir during such storage.

In another embodiment of the push mode invention, a droplet delivery device **10** having a membrane **25** that cooperates with a mesh **22** includes a pre-assembly step of removing a sealed packaging including aluminum and/or aluminum coating that contains the reservoir with a fluid, preferably wherein the reservoir is included in the container assembly that is also packaged for storage in the sealed packaging.

Decreasing Large Droplets in Aerosol

In embodiments of the push mode invention, it is desirable to decrease large droplet formation and encourage smaller droplet sizes to be delivered out of the droplet delivery device and in the aerosol stream.

In one embodiment, a hydrophilic wicking material may be provided to line the mouthpiece of the droplet delivery device. Droplets formed on the outer perimeter of a mesh

exit are absorbed by the hydrophilic wicking material and decrease the likelihood of large droplets propelling off the surface of the mesh exit. This wicking material absorption of large droplets increases MMAD repeatability and prevents pooling.

In another embodiment, a one-dimensional hydrophilic lattice (see laminar flow element **1600** but taking such as a cross section), or a series of one dimensional hydrophilic lattices, may be used to absorb large droplets that might “pop” off the mesh due to pooling.

It has been noticed in tests of push mode droplet production that a fog of aerosol may remain within the mouthpiece tube after inhalation. This fog could lead to pulling on the mesh and along the outer perimeter. This pulling happens due to no entrained air pulling the tail end of the aerosol ejection out. Via electronic programming and monitoring through a microcontroller or microchip integrated or coupled in the droplet delivery device, the droplet device can be programmably controlled to start spraying when the air flow rate reaches a threshold and then the droplet delivery device detection controller records your maximum air intake every 2 ms. The droplet delivery device is programmed to stop spraying when the flow rate recedes to a percentage of the maximum flow rate achieved during inhalation. In embodiments, a parameter labeled “pressure cutoff” can be added to a graphical user interface (GUI) for control/programming of the droplet delivery device so that a manufacturer or other device operator and alter the stop condition parameter for the spray.

Referring to FIGS. **111A-111C**, in another embodiment a baffle **4000** is inserted into the aerosol path. The baffle **4000** may comprise a plastic piece with fins **4050** to hold it in place in the aerosol tube of the droplet delivery device. The plastic piece has a cylindrical cavity which holds an absorbent plug **4100** (e.g., porous polyester or other wicking materials). The plug **4100** is inserted into the baffle cavity and is long enough to extend beyond the opening of the cavity. The absorbent plug faces the ejector mesh **22**. On the side of the baffle opposite the mesh **22**, the plastic baffle **4000** has a teardrop shape to direct airflow and prevent eddies from forming. The baffle **4000** is designed to inertially filter the aerosol by capturing large droplets in the absorbent plug **4100** upon ejection. Initial data using 3 ejectors is shown in the table below. As seen in Table 10, the baffle **4000** decreased the MMAD by approximately 0.1-0.2 μm for each ejector. This inertial filtering creates a smoother inhalation experience with less irritation. The plastic piece of the baffle **4000** and the absorbent plug **4100** may be various lengths and/or diameters.

TABLE 10

Baffle Inertial Filtering		
Sample	MMAD (μm) without baffle	MMAD (μm) with baffle
1	0.83	0.69
2	0.86	0.67
3	0.82	0.75

As described, it is important to get all the small droplets out of the mouthpiece. The small droplets have a very small stopping distance; therefore, the airflow must be close enough to the ejector plate to carry the small droplets. One design was tested wherein airflow directors were used to point the airflow towards the end of the mouthpiece and away from the mesh. As shown in FIG. **112**, the airflow path with the airflow directors caused backwards eddies causing

the small droplets to stay down by the ejector plate. Taking the airflow directors out helped the airflow catch some of the small droplets; however, the airflow was still leaving behind some of the small droplets. The holder for the ejector plate was sloped to help guide the airflow to the ejector plate. This encourages the air to catch most of the small droplets and send the droplets down the middle of the mouthpiece tube, but the ejector still produces larger unwanted droplets.

FIG. **113** illustrates the results when an insertable baffle **4000** was placed in the middle of the mouthpiece tube. This baffle holds a wicking material. As the airflow is pulled down the middle of the mouthpiece tube, the air flows around the baffle. The droplets follow the airflow; however, the larger droplets carry too much momentum and cannot make the turn to flow around the baffle. The larger droplets smash into the wicking material. The wicking material holds the liquid to keep the liquid from falling back onto the ejector plate. The liquid can then evaporate from the wicking material.

FIG. **114** illustrates additional results when an insertable baffle **4000** was also used with airflow directors. This test resulted in airflow coming from the airflow directors and shooting down the sides of the baffle. Eddies were still formed in the middle of the mouthpiece tube and pushed small droplets back onto the ejector plate. These eddies also caused the large droplets to flow around the baffle and resulted in no inertial filtering.

While the push mode invention has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the push mode invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings without departing from the essential scope thereof. Therefore, it is intended that the push mode invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the push mode invention will include all embodiments falling within the scope of the appended claims.

What is claimed:

1. A droplet delivery device comprising:

an inhaler outlet assembly including an airflow outlet aperture of a mouthpiece or nostril insertion element; an ejector bracket including an aperture plate with a plurality of openings formed in the aperture plate, wherein the aperture plate is operably coupled to a vibrating member coupled to an electronic transducer and an exit side of the aperture plate is positioned in or adjacent an airflow channel that exits at the airflow outlet aperture;

a membrane operably coupled to the vibrating member between the vibrating member and the aperture plate, wherein the membrane is a different material from the vibrating member, and wherein the transducer is coupled to a power source and is operable to oscillate the vibrating member and the membrane and generate an ejected stream of droplets through the aperture plate; and

a fluid flow path external to a fluid supply reservoir in fluid communication between a seal of the fluid supply reservoir and the membrane.

2. The droplet delivery device of claim 1, wherein the inhaler outlet assembly is releasably detachable from the ejector bracket or releasably detachable together with the ejector bracket relative to one or more other detachable parts of the delivery device.

37

3. The droplet delivery device of claim 1, wherein the reservoir includes a seal that opens upon connection of the ejector bracket.

4. The droplet delivery device of claim 1, wherein the membrane is configured not to contact the aperture plate.

5. The droplet delivery device of claim 4, wherein the membrane includes a slanted upper surface configured to contact fluid supplied from the reservoir.

6. The droplet delivery device of claim 1, wherein the aperture plate has a top surface in a parallel configuration with a flat surface of a tip of the vibrating member.

7. The droplet delivery device of claim 1, wherein the vibrating member includes a rod-shaped tip.

8. The droplet delivery device of claim 1, wherein the aperture plate has a bottom surface in a non-parallel configuration with an upper surface of the membrane.

9. The droplet delivery device of claim 1, further comprising a central axis of the droplet delivery device passing through the ejection channel and the membrane, and wherein the vibrating member includes a tip coupling to the membrane at a position offset from the central axis.

10. The droplet delivery device of claim 1, wherein the aperture plate comprises a material of at least one of palladium nickel, polytetrafluoroethylene, polyimide, poly ether ketone, polyetherimide, polyvinylidene fluoride, ultra-high molecular weight polyethylene, Ni, NiCo, Pd, Pt and metal alloy.

11. The droplet delivery device of claim 1, wherein the membrane comprises a material of at least one of polyethylene naphthalate, polyethylenimine and poly ether ketone.

12. The droplet delivery device of claim 1, wherein the membrane comprises a material of at least one of metal membranes, metalized polymers, threaded polymers, threaded nylon, threaded polymers that are coated with

38

polymers or metal, threaded nylon coated with polymers or metal, threaded metals, threaded SiC, threaded graphite composites, metalized graphite composites, graphite composites coated with polymers, polymer sheets filled with carbon fibers, poly ether ketone filled with carbon fibers, polymer sheets filled with SiC fibers, polymer sheets filled with ceramic or metal fibers, ULPA filter media, PTFE filter media, PTFE polymer sheets, threaded polymers bonded to a polymer sheet, nylon weave bonded to poly ether ketone or polyimide, graphite composites bonded to polymer sheets, polymer fiber weave with metalized coating, and nylon with sputtered on Al or vapor deposited Al.

13. The droplet delivery device of claim 1, wherein the electronic transducer is coupled to a vibrating member including a tip portion comprised of at least one of Grade 5 titanium alloy, Grade 23 titanium alloy, and about 99% or higher purity titanium.

14. The droplet delivery device of claim 1, wherein an exterior surface of the membrane in fluid communication with the reservoir includes a hydrophobic coating.

15. The droplet delivery device of claim 1, wherein an exterior surface of the membrane in fluid communication with the reservoir includes a hydrophilic coating.

16. The droplet delivery device of claim 1, wherein the aperture plate includes a hydrophobic coating on one or more surfaces of the aperture plate.

17. The droplet delivery device of claim 1, wherein the aperture plate includes a hydrophilic coating on one or more surfaces of the aperture plate.

18. The droplet delivery device of claim 1, wherein the aperture plate includes a hydrophobic coating on a first surface of the aperture plate and a hydrophilic coating on a second surface of the aperture plate.

* * * * *